Slot-Chain: A Preconfirmation Protocol

Implementation-Freeze Candidate — v2.27

September 2026

Abstract

Slot-Chain assigns one-second L2 slots to bonded builders. Builders sign parent-linked blocks off chain; any party can prove and submit a batch to L1. Normal candidates compete for a fixed settlement interval. After an objective finality breach, or when a forced L1 message becomes due, anyone can prove an unsigned deterministic escape block. Recovery is an episode of renewable, objectively expiring rounds, so a prover outage does not make its target stale; as with every validity rollup, producing the next proof still requires a reconstructible canonical prestate. This revision separates proved outputs from L1 close metadata, replaces the unauthenticatable queue hash chain with a range-provable Merkle vector, makes forced-message execution Ethereum-valid, and specifies bounded admission, data, reward and migration state. It is a multi-pass-reviewed implementation-freeze candidate, but not production authorization: the missing initial executable profile/build bundle and the proving, gas, economics and adversarial gates in §14 remain mandatory before production release. The control- plane, migration journals, canonical execution-profile codec and exact Settlement–Market mutation wire in this revision are normative implementation inputs. The bridge-route preparation path now uses only fixed Router/PVM primitive records and exact raw ABI reads; it no longer treats an in-process `SettlementRegistration` object as production authority. The executable address-indexed component world, root Source-factory/terminal deployment certificate, storage-derived restart path and bounded historical indexes close the design-level blockers found in the final freeze audit. No known Critical or High document-level defect remains. The exact ICV2 Inbox-credit verification boundary is also closed in this specification, but every path still requires compiled-contract, circuit and gas conformance evidence.

Contents

1 Status, claims and exclusions

Normative words MUST, MUST NOTand SHOULDhave their usual RFC meaning. Where prose and either executable model disagree, deployment is blocked until the disagreement is resolved; neither source silently overrides the other.

1.1 Claims

Subject to L1 safety/liveness, a sound validity-proof system, availability of a canonical prestate reconstruction package as defined below, and—for kind-1 credits only—safety/finality of the supported source domain and its profile-authenticated, append-only Bridge registry, the protocol claims:

1. **Finalized-state safety.** L1 accepts only a proved state transition extending the stored canonical tuple. Builder signatures never substitute for execution validity.
2. **Bounded protocol recovery.** A builder cartel, an absent aggregator, an empty builder registry, and missing standbys cannot prevent an unsigned escape candidate from advancing the canonical state while the canonical prestate and any unexpired forced bytes remain reconstructible.
3. **Forced transaction inclusion.** A correctly prepaid L1 envelope at the queue head is consumed by the next canonical block after it becomes due; kind 0 is executed if valid bytes are supplied before its bounded validity or consumed-and-discarded otherwise, while durable kind 1 is always applied idempotently.
4. **Permissionless landing and recovery.** No allowlist, builder key, aggregator seat or payment recipient is required to submit a proof or an escape candidate.
5. **Deterministic lookahead.** Every implementation derives the same schedule from an authenticated finalized snapshot and the byte encoding in §4.

1.2 Non-claims

- A preconfirmation is not finality or insurance. Before normal close it may be revoked by builder equivocation, an unprovable skip, a withheld body, or an escape commit. Applications must price this as probabilistic/economic assurance and publish their own exposure limit.
- A bond cannot cap aggregate off-chain promises: the protocol has no reservation ledger and a builder can sign mutually exclusive promises. `L_LEASE` is deterrence and a recovery source, not guaranteed coverage.
- Protocol liveness is a capability claim. Economic liveness additionally assumes at least one actor can fund proof generation and an L1 transaction. A payment failure never invalidates an otherwise valid canonical transition.
- Censorship by L1 for longer than `T_INCLUDE_MAX_SECONDS`, failure of L1 finality, or a broken proof system is outside the model.
- A genesis campaign creates only a finite admission fence, not a persistent checkpoint. During its inclusive block window the legacy proxy rejects new proposals and forced inputs before retaining value, but existing proof-finalization and withdrawal paths remain live. Capped exact scans bind any suffix and forced range that successful migration will abandon, plus all data and time horizons needed for safe resume. Proof verification,

legacy checkpoint/freeze and target adoption then occur in one reverting transaction. If no proof lands by the campaign deadline, admissions reopen automatically and the exact legacy state resumes. A total proving-system outage can delay migration but cannot newly halt a legacy state that was processable before the fence. Resume does not cure a pre-existing invalid proposal or missing archive; such a deployment may need the separate state-migration path.

- A supported source domain finality failure or unauthorized change to its profile-pinned Bridge registry is outside the kind-1 safety claim. It cannot authorize a kind-0 user envelope.
- A source message is not yet an admitted forced envelope. Any preparer must supply its full Message preimage and queue deposit to the same-L1 adapter; source emission alone is not reported as forced-queue acceptance. If nobody enqueues before the record's bounded `enqueueBy`, permissionless source cancellation returns its ETH escrow and permanently forbids later enqueue. Launch V2 is DIRECT-only. ERC20, ERC721, ERC1155 and every official Vault flow remain byte-for-byte V1 and cannot originate a V2 credit. A future asset protocol requires a new kind, domain, release, codec and audit.
- The immutable V2 custody kernel is not self-healing. A defect in its compiled SourceBridge, DestinationBridge, inbox store, accumulator, history router, quota manager, liquidity pool or fixed Merkle verifier cannot be patched for an already-live credit without introducing an authority able to forge delivery/failure and take reserved value. This design chooses fail-closed immutability; production status therefore requires the compiled-code, formal-invariant and differential-test gates in §14, not merely this document.
- A state root is a binding commitment, not an erasure code. If every copy of the canonical state trie, executed bodies, runtime-bytecode preimages and required witness data is destroyed after Ethereum's applicable data-retention horizon, a new prover cannot reconstruct the preimage from L1 roots. Arbitrary-duration recovery after total canonical-data loss is therefore not claimed.

2 Actors and trust model

Builder.
A registered address eligible for scheduled slots. Builders may equivocate, skip parents, withhold bodies, collude, or disappear.

Aggregator.
The current auction seat, expected to collect data, prove and submit normal candidates. It has no exclusive consensus authority.

Prover/lander.
Any address. It supplies a validity proof and names an optional reward beneficiary in the proof statement. Proof validity, not identity, authorizes transition.

Canonical archive.
Any untrusted, replaceable source of canonical L2 bodies, state-trie nodes and runtime-bytecode preimages sufficient to reconstruct the current prestate. Each bytecode object is addressed by its legacy-Keccak `codeHash`; this includes code reached through proxy, beacon or delegation indirection. Content is verified against the L1-canonical block/state roots, so correctness does not require trusting the source; liveness requires at least one complete copy to remain reachable.

Forced-message sender.
Any L1 account that submits a gas-bounded, prepaid L2 transaction envelope.

Bridge source domain.
For kind 1 only at launch, settlement Ethereum, identified by chain ID, genesis hash, fresh immutable non-proxy SourceBridge, non-proxy credit registry, immutable terminal verifier and namespace. The adapter direct-reads its append-only credit record on the same L1. Registry governance is an external safety dependency; a current resolver lookup never substitutes for the emission record.

Protocol governance.
Delayed governance selects each validity verifier, execution profile and Settlement implementation. It is already a safety root: an authorized malicious verifier can finalize an arbitrary state transition. V2 recovery adds no stronger writer. Each immutable Settlement records its own proof-bound canonical history internally; the protocol-lifetime router only registers exact version/code identities and reads those histories. It has no governor, canonical-write callback or caller-supplied-root setter. This root includes the exact authorized creation and runtime bytecode. A malicious or compromised governance that deliberately approves code whose DSV1/TCS2 getters lie, or a validity verifier that accepts false transitions, is outside the protocol safety claim; either capability can already choose an arbitrary state machine. EXTCODEHASH and fixed getters prevent substitution and detect unexpected live state, but are not claimed to prove semantic honesty of root-authorized bytecode. The pinned reproducible-build and constructor- trace verifier below is a mandatory, independently reproducible release gate for honest governance, not a self-attesting on-chain capability. Its complete input bundle, executable hash, report and output hashes are public before the delay starts. Permissionlessness applies to auction participation, proving, landing, recovery and execution of a mature typed operation; it does not mean that arbitrary accounts may select a new verifier or implementation.

Bridge message archive.
Any untrusted, replaceable source of the full Message bytes emitted on L1. Correctness follows from `msgHash`; successful destination execution needs one copy until processing. Loss before `enqueueBy` leads to source cancellation; loss after the L2 pin leads to permissionless destination expiry and failure. Source DIRECT ETH remains recoverable by credit ID from the SourceBridge's pull-refund record, without reconstructing Message bytes.

L1.
Ethereum provides canonical ordering, timestamps, EIP-4788 beacon roots, EIP-2935 historical block hashes, blob commitments and the KZG point-evaluation precompile.

The safety threshold contains no honest-builder assumption. Honest builders improve normal throughput and preconfirmation reliability; the unsigned escape lane provides state and forced-queue progress when all of them collude. The L1-liveness assumption includes transaction inclusion and at least 64 canonical L1 blocks becoming available within `T_DEPTH_MAX=900s`; a longer L1 stall suspends, rather than violates, the L2 recovery clock. Archive operators have no protocol key, seat or exclusive API. Anyone may mirror or serve the same content-addressed data. This is an availability assumption, not an authority assumption; it limits long-horizon cold-start recovery but does not weaken proof soundness or permissionless submission.

3 Clock domains and canonical state

Before activation, Settlement is `PREACTIVE`. Candidate submission, session mutation, reward funding/claim and every other ordinary local mutation are disabled; slot arithmetic saturates. The sole exception is the immutable Router-only `adoptMigrationCanonicalV1` callback during lifecycle `ACTIVATING` under the exact nonzero activation context in §12. Off-chain data preparation may proceed but cannot create a target session. L2 slot zero is fixed by `GENESIS_TIMESTAMP`, with

`currentL2Slot=max(0,block.timestamp-GENESIS_TIMESTAMP).`

L2 slot and every deadline are timestamp seconds. L1 confirmation depth is counted in L1 block numbers. Beacon slot is used only in schedule authentication. Implementations MUST NOTsubstitute one clock for another. Every strict lag predicate uses the ordered form

```
lagExceeded(current, reference, limit) =
  current > reference && current - reference > limit.
```

The order check precedes subtraction. This is consensus-relevant when a proved tip is ahead of the current wall slot by up to `CLOCK_SKEW`; a Solidity underflow/revert is not an allowed interpretation of a zero lag. `settlementChainId` is the L1 chain executing Settlement and domains every L1 contract commitment; `l2ChainId` is the EVM chain ID enforced for L2 transactions and execution. They are independent immutable launch constants and are both explicit in every cross-domain proof statement and builder header.

The L1 contract stores a proved core and local L1 metadata:

```
CanonicalCore = (l2BlockNumber, tipHash, tipSlot, stateRoot, messageCursor,
                 winningDataCommitment, nextBaseFee, nextExcessBlobGas,
                 terminalRoot, terminalCount)
Canonical = (CanonicalCore core, uint64 canonicalizedAtBlock)
```

The proof binds `baseCanonicalHash`, the hash of both stored objects, but outputs only a new `CanonicalCore`. Settlement compares every base field with storage, verifies the proof, writes the explicit proved core, and only then sets `canonicalizedAtBlock=block.number`. A prover is therefore never asked to predict the L1 block in which its transaction will land. No code attempts to read or store `blockhash(block.number)`, which does not exist during that block.

`l2BlockNumber` is the canonical EVM height, distinct from the one-second slot because slot gaps are legal. A candidate with `blockCount=n` proves `endL2BlockNumber=base.l2BlockNumber+n`; every executed header uses the next consecutive EVM block number. While the deployed `ICheckpointStore` uses `uint48`, launch and every transition additionally require `endL2BlockNumber<=UINT48_MAX`; changing that interface and bound is a protocol-version upgrade. The canonical event always emits `(l2BlockNumber,tipHash,stateRoot,terminalRoot,terminalCount)`. The proof constrains the latter pair to the exact post-state slots of the protocol-lifetime `TerminalAccumulatorV2`; Settlement never accepts them as unverified auxiliary calldata. In the same internal storage transition that writes `Canonical`, Settlement increments a checked `uint64canonicalSequence` and writes the complete sequence-tagged canonical read tuple to its 256-cell history ring. A second ordinary canonical transition requires `block.number>lastCanonicalCommitBlock`; it cannot overwrite two cells in one L1

block. FROZEN instances cannot execute any canonical path. PREACTIVE instances reject every ordinary canonical path; their one activation callback writes sequence zero for GENESIS_IMPORT or checked `sourceCanonicalSequence+1` for VERSION_MIGRATION, the complete history cell, `canonicalizedAtBlock=lastCanonicalCommitBlock=block.number`, and the ordinary canonical event in one transition. Recording this ring is an internal Settlement write, not a router storage write, reward dependency or best-effort publication. `nextBaseFee` and `nextExcessBlobGas` are the exact fork-rule outputs needed to construct the next header without retrieving a prunable historical parent body/header. The proof derives them from the last executed header; Settlement stores them but does not calculate fork arithmetic.

A candidate has one prior-block anchor. Normal activation stores the hash of its earlier arm block using native `blockhash`; candidate submissions hash the supplied execution header to that stored value and use MPT proofs, including the historical forced-queue root/count. EIP-2935 is used later for schedule-carrier authentication. Equivocation evidence instead treats the signed context ID as an opaque, release-domain-scoped builder promise and does not attempt to reconstruct overwritten Settlement context storage. A recovery round instead stores `(block.number-1,blockhash(block.number-1))`, directly stores the current queue root/count, and requires that anchor to reach depth `F_L1`. The EVM does not expose the parent timestamp. At candidate submission Settlement hashes and parses the supplied RLP parent header, requiring its number/hash to equal the frozen pair; its state root and timestamp become verifier-statement fields but are not separately frozen scalars. Current queue state is never claimed to be part of the parent's historical state root. An L1 reorg rolls the stored canonical state and anchor back together. Initial values and bridge cursors are migration inputs in §12.

Every canonical or ordinary new-work state-mutating entry point begins with `sync()`, except the exact DataSession open/post/seal sidecars, claims, surplus sweep and mode-specific maintenance frozen in §6. If `sync()` changes mode or commits a mature normal candidate, a wrapper that uses this rule MUSTreturn `SYNCED` successfully before parsing or validating the caller's candidate. The caller resubmits against the new version. This rule is necessary because a later Solidity revert would otherwise roll back the sync transition in the same transaction.

4 Builder registry and exact lookahead

4.1 Registry lifecycle

A registration contains a nonzero address, a `uint192` base bond in the closed interval `[L_LEASE,BOND_MAX]`, a monotonic `uint64registrationIndex`, an `effectiveL2Slot`, and separately reserved `L_LEASE[window]` tranches. Address zero is permanently `VACANT`. Duplicate live addresses and reuse while any prior generation remains active or liable are forbidden. Once the old generation's last evidence/reorg deadline has passed, all tranches are terminal, its liability leaf is safely cleared and no evidence can still land, the address may register again with a fresh `registrationIndex`. No permanent address tombstone mapping exists. With `currentWindow=floor(currentL2Slot/384)`, a reservation is accepted only for `currentWindow<=W<=currentWindow+MAX_TRANCHE_AHEAD_WINDOWS` and `firstManagedWindow<=W<=lastManagedWindow`, where the initial ahead bound is 16; an exit request permanently disables new reservations for that generation. Only a present *active* generation with no exit request and no tombstone may execute `FREE->RESERVED`; a replacement or exit atomically closes reservations before moving the old generation into liability. Liability records may only be slashed or released, never extended with

a new reservation. Two authoritative reverse indexes make these lifecycle checks independent of history. `generationLocation[registrationIndex]` stores the exact ACTIVE or LIABILITY position, and `liveRegistrationIndexPlusOne[builder]` is a `uint256`. The latter is nonzero for either location and equals `uint256(registrationIndex)+1`, so even `registrationIndex=UINT64_MAX` is represented without wrapping. An ACTIVE-to-LIABILITY move changes only the first locator; the address row is cleared only by final generation release. Every lookup exact-checks the indexed cell's builder and registration index and fails closed on disagreement; no fallback scan of active or liability history is permitted.

The active table has exactly 64 indexed cells. On accepted registration, `effectiveL2Slot= currentL2Slot+384*ENTRY_DELAY_WINDOWS`; the stored value, not a later recomputation from a window boundary, is authoritative. Thus a registration becomes effective only after `ENTRY_DELAY_WINDOWS=8`, and registration rejects unless `floor(effectiveL2Slot/384) <=lastManagedWindow`. This exact-slot check admits the final builder only if it can become effective in a real managed window. A non-full table always writes the lowest vacant active-cell index: the first registration therefore uses cell 0, the 64th uses cell 63, and a later hole is filled before a greater index. If the table is full it may replace only the smallest-bond live, non-tombstoned entry whether or not that entry is effective, breaking ties by greatest registration index, and only with a strictly greater bond. Full-table registration first rejects with `MaintenanceRequired` if any active tombstone or oldest live matured exit exists; it never evicts a healthy entry while a tombstone occupies a cell. Including ineffective entries prevents 64 fresh minimum-bond registrations from creating an eight-window replacement fence. At most `MAX_GENERATION_MOVES_PER_WINDOW=4` replacements or effective exits are processed. Exit requests receive a monotonic sequence and mature at `requestWindow+EXIT_DELAY_WINDOWS`. Every replacement path first processes the oldest matured exits by `(matureWindow,exitSequence)` within the window's remaining movement budget; if any matured exit remains, replacement rejects. Anyone, including the exiting builder, may run this bounded maintenance. Thus transaction ordering cannot starve an eligible exit. A displaced generation is moved, with its tranche root and bond, into the fixed `MAX_LIABILITY_GENERATIONS=1,072` liability ring; an occupied ring cell is never overwritten before its release predicate holds. Registration therefore rejects, rather than losing slashable history, when the replacement budget or ring is full.

The six-level `registryRoot` over the 64 active cells commits each cell's mutable tranche root and therefore changes on reservation transitions. It is used only by schedule sealing. A separate eleven-level `admissionRoot` over 2,048 fixed positions contains stable generation identity for the 64 active and 1,072 liability records plus canonical empty padding; its leaves do *not* contain tranche roots. Only generation admission, movement, first tombstoning or generation release increments `admissionVersion` and updates `admissionRoot`. A `FREE/RESERVED/LIABLE` tranche transition updates the tranche and registry roots but cannot invalidate signed blocks or proofs by changing the admission pair. The admission root retains displaced scheduled keys. A healthy replacement changes such a key from active to liability but does not tombstone it: already-sealed tickets can still prove the same generation, while removal from the active registry prevents every later snapshot or reservation. Normal activation and every recovery round freeze the pair `(admissionVersion,admissionRoot)`. Each scheduled signer carries an inclusion proof for a non-tombstoned record effective at that L2 slot; a slash cannot retroactively invalidate an already frozen window. Outside an already-open normal window or live recovery round, a slash tombstone makes sealed future tickets at or after its effective slot return

VACANT; inside that frozen candidate context the stored admission pair remains authoritative until close/commit/expiry.

Admission positions are canonical: active cell i is always position i for $0 \leq i < 64$; liability ring slot j is always position $64 + j$ for $0 \leq j < 1,072$. A generation move uses j= movementSequencemod1072. The transaction first proves/releases the old occupant, writes the moved generation at that same slot, clears its active cell or writes the replacement, increments movementSequence, then updates both affected roots and increments admissionVersion; the new admission pair is published atomically. Reservation-only mutations publish only the registry root and leave that pair byte-for-byte unchanged. There is no caller-chosen active or liability position and no compaction. The bounded 64-cell lowest-vacancy scan above is the sole active placement rule.

Eligibility is evaluated before ranking. A cell's effective L2 slot must be no later than the source-derived L2 slot, its fully reserved tranche for W must be proven, and its tombstone-effective L2 slot must be later than the source. All 64 registry cells are supplied; only present cells supply tranche-ring proofs at Wmod512. A canonical EMPTY tranche leaf is valid input and makes the cell ineligible; it need not claim window W. A nonempty leaf with a different stored window is likewise ineligible, not a seal revert. Only an exact RESERVED leaf with stored window W and the full L_LEASE amount is eligible. The 64 eligible entries with greatest bond, then smallest registration index, form the snapshot set. Supplying proofs only for purported winners is invalid because it cannot prove omission-free ranking.

An exit request becomes eligible for processing only after EXIT_DELAY_WINDOWS=MAX_LIVE_WINDOWS; it may wait behind the same bounded FIFO of older matured exits, and the bond remains escrowed until every generation and tranche is releasable. A tranche follows FREE->RESERVED(W)->LIABLE(W)->RELEASED or RESERVED/LIABLE->SLASHED. Every candidate block slot must be at least the normal window's frozen minimum (or the recovery round start), not merely its tip. Let windowEndSlot(W)=384*(W+1)-1. Release requires windowEndSlot(W)<globalMinReferencedSlot, no stored best or active round references it, and the absolute evidence and L1-reorg deadlines have expired. At that point ScheduleOracle maintenance atomically marks the schedule cell EXPIRED; BuilderRegistry may release a tranche only after an exact authenticated read of that state. An unexpired sealed record is never overwritten. evidenceDeadline(W)=GENESIS_TIMESTAMP+384*(W+1)+EVIDENCE_DELAY_SECONDS; evidenceReplayDeadline(W)=evidenceDeadline(W)+REORG_MARGIN_SECONDS is the contract's inclusive last admissible evidence time. Release is legal only at a strictly later timestamp. Evidence intended to survive a reorg must first be submitted by evidenceDeadline(W). globalMinReferencedSlot is the minimum of the current dynamic floor, an open normal window's frozen floor, an active round's start and the first slot of any stored best (absent values are ignored). Slot indices are never compared with timestamps or window numbers.

The schedule domain contains only complete windows whose slot end and absolute evidence/reorg deadline leave a representable strictly later release time in the uint64 ABI. Define, with checked unsigned arithmetic,

```
LAST_FULL_SLOT_WINDOW = floor((UINT64_MAX - 383) / 384)
                      = 48038396025285289
lastManagedWindow = min(
  LAST_FULL_SLOT_WINDOW,
  floor((UINT64_MAX - 1 - genesisTimestamp
         - evidenceDelaySeconds - reorgMarginSeconds) / 384) - 1)
```

The fixed bound's end slot is 18446744073709551359, strictly below UINT64_MAX; the following 256-slot suffix of the numerical uint64 domain is deliberately outside the protocol schedule. This first bound prevents the release cursor from entering a window whose end cannot be strictly below the 64-bit globalMinReferencedSlot. The deployment-derived second bound additionally guarantees evidenceReplayDeadline(lastManagedWindow)<=UINT64_MAX-1, so the strictly later release instant is representable. Constructor validation rejects underflow, a result below firstManagedWindow, or a supplied peer value unequal to this derivation. Neither bound is independently configurable. UINT64_MAX remains the distinct terminal cursor sentinel.

The liability-capacity proof is a launch invariant, not an assertion by inspection. A generation moved in window *C* can have no reservation beyond C+MAX_TRANCHE_AHEAD_WINDOWS. Therefore

```
MAX_LIABILITY_RESIDENCE_WINDOWS =
  MAX_TRANCHE_AHEAD_WINDOWS + 1 +
  ceil((EVIDENCE_DELAY_SECONDS + REORG_MARGIN_SECONDS) / 384) + 2
```

must be strictly less than MAX_LIVE_WINDOWS=268. The two extra windows cover boundary ordering and bounded maintenance. At four moves per window, the ring position reused 268 windows later is consequently releasable. Each generation maintains an exact unreleased-tranche counter, so release and reuse do not scan 512 leaves. unreleasedTrancheCount is incremented exactly once on first reservation and decremented exactly once on release or slash; zero is the sole tranche-terminality predicate for generation release. The contract never supplements it with a scan of retained tranche records.

Protocol-lifetime terminal drain. No generation may be admitted if its exact effectiveL2Slot falls after lastManagedWindow, and no reservation, schedule seal or fork boundary may name a greater window. Once ScheduleOracle has expired lastManagedWindow, its authenticated next-release cursor is the exact UINT64_MAX sentinel and every greater-window SWR1 read remains UNSEALED. At that point ordinary movement from active into liability is disabled; there is no future schedule for which such a move is useful. An exit request after lastManagedWindow likewise rejects.

Instead, existing permissionless normalization closes one active generation's at-most-17 reservation leaves in place. Its terminal branch first exact-reads SWR1(lastManagedWindow), requires EXPIRED and the exact UINT64_MAX cursor, and executes before narrowing or storing a derived current window. Since every managed window is then stale, it converts every represented RESERVED leaf to LIABILITY, clears the bitmap, and retains the already-representable bitmap base; even a first call after genesisTimestamp+UINT64_MAX therefore cannot overflow a stored clock. It preserves reservationsClosed: the terminal window bounds already make every new reservation impossible, and a zero-leaf normalization remains the canonical no-root-change 0x00 case. The generation remains in its canonical active admission position, so evidence is still admissible through each inclusive replay deadline. After its exact unreleased-tranche count is zero and block.timestamp>maximumLiableUntil, anyone may release that active generation directly. The call exact-reads SWR1(lastManagedWindow), requires EXPIRED and nextReleaseWindow=UINT64_MAX, clears the six-level registry leaf and eleven-level active admission leaf atomically, resolves any exit row, and creates the builder's pull credit. It consumes neither a liability-ring cell nor the four-move allowance. One call releases one derived generation, so an arbitrarily late caller can drain all 64 positions without an early-maintenance

assumption, and a blocked liability-ring occupant cannot strand the first or last active builder. This terminal rule does not accelerate evidence or tranche release: equality remains slashable and release remains strictly later. The same authenticated sentinel permits an already-LIABILITY generation to release after its tranche/deadline predicates even if its conservative `releaseWindow` lies beyond the last representable current-window value. This matters for valid zero-delay deployments where the fixed complete-slot bound, rather than the timestamp bound, selects `lastManagedWindow`; it cannot bypass `maximumLiableUntil` or clear a live tranche. Terminal finalization, tranche/generation release and pull-credit claims compare the `uint64` stored deadline against the native `uint256` `block.timestamp`; they do not narrow the current timestamp or require it to remain at most `UINT64_MAX`. Thus the derived one-second minimum strict-release point is not a finite processing window: all 64 generations and all existing liability cells remain permissionlessly releasable later.

4.2 BuilderRegistry implementation freeze

This subsection is the implementation authority for the registry lifecycle. Where earlier explanatory prose is less specific, the rules below control. `BuilderRegistry` is one protocol-lifetime, non-proxy, ownerless and unpausable contract. It does not inherit an administrative base contract and has no setter, redirect, arbitrary call or emergency withdrawal. It pins the builder-token and Router addresses/runtime/configuration, plus the ScheduleOracle address/runtime and bounded read gas in constructor-only writerless state. Scalar/local values may use Solidity immutables, but a peer runtime hash must not be bytecode-patched when that peer also binds the Registry: those cross-hashes remain ordinary constructor-written storage with no reachable writer, avoiding a cryptographic runtime-hash fixed point. The builder token must be an exact-balance ERC-20. It is a release-audited, immutable non-proxy artifact with no delegatecall, beacon, replaceable implementation, fee switch, rebase, blacklist/confiscation authority, privileged balance mutation, transfer callback or external hook. Its complete runtime and reachable code dependencies are manifest-pinned; pinning only a proxy shell's `EXTCODEHASH` is invalid. Every token-moving path is non-reentrant and writes pull credit before an external transfer.

The actual Solidity constructor argument list begins with the three root- activation fields and continues with every BuilderRegistry-local field below. Their canonical ABI encoding and init code are frozen in the protocol-root deployment paragraphs; the local field suffix is

```
BuilderRegistryConstructorV1(
 uint256 settlementChainId,
 address builderLeaseToken,bytes32 builderLeaseTokenRuntimeHash,
 uint8 builderLeaseTokenDecimals,uint192 leasePerWindowAtomic,
 uint192 maximumBondAtomic,uint192 reporterRewardCapAtomic,
 uint64 genesisTimestamp,uint64 evidenceDelaySeconds,
 uint64 reorgMarginSeconds,uint64 firstManagedWindow,
 address builderPenaltySink,uint64 rewardClaimWindowSeconds,
 address activeSettlementRouter,bytes32 routerRuntimeHash,
 bytes32 routerConfigurationHash,address scheduleOracle,
 bytes32 scheduleOracleRuntimeHash,
 BuilderRewardClassConfigV1[3] rewardClasses)

BuilderRewardClassConfigV1 =
```

```
 (uint8 classId,bytes32 nameHash,uint256 fixedWei,
  uint256 perExecutionGasWei,uint256 perPublishedByteWei,uint256 capWei)
```

It requires `settlementChainId=block.chainid`, every address/runtime/config word nonzero, and all of

```
0 < leasePerWindowAtomic <= maximumBondAtomic <= UINT192_MAX
reporterRewardCapAtomic <= UINT192_MAX
5 * reporterRewardCapAtomic <= leasePerWindowAtomic
```

with checked arithmetic, and class IDs exactly 1, 2 and 3 in that order with nonzero name hashes. It checks the token's runtime, exact 50,000-gas `decimals()` return and configured decimal value, and recomputes the economic `builderRegistryConfigurationHash` from these values, the fixed geometry `(64,76,8,268,16,1072,4)` and the three supplied name hashes/rate rows. The final hash is never a constructor input. `componentConfigHashV2()` returns exactly that derived 32-byte hash. The immutable `firstManagedWindow` and derived immutable `lastManagedWindow` are deployment topology, not part of the economic hash; release registration must derive and cross-check both together with the ScheduleOracle configuration below. The constructor derives `lastManagedWindow` by the formula above from its genesis/evidence/reorg values and rejects if it is below `firstManagedWindow`.

Because the profile's historical `builderRegistryConfigurationHash` is specifically the economic hash, topology has a separate noncircular identity. Let B be

```
u256(settlementChainId) || address20(builderLeaseToken) ||
builderLeaseTokenRuntimeHash || u8(builderLeaseTokenDecimals) ||
u64(genesisTimestamp) || u64(evidenceDelaySeconds) ||
u64(reorgMarginSeconds) || u64(firstManagedWindow) ||
u64(lastManagedWindow) ||
address20(builderPenaltySink) || u64(rewardClaimWindowSeconds) ||
address20(activeSettlementRouter) || routerRuntimeHash ||
routerConfigurationHash || address20(scheduleOracle) ||
scheduleOracleRuntimeHash || builderRegistryConfigurationHash
```

and

```
builderRegistryTopologyHash = H(
 "slot-chain-builder-registry-topology-v1" || u16(byteLength(B)) || B).
```

Here `byteLength(B)=321`; any other width or field order is a different topology. Both hashes are derived, stored writerlessly and exposed by the exact deployment read

```
builderRegistryConfigV1() returns
 (bytes4 magic,uint256 settlementChainId,address builderLeaseToken,
  bytes32 builderLeaseTokenRuntimeHash,uint8 builderLeaseTokenDecimals,
  uint192 leasePerWindowAtomic,uint192 maximumBondAtomic,
  uint192 reporterRewardCapAtomic,uint64 genesisTimestamp,
  uint64 evidenceDelaySeconds,uint64 reorgMarginSeconds,
  uint64 firstManagedWindow,uint64 lastManagedWindow,
  address builderPenaltySink,
```

```
  uint64 rewardClaimWindowSeconds,address activeSettlementRouter,
  bytes32 routerRuntimeHash,bytes32 routerConfigurationHash,
  address scheduleOracle,bytes32 scheduleOracleRuntimeHash,
  bytes32 economicConfigurationHash,bytes32 topologyHash)
// selector 0x66f0bd82; exact 704 bytes; magic BRC1=0x42524331

builderRegistryTopologyHashV1() returns (bytes32 topologyHash)
// selector 0xa571e34b; exact 32 bytes
```

Release registration exact-reads BRC1, recomputes the topology hash and compares the economic hash to the manifest-pinned calibrated profile and the exact `componentConfigHashV2()` result. BRC1 deliberately does not repeat the three reward-class rows and therefore is not an alternate economic-hash preimage. The one-word topology getter exists for bounded operational authentication; it is not a caller-supplied substitute for the full deployment check.

Before every Router functional read, Registry rechecks the pinned `EXTCODEHASH`, exact 50,000-gas `componentConfigHashV2()` result and only then performs ASR1 or RTR2. Before SWR1 it rechecks the pinned ScheduleOracle `EXTCODEHASH`; no Schedule configuration hash is a Registry constructor field. ProtocolRootFactory finalization has already exact-read and cross-validated that writerless Schedule instance before activating the root. Later REGISTER_RELEASE validation repeats the same join as a secondary release check; it is not root publication authority. Schedule's address and runtime cannot change, and its own configuration binds the Registry in the opposite direction. A code/config mismatch, call fault or noncanonical return reverts before token movement or a root write. Every token path similarly rechecks the token runtime. `balanceOf` uses selector-only canonical ABI, 50,000 gas and exactly one 32-byte return; `transfer`/`transferFrom` accept only empty return or one canonical true word. An incoming transfer must increase Registry balance by exactly its nominal amount. An outgoing transfer rejects a zero or Registry recipient and must both decrease Registry balance and increase that recipient's balance by exactly the same nominal amount; checking only the sender delta is forbidden. After every token call, and before any later token-dependent state transition, Registry re-reads its balance and requires it to cover `accountedBuilderToken`. A mismatch reverts the entire operation. These runtime checks detect transfer fees and transaction-local balance mutation; the immutable-asset release rule excludes asynchronous rebases, confiscation and proxy upgrades that a local delta check cannot make safe. Registry and Schedule are protocol-root objects, not children of a later REGISTER_RELEASE call. A monolithic root constructor is forbidden: its child creation code cannot fit the EIP-3860 49,152-byte init-code bound and aggregate code-deposit gas is not assumed to fit one L1 block. Root deployment instead uses the immutable, non-proxy `ProtocolRootFactoryV1`. The factory pins the settlement chain, manifest namespace and delayed root-migration executor's address/runtime/configuration. It contains one reviewed no-argument `ProtocolRootCreate3ProxyV1` creation artifact; its hash is in the factory configuration and no caller supplies or replaces proxy code. The proxy constructor writes `msg.sender` to storage slot zero as the factory and initializes its one-shot flag in storage slot one. Its creation and runtime bytecode are fixed and identical for every factory; neither uses an immutable or bytecode patch containing the factory address. Its sole runtime entry accepts nonempty init code only from that factory, sets `used=true` before CREATE nonce 1, returns the child address, and thereafter rejects every call. It has no arbitrary target, second CREATE, withdrawal, upgrade or selfdestruct. That entry is exactly `deployV1(bytes)` (selector `0x5d5fb6bc`), nonpayable, canonical offset `0x20`, total calldata length `68+ceil32(initCodeLength)`, zero

padding and no gap, alias or suffix. Success returns exactly one canonically padded nonzero address word equal to the nonce-1 child; short/trailing/dirty return rejects. Factory first builds the complete proxy calldata in memory and warms the already-deployed proxy. Factory expands and touches the exact 32-byte output region before measuring `g0=gasleft()` immediately before CALL. It requires `g0>510000`, requests exactly `g0-510000` gas, supplies zero output space, checks success and exact returndata size, copies exactly the bounded 32-byte return, and then requires `gasleft()>=500000` before any runtime/configuration/activation postcheck; otherwise the whole deployment reverts. Unexpected-size returndata is never copied. The per-role deployment certificate must succeed at its published transaction gas limit, while the boundary vector whose exact copy leaves 499,999 gas rejects immediately after that copy and before any postcheck or state write. A transaction one gas below the synthetic minimum may fail either at the preflight or atomically inside the callee, depending on EIP-150 forwarding. The 10,000-gas gap is the fixed warm-CALL and return-envelope reserve; release measurement must prove it covers that path.

Staging does not waive EIP-170 or EIP-3860. Before the DAO may queue the root operation, the reproducible compiled-artifact gate covers all eighteen deployment artifacts: RootMigrationExecutor, ProtocolRootFactory, the fixed CREATE3 proxy, all nine component roles, and the six additional artifacts reachable from role 9 (bundle deployer, BridgeInboxAdapter, SourceBridgeV2, BridgeCreditRegistryV2, source native QuotaManager and SourceTerminalVerifier). The SourceBundleFactory artifact occurs once in this count and must be byte-identical in the root-role and Source-artifact gates. Every runtime byte array must be at most 24,576 bytes and every complete init code must be at most 49,152 bytes, with the committed hashes equal to those exact bytes. All eighteen artifacts are compiled by one release-pinned compiler build and artifact-owner profile; Factory, proxy and role artifacts from mixed IR/non-IR or optimizer profiles are not a root. In particular the Factory-embedded proxy creation/runtime hashes and every role's independently derived proxy hash must come from those same canonical bytes. BuilderRegistry and ScheduleOracle are specifically blocking until their real optimized artifacts pass both bounds. ProtocolRootFactory is also specifically blocking because it owns strict manifest parsing, CREATE3 and the complete BRC1/SOC1/PCT1/PVM1 postvalidation graph; V1 currently authorizes no Factory helper. If any artifact exceeds EIP-170, the design must first specify an exact immutable helper topology: helper addresses, runtimes, configurations, transitive reachable-code closure, STATICCALL ABI and gas must enter that artifact's configuration and the root DAG. A linked library using DELEGATECALL, an uncommitted parser/verifier, or silently split mutable code is forbidden. No manifest may be queued on the assumption that a later compiler or implementation will make the size problem disappear.

The sole bootstrap authority is the immutable non-proxy `RootMigrationExecutorV1`. Its reviewed executor artifact is permissionlessly deployed through the canonical ERC-2470 singleton factory; its CREATE2 init code binds only settlement chain and the release-pinned genesis DAO proposer, never the ProtocolRootFactory or a root component. Its exact configuration is

```
rootMigrationExecutorConfigurationHash = H(
 "slot-chain-root-migration-executor-config-v1" ||
 u256(settlementChainId) || address20(daoProposer) ||
 u64(604800) || u64(604800) ||
 u64(100000) || u64(50000) || u64(1500000) || u64(300000))
rootMigrationExecutorConfigV1() // selector 0xe9d1d099; 320 bytes; RME1
 returns (bytes4,uint256 settlementChainId,address daoProposer,
```

```
          uint64 minimumDelay,uint64 executionWindow,
          uint64 factoryConfigReadGas,uint64 componentConfigReadGas,
          uint64 factoryStageCallGas,uint64 postStageReserveGas,
          bytes32 configurationHash)
componentConfigHashV2() // selector 0xf6c0f7d2; exact 32 bytes
 returns (bytes32 configurationHash)
rootMigrationAuthorityV1() // selector 0xccf788d5; exact 288 bytes; RMA1
 returns (bytes4 magic,uint8 state,address candidateFactory,
          bytes32 candidateOperationId,bytes32 candidateCampaignKey,
          address activeFactory,bytes32 activeOperationId,
          bytes32 activeCampaignKey,bytes32 activeRootReceipt)
```

The DAO alone calls `queueRootMigrationV1(address,bytes32,bytes32,bytes32)` (selector `0x3499bb4a`) with exact Factory address, root-manifest hash, expected Factory runtime hash and expected Factory configuration hash; calldata is the canonical 132 bytes. The checked monotone `nextOperationNonce` starts at one, must be strictly below `UINT64_MAX` when queued and increments after storing the row. `UINT64_MAX` is its terminal unused exhaustion sentinel. The call returns exactly one `bytes32operationId`, where

```
operationId = H("slot-chain-root-migration-operation-v1" ||
 u256(settlementChainId) || address20(executor) || u64(nonce) ||
 address20(factory) || manifestHash ||
 factoryRuntimeHash || factoryConfigurationHash)
```

The Executor must be IDLE when queueing. Queue requires `block.timestamp<=UINT64_MAX-604800-604800`, stores a distinct row before incrementing the nonce and cannot replace or accelerate an earlier row. The two queued Factory hashes are execution predicates, not self-reported labels: the release gate rejects a proxy, beacon, delegatecall, replaceable target, selfdestruct or mutable reachable dependency, and execution requires the live `EXTCODEHASH` and exact `componentConfigHashV2()` to equal the queued words. Binding them per operation creates no Executor/Factory address cycle. After 604,800 seconds and through the inclusive following 604,800-second execution window, anyone calls `executeRootMigrationV1(bytes32,address,bytes)` (selector `0x9987f8ae`) with that operation ID, factory and canonical manifest ABI. The dynamic offset is `0x60`; the address word is canonically padded; the manifest length is 969; total calldata length is 1,124; and zero tail padding has no gap, alias or suffix. It recomputes the manifest hash and operation ID from the complete stored row, requires exact queued runtime/configuration, IDLE authority and QUEUED state, then tentatively writes CANDIDATE/EXECUTING before any external call. It exact-reads the 736-byte PRF1 and 32-byte configuration result with the fixed stipends, requires the configured delayed executor to be itself, then calls the Factory stage entry with exactly 1,500,000 gas while preserving 300,000 gas for receipt/poststate checks. It accepts only the exact STG1 return and exact 288-byte PRC1 STAGED postread frozen below before changing the operation to STAGED and publishing the candidate key. Call failure, OOG, short/trailing or dirty data, wrong poststate, or insufficient reserve reverts the Factory stage, candidate and operation together. A bare successful no-op can never select a candidate. The DAO may call `cancelRootMigrationV1(bytes32)` (selector `0xfcb20098`) only while the row is QUEUED; after `block.timestamp>executeBefore`, anyone calls `expireRootMigrationV1(bytes32)` (selector `0xe55926d1`) to mark a still-QUEUED row EXPIRED. Cancellation is legal before an execution transaction begins,

including during the execution window; execution at both inclusive endpoints is legal, and expiry at the upper endpoint is not. Each successful terminal transition emits respectively `RootMigrationCancelled(operationId)` or `RootMigrationExpired(operationId)`; reverts emit nothing. The exact 320-byte `rootMigrationOperationV1(bytes32)` view (selector `0xdade63bb`, magic RMO1) returns state, nonce, factory, manifest hash, Factory runtime/configuration hashes, queuedAt, executeAfter and executeBefore. NONE=0, QUEUED=1, EXECUTING=2, STAGED=3, ACTIVATED=4, CANCELLED=5, EXPIRED=6 and ABORTED=7; NONE zeroes every other word. RMA1 authority states are IDLE=0, CANDIDATE=1 and ACTIVE=2. IDLE zeroes all authority fields; CANDIDATE has only the three candidate fields; ACTIVE has only nonzero active Factory, operation, campaign and root-receipt fields. While CANDIDATE or ACTIVE, no operation may queue or execute. An independently queued operation cannot create a second candidate.

Factory finalization calls `confirmRootMigrationV1(bytes32,bytes32,bytes32)` (selector `0xb1372f22`; canonical 100-byte calldata) with operation ID, campaign key and root receipt. Executor accepts only its current queued-hash-authenticated candidate Factory, exact-reads that Factory's PRC1 ACTIVE row, and atomically changes STAGED to ACTIVATED and RMA1 CANDIDATE to ACTIVE. It returns exactly one padded `RAC1` magic word. If the Factory instead reaches authenticated ABORTED, anyone calls `clearAbortedRootMigrationV1(bytes32, bytes32)` (selector `0x95b71205`; canonical 68 bytes). Exact Factory code/config and PRC1 operation/key/ABORTED checks change STAGED to ABORTED and return RMA1 to IDLE; deadline alone is insufficient. Thus a failed or abandoned campaign can retry even with another DAO-approved immutable Factory, while an activated root can never be replaced or coexist with a second root through this Executor. When the STAGED receipt is accepted, Executor stores its exact generation and expiry as internal operation metadata. Every later PRC1 decode validates all narrow-word padding and compares those two values exactly; no terminal integer is a wildcard. ACTIVE additionally requires `deployedBitmap=0x01ff` and the exact nonzero confirmation root receipt. ABORTED requires that `deployedBitmap` has no bit outside `0x01ff`, and requires a zero root receipt. Executor has no generic target call. ERC-2470 address/runtime, executor creation/runtime/configuration, DAO and salt are genesis-deployment receipt constants. Because executor address is derived before and independently of Factory init code, Factory may safely bind it without an address cycle. The exact executor event declarations are

```
event RootMigrationQueued(
 bytes32 indexed operationId,address indexed factory,
 bytes32 indexed manifestHash,bytes32 factoryRuntimeHash,
 bytes32 factoryConfigurationHash,uint64 nonce,
 uint64 executeAfter,uint64 executeBefore);
event RootMigrationStaged(bytes32 indexed operationId,
 bytes32 indexed campaignKey,uint64 indexed generation,uint64 expiresAt);
event RootMigrationActivated(bytes32 indexed operationId,
 bytes32 indexed campaignKey,bytes32 indexed rootReceipt);
event RootMigrationCandidateCleared(bytes32 indexed operationId,
 bytes32 indexed campaignKey);
event RootMigrationCancelled(bytes32 indexed operationId);
event RootMigrationExpired(bytes32 indexed operationId);
```

The factory constructor is exactly

```
ProtocolRootFactoryConstructorV1(
 uint256 settlementChainId,bytes32 manifestNamespace,
 address delayedRootMigrationExecutor,bytes32 executorRuntimeHash,
 bytes32 executorConfigurationHash)
```

It requires the live chain, all nonzero identities and exact executor runtime/`componentConfigHashV2()`. Because the proxy has neither constructor arguments nor immutables, Factory derives `create3ProxyCreationCodeHash=H(type(ProtocolRootCreate3ProxyV1).creationCode)` and `create3ProxyRuntimeHash=H(type(ProtocolRootCreate3ProxyV1).runtimeCode)`; neither hash is a constructor input. The executable model likewise accepts the compiler-produced proxy code bytes only as an external deployment artifact, enforces the EIP-3860/EIP-170 byte limits and hashes those bytes internally; it does not model either hash as a Factory constructor field. Factory also derives, rather than accepts,

```
protocolRootFactoryConfigurationHash = H(
 "slot-chain-protocol-root-factory-config-v1" ||
 u256(settlementChainId) || manifestNamespace ||
 address20(delayedRootMigrationExecutor) || executorRuntimeHash ||
 executorConfigurationHash || create3ProxyCreationCodeHash ||
 create3ProxyRuntimeHash || sourceArtifactRoot ||
 bytes4(0xf95ec7e4) || bytes4(0x43b610d6) ||
 u64(2000000) || u64(300000) || u64(75000) ||
 u64(2592000) || u64(500000) || u8(18) ||
 u64(100000) || u64(50000) || u64(50000) || u64(100000) || u64(500000))
```

Here `sourceArtifactRoot` is `H("slot-chain-root-source-artifact-root-v1"||creationHash||runtimeHash)` over, in order, the SourceBundleFactory, bundle deployer, BridgeInboxAdapter, SourceBridgeV2, BridgeCreditRegistryV2, source native QuotaManager and SourceTerminalVerifier creation/runtime hash pairs. Every pair comes from the same release-pinned compiler artifacts; none is a caller or manifest input. Factory exposes its configuration through `componentConfigHashV2()` and the exact 736-byte `protocolRootFactoryConfigV1()` view (selector `0xd7b40838`, magic PRF1). To remain compilable under the release's non-IR Solidity profile, the interface returns one static `ProtocolRootFactoryConfigV1` struct whose 23 fields, in exact wire order, are

```
(bytes4 magic,uint256 settlementChainId,bytes32 manifestNamespace,
 address delayedRootMigrationExecutor,bytes32 executorRuntimeHash,
 bytes32 executorConfigurationHash,bytes32 proxyCreationCodeHash,
 bytes32 proxyRuntimeHash,bytes32 sourceArtifactRoot,
 bytes4 ensureSourceInfrastructureSelector,
 bytes4 sourceInfrastructureViewSelector,uint64 campaignLifetime,
 uint64 deploymentPostcheckReserve,
 uint8 minimumFirstManagedRunwayWindows,uint64 executorConfigReadGas,
 uint64 componentConfigReadGas,uint64 activationCallGas,
 uint64 externalReadGas,uint64 executorConfirmCallGas,
 uint64 sourceTerminalDeploymentGas,
 uint64 sourceInfrastructurePostcheckGas,
 uint64 sourceInfrastructureCallOverheadGas,
 bytes32 configurationHash)
```

A single static tuple has the same 736-byte ABI returndata and selector as the flat list; outputs never enter the selector. The named struct is normative so the generated ABI remains usable without a 23-output non-IR wrapper. Its fields are those constructor fields, derived proxy hashes, Source artifact root, the exact selectors `(0xf95ec7e4,0x43b610d6)`, lifetime, deployment postcheck reserve, `minimumFirstManagedRunwayWindows=18`, then Executor-config read, component-config read, activation-call, external-read and Executor-confirm-call gas constants `(100000,50000,50000,100000,500000)`, then Source-terminal deployment, Source-infrastructure postcheck and call-overhead gas constants `(2000000,300000,75000)`, and the derived hash. The Executor-confirm value is a separate outer-call stipend: it leaves enough EIP-150 headroom for Executor's nested 50,000-gas configuration read, 100,000-gas PRC1 read, state writes and return. Release boundary tests must show 500,000 succeeds and the measured minimum successful nested-call budget plus safety margin remains below that stipend; an OOG at any smaller test budget rejects atomically. The factory is permissionlessly deployed through ERC-2470 with release-pinned creation/runtime/configuration and salt after the independently addressed executor; its address/runtime/configuration do not depend on any root component, manifest or generation.

One campaign has the exact 969-byte packed `ProtocolRootManifestV1`:

```
u8(schema=1) || u256(settlementChainId) || u64(generation) ||
bytes32(manifestNamespace) || bytes32(predecessorRootReceipt) ||
for role=1..9 in exact order:
  bytes32(initCodeHash) || bytes32(runtimeHash) || bytes32(configurationHash)

roles = 1 BuilderRegistry, 2 ScheduleOracle, 3 ProtocolChangeTimelock,
        4 ProtocolVersionManager, 5 ActiveSettlementRouter, 6 ForcedQueue,
        7 AggregatorSeatMarket, 8 BridgeDomainRegistry,
        9 SourceBundleFactory
campaignKey = H("slot-chain-protocol-root-campaign-v1" ||
  u256(settlementChainId) || address20(factory) || u64(generation) ||
  manifestNamespace || predecessorRootReceipt)
componentSalt(role) = H(
  "slot-chain-protocol-root-component-v1" || campaignKey || u8(role))
proxy(role) = CREATE2(factory,componentSalt(role),fixedProxyCreationCodeHash)
component(role) = low20(H(0xd6||0x94||address20(proxy(role))||0x01))
manifestHash = H("slot-chain-protocol-root-manifest-v1" ||
                 u16(969) || packedManifest)
```

For compile-time role contract `RoleV1`, the first committed word is exactly

```
constructorArgs = abi.encode(
 protocolRootFactory,factoryRuntimeHash,campaignKey,
 <every RoleV1-local constructor field in its displayed order>)
initCode = type(RoleV1).creationCode || constructorArgs
initCodeHash = H(initCode)
```

The activation fields are the first three constructor arguments, not an independent byte prefix or nested ABI tuple. Solidity's canonical ABI defines every static word and any role-local dynamic offset/tail; a release manifest pins the resulting complete init-code bytes, and

no initializer or suffix may follow them. All 27 component hashes are nonzero. Generation starts at zero and is the Factory's exact next generation; staging additionally requires `nextGeneration<UINT64_MAX`. `UINT64_MAX` is the terminal unused generation sentinel: generation `UINT64_MAX-1` may finalize or abort and advance to it, but no campaign can then stage. Namespace and chain equal its configuration; predecessor is canonical zero. ProtocolRootFactoryV1 installs exactly one genesis root; it does not implement root succession. No manifest word, component init code or configuration hash enters the campaign key, proxy salt or component address, so code/configuration commitments may safely contain all nine already-derived addresses without an address fixed point.

Only the pinned delayed executor may call `stageProtocolRootV1(bytes32,bytes)` (selector `0x53c99a5f`) with the nonzero operation ID and canonical ABI wrapping exactly those 969 bytes: dynamic offset `0x40`, total calldata length 1,092, zero tail padding and no gap, alias or suffix. Factory rechecks executor EXTCODEHASH, its exact 320-byte configuration, 32-byte configuration hash and RMA1 CANDIDATE fields against this Factory, operation and derived campaign key before storing the operation/key/hash, a checked `expiresAt=block.timestamp+2592000`, zero deployment bitmap and STAGED state. Generation begins at zero. It accepts no second live campaign or same generation. The call returns exactly 192 canonical bytes

```
(STG1,operationId,campaignKey,manifestHash,generation,expiresAt)
```

with no short/trailing/dirty word. After staging, anyone may call

```
deployProtocolRootComponentV1(bytes32,uint8,bytes)
selector 0x31f4d140
```

The call requires the exact campaign key, role 1–9, unset role bit, live deadline, nonempty init code of at most 49,152 bytes and its committed `initCodeHash`. Its dynamic offset is `0x60`, role has canonical uint8 padding, total calldata length is `132+ceil32(initCodeLength)`, and tail padding is zero with no gap, alias or suffix. Factory CREATE2-deploys the fixed proxy and calls it in the same frame only after its EXTCODEHASH equals the configured proxy runtime hash. It requires the returned derived component address, committed EXTCODEHASH, the exact 32-byte `componentConfigHashV2()` under 50,000 gas and canonical 128-byte inactive activation view under 100,000 gas before setting exactly `deployedBitmap|=uint16(1)<<(role-1)`. No other bit may be set. Constructor failure, wrong code/config/view or late failure reverts the proxy, child and bit together; no adversary can consume the proxy nonce with another init code. At both deterministic addresses, a balance-only prefund with zero nonce and no code is nonblocking and its balance survives successful deployment. Nonzero nonce or existing code at the proxy rejects before its role bit; nonzero nonce or existing code at the child rejects and reverts the transient proxy. After success the proxy has the exact pinned runtime and nonce 2 (contract creation plus its single child CREATE), while the child has nonce 1 and the committed runtime/configuration. A fault as late as verifier or activation postcheck restores both addresses to their pretransaction nonce/code/storage while retaining pre-existing balances. Boundary conformance deploys role 1 first and role 9 last, then repeats with role 9 first and role 1 last, and injects proxy/child code, nonce and balance-only states in each position.

After all nine bits are set but before finalization, anyone must ensure the single root-lifetime Source terminal artifact through the same Factory:

```
ensureSourceInfrastructureV1(bytes32 campaignKey,uint8 kind,bytes initCode)
  selector 0xf95ec7e4; kind=1 only; value=0; minimum gas=2375000
protocolRootSourceInfrastructureV1(bytes32 campaignKey,uint8 kind)
  selector 0x43b610d6; calldata 68; gas=100000; return 320 bytes / PSV1
```

The ensure ABI has head words `(campaignKey,kind,0x60)`, one canonical bytes length, contiguous data and zero tail padding. It requires the current STAGED campaign, its live inclusive deadline, `deployedBitmap=0x01ff` and no infrastructure call in progress. Kind 1 is the immutable SourceTerminalVerifier whose constructor is `abi.encode(settlementChainId, activeSettlementRouter)` and whose salt is

```
H("slot-chain-root-source-terminal-salt-v1" ||
  u256(settlementChainId) || manifestNamespace || address20(router)).
```

Factory calls only the canonical ERC-2470 singleton at `0xce0042B868300000d44A59004Da54A005ffdcf9f`; its runtime hash, CREATE2 address, complete init-code hash and compiled runtime/configuration are exact. If the address has no code, `initCode` must equal the complete derived bytes and CREATE2 must succeed. A balance-only prefund is retained; a nonzero nonce/code collision fails under EIP-684. If exact code already exists, `initCode` must instead be empty. Reuse with nonempty bytes and every inexact code, configuration, selector-only STF1 return or ERC-2470 returned address fail. The deployment call uses 2,000,000 gas and reserves 300,000 gas for postchecks within a fixed 75,000-gas outer envelope; any failure rolls back the account, in-progress word, bitmap and receipt together.

Success returns exactly 256 bytes

```
PSI1 = (bytes4 magic,bytes32 campaignKey,uint8 kind,address artifact,
        uint8 created,bytes32 runtimeHash,bytes32 configurationHash,
        bytes32 infrastructureReceipt)
infrastructureReceipt = H(
 "slot-chain-root-source-infrastructure-receipt-v1" || campaignKey ||
 u8(kind) || address20(artifact) || runtimeHash || configurationHash ||
 H(expectedInitCode) || sourceArtifactRoot)
```

where `created` is 1 only for this call's CREATE2 and otherwise 0. PSV1 is exactly `(magic,key, kind,state,address,runtimeHash,configurationHash,initCodeHash,sourceArtifactRoot, receipt)` with ABSENT=0, IN_PROGRESS=1 and READY=2. An unknown campaign returns its queried canonical kind and zeros the remaining seven words; another kind rejects. Finalization requires exactly kind 1 READY, no call in progress and its nonzero stored receipt, then repeats the singleton, address, code, configuration and 192-byte STF1 checks. The terminal's exact selector-only configuration surface is

```
sourceTerminalVerifierConfigV1()
  selector 0x1f683f3b; value=0; gas=100000; return 192 bytes / STF1
 returns (bytes4 magic,uint256 settlementChainId,
          address activeSettlementRouter,bytes32 creationCodeHash,
          bytes32 runtimeHash,bytes32 configurationHash)
sourceTerminalVerifierConfigurationHash = H(
 "slot-chain-component-config-v1" || u8(3) || u16(21) ||
 address20(activeSettlementRouter) || u8(64))
```

The magic is `0x53544631`; the address and every fixed-width value use canonical ABI padding. The Factory requires the returned chain and Router to equal this campaign, both code hashes to equal its pinned compiler artifacts, and the returned configuration hash to equal both the displayed preimage and the 32-byte `componentConfigHashV2()` result. Revert/OOG, caller-dependent, short/trailing or dirty STF1 data fails before activation.

Every component-local constructor displayed for one of the nine roles is the local suffix of the actual Solidity constructor argument list whose first three fields are the exact tuple

```
RootActivationConstructorV1 = (
  address protocolRootFactory,
  bytes32 factoryRuntimeHash,
  bytes32 campaignKey
)
```

The factory address may be an immutable address; the cross-runtime hash and key are constructor-written writerless storage. Constructor deployment requires the pinned factory code, the exact campaign key, `msg.sender=proxy(role)` and `address(this)=component(role)` derived from its compile-time role, but does not read not-yet-deployed peers. The committed init-code hash covers the compiled creation code plus canonical ABI encoding of that complete argument list. The activation prefix is deployment topology and is excluded from every previously defined component-local economic or configuration hash unless that hash's explicit preimage says otherwise. In particular it does not change the Builder Registry economic/topology formulas or Schedule configuration constructor order. No optional initializer suffix or later setter exists.

Every one of the nine root components is nonfunctional until its sole `activateProtocolRootV1(bytes32)` entry (selector `0x74e3aa45`) is called by the pinned factory with its constructor-stored campaign key. The key is independent of the manifest and is constructor-written writerless storage, not a bytecode immutable. Before activation every functional or mutating entry, including a functional read, reverts `RootInactive`. The complete pre-activation read allowlist is: every role's `componentConfigHashV2()` and `protocolRootActivationV1()`, plus `builderRegistryConfigV1()` for role 1, `scheduleOracleConfigV1()` for role 2 and `protocolChangeTimelockConfigV1()` (PCT1) for role 3, and `protocolVersionManagerConfigV1()` for role 4, and the exact 640-byte `sourceBundleFactoryConfigV1()` (SBF1, selector `0x74bab161`) for role 9. Role 9's SBD1 and SAD1 deployment entries are not allowlisted and remain `RootInactive` until atomic root activation. Roles 2 and 4 also permit the exact `scheduleForkRouteStateV1()` (FRS1) view and `forkVerifierRegistrationV1(bytes4)` (FVR1) view before activation solely so Factory can authenticate their independently constructed initial fork-route mirrors. At that point the only accepted digest is the constructor-installed initial digest and every other FVR1 query reverts. The initial verifier's selector-only SFV1 view is an ordinary immutable verifier configuration read, not a root-component activation surface. No other role-specific view is callable before activation; in particular SWR1, SWV1, all other operational topology getters and all mutation surfaces revert. The exact zero-argument view `protocolRootActivationV1()` (selector `0xda930d69`) returns exactly 128 bytes `(PRA1,protocolRootFactory,campaignKey,state)`, with INACTIVE=0 and ACTIVE=1 and canonical padding. Activation only changes that one local state, emits no external call and returns exactly one padded `RAA1` magic word. Factory supplies exactly 50,000 gas and then exact-reads PRA1 with 100,000 gas; fault, OOG, short/trailing/dirty return or wrong state reverts the full finalization.

Once all nine bits are set, anyone may call `finalizeProtocolRootV1(bytes32)` (selector `0x95f759ad`) before or at the deadline. Factory requires that no root was previously installed and rechecks the zero predecessor, all nine addresses, runtimes, exact configuration/activation views, kind-1 Source infrastructure READY state, and the complete BRC1/SOC1/PCT1/PVM1/SBF1/STF1 cross-graph. This is a strict word-by-word decode, not merely a comparison of the three outer configuration hashes: BRC1 requires a nonzero token runtime, canonical words 5–7 and `0<leasePerWindowAtomic<=maximumBondAtomic` with `5*reporterRewardCapAtomic<=leasePerWindowAtomic`; SOC1 requires canonical beacon-genesis/lookahead words, `lookaheadSeconds=768`, and the first managed-window start numerically at least 768 and at least 3,072 seconds after beacon genesis. Factory exact-reads role 3's 160-byte PCT1, role 9's 640-byte SBF1 and the root-lifetime terminal's 192-byte STF1. SBF1 must commit the exact Source artifact hashes, deployment selectors/gas and PVM address; STF1 must commit the settlement chain, role-5 Router and terminal creation/runtime hashes. Factory requires PCT1's PVM to be role 4, delay 604,800 and operation domain `H("slot-chain-protocol-change-operation-v1")`, and recomputes the governance descriptor from the actual chain, role-3 address/runtime, returned DAO and role-4 address. The returned DAO must equal the genesis DAO proposer committed by the Executor's exact RME1/configuration row and be distinct from Factory, Executor and all nine component addresses. That descriptor must equal both the role-3 component configuration and PVM1 word 12. PVM1 words 14–16 are exactly 604,800, word 17 is exactly 64, and all four release gas caps in words 18–21 are nonzero. Every address, narrow integer and bytes4 word in the three rows has canonical padding. Component configuration calls use 50,000 gas and every larger activation/cross-view uses 100,000 gas; every call has exact selector-only calldata and exact return length/padding. It requires BRC1 and SOC1 to agree on `genesisTimestamp`, `evidenceDelaySeconds`, `reorgMarginSeconds`, `firstManagedWindow` and `lastManagedWindow`; independently recomputes `lastManagedWindow` by the checked formula above; derives `currentWindow=max(0,(block.timestamp-genesisTimestamp)/384)` with checked arithmetic, and requires both

```
firstManagedWindow <= lastManagedWindow <= LAST_FULL_SLOT_WINDOW
firstManagedWindow >= currentWindow + 18.
```

Before activating any role, Factory additionally makes 50,000-gas FRS1 reads from the actual role-2 and role-4 addresses and requires byte-identical 160-byte returns with counts exactly `(1,1,1)`. It then makes 100,000-gas FVR1 reads from both at the SOC1 initial digest and requires byte-identical 320-byte rows with zero successor. From that row it obtains the actual verifier address and runtime hash, checks `EXTCODEHASH`, and makes the selector-only 100,000-gas SFV1 read directly from that verifier. Factory strictly combines FVR1 and SFV1 into the complete canonical 480-byte initial row, recomputes its verifier configuration hash, checks its digest and first/last bounds against SOC1, and independently recomputes the singleton order prefix, registration sparse root, used-digest sparse root and exact FRS1 return. Thus neither a blank/divergent PVM mirror nor two colluding view values that omit the actual singleton row can cross root activation. Revert/OOG, wrong caller-independent runtime, short/trailing/dirty return, noncanonical padding, a second mapping/order/tombstone, or any Oracle–PVM–verifier mismatch reverts before the first activation call. This finalization-time check, rather than a stale constructor-time promise, allows one complete boundary window after activation before registration, eight entry-delay windows, the eight-window historical snapshot offset and one additional boundary window. It also leaves time to reserve the first managed window under the 16-window ahead cap: an activation at the end of the current

window can register in the next, reserve at most by the second, and is effective strictly before the first target snapshot. Overflow or a smaller runway rejects before any activation call. Factory then calls the nine activation entries in role order, exact- reads all nine ACTIVE views, and only then stores

```
rootReceipt = H("slot-chain-protocol-root-receipt-v1" || campaignKey ||
  manifestHash || address20(component(1)) || ... || address20(component(9)) ||
  address20(sourceTerminalVerifier) || sourceInfrastructureReceipt)
```

as its sole root receipt and marks the campaign ACTIVE. It then calls the pinned Executor's `confirmRootMigrationV1` with exactly 500,000 gas and requires the exact padded `RAC1` word. Only after that confirmation does it increment generation and permanently close staging/finalization. All activation calls, local publication, Executor ACTIVE confirmation and final Factory state share one revert domain. The final call contains no creation code and production requires a measured worst-case gas certificate below the supported L1 block gas limit. After the deadline, `abortProtocolRootCampaignV1(bytes32)` (selector `0x3774c8a5`) is permissionless, marks the generation ABORTED and advances generation without activating or deleting anything. Partial children remain permanently INACTIVE; after the separate authenticated Executor candidate-clear, a retry before the first successful root uses a new generation, salts and addresses. Staging, withholding and abandonment therefore cannot install a partial root. Root succession is deliberately absent: a future incompatible root requires a separately specified cutover that preserves old historical/claim surfaces and prevents dual mutable authority; this factory cannot perform or approximate that migration.

The exact bounded reads are

```
protocolRootCampaignV1(bytes32 campaignKey) // 288 bytes, magic PRC1
 returns (bytes4,bytes32 operationId,bytes32 key,bytes32 manifestHash,uint8 state,
          uint64 generation,uint64 expiresAt,uint16 deployedBitmap,
          bytes32 rootReceipt)
protocolRootComponentV1(bytes32 campaignKey,uint8 role) // 224 bytes, PRD1
 returns (bytes4,bytes32 key,uint8 role,address component,
          bytes32 initCodeHash,bytes32 runtimeHash,
          bytes32 configurationHash)
```

with selectors `0xe6139cad` and `0x007f4b23`; NONE=0, STAGED=1, ACTIVE=2 and ABORTED=3 are the only campaign states, and NONE requires every other word zero. Successful state changes emit exactly

```
event ProtocolRootCampaignStaged(
 bytes32 indexed campaignKey,bytes32 indexed manifestHash,
 uint64 indexed generation,bytes32 operationId,uint64 expiresAt);
event ProtocolRootComponentDeployed(
 bytes32 indexed campaignKey,uint64 indexed generation,
 uint8 indexed role,address component);
event ProtocolRootActivated(
 bytes32 indexed campaignKey,bytes32 indexed rootReceipt,
 uint64 indexed generation);
event ProtocolRootCampaignAborted(
 bytes32 indexed campaignKey,uint64 indexed generation);
```

Reverts and repeated calls emit nothing.

Registry and ScheduleOracle may therefore receive each other's addresses before either exists. Peer runtime hashes live in writerless constructor storage rather than Solidity immutable bytecode. Configuration binding is deliberately acyclic: Registry's derived configuration commits the Schedule address/runtime but not its configuration hash; Schedule's derived configuration commits the Registry address/runtime/configuration. ProtocolRootFactory postvalidates every child address, runtime/configuration and this one-way cross-pin atomically before root activation/publication. REGISTER_RELEASE later repeats that validation only as a secondary release gate. Thus neither a runtime-hash nor configuration-hash fixed point exists, and no independently deployed or partially validated pair is accepted.

The bytecode dependency rule is exact. Let `RUNTIME_R` and `RUNTIME_S` be the Registry and Schedule runtime hashes, `E` the Registry economic hash, `T` its topology hash and `C` the Schedule configuration hash. Both runtimes may contain fixed protocol constants and the consecutively precomputed peer address, but their compiler `immutableReferences` MUST exclude `E`, `T`, `C`, every peer runtime/configuration hash and every value derived from those words. Those values exist only in constructor-written storage with no reachable writer. If `R` is the independently derived Router configuration hash, the only relevant hash-dependency edges are `(RUNTIME_R,RUNTIME_S,E,R,otherRegistrytopologyfields)->T` and `(RUNTIME_R,RUNTIME_S,E,T,R,otherSchedulefields)->C`; there is no edge from `T` or `C` back into either runtime. Release tooling rejects runtime metadata that violates this DAG even when the resulting addresses would otherwise match.

The same rule applies to all nine staged root roles. Their runtime bytecode may patch only code-independent CREATE3 addresses and local scalar constants. Compiler `immutableReferences` MUST exclude the campaign key, the component's own configuration hash, every peer runtime/configuration hash and every value derived from those hashes. All such words are constructor-written storage with no post-activation writer. Configuration derivation has the following exact levels. Level 0 contains all code-independent component and root-terminal addresses, runtime hashes and the release-pinned Source artifact hashes. Level 1 derives the true leaves: Builder economic identity E and the local Timelock, Router, SourceBundleFactory and SourceTerminalVerifier configurations from profile scalars plus Level-0 identities only. In particular Router configuration cannot depend on any peer configuration, E, T or C. Level 2 derives Registry topology T from E, Router configuration and the explicit Level-0 identities in B. It also derives the ForcedQueue, Market and protocol-lifetime BridgeDomainRegistry configurations from their already displayed exact formulas; those formulas may consume Router configuration (Market does through profile word 25) but must not consume T, C, PVM configuration or another Level-2 configuration. Level 3 derives Schedule configuration C from E, T, Router configuration and its displayed local fields. Level 4 derives PVM configuration as the sole final root over the other eight committed role configurations plus the SourceTerminalVerifier identity/configuration. No edge within a level or from a lower-numbered level to a higher-numbered consumer may point backward. The reproducible-build/root-artifact gate inspects compiler metadata, full constructor preimages and this graph and refuses to queue a manifest with a forbidden edge; these are not facts an on-chain getter can prove. On-chain Factory finalization checks the committed runtime hashes, exact configuration and explicit cross-views. Thus staged deployment removes address cycles without replacing them with runtime- or configuration-hash fixed points.

Generation and custody state. Each generation stores the builder, `uint192baseBondAtomic`, registration index, effective and tombstone L2 slots, tranche root, a reservation base window, a 17-bit reservation bitmap, unreleased-tranche count, maximum liable deadline, one-shot exit record and its active/liability locator. The live tombstone sentinel is `UINT64_MAX`. Storage consists of exactly 64 active cells, 1,072 liability cells, the authoritative `mapping(uint64=>GenerationLocation)generationLocation` and `mapping(address=>uint256)liveRegistrationIndexPlusOne`, a monotonic `uint256nextRegistrationIndex`, a monotonic `movementSequence`, a per-window move counter and a global append-only exit FIFO. Default all-zero Solidity storage is never decoded as a tranche: canonical `EMPTY` is `(window=UINT64_MAX,state=EMPTY,amount=0,liableUntil=0)`.

Registration is self-consenting: `builder=msg.sender`. Reservation and exit likewise require that same ECDSA address and registration generation. There is no `registerFor`, `tx.origin`, EIP-1271 delegation or caller-selected refund owner. The caller supplies only race guards and proofs; the contract derives the builder, active index, replacement victim, liability index, tranche index, current slot/window, tombstone slot and every deadline. The supplied `baseBondAtomic` must satisfy `leasePerWindowAtomic<=baseBondAtomic<=maximumBondAtomic`; both inclusive boundaries are checked before proof mutation, token movement or state writes. `nextRegistrationIndex` is accepted while it is at most `UINT64_MAX`; admitting that last external `uint64` registration ID increments the internal counter to the sole exhausted value `UINT64_MAX+1`. Every later registration rejects before a cast, transfer or write. This stored monotonic counter is never recomputed from the active or liability sets: registrations already released from both sets leave permanent historical holes, including across restart, and no such identifier is reused. The entry delay is `effectiveL2Slot=currentL2Slot+8*384`, not rounded to a window. The addition is checked and registration rejects unless `floor(effectiveL2Slot/384)<=lastManagedWindow`; this applies equally to vacant-cell and competitive registration.

Base-bond and per-window escrow are disjoint. Registration pulls exactly the immutable base bond; each first reservation of a window separately pulls exactly `L_LEASE`. The base bond ranks the generation and cannot be changed in place. Moving active to liability moves both obligations without a credit. Tranche release debits tranche escrow and credits the builder. A slash debits one tranche once, credits `min(reporterRewardCapAtomic,L_LEASE)` to the evidence submitter and the remainder to the immutable builder-penalty sink. Generation release debits base escrow and credits the builder only after every tranche is terminal and all evidence/reorg predicates are safe. Therefore at every external boundary

```
accountedBuilderToken = baseBondEscrow + trancheEscrow
                      + totalBuilderTokenPullCredits
builderToken.balanceOf(BuilderRegistry) >= accountedBuilderToken.
```

`totalBuilderTokenPullCredits` is an authoritative stored `uint256`: each credit creation increments it and each successful claim decrements it in the same rollback domain as the owner row. Solvency checks read this scalar and never sum the lifetime-growing pull-credit mapping. An exhaustive recount is an off-chain/model invariant only. Unsolicited token transfers are surplus and are permanently unclaimable in V1; they cannot create a credit or block ordinary claims.

Tranche transition and wrap. The only funded transition is an atomic `EMPTY/safe-old-FREE/RELEASED/SLASHED->FREE(W)->RESERVED(W)`; the intermediate FREE value is not published by a public V1 operation. `RESERVED(W)->LIABLE(W)`, `RESERVED/LIABLE->SLASHED` and

`LIABLE->RELEASED` are the other legal transitions. RESERVED and LIABLE carry exactly `L_LEASE` and `evidenceReplayDeadline(W)`; terminal leaves carry zero amount but retain their original window and deadline. The leaf index is derived as `Wmod512`. Reuse requires the old leaf to be EMPTY or terminal, its stored window strictly smaller than W and the same modulo index; a nonterminal old generation is never overwritten. Thus W=511 uses leaf 511 and W=512 uses leaf 0. Every multiplication, deadline addition and `uint64` boundary is checked before a token transfer or state write. Reservation additionally requires `firstManagedWindow<=W<=lastManagedWindow`; idempotent re-reservation cannot bypass either bound. In particular, the activation runway cannot fund a pre-first tranche that SWR1 would permanently classify UNSEALED.

The bitmap is always relative to a stored `reservationBaseWindow` and may set only offsets 0 through 16. Before reserving, the transaction closes every set bit below the new current window in ascending W order using exact nine-sibling proofs, then shifts the base. Active-to-liability movement closes all remaining RESERVED bits in the same order. Consequently one operation touches at most 17 tranche leaves. The implementation enumerates only set bits of that generation's stored bitmap and exact-checks the derived tranche rows; it never enumerates the lifetime tranche mapping, reservation sets or all 512 leaf positions. A witness must contain the complete bitmap popcount for movement or the exact stale-mask popcount for reservation/normalization, consume all bytes and contain no caller-selected height, orientation or leaf index. The counter transition is exact: first successful creation of `RESERVED(W)` increments `unreleasedTrancheCount` once; idempotent re-reservation and `RESERVED->LIABLE` leave it unchanged; `LIABLE->RELEASED` and either slash transition decrement it once; terminal reuse for a different W increments once only after the old terminal leaf passes its overwrite predicate. No other path changes the counter. Every increment or decrement is checked, and an underflow/overflow or leaf/counter disagreement reverts before a root, escrow or credit write.

Exit, tombstone and movement. An exit request is one-shot, closes future reservation immediately, stores `requestWindow=currentWindow` and matures at equality with `requestWindow+268`. It rejects once `currentWindow>lastManagedWindow`; terminal direct release below is then the only active-generation exit. Records are appended by checked sequence. Requests created in one window have the same maturity and are processed by sequence, so the global queue is strict FIFO. Before a full-table registration, resolved heads may be skipped, but that path inspects at most 64 FIFO records; if a 65th head would be needed it reverts `MaintenanceRequired`. Any active tombstone or an oldest live matured exit causes `MaintenanceRequired` before any token pull. The permissionless bounded maintenance entry inspects at most 64 FIFO records and performs at most the caller's requested number of moves, capped by the remaining four-move window budget. It moves matured exits first, then active slash tombstones ordered by `(tombstonedAtL2Slot,registrationIndex)`. A not-yet-mature exit may still be competitively replaced and its queue record becomes resolved atomically. When `currentWindow>lastManagedWindow`, this ordinary movement entry is a canonical no-op; terminal active release does not use it or its budget.

Healthy competitive replacement or matured exit preserves the moved generation's live tombstone sentinel in liability. It closes reservations and removes the generation from every later active snapshot, but its already-sealed tickets remain provable against the current admission root throughout the newcomer's entry delay. The first valid slash of either an active or liability generation sets its tombstone to the current L2 slot; an existing tombstone is never postponed.

Slash itself does not consume the movement budget, and later maintenance performs the move. Thus many evidence submissions cannot bypass the four-move capacity proof or overwrite the liability ring. A tombstone at slot S excludes new schedule eligibility for source slots at or after S but cannot mutate an already-frozen admission pair.

For movement sequence q the liability position is `64+(qmod1072)`. The contract first closes tranches and derives the victim's retained liability state, then uses the sole six-level registry proof to replace the old victim active leaf directly by the replacement or canonical empty leaf. It then verifies/releases the old liability occupant against the pre-root and writes the moved generation, producing intermediate admission root R1. The active-position proof is valid only against R1 and writes the replacement or canonical empty leaf, producing R2. Only R2 is the next record's pre-root; the external call publishes only its final R2 and increments `admissionVersion` exactly once. Two proofs against the pre-root, the reverse order, an explicit witness index or unused suffix bytes reject. Sequence 0 maps to position 64, 1,071 to 1,135 and 1,072 wraps to 64 only after its former occupant satisfies generation release.

Exact registry surface. The fixed reads are

```
admissionStateV1() returns
 (bytes4 magic,uint64 admissionVersion,bytes32 admissionRoot)
// selector 0x4a9dfa3f; exact 96 bytes; magic ADS1

scheduleRegistryStateV1() returns
 (bytes4 magic,uint8 schema,uint8 activeCount,
  uint64 registryMutationVersion,uint64 admissionVersion,
  uint256 nextRegistrationIndex,bytes32 registryRoot)
// selector 0xad95cea1; exact 224 bytes; magic BRS1; schema=1
```

The mutation signatures, parameter meanings and computed selectors are

```
registerBuilderV1(uint192 baseBondAtomic,
 uint64 expectedNextRegistrationIndex,uint8 expectedActiveIndex,bytes witness)
 returns (bytes4 magic,uint64 registrationIndex,uint8 activeIndex,
          uint64 effectiveL2Slot,uint64 admissionVersion,bytes32 admissionRoot)
                                                          0x5fc42c69 BRG1/192
reserveBuilderWindowV1(uint64 expectedRegistrationIndex,uint64 window,
 bytes witness)
 returns (bytes4 magic,uint64 registrationIndex,uint64 window,
          uint64 registryMutationVersion,bytes32 registryRoot)
                                                          0x46a53315 BRV1/160
requestBuilderExitV1(uint64 expectedRegistrationIndex)
 returns (bytes4 magic,uint64 registrationIndex,uint64 matureWindow,
          uint8 activeIndex)                             0xc8f20b55 BRE1/128
processBuilderMaintenanceV1(uint8 maxMoves,bytes witness)
 returns (bytes4 magic,uint8 inspected,uint8 moved,
          uint64 admissionVersion,bytes32 admissionRoot)    0x0e1ffc68 BRM1/160
normalizeBuilderTranchesV1(address builder,
 uint64 expectedRegistrationIndex,bytes witness)
 returns (bytes4 magic,uint64 registrationIndex,uint8 closedCount,
```

```
          uint64 registryMutationVersion,bytes32 registryRoot)
                                                              0x5e7c8afe BRN1/160
releaseBuilderTrancheV1(address builder,uint64 expectedRegistrationIndex,
 uint64 window,bytes witness)
 returns (bytes4 magic,uint64 registrationIndex,uint64 window,address builder,
          uint256 creditedAmount,uint64 registryMutationVersion,bytes32 registryRoot)
                                                              0xf8668bb9 BTR1/224
releaseBuilderGenerationV1(address builder,
 uint64 expectedRegistrationIndex,bytes witness)
 returns (bytes4 magic,uint64 registrationIndex,address builder,
          uint256 creditedBond,uint64 admissionVersion,bytes32 admissionRoot)
                                                              0xe7bae370 BGR1/192
claimBuilderLeaseCreditV1(address recipient)
 returns (bytes4 magic,address recipient,uint256 paidAmount) 0x8f73793c BCL1/96
submitBuilderEquivocationV1(bytes evidence)
 returns (bytes4 magic,uint64 registrationIndex,uint64 window,address builder,
          uint256 reporterAmount,uint256 penaltyAmount,
          uint64 admissionVersion,bytes32 admissionRoot)
                                                              0x979c1f72 BEV1/256
```

Normalization derives `currentWindow` in `uint256`. At or below `lastManagedWindow` it uses the ordinary 17-bit sliding-window rule and stores the safely bounded base. Above that bound it rejects unless the exact terminal SWR1 read described above succeeds, then closes every set bit without storing or narrowing the derived window. A malformed, substituted or nonterminal Schedule read reverts before any tranche or root write. The complete event ABI is

```
BuilderRegistered(address indexed builder,uint64 indexed registrationIndex,
 uint8 activeIndex,uint192 baseBondAtomic,uint64 effectiveL2Slot)
BuilderWindowReserved(address indexed builder,uint64 indexed registrationIndex,
 uint64 indexed window)
BuilderExitRequested(address indexed builder,uint64 indexed registrationIndex,
 uint64 requestWindow,uint64 matureWindow,uint64 exitSequence)
BuilderGenerationMoved(address indexed builder,uint64 indexed registrationIndex,
 uint8 indexed formerActiveIndex,uint16 liabilityPosition,uint64 movementSequence)
BuilderTrancheReleased(address indexed builder,uint64 indexed registrationIndex,
 uint64 indexed window,uint256 amount)
BuilderGenerationReleased(address indexed builder,uint64 indexed registrationIndex,
 uint256 baseBondAtomic)
BuilderEquivocationSubmitted(address indexed builder,
 uint64 indexed registrationIndex,uint64 indexed window,address reporter,
 uint256 reporterAmount,uint256 penaltyAmount)
BuilderLeaseCreditClaimed(address indexed owner,address indexed recipient,
 uint256 amount)
```

Events are observations of already-committed state and never substitute for a root, locator, credit or custody write. Exactly one corresponding event is emitted for each successful state-changing result; canonical no-ops emit none. All dynamic tails are minimal canonical ABI tails and their internal parsers consume every byte. The column suffix is selector, magic and exact ABI return length. The unnamed return fields occur in the same order as the prose state above;

all IDs, windows, versions, beneficiaries, amounts and roots are echoed from stored or derived state, and every narrow word has zero high padding. Golden vectors freeze selector, calldata, return, padding and one-field substitutions. A mutation never uses a return from an untrusted external caller as an authority fact.

The contract stores every nonempty tranche leaf preimage by `(registrationIndex,leafIndex)`. Proofs therefore carry siblings, not a caller-supplied old leaf. The three primitive paths are exactly `TranchePath=bytes32[9]` (288 bytes), `RegistryPath=bytes32[6]` (192 bytes), and `AdmissionPath=bytes32[11]` (352 bytes), with siblings bottom-up. A close record is exactly `u64(window)||TranchePath` (296 bytes). Close records are strictly ascending by window; record zero proves against the pre-tranche root and every later record proves against the root produced by its predecessor.

The canonical move witness is

```
u8 closeCount
repeat closeCount in ascending W: u64(W) || bytes32 trancheSibling[9]
bytes32 registrySibling[6]
bytes32 liabilityAdmissionSibling[11] // against pre-root
bytes32 activeAdmissionSibling[11]    // against intermediate R1
```

where `closeCount` equals the complete reservation-bitmap popcount and is at most 17. Its exact length is `897+296*closeCount`. The registry proof anchors the original victim active cell under the pre-registry root and replaces it directly with the caller-independent replacement generation for a full-table registration or canonical empty for maintenance. There is no intermediate registry root containing the victim's post-close tranche root; that root is retained only in the moved liability generation. The liability admission proof anchors the pre-admission root and produces R1; the active proof anchors R1 and produces R2.

Every other dynamic registry tail has exactly one prestate-derived grammar:

- Vacant registration is `RegistryPath||AdmissionPath`, exactly 544 bytes. The two paths prove the derived lowest vacant active cell under the two independent pre-roots. A full-table competitive registration uses the move witness above. `expectedActiveIndex` must equal that lowest vacancy or, when full, the deterministically ranked victim. The registration race guard must equal `nextRegistrationIndex` before any transfer.
- Normalization derives the exact stale bitmap set, namely every set window strictly below the current window. It is `u8(closeCount)||CloseRecord[closeCount]||RegistryPath`, exactly `193+296*closeCount` bytes when nonempty. The canonical no-op is the sole byte `0x00`; it changes no root or version.
- Reservation first closes that same exact stale set. If the target is already the exact live RESERVED window, zero closures use sole byte `0x00`; nonzero closures use the normalization grammar and do not pull a second tranche. Otherwise the tail is `u8(closeCount)||CloseRecord[closeCount]||TranchePath||RegistryPath`, exactly `481+296*closeCount` bytes. The target path proves against the post-close tranche root and the registry path replaces the original cell by the final cell under the pre-registry root.
- Tranche release derives location from the reverse locator. Liability is exactly `TranchePath` (288 bytes); active is `TranchePath||RegistryPath` (480 bytes). The current leaf must be exact LIABLE for the requested window and the ScheduleOracle predicate below must be safe. No location bit or leaf index is supplied.

- Generation release derives location from the reverse locator. LIABILITY release is exactly `AdmissionPath` (352 bytes). It uses the ordinary derived release-window guard when reachable, or the terminal sentinel guard below. Terminal ACTIVE release is exactly `RegistryPath||AdmissionPath` (544 bytes). Either terminal variant additionally exact-authenticates EXPIRED plus the `UINT64_MAX` cursor from `SWR1(lastManagedWindow)`. All variants require the unreleased-tranche count to be zero and `block.timestamp> maximumLiableUntil`; zero is the sole tranche-terminality predicate and is never supplemented by a retained-leaf scan. All tranche leaves are already terminal through authenticated tranche release or slash. Liability release clears its admission leaf; terminal active release clears both its registry and active admission leaves and resolves any exit row. Neither scans the tranche ring, and no caller supplies a location or position.

For every path, the expected registration index must equal the stored reverse locator and all exact byte lengths are checked before a token call or state write. `registryMutationVersion` increments exactly once iff the registry root changes; `admissionVersion` increments exactly once iff the admission root changes. These are per-external-call increments: a maintenance call that performs two through four sequential moves still increments each changed-root version by exactly one, not once per move.

A maintenance tail is the concatenation `u32(moveLength)||MoveWitness` for each move; empty bytes is canonical when no move is eligible or the current window's four-move budget is already exhausted. `maxMoves` is 1–4. A budget-exhausted call returns `moved=0` with empty witness even when an otherwise eligible record exists; it cannot inspect or skip that record. After inspecting at most 64 FIFO records, the contract derives FIFO-matured exits first and then active tombstones in their canonical order. The record count must equal the complete derived prefix until `maxMoves`, the remaining per-window move budget, or no eligible item is reached; a caller cannot omit an eligible derived record. Each record anchors the roots produced by the prior record, every length prefix must equal that record's canonical length, and no suffix is accepted.

The equivocation tail is exactly 2,366 bytes:

```
canonicalPackedSlotChainBlockA[521] || canonicalLowS65SignatureA ||
canonicalPackedSlotChainBlockB[521] || canonicalLowS65SignatureB ||
u16(historicalAdmissionPosition) || bytes32 historicalAdmissionSibling[11] ||
u64(window) || bytes32 currentTrancheSibling[9] ||
bytes32 currentAdmissionSibling[11] || bytes32 currentRegistrySibling[6]
```

Both block encodings use the fixed widths and field order of the signed-header tuple in §5.1. They must have equal `settlementChainId`, `l2ChainId`, `protocolVersion`, verifying contract, slot, context, admission pair and recovered builder but also the same tier, anchor number/hash, force root/cutoff, episode, recovery revision and recovery ID. Define each `hashStruct` as the exact `Keccak256(abi.encode(TYPEHASH,...))` from Section 5.1; these two values must be distinct and occur in increasing byte order. Signature recovery uses the corresponding full `Keccak256(bytes2(0x1901)||domainSeparator||hashStruct)` signing digest. Because the enumerated domain fields are equal, either distinctness test is equivalent, but implementations MUST order and compare `hashStruct`, not choose another field predicate. Their common `contextId` must be nonzero. Parent, body, state, message, data and coinbase fields may differ and are precisely the conflicting block claims. Only tiers 1 and 2 are signed evidence tiers. Tier 1 requires exact zero episode, recovery revision and recovery ID. Tier 2 requires nonzero recovery ID, `contextId=recoveryId`, and nonzero episode and recovery

revision. These are the complete header-local semantic checks. Evidence does not claim that overwritten Settlement context storage is still readable: the same constrained `contextId` is an opaque promise namespace inside the authenticated registered Settlement EIP-712 domain. The two block hashes may happen to be equal only when some other signed claim differs; that is still equivocation, for example two data-manifest promises for one execution header. A builder must therefore never sign two distinct struct digests for the same such namespace and slot, whether or not either claim was eventually submitted or scheduled. Window is derived again from slot.

Before signature recovery the Registry exact-reads, with 100,000 gas, Router's 416-byte `targetReleaseRegistrationV2(protocolVersion)` (RTR2), requires canonical padding and a nonzero registration hash, and requires its returned version and target Settlement to equal the signed protocol version and verifying contract. Revert, OOG, unknown version, short/trailing data or any substitution rejects. Thus arbitrary EIP-712 contracts are not slash domains; every accepted signature domain is an immutable Router-registered release.

The historical admission position is witness data only and must be below 1,136. It reconstructs, rather than supplies, the exact historical leaf as

```
location = historicalAdmissionPosition < 64 ? ACTIVE(1) : LIABILITY(2)
historicalLeaf = H("slot-chain-admission-leaf-v1" ||
 u16(historicalAdmissionPosition) || u8(1) || u8(location) ||
 address20(currentRetainedBuilder) || u192(currentRetainedBond) ||
 u64(currentRetainedRegistrationIndex) ||
 u64(currentRetainedEffectiveL2Slot) || u64(UINT64_MAX))
```

Builder, bond, registration index and effective slot are immutable identity fields read from the retained current generation; the historical tombstone is canonical `UINT64_MAX`, because a signer eligible in that frozen signed context had not yet been tombstoned. It must not copy the current location or current tombstone into the historical leaf. This exact leaf and the eleven siblings prove the generation against the signed admission root, including when a healthy generation moved or a first slash tombstoned it after that root was frozen. The current registration reverse locator independently derives the live ACTIVE or LIABILITY position. Its current admission path proves that exact position against the current pre-admission root, and its exact RESERVED or LIABLE tranche leaf is proven at the derived tranche index. For ACTIVE, the registry path proves the same generation's current cell against the pre-registry root and atomically updates its tranche root and first tombstone; for LIABILITY all six registry sibling words must be zero and the registry root does not change. The current admission path atomically installs the first tombstone at its derived live position; a later per-window slash proves the already-tombstoned leaf and does not postpone it. All proofs and exact lengths are checked before either root or escrow changes. Thus a historical position, current position, or padding word cannot select a generation or redirect a slash.

Migration and release authority. Only new reservations read Router's exact 50,000-gas ASR1 view. ACTIVE is accepted only with zero target tuple; ARMED, READY, malformed data, wrong code, magic, length or padding reject before token movement. Registration, replacement, exit, maintenance, evidence, normalization, release and claims remain live in every Router phase. Existing RESERVED bytes are not rewritten by arm, abort or cutover. Evidence's historical RTR2 authentication is the sole additional Router read. No caller supplies a phase or Settlement address.

ScheduleOracle owns a 268-cell schedule ring, immutable `firstManagedWindow`, derived immutable `lastManagedWindow` and monotonic `nextReleaseWindow`, initialized to the first bound at construction. Reservations and seals outside the inclusive managed interval reject and migration never resets any of these values. Its actual Solidity constructor argument list is exactly the three activation fields followed by every field of `ScheduleOracleConstructorV1`; the exact Schedule-local suffix is:

```
ScheduleOracleConstructorV1(
 uint256 settlementChainId,address protocolVersionManager,
 address activeSettlementRouter,bytes32 routerRuntimeHash,
 bytes32 routerConfigurationHash,address builderRegistry,
 bytes32 builderRegistryRuntimeHash,bytes32 builderRegistryConfigurationHash,
 bytes32 builderRegistryTopologyHash,
 uint64 firstManagedWindow,uint64 genesisTimestamp,
 uint64 evidenceDelaySeconds,uint64 reorgMarginSeconds,
 uint64 beaconGenesisTime,
 uint64 lookaheadSeconds,uint192 leasePerWindowAtomic,
 bytes32 beaconRootsRuntimeHash,uint64 beaconRootsFirstSupportedTimestamp,
 bytes32 historyStorageRuntimeHash,uint64 historyFirstSupportedBlock,
 ScheduleForkConstructorV1 initialFork)

ScheduleForkConstructorV1 =
 (bytes4 forkDigest,uint64 lastParentSlotExclusive,
  address verifier,bytes32 runtimeHash,
  uint64 beaconSlotGindex,uint64 executionPayloadGindex,
  uint64 stateRootGindex,uint64 prevRandaoGindex,
  uint64 timestampGindex,uint64 blockHashGindex,
  bytes32 witnessSchemaHash,bytes32 configurationHash,
  bytes4 selector,uint64 gasLimit)
```

The two system addresses are protocol constants:

```
BEACON_ROOTS_ADDRESS = 0x000F3df6D732807Ef1319fB7B8bB8522d0Beac02
HISTORY_STORAGE_ADDRESS = 0x0000F90827F1C53a10cb7A02335B175320002935
```

Neither is a constructor argument. Every address/hash is nonzero, `settlementChainId=block.chainid`, `leasePerWindowAtomic` equals the BuilderRegistry economic value, and the checked window bound is

```
firstManagedWindow <= lastManagedWindow <= LAST_FULL_SLOT_WINDOW.
```

The initial fork row has `firstParentSlot=0` and an explicit nonzero `lastParentSlotExclusive`; its constants/configuration obey the fork formula below. At both construction and root finalization, checked `targetSlot(currentWindow+FORK_CHANGE_RENEWAL_RUNWAY_WINDOWS)+64<lastParentSlotExclusive` must hold. The runway is 1,811 windows: the ceiling of the 900-second queue-inclusion allowance, seven-day protocol-change delay and one-day fork-change execution window divided by 384 seconds, plus the eight-window early-seal bound. Equality rejects. Thus a follow-up change queued immediately remains executable through its last valid second even when its queue transaction lands at the inclusion bound and

crosses a window boundary, without a sealing outage. The Registry genesis/first-managed values must equal the Schedule values; its evidence/reorg values and independently derived `lastManagedWindow` must also agree; `lookaheadSeconds=768`; and the first managed window start must be numerically at least `lookaheadSeconds` and at least 256 beacon slots after `beaconGenesisTime`; these checked constructor inequalities make its first deadline and target nonnegative. The beacon-roots support boundary must be no later than the first derived query; the historical block boundary is enforced against the authenticated carrier number per call. Every later multiplication/addition is checked before narrowing. The protocol constants are window size 384, ring size 268, expiry batch cap 8, carrier scan 64, assignment cap 76, early-seal bound 8, and configuration/ASR1/SWR1/SWV1/history read gas 50,000. RTR2 and fork-config reads use 100,000 gas, beacon-roots uses 50,000, and SSR1/STS1 share a 100,000-gas limit.

Let `S` be the exact packed concatenation, in constructor order, of all scalar constructor fields above; replace each nested initial-fork value by its fixed-width fields in displayed order, and insert the two fixed system addresses immediately before their runtime/boundary pairs. Append

```
u16(384)||u16(268)||u8(8)||u8(64)||u8(8)||u16(76)||
u32(604800)||u32(86400)||u16(900)||u16(1811)||
u8(12)||u8(32)||u16(256)||u8(64)||u16(8191)||
u16(2048)||u8(32)||u8(65)||u16(600)||u32(117393)||
u32(131072)||u32(280000)||
u64(50000)||u64(50000)||u64(100000)||u64(50000)||u64(50000)||
u64(50000)||u64(50000)||u64(100000)||u64(100000)||
u64(4000000)||u64(50000)
```

where the added constants are, in order, window/ring/expiry/scan/early-seal geometry, assignment cap, protocol-change delay, fork-change execution window, fork-change queue-inclusion allowance, renewal-runway windows, beacon seconds/epoch slots, eight-epoch snapshot lag, carrier finality, history serve window, header-byte and header-field bounds, MPT path/node/aggregate bounds, fork-witness bound and whole seal-witness bound. The final eleven values are configuration, ASR1, RTR2, SWR1, SWV1, history, beacon-roots, fork-config, the shared SSR1/STS1 gas, fork mutation gas and FRS1 read gas. Then

```
scheduleOracleConfigurationHash = H(
 "slot-chain-schedule-oracle-config-v1" || u32(byteLength(S)) || S).
```

The constructor derives this hash; it is never supplied. The universal `componentConfigHashV2()` returns exactly that 32-byte value. The exact peer runtime/configuration hashes, the Builder topology hash and this Schedule configuration hash are constructor-written writerless storage, not Solidity immutable bytecode, under the exact dependency DAG in Section 4.2. The exact zero-argument deployment read is

```
scheduleOracleConfigV1() returns
 (bytes4 magic,uint256 settlementChainId,address protocolVersionManager,
  address activeSettlementRouter,address builderRegistry,
  uint64 firstManagedWindow,uint64 lastManagedWindow,
  uint64 genesisTimestamp,uint64 evidenceDelaySeconds,
  uint64 reorgMarginSeconds,uint64 beaconGenesisTime,
```

```
  uint64 lookaheadSeconds,uint256 leasePerWindowAtomic,
  bytes4 initialForkDigest,uint64 initialForkFirstParentSlot,
  uint64 initialForkLastParentSlotExclusive,
  bytes32 builderRegistryTopologyHash,
  bytes32 configurationHash)
// selector 0xe3d91a33; exact 576 bytes; magic SOC1=0x534f4331
```

Release registration exact-reads this SOC1 view, BRC1 and both components' `componentConfigHashV2()` values. It recomputes `T` from the complete BRC1 topology preimage; compares `E` and `C` to the manifest-pinned calibrated profile and component results; checks both pinned runtime hashes; and rejects any cross-pin, timing-bound, first/last-window, initial-fork boundary, renewal-runway or lease disagreement before publishing the two protocol-lifetime objects. Root finalization separately authenticates the full initial row and both live route mirrors through FRS1/FVR1/SFV1 as specified above; SOC1 is a bounded deployment join, not a second complete Schedule-configuration preimage.

It further exposes

```
scheduleWindowReleaseStateV1(uint64) returns
 (bytes4 magic,uint64 returnedWindow,uint8 state,
  uint64 nextReleaseWindow,bytes32 entryRoot)
// selector 0xf4cd9a5e; exact 160 bytes; magic SWR1
// state UNSEALED=0, SEALED=1, VACANT=2, EXPIRED=3

scheduleWindowV1(uint64) returns
 (bytes4 magic,uint64 returnedWindow,uint8 state,
  bytes32 entryRoot,bytes32 seed)
// selector 0x15fadbaa; exact 160 bytes; magic SWV1=0x53575631

expireScheduleWindowsV1(uint8 maxWindows) returns
 (bytes4 magic,uint8 expired,uint64 nextReleaseWindow)
// selector 0xb1357479; exact 96 bytes; magic SWE1; 1<=maxWindows<=8

finalizeScheduleExpiryV1() returns
 (bytes4 magic,uint64 firstExpiredWindow,uint64 lastManagedWindow,
  uint64 nextReleaseWindow)
// selector 0x774167d9; exact 128 bytes; magic SWT1=0x53575431

sealScheduleWindowV1(uint64 window,bytes4 forkDigest,bytes witness) returns
 (bytes4 magic,uint64 returnedWindow,bytes32 entryRoot,bytes32 seed)
// selector 0x68949048; exact 128 bytes; magic SSW1=0x53535731
```

Seal calldata is the sole canonical ABI encoding: the two narrow fixed words have zero padding, the dynamic offset is exactly `0x60`, the tail is minimal and contiguous, its final word padding is zero, and no suffix follows. The total calldata length is therefore `132+ceil32(byteLength(witness))`; the parser rejects an ABI-equivalent alternative before an external call. The nonempty packed witness is at most 280,000 bytes and has exactly this version-one grammar:

```
u8(1)
```

```
u32(forkWitnessBytes) || bytes[forkWitnessBytes] forkWitness
u16(carrierHeaderBytes) || bytes[carrierHeaderBytes] carrierHeaderRlp
MptPath accountPath
MptPath headerSlotPath
MptPath rootSlotPath
repeat activeIndex=0..63: RegistryCell
repeat each present RegistryCell in ascending activeIndex: TrancheRecord

MptPath = u8(nodeCount) ||
          repeat nodeCount: u16(nodeBytes) || bytes[nodeBytes] canonicalRlpNode
RegistryCell = u8(present) || address20(builder) || u192(bond) ||
               u64(registrationIndex) || u64(effectiveL2Slot) ||
               bytes32(trancheRoot) || u64(tombstonedAtL2Slot) // 101 bytes
TrancheRecord = u64(storedWindow) || u8(state) || u192(amount) ||
                u64(liableUntil) || bytes32 trancheSibling[9] // 329 bytes
```

Here `1<=forkWitnessBytes<=131072`, `1<=carrierHeaderBytes<=2048`, every path has `1<=nodeCount<=65`, every framed path node is one complete canonical RLP list of at most 600 bytes, and the three path encodings together are at most 117,393 bytes (exactly `3*(1+65*(2+600))`), and the carrier header is one complete canonical RLP list with 20–32 fields. The first 20 header fields have the post-Cancun order through `parentBeaconBlockRoot`; appended fields are hashed and must be canonical but are otherwise uninterpreted. Every path traversal must consume every listed node, enforce canonical hex-prefix and embedded-versus-hashed child rules, and end at exactly the requested value; missing, unused or trailing nodes reject. The account key is `keccak256(address20(builderRegistry))`. Its four-field account leaf must have the pinned BuilderRegistry code hash and yields the sole storage root. The two storage keys are `keccak256(bytes32(HEADER_SLOT))` and `keccak256(bytes32(ROOT_SLOT))`; both paths use that same storage root and must end in exactly one canonical nonzero RLP byte string of length 1–32 with no leading zero byte. The verifier left-pads that value to 32 bytes. The decoded HEADER value must already be 32 bytes and start with BRH1; the decoded ROOT value may be shorter before left-padding. Zero, absent, branch-value, non-canonical integer/string or alternative RLP encodings reject. This rule matches Ethereum's leading-zero-stripping storage-trie encoding without making roots whose first byte is zero unprovable.

The 65-node limit is complete for a 64-nibble hashed Ethereum key: every nonterminal branch consumes one nibble, every extension consumes at least one, and at most one terminal leaf remains. Thus the verifier accepts every canonical membership-path depth and rejects 66 before traversal; no probabilistic prefix-grinding assumption is used for proof availability. At a branch, the first 16 direct RLP items must have canonical outer framing and an allowed empty/hash/inline reference type; the seventeenth item must be the canonical empty byte string. This is complete for fixed-length hashed keys, none of which can be a prefix of another, and rejects the unused branch-value representation. The selected inline child must equal the next framed path node, which is fully decoded and semantically validated. An unselected inline child is already authenticated by the current node hash and cannot affect the selected membership statement, so its nested payload is opaque and must not be recursively traversed.

Snapshot evaluation reparses this exact witness byte string; it never accepts caller-supplied framing offsets that could refer to another calldata slice. The ordinal supplies each RegistryCell's leaf index. An absent cell has `present=0`, all remaining 100 bytes zero and no

TrancheRecord. A present cell has `present=1`, nonzero builder and tranche root, bond at least the lease, registration index below the authenticated next index and exactly one following record. The record carries no index: both its leaf index and proof orientation are derived only as `windowmod512`. Thus the derived two-byte index plus its 41-byte supplied leaf payload and nine bottom-up siblings must reconstruct that cell's tranche root. The number of records is derived solely from the 64 presence bits; any missing byte or suffix rejects. States outside 0–5 reject and the EMPTY encoding is canonical `(UINT64_MAX,EMPTY,0,0)`. FREE requires zero amount and deadline; RESERVED/LIABLE require exactly `leasePerWindowAtomic`; and RELEASED/SLASHED require zero amount while preserving their authenticated deadline. Every non-EMPTY record's stored window must lie in `[firstManagedWindow,lastManagedWindow]` and have the same modulo-512 index as the requested window. Only `(window,RESERVED,leasePerWindowAtomic,liableUntil)` is eligible; every other authenticated leaf is ineligible irrespective of caller claims.

Its complete event ABI is

```
ScheduleWindowSealed(uint64 indexed window,bytes32 entryRoot,
 bytes32 seed,bytes4 forkDigest,bytes32 carrierStatementHash)
ScheduleWindowsExpired(uint64 indexed firstWindow,uint8 expired,
 uint64 nextReleaseWindow)
ScheduleExpiryFinalized(uint64 indexed firstExpiredWindow,
 uint64 indexed lastManagedWindow)
ScheduleForkVerifierInstalled(bytes4 indexed forkDigest,
 uint64 indexed firstParentSlot,uint64 lastParentSlotExclusive,
 address verifier,bytes32 runtimeHash,
 bytes32 configurationHash,bytes4 selector,uint64 gasLimit)
SchedulePendingForkVerifierReplaced(bytes4 indexed oldForkDigest,
 bytes4 indexed newForkDigest,uint64 newFirstParentSlot,
 uint64 newLastParentSlotExclusive)
```

A successful nonempty expiry batch emits one event for the whole committed prefix; a zero-length safe stop emits none. Seal and fork-install events are emitted after their corresponding ring/registry state is complete and never serve as authority for a later read. `maxWindows=0` rejects. Expiry advances strictly in W order, so `firstManagedWindow<=W<=lastManagedWindow` and `W<nextReleaseWindow` proves an overwritten managed ring generation expired. A call commits the maximal releasable prefix up to `maxWindows` and stops, without reverting that prefix, at the first authenticated but not-yet-safe window. An external-read authentication failure reverts the whole call. A cursor equal to `UINT64_MAX` is terminal and is never incremented or wrapped. Releasing `lastManagedWindow` sets the cursor directly to that sentinel rather than to the intervening invalid partial-window index; a call reaching it returns its already-completed prefix. At the start of the call Oracle rechecks Router's pinned runtime and configuration, then exact-reads ASR1. For each candidate W it exact-reads RTR2(activeProtocolVersion), requires RTR2's version/target to equal ASR1 and a nonzero registration hash, rechecks the returned target runtime and its exact 50,000-gas `componentConfigHashV2()` against RTR2, and only then calls the active Settlement's

```
settlementScheduleReleaseStateV1(uint64) returns
 (bytes4 magic,uint64 returnedWindow,uint64 protocolVersion,
  uint64 currentL2Slot,uint64 globalMinReferencedSlot,
  uint8 referenceMaskForWindow)
```

```
// selector 0xa4574c77; exact 192 bytes; magic SSR1

settlementScheduleTerminalStateV1(uint64) returns
 (bytes4 magic,uint64 returnedWindow,uint64 protocolVersion,
  uint64 cappedGlobalMinReferencedSlot,uint8 referenceMaskForWindow)
// selector 0x9338be7d; exact 160 bytes; magic STS1=0x53545331
```

with exactly 100,000 gas. The returned version must equal ASR1/RTR2, `windowEndSlot(W)<globalMinReferencedSlot`, and the mask must be zero. Canonical ASR1 ACTIVE, ARMED and READY are all accepted: the returned active Settlement/version is always the authority, while ARMED/READY merely carry a nonzero target tuple. Only a stored SEALED record or an objectively late unsealed VACANT record advances.

SWR1 has one exact root rule. A stored unexpired SEALED cell returns its stored entry root. UNSEALED returns zero. An objectively late unsealed VACANT cell returns the canonical all-empty ranked-entry root. Every `firstManagedWindow<=W<=lastManagedWindow` with either `W<nextReleaseWindow` or terminal `nextReleaseWindow=UINT64_MAX`, including a just-expired cell and an already-pruned managed generation, returns EXPIRED with zero entry root. Every `W<firstManagedWindow` and every `W>lastManagedWindow` returns UNSEALED with zero root and can never authenticate release, including after the cursor reaches its terminal sentinel. BuilderRegistry releases W only after an exact 50,000-gas SWR1 read authenticates EXPIRED and `block.timestamp>evidenceReplayDeadline(W)`. Evidence remains admissible through equality, leaving no boundary gap.

`finalizeScheduleExpiryV1` is the O(1) permissionless terminal path and is valid from any nonterminal cursor position. It records the pre-call cursor as `firstExpiredWindow`, performs the same exact Router, RTR2 and target runtime/configuration authentication, then exact-reads STS1 once for `lastManagedWindow`, and requires

```
windowEndSlot(lastManagedWindow) < cappedGlobalMinReferencedSlot
referenceMaskForWindow == 0
block.timestamp > evidenceReplayDeadline(lastManagedWindow).
```

Window ends and replay deadlines are monotonic in W, so these last-window predicates prove every earlier managed window safe without scanning or assuming prior cursor maintenance. At that timestamp an unsealed last window is necessarily past its earlier seal deadline and is objectively VACANT; no stored-state claim is needed. The call atomically writes `nextReleaseWindow=UINT64_MAX`, emits its single event and returns SWT1 with the echoed bounds and sentinel. An already-terminal call, malformed external read, nonzero reference or failed strict inequality reverts without an event or cursor change. This terminal jump affects only Schedule classification and reclamation: it cannot admit a new window, erase evidence at equality or create a builder credit.

STS1 exists specifically so this terminal proof never acquires SSR1's finite `uint64currentL2Slot` encoding deadline. Settlement computes the same actual global minimum over its dynamic floor, open normal floor, active-round start and stored-best first slots, but returns `min(actualGlobalMinReferencedSlot,UINT64_MAX)` and omits the current slot. Saturation is a conservative lower bound; it cannot turn a false strict inequality into a true one. All retained reference slots and the mask remain exact `uint64` state. STS1 derives and echoes the caller's exact window, requires the caller to be Settlement's release-pinned ScheduleOracle; the Oracle itself supplies and verifies `window=lastManagedWindow`. STS1 uses canonical narrow-word

padding and remains callable after `block.timestamp>genesisTimestamp+UINT64_MAX`. It is read-only, unpausable, contains no callback, and is assigned the same exact 100,000-gas stipend as SSR1.

SWV1 is the candidate/proof authentication view and uses the same state classification at the same block. SEALED returns the stored entry root and seed; VACANT returns the canonical all-empty ranked-entry root and zero seed; UNSEALED and EXPIRED return two zero words. It also echoes the exact window and uses canonical padding. Settlement exact-reads SWV1 with 50,000 gas for every strictly increasing schedule-list row, accepts only SEALED or VACANT, and requires the caller's entry root and seed to equal the return before verifier dispatch. It rechecks the pinned Schedule runtime/configuration before the first read of each candidate batch; a later row fault reverts the entire candidate submission. Thus an event, seal transaction return or caller-supplied seed is never authority, and expiry cannot erase a schedule while Settlement still reports a reference to that window.

Historical registry proof slots. BuilderRegistry exclusively writes the following two raw storage slots in the same transaction as their ordinary mirrors:

```
HEADER_SLOT =
  0xe7ce7a505bf18b9ed57a0785851385323c9487991c55b422861381f92e5c245a
ROOT_SLOT =
  0x4dc6f1bf199f7518c646d40ed35ca04703f646273e6de1d7eace0746cc7100a6
keccak256(HEADER_SLOT) =
  0x3dc336ad17079f9525d2b0708a8773321ab29957d522a6f2d10fc522b7362aec
keccak256(ROOT_SLOT) =
  0x04bafd6333ae949248e59a002ce1fbccb8a4bc8da3a0c3228ae523d727170880
```

The header is the exact big-endian 32 bytes `"BRH1"||u8(1)||u8(activeCount)||u8(registrationExhausted||u8(0)||u64(registryMutationVersion)||u64(admissionVersion)||u64(nextRegistrationIndexLow64`. An ordinary row is canonical only as `registrationExhausted=0` with the complete next index in the final `u64`; an exhausted row is canonical only as `registrationExhausted=1, nextRegistrationIndexLow64=0` and reconstructs the internal `uint256` value `UINT64_MAX+1`. A flag outside `\{0,1\}`, a nonzero reserved byte or any other exhausted flag/tail combination rejects. Thus every pre-exhaustion BRH1 encoding is byte-for-byte unchanged. ROOT_SLOT is the raw registry root. A schedule witness authenticates the pinned registry account and code hash from the fork-verifier-returned parent payload state root, proves both keys against the same account storage root, requires canonical RLP without unused/trailing nodes, checks header reserved bytes, matches present-cell count to `activeCount`, and recomputes the six-level root from all 64 serialized cells. Getter/raw-slot conformance, cell 0/cell 63 orientation and both liability wrap boundaries are deployment tests, not operational assumptions.

4.3 Authenticated snapshot

For aligned window W, let its start timestamp be `GENESIS_TIMESTAMP+384W`. The target is the greatest `BEACON_GENESIS_TIME+12k` not later than eight L1 epochs before that start; L2-genesis alignment is irrelevant. Define `sealDeadline(W)=windowStart-H_LOOK`. A seal is valid only when its carrier execution block has `F_FINAL` L1-block depth and `block.timestamp<sealDeadline(W)`. Beacon slots are not used as a surrogate for execution-block depth. At or after the deadline an unsealed window is permanently all-`VACANT`; a

late transaction cannot change it. Before touching the witness, either system contract, or a fork verifier, the entry point uses checked arithmetic to derive `currentWindow=max(0, block.timestamp-GENESIS_TIMESTAMP)/384` and rejects unless `W>=firstManagedWindow`, `W>=nextReleaseWindow`, `W<=lastManagedWindow`, `currentWindow<=W<=currentWindow+8`, and the strict deadline holds. It also derives `targetSlot(W)` and requires checked `targetSlot(W)+64<latestLastParentSlotExclusive`, where the latter is the latest row's authoritative reviewed-schema horizon and 64 is the maximum carrier-query offset; equality rejects before any system call. This margin protects the fixed EIP-4788/EIP-2935/header grammar even when the authenticated parent remains in an older route at a missed fork boundary. `currentWindow` is recomputed from `block.timestamp` inside every seal, append, extension and replacement call; it is not cached in storage or supplied by the caller. A stored seal for W rejects. A different still-unexpired generation in cell `Wmod268` rejects rather than being overwritten; a generation below `nextReleaseWindow` is already authenticated EXPIRED and may be replaced. At call start Oracle rechecks the pinned BuilderRegistry, EIP-4788 and EIP-2935 runtime hashes and exact-reads the Registry's 50,000-gas one-word topology hash against its constructor pin. That topology hash commits the Registry economic hash and its Schedule address/runtime; the release-time BRC1 check already proved the full preimage. Every subsequent external operation is a bounded STATICCALL. No ring write or event occurs until all checks and all root computations below complete. The contract performs this procedure:

1. The contract itself, not caller data, queries EIP-4788 for $j = 1, \ldots, 64$ at `q=targetTimestamp+12j`; missing timestamps revert and are skipped. The base is the aligned `targetTimestamp=BEACON_GENESIS_TIME+12*targetSlot(W)`, not the raw possibly unaligned L2-window expression. The first successful query identifies the *carrier* execution block. EIP-4788 returns its parent beacon-block root, not the carrier root.
2. For the first successful query the caller supplies the carrier execution header. EIP-2935 authenticates its hash and block number; the header timestamp equals q and its `parent_beacon_block_root` equals the EIP-4788 result. Fork-specific SSZ branches in that parent authenticate its slot and one execution payload's `blockHash`, `stateRoot`, `prevRandao` and timestamp. The parent slot must be at or before target, its payload timestamp must equal its beacon-slot time, and its payload block hash must equal the carrier header's parent hash. No carrier observed by all 64 contract calls, a malformed header, a later parent, or a fork/gindex mismatch reverts without recording state. Before the deadline only a completely valid seal writes; adversarial query timing or witness bytes can never manufacture a vacant seal. At or after the deadline, consumers deterministically synthesize permanent all-`VACANT` only from the objective absence of a stored valid seal.
3. Verify Merkle–Patricia storage proofs from that authenticated parent-payload state root for the registry header and `registryRoot`. The caller supplies all 64 canonical serialized cells, including empty cells; recompute their six-level Keccak tree and require equality to `registryRoot`. Filter eligibility and order the entries, then store the byte-exact 64-leaf `entryRoot` and seed. Every *present* active cell supplies its tranche leaf and nine-sibling proof at index `Wmod512`. An absent registry cell has `trancheRoot=0`, supplies no tranche proof and is synthesized ineligible. For a present cell, canonical `EMPTY`, `FREE`, or a nonempty leaf for a different stored window is valid but ineligible; only exact `RESERVED(W)` with `amount=L_LEASE` is eligible. Proofs occur in ascending cell-index order, and extra/missing proofs reject. The fixed cell count proves completeness with one registry-root storage path rather than 64 independent storage paths. The carrier

satisfies `currentExecutionBlock-carrierBlockNumber>=F_FINAL`; the source payload is necessarily older.

`sealScheduleWindowV1` is permissionless. An optional maintenance distributor may pay its event in a later transaction. At or after its deadline a missing stored seal is consumed as `VACANT`; a failing pre-deadline seal attempt never stores that result and cannot give an address an unauthorized ticket. The snapshot age plus 64-slot carrier scan, finality and seal margin fits the eight-epoch geometry and remains far below EIP-4788's 8,191-slot history window. Consensus constants include the Deneb SSZ generalized indices. Every route in this V1 Schedule epoch authenticates `BeaconBlock.slot` at generalized index exactly 8. An L1 fork that moves that field requires a new schedule epoch, not another verifier row and never runtime guessing.

For Deneb, Electra and Fulu, the generalized index from `BeaconBlock` root to `slot` is 8 and to `body.execution_payload` is 201. Within the composed tree, payload `state_root`, `prev_randao`, `timestamp` and `block_hash` are 6,434, 6,437, 6,441 and 6,444 respectively. The seal input names a protocol-local route identifier in the historically named `forkDigest` field; only configured identifiers are accepted. It is not the Ethereum consensus fork digest and is never inferred from one. The immutable review bundle names the actual consensus fork version/digest and the exact parent/carrier schemas to which the route applies. A fresh route ID may therefore repair a verifier inside the same consensus fork without lying about consensus identity. The protocol-lifetime ScheduleOracle holds a parent-slot-routed `forkDigest->(firstParentSlot,lastParentSlotExclusive,verifier,codeHash,configHash,gasLimit)` registry controlled only by a delayed ProtocolVersionManager manifest. Seal dispatch uses a bounded `STATICCALL`, exact return length and rechecked `EXTCODEHASH`; failure reverts and leaves the window unsealed; it cannot mutate an old seal. Gloas or any fork that preserves the V1 slot leaf but changes the remainder of the container requires a reviewed verifier, constants, fixtures and delayed registration before its activation. That registration may change witness parsing only: it cannot change the protocol-lifetime seed, eligibility, allocation or placement rules.

For both the constructor-installed row and every later installation, "reviewed verifier" is an enforceable release property. The verifier is an immutable non-proxy artifact with no `DELEGATECALL`, beacon, replaceable implementation/target, target setter, arbitrary call, selfdestruct or mutable output-affecting dependency. Release tooling closes the complete reachable code graph for every CALL/STATICCALL/CREATE dependency that can influence SFV1/SFC1, pins every runtime and constructor configuration transitively, and rejects a proxy-shell-only identity even when its shell EXTCODEHASH and getter remain stable. The initial root release certificate and every delayed change package record that closed graph; production tooling refuses to build the root manifest or queue REGISTER_FORK_VERIFIER unless the certificate matches the row's runtime/configuration identity. PVM enforces the queued row words. The graph-closure judgment is necessarily an external release/governance validity rule, not a commitment hidden in the fixed row or something EVM introspection can prove. ScheduleOracle's per-call EXTCODEHASH/SFV1 checks are defense in depth, not permission for mutable reachable code. An artifact that cannot satisfy this closure is not registrable in a production release.

The fork registry and verifier wire format are exact. The deployment constructors of Schedule Oracle and PVM each receive and independently decode, validate and store the same canonical 480-byte reviewed current-fork row; one contract never initializes its mirror by reading or copying the other's storage. Each constructor installs that interval

`[0,lastParentSlotExclusive)`, inserts its digest into the permanent used-digest tombstone set, and rejects empty or prepopulated route state. There is therefore no deployed state in which the route registry is empty. Factory's pre-activation FRS1/FVR1/SFV1 join described in the root section is the on-chain proof that those independent constructor results agree before the root becomes usable. Every later interval is contiguous with its predecessor. The typed install operation is:

```
installForkVerifierV1(
 bytes4 forkDigest,uint64 firstParentSlot,uint64 lastParentSlotExclusive,
 address verifier,bytes32 runtimeHash,
 uint64 beaconSlotGindex,uint64 executionPayloadGindex,
 uint64 stateRootGindex,uint64 prevRandaoGindex,uint64 timestampGindex,
 uint64 blockHashGindex,bytes32 witnessSchemaHash,
 bytes32 configurationHash,bytes4 selector,uint64 gasLimit)
 returns (bytes4 magic,bytes4 returnedForkDigest,
          uint64 returnedFirstParentSlot)
forkVerifierRegistrationV1(bytes4 forkDigest) returns
 (bytes4 magic,bytes4 returnedForkDigest,uint64 firstParentSlot,
  bytes4 successorForkDigest,uint64 lastParentSlotExclusive,address verifier,
  bytes32 runtimeHash,bytes32 configurationHash,bytes4 selector,
  uint64 gasLimit)
scheduleForkVerifierConfigV1() returns
 (bytes4 magic,bytes4 forkDigest,uint64 beaconSlotGindex,
  uint64 executionPayloadGindex,uint64 stateRootGindex,
  uint64 prevRandaoGindex,uint64 timestampGindex,uint64 blockHashGindex,
  bytes32 witnessSchemaHash,bytes32 configurationHash)
scheduleForkRouteStateV1() returns
 (bytes4 magic,bytes32 routeStateHash,uint64 orderedRouteCount,
  uint64 mappedRouteCount,uint64 usedDigestCount)
```

The 15-word row is Oracle's exact install input and persistent registration. The delayed `REGISTER_FORK_VERIFIER` operation payload is instead the exact 576-byte concatenation `row||review`, where

```
ForkReviewEnvelopeV1 =
 (uint64 reviewedThroughParentSlotExclusive,
  bytes32 reviewEvidenceHash,bytes32 reviewCertificateHash)
reviewCertificateHash = H(
 "slot-chain-schedule-fork-review-certificate-v1" ||
 u256(settlementChainId) || address20(scheduleOracle) || u8(operationKind) ||
 u16(changeRowsBytes) || H(changeRows) ||
 u64(reviewedThroughParentSlotExclusive) || reviewEvidenceHash)
```

`reviewEvidenceHash` is the nonzero content hash of the immutable review bundle. PVM recomputes the certificate, requires the reviewed-through value to cover the new latest horizon, and the operation ID commits all 576 bytes. The selectors are `0x9bb6fe73`, `0xc614591c` and `0x44efa773`; the route-state selector is `0x7e9f3c0d`. Return lengths/magics are 96/FVI1 `0x46564931`, 320/FVR1 `0x46565231` and 320/SFV1 `0x53465631`, plus 160/FRS1 `0x46525331`. Install, replacement and split are nonpayable PVM-only calls made with exactly 4,000,000 gas.

Oracle authorization is exactly `msg.sender` equal to the constructor-stored PVM address. Oracle neither calls back to PVM nor reads PVM lifecycle, operation or route storage. The root-pinned PVM exposes no generic call surface, so its immutable caller identity cannot be used to invoke an unapproved Oracle transition. FRS1 and SOC1 are exact 50,000-gas STATICCALLs. FVR1 is an exact 100,000-gas STATICCALL whose calldata is the 4-byte selector, the queried `bytes4` digest and 28 zero bytes. SFV1 is an exact selector-only 100,000-gas STATICCALL. Every call has zero value and requires its exact return length, magic and canonical ABI padding.

Before any fork mutation, PVM performs one ordered pretransition join: it reads SOC1, then FRS1, then each affected prestate FVR1 in oldest-to-newest route order, then the corresponding live verifier EXTCODEHASH/SFV1 pairs in that same order. It compares SOC1 to its immutable chain, Oracle, initial-fork and root-pinned configuration commitments; it compares FRS1 and every FVR1 byte for byte to its independently maintained route. Only after that join does it validate the candidate verifier and execute the Oracle mutation. A failed, short, dirty, reverted or out-of-gas observation reverts before the Oracle call. PVM and Oracle independently check runtime, getter, configuration formula, `beaconSlotGindex=8`, and the checked lead

```
targetSlot(currentWindow + MAX_EARLY_SEAL_WINDOWS)
 + MAX_SCHEDULE_CARRIER_SCAN_SLOTS < new.firstParentSlot
```

with equality rejecting, and requires `100000<=gasLimit<=5000000`. An append uses a globally unused digest and requires `new.firstParentSlot=latest.lastParentSlotExclusive`; it sets the old latest successor digest and appends the nonempty new interval. A same-row horizon extension uses this same 15-word function and selector: its digest, first slot, verifier, runtime, all gindices, schema/configuration, selector and gas are byte-identical to the latest row, its last slot strictly increases, and the stronger execution-time checks require both `targetSlot(currentWindow+8)+64<old.lastParentSlotExclusive` and `targetSlot(currentWindow+1811)+64<new.lastParentSlotExclusive`. An append additionally requires the latter protected target to remain below its new first slot, preserving a full correction runway. No other REGISTER operation replaces, shortens or cancels a row.

PVM and Oracle execute separate implementations of the descriptor, timing, interval and tombstone checks. Each independently reads the verifier address's live `EXTCODEHASH` and exact 320-byte SFV1 return; neither derives that observation from an Oracle-stored registration row. Their post-transition join exact-reads every changed FVR1 followed by FRS1. PVM reconstructs this poststate locally before comparison; it never introspects Oracle storage. Define

```
orderPrefix[0] = H("slot-chain-schedule-fork-order-empty-v1")
orderPrefix[i+1] = H("slot-chain-schedule-fork-order-step-v1" ||
                     orderPrefix[i] || u64(i) || forkDigest[i])
registrationLeaf[d] = H("slot-chain-schedule-fork-registration-smt-v1" ||
                        0x02 || d || registrationHash[d])
usedLeaf[d] = H("slot-chain-schedule-fork-used-smt-v1" || 0x02 || d ||
                H("slot-chain-schedule-fork-used-leaf-v1" || d))
routeStateHash = H(
 "slot-chain-schedule-fork-route-state-v1" ||
 orderPrefix[orderedRouteCount] || registrationRoot || usedRoot ||
 u64(orderedRouteCount) || u64(mappedRouteCount) || u64(usedDigestCount))
```

The two roots are independent fixed-depth 32-level sparse Merkle trees keyed by the big-

endian `bytes4` digest. At level zero the absent hash is `H(domain||0x00)`; the absent hash at every next level is `H(domain||0x01||child||child)`. A branch node is `H(domain||0x01||left||right)`. Each contract stores nondefault nodes by `(level,prefix)` and updates exactly 32 sibling levels per changed leaf; it never enumerates or sorts a mapping. The ordered prefix hash is stored per route index. A bounded `indexByDigest` reverse map makes every FVR1 successor read constant-time; no getter scans the order. Append and split write the new index, while replacement deletes the old latest index when its ID changes and writes the replacement at that same index. Append advances one prefix step, replacement recomputes only the last step from its stored predecessor prefix, and split advances one step. Therefore FRS1 and FVR1 are constant five-/ten-word reads and append, extension, replacement or split performs a fixed maximum number of hashes and storage writes independent of protocol age. The mapping root includes rows not reachable through the order, so a retained replaced mapping changes it. The used root is append-only and includes every constructor, append, replacement and split digest. PVM reconstructs its own poststate and requires exact FRS1 equality after every transition; replacement therefore proves the rewritten predecessor, old-map absence and permanent old-digest tombstoning. It exact-reads every FVR1 whose row or successor changed, so a wrong reverse index or successor cache cannot survive a transition. Any mismatch reverts both authorities to their pre-operation state.

The 320-byte FVR1 words are exactly magic, digest, first parent slot, successor digest, last parent slot exclusive, verifier, runtime hash, configuration hash, selector and gas. A latest row has zero successor digest but retains its finite nonzero last slot, which is the authoritative global `latestLastParentSlotExclusive`. A nonlatest row's last slot equals its successor's first. Each initial, appended, extended or replacement last slot is no later than the earliest not-yet-reviewed consensus-schema boundary attested by the release/change certificate. A certificate may conservatively choose an earlier finite horizon. If governance does not extend it in time, targets whose checked 64-slot carrier margin reaches it fail closed and the ordinary post-deadline unsigned/VACANT escape remains available; no unreviewed verifier is guessed.

A queued but still unreachable latest noninitial row has one narrow correction path. Protocol operation `REPLACE_PENDING_FORK_VERIFIER=5` carries exactly 1,536 bytes: the canonical current predecessor, current-old latest and replacement 15-word rows, followed by the same 96-byte review envelope. For this operation `changeRows` is the exact 1,440-byte predecessor/old/replacement concatenation. Define

```
registrationHash = H("slot-chain-schedule-fork-registration-v1" ||
                     u16(480) || exact480ByteRow)
replacePendingForkVerifierV1(bytes32 expectedPredecessorRegistrationHash,
 bytes32 expectedOldRegistrationHash,
 <the 15 new-row fields>)
 returns (bytes4 magic,bytes4 oldDigest,bytes4 newDigest,
          uint64 newFirstParentSlot,uint64 newLastParentSlotExclusive)
// selector 0xb48bbef1; exact 160 bytes; magic FVP1
```

PVM exact-decodes all three rows, proves the predecessor and old row reconstruct the current last two routes through FVR1, proves the old verifier through SFV1, and passes both registration hashes. Oracle independently reconstructs and hashes the same predecessor and full latest row. The latest must be noninitial and have no successor, and `targetSlot(currentWindow+8)+64<min(old.firstParentSlot,new.firstParentSlot)`; equality rejects, so no carrier reach-

able by an already early-sealable window is mutable. The first-slot test deliberately does not repeat the original append runway: by execution, the queued change has already consumed that delay. The strict comparison using `currentWindow+1811` applies only to `new.lastParentSlotExclusive`, so a replacement made at the last valid execution second still leaves one full future correction/renewal runway. The new interval is nonempty, its first slot is strictly after the predecessor's first, and the transaction atomically changes the predecessor's last/successor, latest row and digest mapping, and the authoritative horizon. The recomputed delayed change certificate commits the exact replacement boundary, new reviewed-through horizon, predecessor/old/new rows and code/config identities; its evidence must prove both the rewritten predecessor interval and the new interval remain within reviewed parent and carrier/header schemas. The new digest is either the old digest or globally unused. Every digest ever installed is permanently tombstoned: when a changed old digest becomes unknown it can never be reused. Exact-old mismatch or any failed post-read rolls the whole operation back. There is no general replacement and no cancellation.

An overlong reviewed horizon on the current latest row, including the initial row, has a separate forward-only repair. Protocol operation `SPLIT_LATEST_FORK_VERIFIER=6` carries exactly 1,056 bytes `exactOldLatestRow||exactSuccessorRow||review`. Its `changeRows` is the exact 960-byte two-row prefix. Define

```
splitLatestForkVerifierV1(bytes32 expectedOldRegistrationHash,
 <the 15 successor-row fields>)
 returns (bytes4 magic,bytes4 oldDigest,bytes4 successorDigest,
          uint64 boundaryParentSlot,uint64 successorLastParentSlotExclusive)
// selector 0x455b4ff5; exact 160 bytes; magic FVS1
```

Both authorities require the supplied old row to equal the current latest, the fresh successor route ID to be globally unused and different, and `old.firstParentSlot<successor.firstParentSlot<old.lastParentSlotExclusive`. Both strict-check `targetSlot(currentWindow+8)+64` below the successor first slot, and `targetSlot(currentWindow+1811)+64` below the successor last slot. Thus the boundary stays outside every already early-sealable carrier range without charging the original governance delay twice, while the new latest retains a full future correction/renewal runway. The operation atomically shortens the old latest to the successor's first slot, appends the successor, updates both registration paths, the used path, reverse index and order prefix, and exact-postreads both FVR1 rows plus FRS1 after the candidate's pre-mutation SFV1 validation. It cannot rewrite any parent reachable by the full correction runway, cannot lengthen the old route, cannot reuse an identifier, and cannot change any earlier interval. This is the only way to shorten the current latest horizon; it prevents a mistaken early horizon extension from becoming irreversible while preserving monotone history.

The exact configuration layers are

```
forkConstantsHash = H(
 "slot-chain-schedule-fork-constants-v1" || u64(beaconSlotGindex) ||
 u64(executionPayloadGindex) || u64(stateRootGindex) ||
 u64(prevRandaoGindex) || u64(timestampGindex) || u64(blockHashGindex))
scheduleForkOutputSchemaHash = H(
 "ScheduleCarrierOutputV1(bytes32 statementHash,uint64 parentSlot,"
 "uint64 parentExecutionBlockNumber,uint64 payloadTimestamp,bytes32 blockHash,"
```

```
 "bytes32 stateRoot,bytes32 prevRandao)")
configurationHash = H(
 "slot-chain-schedule-fork-verifier-config-v1" || forkDigest ||
 forkConstantsHash || witnessSchemaHash || scheduleForkOutputSchemaHash ||
 bytes4(0x7e981e0b) || u64(gasLimit))
```

For Deneb, Electra and Fulu the six gindices are exactly 8, 201, 6,434, 6,437, 6,441 and 6,444 in that order. A later fork supplies its reviewed constants and witness schema through a new delayed row; an opaque configuration hash is not accepted.

The initial row's fork witness is not opaque. Its exact current-fork schema is

```
ScheduleSszMultiproofV1 =
 u64(window) || u64(parentSlot) || u64(parentExecutionBlockNumber) ||
 u64(payloadTimestamp) || bytes32(blockHash) || bytes32(stateRoot) ||
 bytes32(prevRandao) || bytes32 helperNode[17]
```

and is exactly 672 bytes. The four uint64 values in this packed outer format are big-endian. For the two SSZ-proven values, parent slot and payload timestamp, the verifier reconstructs the canonical little-endian uint64 followed by 24 zero bytes; window and carrier number remain context fields. The 17 helper generalized indices, in the only accepted descending order, are

```
6445, 6440, 6436, 6435, 3223, 3221, 3219, 3216,
403, 200, 101, 51, 24, 13, 9, 7, 5
```

and are derived from the five proved leaves under the displayed fork constants, not supplied in the witness. The four execution-payload leaves must all be descendants of gindex 201. The exact initial `witnessSchemaHash` is

```
H("slot-chain-schedule-ssz-multiproof-v1" || forkConstantsHash ||
  u32(672) || u16(17) ||
  u64(6445)||u64(6440)||u64(6436)||u64(6435)||u64(3223)||
  u64(3221)||u64(3219)||u64(3216)||u64(403)||u64(200)||
  u64(101)||u64(51)||u64(24)||u64(13)||u64(9)||u64(7)||u64(5))
```

which is `0x4d3c1d3a7f2921c2f8e7526a2e8aa2c47dc6bdcd3dc9e2b14c58f2b9d39ed30f`. Verification uses SSZ `hash_tree_root` binary compression exactly as `SHA256(leftChild||rightChild)`. The verifier seeds the five fixed target gindices with the canonical chunks above and the 17 fixed helper gindices with their witness words. A repeated, missing or ancestor/descendant-overlapping position rejects. It repeatedly combines a complete sibling pair exactly once, taking the even gindex as the left child and the odd gindex as the right, and rejects an already-present or previously derived parent. Verification succeeds only when every supplied position has been consumed into the sole gindex-1 root and that root equals the exact nonzero EIP-4788 word. Keccak, reversed child order, a separately supplied helper index and an unconsumed node are invalid. The constructor rejects any different initial value. A future delayed row may install a different reviewed schema hash and verifier only at its protected parent-slot boundary; it cannot reinterpret a current-fork row. Every V1 schema must still prove `BeaconBlock.slot` at gindex 8.

ScheduleOracle obtains the beacon root through the exact contract-selected EIP-4788 path; the root itself is never a witness field. Each query is exactly the 32-byte timestamp, 50,000 gas

and zero value. Revert means absent and is the only result that advances the scan; a successful call must return exactly one nonzero word. Oracle chooses the first success, so no caller supplies a query index or timestamp. For target T satisfying the checked $T + 64 < H$ reviewed-horizon margin, the contract-selected first successful carrier authenticates the canonical parent beacon root. The reviewed verifier's mandatory slot-at-gindex-8 proof yields its authenticated parent slot P, rather than a merely asserted slot, and the timestamp join below proves $P \leq T$. Therefore $P < H$ after the selected route's reviewed proof and independently of the caller-selected digest. Before the call a caller-selected row may be rejected when `row.firstParentSlot>targetSlot`; Oracle must not compare the row's upper bound with targetSlot, because a missed boundary slot may legitimately leave the canonical parent in the predecessor interval. It then rechecks the selected verifier's code and exact SFV1 configuration before STATICCALLing it using only

```
verifyScheduleCarrierV1(bytes witness,bytes32 beaconBlockRoot)
 returns (bytes4 magic,bytes32 statementHash,uint64 parentSlot,
          uint64 parentExecutionBlockNumber,uint64 payloadTimestamp,
          bytes32 blockHash,bytes32 stateRoot,bytes32 prevRandao)
```

Selector is `0x7e981e0b`, return is exactly 256 bytes and magic is SFC1 `0x53464331`. Calldata has selector, a two-word head with bytes offset `0x40`, one minimal contiguous witness tail, zero padding and no suffix; witness length is nonzero and bounded by the row's reviewed schema and `gasLimit`, with an absolute 131,072-byte protocol cap. Immediately after authenticating exact length and SFC1, before accepting any other returned field or performing a write, Oracle requires `row.firstParentSlot<=parentSlot<row.lastParentSlotExclusive` and `parentSlot<=targetSlot`. Both endpoints are exact; a boundary parent uses its successor row, while a missed-slot predecessor remains provable with the predecessor row. Oracle then recomputes

```
statementHash = H(
 "slot-chain-schedule-carrier-statement-v1" || u256(settlementChainId) ||
 address20(scheduleOracle) || forkDigest || u64(W) || beaconBlockRoot ||
 u64(parentSlot) || u64(parentExecutionBlockNumber) || u64(payloadTimestamp) ||
 blockHash || stateRoot || prevRandao)
```

from the six returned payload fields and requires exact equality before using any leaf. Every reviewed fork-witness schema binds the outer `u64(W)`; the verifier uses that bound value, `block.chainid`, `msg.sender` and its configured protocol-local route ID when constructing the statement. Oracle's independent recomputation with the public W makes a substituted inner value fail. The returned `parentExecutionBlockNumber` means the parent execution-payload block number; it is not claimed to be one of the six SSZ leaves. Header field 8 is the carrier execution-block number. Oracle requires both minimally encoded values to fit `uint64` and requires checked `parentExecutionBlockNumber+1=header.blockNumber`; a parent value of `UINT64_MAX`, equality or any other off-by-one rejects. Field 0 is the exact 32-byte parent hash, field 11 is the minimally encoded uint64 query timestamp, and field 19 is the exact 32-byte parent-beacon root; wrong widths reject. The number must lie in `[max(historyFirstSupportedBlock,block.number-8191),block.number-1]`. Oracle authenticates `keccak256(carrierHeaderRlp)` with the exact EIP-2935 read, and checks `block.number-header.blockNumber>=64`. The returned payload block hash must equal the header parent hash; the returned parent payload timestamp must equal `BEACON_GENESIS_`

`TIME+12*parentSlot`, with `parentSlot<=targetSlot`. The header timestamp is the contract-selected query timestamp and its parent-beacon-root field is the EIP-4788 word. Oracle requires nonzero returned block hash, parent state root and prevRandao, and uses only the returned parent payload state root and prevRandao after all these joins hold. Revert, OOG, wrong interval/code/config/magic/hash, short/trailing returndata, bad padding or inconsistent header reverts with no write; only a consumer at or after `sealDeadline(W)` interprets the still-unsealed window as VACANT, while a valid pre-deadline retry remains possible.

The parent payload timestamp derives `sourceL2Slot=max(0,payloadTimestamp-GENESIS_TIMESTAMP)`. After the MPT and all 64 tranche proofs verify, Oracle keeps exactly those present cells with `effectiveL2Slot<=sourceL2Slot<tombstonedAtL2Slot` whose tranche is the exact eligible RESERVED tuple. The live tombstone sentinel `UINT64_MAX` obeys the same strict inequality; it is not a special-case bypass at the uint64 boundary. Oracle sorts them by decreasing bond then increasing registration index, hashes these entries at ranks 0 upward and fills every remaining rank through 63 with its position-bound canonical empty leaf. It derives `seed_W` from the authenticated prevRandao, writes `(W,SEALED,entryRoot,seed)` once, emits the one seal event including the verified statement hash, and returns SSW1. Duplicate builders or registration indices, a present registration index not below BRH1's next index, an active count different from BRH1, a recomputed 64-cell root mismatch or a noncanonical empty rank reject before that write.

4.4 Consensus byte encoding and allocation

All hashes use Ethereum legacy Keccak-256. Concatenation below is packed fixed width; strings are the literal ASCII bytes without a length prefix:

$$
\begin{aligned}
seed_W &= H(\texttt{"slot-chain-seed-v2"} \| u256(settlementChainId) \\
&\quad \| u256(W) \| bytes32(prevRandao)), \\
qTie_i &= H(\texttt{"slot-chain-quota-tie-v1"} \| seed_W \| u64(regIndex_i)), \\
sTie_{p,i} &= H(\texttt{"slot-chain-slot-order-v1"} \| seed_W \| u16(p) \| address20(i)).
\end{aligned}
$$

Starting from zero, capped D'Hondt awards each of 384 tickets to the unsaturated entry maximizing `bond/(allocation+1)`, with `Q_MAX=76`. Quotients are compared by cross multiplication; a `uint192` bond times `Q_MAX+1` cannot overflow `uint256`. Equal quotients use smallest `qTie`. Remaining tickets are `VACANT`; six addresses are required to fill a window.

Placement processes positions in order within one window. Position zero has no predecessor; there is no dependency on the prior window. At every later position a real address is feasible if it has a ticket remaining, differs from the immediately preceding position when that position is real, and removing it preserves `remaining[a]<=otherRemaining+1` for every real address. Choose the feasible value with smallest `sTie`; `VACANT` may repeat. Consequently one registration may occupy both slot 383 of W and slot 0 of $W+1$, but its global self-run is at most two. This boundary case is not a safety or liveness assumption: different registered keys may collude at any adjacent positions, every accepted block remains execution-proved, and separate windows retain separate equivocation tranches. The executable model and its golden vectors are normative test fixtures.

The seed and allocation/placement algorithm are protocol-lifetime scheduling rules, not release-profile rules. In particular, neither a seal nor its seed contains a Settlement protocol

version. A sealed future window therefore has identical meaning before and after a Settlement migration, matching the protocol-lifetime ScheduleOracle and retained reservations. An incompatible scheduling change cannot ride an ordinary profile migration: it requires a separately specified schedule-epoch transition that waits until every old-epoch seal and liability is outside the retention horizon. This document defines no such transition.

5 Blocks, promises and proof statement

5.1 Signed header

A builder signs exactly:

```
(settlementChainId, l2ChainId, protocolVersion, verifyingContract,
 slot, parentHash,
blockHash, stateRoot, bodyRoot, anchorNumber, anchorHash,
forceRoot, forceCutoff, messageStart, messageEnd, dataManifestRoot,
coinbase, tier, contextId, admissionVersion, admissionRoot,
 episode, recoveryRevision, recoveryId)
```

The normative EIP-712 type string is:

```
SlotChainBlock(
 uint256 settlementChainId,uint256 l2ChainId,uint256 protocolVersion,
 address verifyingContract,
 uint64 slot,bytes32 parentHash,bytes32 blockHash,bytes32 stateRoot,
 bytes32 bodyRoot,uint64 anchorNumber,bytes32 anchorHash,bytes32 forceRoot,
 uint64 forceCutoff,uint64 messageStart,uint64 messageEnd,
 bytes32 dataManifestRoot,address coinbase,uint8 tier,
 bytes32 contextId,uint64 admissionVersion,bytes32 admissionRoot,
 uint64 episode,
 uint64 recoveryRevision,
 bytes32 recoveryId)
```

The hashed ASCII has one space between each field type and field name and no other whitespace; displayed line breaks, indentation and spaces after commas are typography only. The exact single-line literal and a standalone Keccak implementation are in `commitment-model.py`. `TYPEHASH` is legacy Keccak-256 of those ASCII bytes and equals the concatenation of these two fragments:

```
0xee6a8c8e31e8245cd527869508f6e464
 d60848939912038 76f734d1855aed87c
```

The EIP-712 domain is `("SlotChain","2",settlementChainId,verifyingContract)`. The explicit `l2ChainId` is the EVM `CHAINID` used by raw L2 transactions. Signatures are exactly 65-byte `r||s||v`; `v` is 27 or 28, s is low, and zero or recovered-zero addresses reject. EIP-2098 and EIP-155 transaction-style `v` values are not accepted. Tier 1 requires

```
contextId = H("slot-chain-normal-context-v1" || baseCanonicalHash ||
              u64(admissionVersion) || admissionRoot ||
              u64(armBlockNumber) || armBlockHash)
```

and its execution anchor is exactly that arm block. `episode=recoveryRevision=0` and `recoveryId=0`; tier 2 requires the exact live round values and `contextId=recoveryId`. The unsigned tier 3 uses that same recovery context. Consequently semantically unused bytes cannot create false equivocation evidence. The execution `blockHash` excludes the off-chain builder signature, avoiding a circular hash definition. The signed `bodyRoot` covers only canonical discretionary transaction bytes; the proof derives the full Ethereum transactions root after adding the profile-defined system and valid forced transactions.

5.2 Three validity tiers

For every tier and every block, the circuit requires the EVM header `timestamp=GENESIS_TIMESTAMP+slot` with checked addition, and derives `blockHash` from that exact header. The signed relative slot can therefore never disagree with EVM time, forced-expiry classification, or the canonical tip clock. Every signed block also carries the exact frozen `admissionRoot`; the circuit checks it together with `admissionVersion`.

1. **Normal signed.** Every block is signed by its scheduled builder under the normal window's frozen `(admissionVersion,admissionRoot)`. Slots strictly increase, each slot is no later than `currentL2Slot+CLOCK_SKEW`, and the first slot strictly exceeds the canonical base slot. Every block is at or above the window's frozen `minAdmissibleSlot`; parent gaps are at most `G_MAX=3,600`, equal to the objective final-lag trigger and within the 12-window schedule-proof cap.
2. **Recovery signed.** The first block extends the L1-final canonical tuple, binds the active episode and current recovery revision, and may cross an unbounded time gap. Every block still requires its scheduled builder signature under the current recovery round's frozen admission pair. Every slot is at or after that round's `roundStartSlot`.
3. **Escape unsigned.** Exactly one deterministic block extends the L1-final canonical tuple during recovery. It has no builder signature, no beneficiary-controlled coinbase, no discretionary transaction and no blob body. It contains only the protocol anchor and the deterministic maximum prefix of the current round's frozen forced queue. When the queue is empty an SLA-triggered episode may commit an empty anchor-only escape block. Its slot is `max(roundStartSlot+ESCAPE_OFFSET,canonical.tipSlot+1)` and its anchor, queue root/cutoff, block hash and state root are unique functions of the current recovery round and proved execution result.

Every candidate uses exactly one `batchAnchor`; every block repeats that anchor and frozen `forceRoot/forceCutoff`. For a normal candidate the supplied RLP execution header hashes to the EIP-2935 value, its timestamp is no later than the first L2 block timestamp, and MPT proofs under its state root authenticate the historical forced queue and any other contract state. For recovery, the RLP header authenticates the stored parent number/hash and derives its timestamp and state root; the queue pair is authenticated directly by the round fields and is not asserted against that header's state root. Requiring one anchor, not up to 4,096 independent anchors, bounds both circuit and L1 work. Cutoffs therefore do not change within a candidate; the first blocks drain the frozen FIFO prefix and later blocks see an empty prefix.

The circuit verifies parent linkage, strict slot monotonicity, schedules, signatures for tiers 1–2, version binding, anchor/header/MPT authentication, forced-prefix ordering, execution, exact body/data coverage and public outputs. It rejects more than 4,096 blocks, 12 schedule windows, 16 sealed sessions, 2,100 referenced blob records, 256 forced items, 4 MiB forced bytes or

80 million total accounted forced gas per candidate. These candidate-level caps are independent of the per-block caps.

5.3 Verifier statement ABI

Each immutable Settlement accepts proofs through exactly one `SettlementValidityVerifierDescriptorV2`; no caller, tier, Router row or mutable registry selects a verifier. One descriptor serves tiers 1–3. Its verifying key MUST authenticate the statement's tier and implement all three tier rules above; a verifier or key supporting fewer tiers requires a different profile schema and is not silently defaulted or indexed. Rotation therefore requires a new immutable protocol version, while historical Settlements retain their original descriptor.

The descriptor and its acyclic configuration commitment are exact:

```
struct SettlementValidityVerifierDescriptorV2 {
  address verifier; bytes32 runtimeHash; bytes32 configurationHash;
  bytes32 verifyingKeyHash; bytes32 proofSystemId;
  bytes32 publicInputSchemaHash; bytes4 selector;
  uint32 maximumProofBytes; uint64 verificationGasLimit;
  uint64 postVerificationReserveGas;
}
SETTLEMENT_VALIDITY_VERIFIER_CONFIG_TYPEHASH = keccak256(
 "SettlementValidityVerifierConfigV2(bytes32 verifyingKeyHash,"
 "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
 "uint32 maximumProofBytes,uint64 verificationGasLimit,"
 "uint64 postVerificationReserveGas)")
settlementValidityVerifierConfigurationHash = keccak256(abi.encode(
 SETTLEMENT_VALIDITY_VERIFIER_CONFIG_TYPEHASH, verifyingKeyHash,
 proofSystemId, publicInputSchemaHash, selector, maximumProofBytes,
 verificationGasLimit, postVerificationReserveGas))
SETTLEMENT_VALIDITY_VERIFIER_DESCRIPTOR_TYPEHASH = keccak256(
 "SettlementValidityVerifierDescriptorV2(address verifier,"
 "bytes32 runtimeHash,bytes32 configurationHash,bytes32 verifyingKeyHash,"
 "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
 "uint32 maximumProofBytes,uint64 verificationGasLimit,"
 "uint64 postVerificationReserveGas)")
settlementValidityVerifierDescriptorHash = keccak256(abi.encode(
 SETTLEMENT_VALIDITY_VERIFIER_DESCRIPTOR_TYPEHASH, verifier, runtimeHash,
 configurationHash, verifyingKeyHash, proofSystemId, publicInputSchemaHash,
 selector, maximumProofBytes, verificationGasLimit,
 postVerificationReserveGas))
```

Adjacent quoted fragments in the two typehash definitions concatenate with no separator; the type strings contain no whitespace or newline. All ten fields are nonzero; `maximumProofBytes<=65536`, both gas values are at most 30,000,000, and `selector=0x8c6cb224`, the first four bytes of Keccak of `"verify(bytes,uint256[2])"`. The `publicInputSchemaHash` is the fixed value `keccak256("slot-chain-settlement-validity-public-input-schema-v2") = 0xb9313bdbe8b2203bcda0a1d140fb1c0446a7424de6440cde31317eafb654cac4`; it denotes exactly the displayed `SettlementStatementV2`, its `"slot-chain-statement-v2"` domain and the unreduced high/low 128-bit limb transport below. The configuration hash excludes the

verifier address, runtime hash, descriptor hash, Settlement address, execution-profile hash and every target/manifest/registration-derived value. The verifier exposes

```
settlementValidityVerifierConfigHashV2() // selector 0xa8e357ec
  external view returns (bytes32 configurationHash)
```

as an exact four-byte, zero-value, 50,000-gas STATICCALL with exactly 32 return bytes. Before every proof call Settlement checks the nonzero account's exact EXTCODEHASH and this configuration getter; a proxy, empty account, mismatch, revert, OOG, short or trailing return rejects.

Settlement exposes its complete constructor-bound descriptor as

```
settlementValidityVerifierDescriptorV2() // selector 0x26bad12b
  external view returns
  (bytes4 magic,address verifier,bytes32 runtimeHash,
   bytes32 configurationHash,bytes32 verifyingKeyHash,
   bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,
   uint32 maximumProofBytes,uint64 verificationGasLimit,
   uint64 postVerificationReserveGas)
```

with magic `0x53564432` (SVD2) and exactly 352 canonical return bytes. Registration derives and stores the descriptor hash in RTR2; registration exact-compares every SVD2 field with the transient profile. Activation strictly decodes the complete SVD2 row, recomputes its descriptor hash against the stored RTR2 value, and uses the returned address/runtime/configuration for its independent live verifier checks. SVD2 is caller-independent, uses canonical narrow-field padding and makes no external call.

The ordinary verifier is a nonproxy, state-independent release artifact. The release bundle publishes the exact runtime preimage whose Keccak equals the descriptor. The mandatory conformance verifier rejects a runtime containing SLOAD, SSTORE, TLOAD, TSTORE, CREATE, CREATE2, DELEGATECALL, CALL, CALLCODE or SELFDESTRUCT, or whose result/configuration depends on caller, account balance or block/transaction environment. Its verifying key and configuration are embedded in code. STATICCALL is permitted only to the fixed Ethereum precompiles in the reviewed execution policy committed by `proofSystemId`; alternate external accounts or dynamically selected targets reject. The release certificate proves every reachable path obeys that policy and that the configuration getter is the displayed constant. Consequently identical runtime hash means identical proof behavior even if an attacker prefunds the account or its deployment storage is nonzero.

Settlement calls exactly

```
verify(bytes proof,uint256[2] digestLimbs)
  external view returns (bytes32 statementHash)
```

with zero value. The ABI head is `[0x60,high128(statementHash),low128(statementHash)]`, followed by the canonical proof length/data/zero-padding tail; total calldata is `132+ceil32(proof.length)` bytes. The proof is nonempty and no longer than the descriptor bound. Settlement builds the complete calldata before measuring gas and warms the verifier through EXTCODEHASH. It then requires

```
gasleft() >= verificationGasLimit
```

```
          + ceil(verificationGasLimit / 63)
          + postVerificationReserveGas + 10000
```

with checked arithmetic, performs one gas-bounded STATICCALL with zero output area, checks RETURNDATASIZE before copying exactly 32 bytes, and after that copy requires at least `postVerificationReserveGas` remains. Revert, OOG, short/trailing/oversized returndata, a Boolean-shaped value, or a returned word different from the reconstructed statement hash rejects before any canonical state write. Revert data and unexpected returndata are never copied. The 10,000-gas envelope and post-verification reserve are release-calibrated against the complete remaining success and rejection suffix; threshold and threshold-minus-one measurements are release gates. Profile validation also requires the complete checked preflight sum above to be no greater than the same profile's nonzero `supportedL1BlockGasLimit`; independent release measurement must leave the general 30% transaction headroom required below.

Settlement constructs the following `SettlementStatementV2` Solidity struct in exactly this order. It computes `statementHash` as legacy Keccak of the ASCII domain `"slot-chain-statement-v2"` followed by `abi.encode(S)`:

```
S = (uint256 settlementChainId, uint256 l2ChainId, uint256 protocolVersion,
     bytes32 executionProfileHash, address verifyingContract,
     uint64 releaseProtocolVersion, bytes32 releaseManifestHash,
     uint8 tier, bytes32 baseCanonicalHash,
     bytes32 candidateCommitment, uint64 blockCount,
     uint64 firstSlot, bytes32 tipHash, uint64 tipSlot,
     uint64 endL2BlockNumber, bytes32 endStateRoot,
     bytes32 endTerminalRoot, uint64 endTerminalCount, uint64 endCursor,
     bytes32 winningDataCommitment, uint64 nextDueAt,
     uint256 nextBaseFee, uint64 nextExcessBlobGas,
     uint64 anchorNumber, bytes32 anchorHash,
     bytes32 anchorStateRoot, uint64 anchorTimestamp,
     bytes32 forceRoot, uint64 forceCutoff,
     bytes32 forcedDescriptorCommitment,
     uint64 startCursor,
     uint64 admissionVersion,
     bytes32 admissionRoot,
     uint64 episode, uint64 recoveryRevision, bytes32 recoveryId,
     bytes32 scheduleRootsCommitment, uint8 scheduleWindowCount,
     bytes32 sessionRefsCommitment, uint8 sessionCount,
     uint16 dataRecordCount, uint256 rewardExecutionGas,
     uint64 rewardPublishedBytes, bytes32 executionOutputsCommitment,
     address proofBeneficiary)
```

`digestLimbs[0]` and `[1]` are respectively the high and low 128 bits of `statementHash`; neither is reduced modulo the proof field. The circuit recombines them and proves the exact preimage. Settlement independently checks the immutable `protocolVersion->executionProfileHash` mapping, the base canonical state, current/frozen versions and recovery round, recomputes the bounded schedule-root and sealed-session-reference commitments from calldata, recomputes the bounded forced-descriptor commitment from authoritative queue storage,

recomputes `winningDataCommitment` from the candidate and sealed-session commitments, requires `endL2BlockNumber=stored.l2BlockNumber+blockCount`, and constructs the new core from the explicit `(endL2BlockNumber,tipHash,tipSlot,endStateRoot,endTerminalRoot, endTerminalCount,endCursor)` fields plus that recomputed value and proved `(nextBaseFee, nextExcessBlobGas)`. The circuit derives `rewardExecutionGas` as the exact sum of the Ethereum `gasUsed` fields of all candidate blocks and `rewardPublishedBytes` as the exact sum of `chunkByteLength` over the distinct sealed data records consumed by the candidate. Every consumed record is identified by its nonduplicated `(sessionId,recordIndex)` manifest reference; a repeated reference rejects rather than counting twice. Both are reward-only metering outputs: they do not change candidate ordering or validity beyond being equal to the already executed and covered candidate. The reward class is never a statement or caller choice; it is exactly the authenticated `tier` (1, 2 or 3), and every calibrated profile contains exactly those three class IDs. The statement tier must equal the tier in every included `SlotChainBlock`; mixed-tier candidates reject before proof acceptance. It recomputes `executionOutputsCommitment` including the same explicit terminal pair; the circuit proves that pair equals the accumulator's exact post-state slots, so neither calldata nor Settlement can substitute a root. It rejects zero/ill-typed fields, cursor regression, or disagreement with the last proved block, then stamps `canonicalizedAtBlock=block.number` outside the proof. `candidateId=statementHash`. `nextDueAt` is authenticated inside the frozen queue as the boundary leaf's deadline, or `UINT64_MAX` when `endCursor=forceCutoff`. It is only a proof-consistency output. Normal submission additionally verifies the current-root boundary proof described in §7. After every commit and on every activation check, the sole authoritative live head value is `ForcedQueue.dueAt(canonical.messageCursor);` no recovery-round or canonical-core scalar caches it. For the one proof-carrying launch-activation or migration candidate, Settlement requires `releaseProtocolVersion=targetProtocolVersion` and `releaseManifestHash=targetManifestHash` from the authenticated launch or migration gate; every ordinary candidate requires both fields to be zero. The circuit uses the nonzero pair to require the first block's activation Anchor call and the exact manifest, and requires the exact release/domain seals and registered route to be active in every later block of that candidate. The L1 coordinator verifies this proof before final adoption, legacy FROZEN state or queue-authority change, then adopts its output and consumes the target pair in the same atomic cutover. The genesis campaign's bounded QUIESCENT phase changes neither the legacy canonical core nor Queue authority and automatically resumes ACTIVE at expiry; transaction-local READY cannot survive a revert. No ACTIVE target Settlement can be left waiting on a newly activated verifier to clear a pending release. `proofBeneficiary` is a nonzero proof input, never builder-signed block data or coinbase. It receives the consumed queue-fee interval and any separate proof reward; copying proof bytes cannot redirect either payment.

5.4 Preconfirmation semantics

A builder signature proves authorship and makes same-slot equivocation slashable. It does not prove that the body reached a prover or that the next builder saw it. A later scheduled builder may sign a profitable skip that the protocol cannot distinguish from non-receipt. Accordingly:

- before L1 candidate acceptance, a promise is *observed*;
- after acceptance and before close, it is *provisionally proved*;
- after normal close or recovery commit, it is *canonical*; and

- only the underlying L1 finality policy makes it *L1-final*.

Wallets and applications MUST NOTdisplay the first two states as final. An escape commit explicitly revokes every omitted observed or provisionally proved promise and returns omitted transactions to the mempool.

6 Blob data authentication

Data is posted into publisher-owned, bonded sessions. A session ID is

```
H("slot-chain-session-v1" || u256(settlementChainId) || address20(contract) ||
  address20(owner) || u64(sessionSequence))
```

The contract stores one checked `uint64nextSessionSequence`, initially zero. A successful open uses `s=nextSessionSequence`, requires `s<UINT64_MAX`, derives and returns the ID above, then stores `s+1`. The caller supplies neither ID nor sequence, and a rejected or no-cell open does not consume one. Sequence `UINT64_MAX` is the explicit exhaustion sentinel. The sequence is global rather than per owner: a Sybil cannot create permanent nonce keys, while the owner and Settlement address in the preimage prevent cross-owner and cross-release aliases. A fresh migration target starts at zero; the retired source retains its historical sequence.

DataSession is a bounded storage region of the immutable Settlement, and the contract address in the ID above is that Settlement. Settlement accepts native ETH only through the session open/post surfaces and the three-class `fundRewardClassV1` surface; it has no Queue, Market, Bridge or other native liability. Its configuration binds the shared migration gate and immutable top-level `dataRent` sink. Every other payable selector, fallback and receive path rejects, and conformance tests exercise `msg.value>0` against every such selector. This complete custody inventory makes the balance invariant and surplus calculation independent of any cross-contract caller-supplied "pinned" or "boundary closed" flag.

The profile/runtime geometry is exact: 1,024 physical cells, at most two LIVE cells per owner, 2,100 records per LIVE cell, at most six blobs per post, eight maintenance inspections and an 86,400-second maximum TTL. Each cell has one exact semantic tag `FREE=0`, `LIVE=1` or `REFUND=2`. Storage is exactly `cells[1024]`, an ID-to-cell-plus-one mapping with at most one key per non-FREE cell, a live-owner-count mapping containing only owners with at least one LIVE cell, checked `liveCount`, `refundCount`, `occupiedCount=liveCount+refundCount<=1024`, and `gcCursor`. A REFUND cell stores only `(sessionId,owner,bondWei,claimDeadline)`; its old MMR words are stale and ignored. Clearing a cell deletes its ID key, and removing an owner's last LIVE cell deletes its owner key. No refund or historical-owner mapping exists. The three counters and cursor are `uint16`; owner LIVE counts are also `uint16` and never exceed two. PREACTIVE and historical Settlements have zero local LIVE count. OPEN increments local LIVE and occupied by one; LIVE-to-REFUND decrements local LIVE and increments refund without changing occupied; REFUND-to-FREE decrements refund and occupied. No other transition changes those fields. An eligible owner claim directly from LIVE executes those two logical transitions atomically: its net effect decrements local LIVE and occupied, deletes the owner/ID entries, and reduces LIVE liability and balance by the same bond; refund count and liability have net zero change.

There is deliberately no Router/global LIVE counter and therefore no cross-contract write on OPEN, claim or maintenance. The Router authenticates exactly one current Settlement. Inductively, a PREACTIVE target begins with zero LIVE cells; only the current ACTIVE Settlement

may open one; READY requires that source's local LIVE count zero; and cutover freezes that zero-LIVE source before making the zero-LIVE target current. Abort leaves the same source current and never activates the target. The generation-authenticated READY handshake and activation-time static reads below close this induction.

The direct sidecars authenticate that premise through one exact bounded Router read, never through a caller-supplied flag:

```
activeSettlementStateV1() returns
  (bytes4 magic,address activeSettlement,uint64 generation,
   uint64 activeProtocolVersion,uint64 targetProtocolVersion,
   bytes32 targetManifestHash,bytes32 targetRegistrationHash,uint8 phase)
```

The magic is `0x41535231` (ASCII `ASR1`), phases are exactly ACTIVE=0, ARMED=1 and READY=2, and returndata is exactly 256 bytes with canonical ABI padding. OPEN, POST and SEAL read the immutable Router with an exact 50,000-gas static call. They reject revert/OOG/short/trailing or noncanonical data, and require the returned active address to equal this Settlement, the returned active version to equal its immutable protocol version and phase ACTIVE. ACTIVE requires the returned target version, manifest and registration hash to be canonical zero; ARMED and READY require all three nonzero. The finite genesis campaign never adds a public gate phase: the constructor-bound legacy branch remains ACTIVE until the atomic switch. This check and the session mutation share one EVM revert domain.

Payable `openSession(uint16expectedEmptyCell,uint64expiry)` neither calls `sync` nor performs GC. It requires the current non-PREACTIVE component, the shared gate ACTIVE, `expectedEmptyCell<1024`, that exact tag FREE, owner/live/sequence capacity and the exact profile-priced value. It derives and returns the session ID and increments the global sequence only on success. Every failure reverts, returning value and preserving sequence, cursor, cells and liabilities. A cell-hint race therefore costs the winner the full rent and bond but gives no identity authority. Payable `postData` follows the same sidecar rule: it checks current target and ACTIVE directly and reverts on every failure; it does not run a canonical transition while holding caller value.

The native-value split is exact. The symbols below name these exact profile fields:

```
BOND     = refundableBondWei
BASE     = baseRentWei
BYTE     = rentPerPublishedByteWei
BLOB_BPS = blobBaseFeeMultiplierBps
```

The profile requires `BLOB_BPS<=10000`. Opening requires checked `msg.value=BOND+BASE`; only `BOND` increments the LIVE liability and `BASE` is immediately surplus. A post with n manifests requires $1 \le n \le \min(6, 2100 - count)$, nonzero `BLOBHASH(i)` for every $i < n$ and `BLOBHASH(n)=0`, so the manifest array is in one-to-one order-preserving correspondence with every blob in that transaction. Define

```
publishedBytes = sum(manifest[i].chunkByteLength)
blobGasUsed    = 131072 * n
blobWeight     = blobGasUsed * BLOB_BPS
blobSurcharge =
```

```
  mulDivUp(blobWeight, block.blobbasefee, 10000)
postRent       = publishedBytes * BYTE + blobSurcharge
```

over mathematical nonnegative integers; `mulDivUp(a,b,d)` is exactly $\lceil ab/d \rceil$ using a full 512-bit product and reverts if the result does not fit `uint256`. L1 checks the declared `chunkByteLength<=126972`, but cannot read or decode blob contents. The validity circuit later checks the blob's length prefix, zero padding, decoded bytes and computed chunk root against that declaration. The implementation must produce the exact ceiling without an intermediate-width approximation and then require every result and `postRent` to fit `uint256`; otherwise it reverts. The post requires `msg.value==postRent`; underpayment and overpayment both revert. All post value is immediately surplus and never enters either bond liability. Zero-byte padding still consumes a whole blob and therefore its whole blob-gas surcharge. Any blob/KZG/MMR/value/arithmetic failure rolls back the complete call, including the value, record count, peaks and emitted events.

The externally callable surface is frozen below. `DataPostV1` is the static tuple

```
(bytes32 fullBodyRoot, uint16 blockOrdinal, uint16 chunkIndex,
 uint16 chunkCount, uint32 chunkByteLength, bytes32 chunkRoot, bytes32 y,
 bytes32 commitmentHi, bytes16 commitmentLo,
 bytes32 proofHi, bytes16 proofLo)
```

where concatenating each high/low pair yields exactly 48 bytes; `bytes16` ABI right-padding must be zero. The expanded selectors and returns are:

```
openSession(uint16,uint64) payable returns (bytes32 sessionId)
postData(bytes32,(bytes32,uint16,uint16,uint16,uint32,bytes32,bytes32,
  bytes32,bytes16,bytes32,bytes16)[]) payable
  returns (uint16 firstRecordIndex,uint16 newCount,bytes32 newRoot)
sealSession(bytes32) returns (uint16 count,bytes32 root,uint64 expiry)
maintainDataSessions() returns
  (uint8 status,uint8 inspected,uint8 changed,uint16 nextCursor)
claimSessionBond(bytes32,address) returns (uint256 amount)
sweepSessionSurplus() returns (uint256 amount)
```

Open, post and seal are direct session sidecars: none calls `sync` or moves the maintenance cursor, all recheck current Settlement and ACTIVE, and every invalid input or state reverts. POST and SEAL additionally require `block.timestamp<expiry`; equality is already expired and is eligible for ordinary conversion or a direct claim. Seal also requires a nonempty unsealed LIVE session owned by the caller. Maintenance status is exact: NOOP=0, SYNCED=1 and SCANNED=2. Current ACTIVE first invokes `sync`; if it changes state the call returns SYNCED without a separate scan, otherwise it performs an ordinary exact scan and returns SCANNED. Current ARMED returns the result of its sync-owned migration work as SYNCED, or NOOP while a boundary remains open. READY/FROZEN and historical components perform only the overdue-REFUND scan; PREACTIVE returns NOOP. SCANNED reports eight inspections, while the other statuses report zero explicit maintenance inspections even if ARMED `sync` performed its separately accounted migration scan. An invalid claim reverts; a zero-surplus sweep returns zero, while sink failure reverts only that sweep. PREACTIVE claim and sweep revert without a CALL, event or write, and PREACTIVE maintenance is the stated NOOP; forced ETH is the only way its balance can change before activation.

For manifest *i*, the contract derives `recordIndex=oldCount+i`, `versionedHash=BLOBHASH(i)`, `publisher=msg.sender` and `validUntil=session.expiry`; none is caller supplied. It checks `1<=chunkCount<=9`, `chunkIndex<chunkCount`, the declared length bound, field-canonical `y`, the versioned hash/commitment relation and the point-evaluation proof. It derives *z* and the Appendix leaf separately for each tuple, appends exactly *n* leaves in array/blob order, and emits exactly *n* leaf events. One failure at any position reverts all earlier appends, events and value.

The canonical events, with no more than three indexed arguments, are:

```
SessionOpened(bytes32 indexed sessionId,address indexed owner,
  uint16 cell,
  uint64 sequence,uint64 expiry,uint256 bondWei,uint256 baseRentWei)
DataRecordAppended(bytes32 indexed sessionId,uint16 indexed recordIndex,
  bytes32 indexed versionedHash,bytes32 leaf)
SessionSealed(bytes32 indexed sessionId,
  uint16 count,bytes32 root,uint64 expiry)
SessionLiveToRefund(bytes32 indexed sessionId,address indexed owner,
  uint16 cell,
  uint64 claimDeadline,uint64 migrationGeneration)
SessionBondClaimed(bytes32 indexed sessionId,address indexed owner,
  address indexed recipient,uint256 amount)
SessionRefundForfeited(bytes32 indexed sessionId,
  address indexed owner,
  uint16 cell,uint256 amount)
SessionSurplusSwept(address indexed sink,uint256 amount)
DataSessionsMaintained(uint8 mode,
  uint16 startCursor,uint16 nextCursor,
  uint8 inspected,uint8 changed)
```

An ordinary conversion emits generation zero; a migration conversion emits the armed generation. A direct LIVE claim emits only `SessionBondClaimed`; `SessionLiveToRefund` means the cell persistently ended the call as REFUND. Maintenance event mode is ORDINARY=1, MIGRATION=2 or REFUND_ONLY=3. Events are the permanent leaf/discovery oracle, not contract storage.

The exact views are

```
dataSessionCellV1(uint16 cell) returns
  (uint8 tag,bytes32 sessionId,address owner,uint64 sequence,uint64 expiry,
   uint16 count,bool sealed,bytes32 root,bytes32[12] peaks,uint256 bondWei,
   uint64 claimDeadline)
dataSessionByIdV1(bytes32 sessionId) returns
  (uint16 cellPlusOne,uint8 tag,address owner,uint64 sequence,uint64 expiry,
   uint16 count,bool sealed,bytes32 root,uint256 bondWei,uint64 claimDeadline)
dataSessionAccountingV1() returns
  (bytes4 magic,uint16 liveCount,uint16 refundCount,uint16 occupiedCount,
   uint16 gcCursor,uint64 nextSessionSequence,uint256 liveBondLiability,
   uint256 refundBondLiability,uint64 migrationRefundGeneration,
   uint64 migrationRefundClaimDeadline,bool guardEntered,
```

```
bytes32 dataSessionConfigHash,uint256 rewardFundingClass1,
uint256 rewardFundingClass2,uint256 rewardFundingClass3,
uint256 totalRewardFunding)
```

The accounting magic is `0x44535631` (ASCII `DSV1`); narrow words, booleans, addresses and the fixed array are canonically encoded, and returndata lengths are exactly 704, 320 and 512 bytes respectively. FREE cell/by-ID misses return canonical zero for every output. A REFUND view returns only `sessionId`, owner, bond and deadline nonzero; it masks sequence, expiry, count, sealed, root and peaks to canonical zero without clearing their stale physical words. Runtime code hash binds selector, event, tuple and fee-algorithm semantics. The returned configuration hash is the exact immutable-value commitment

```
H("slot-chain-data-session-config-v1" || u32(340) ||
  u256(settlementChainId) || u64(protocolVersion) || address20(settlement) ||
  address20(activeSettlementRouter) || address20(protocolVersionManager) ||
  address20(dataRent) || executionProfileHash ||
  u256(BOND) || u256(BASE) || u256(BYTE) || u16(BLOB_BPS) ||
  u64(dataSessionMaximumTtlSeconds) ||
  u64(dataSessionRefundClaimWindowSeconds) ||
  u16(1024) || u16(2) || u16(2100) ||
  u8(8) || u8(6) || address20(0x0a) || u32(50000) || u256(BLS_MODULUS) ||
  u32(131072) || u32(126972) || u16(9))
```

Every address/hash is nonzero except that the point-evaluation address is the canonical numeric `0x0a`; the BPS/rate fields may be zero. The delayed profile requires both DataSession time values nonzero and `dataSessionRefundClaimWindowSeconds>=dataSessionMaximumTtlSeconds`; the release fixture sets both to 86400. The delayed target registration must carry this exact value, the target Settlement's wider configuration commitment must include it as a named field, and the Router must compare the accounting return against that registered value. The 232-byte deployment descriptor, target-registration hash and shared activation context close that control-plane binding; the DataSession sub-configuration hash remains distinct from the wider `targetConfigurationHash`.

The union-cell transition writes are normative. FREE-to-LIVE writes the new ID, owner, sequence, expiry and bond and explicitly writes `count=0`, `sealed=false` and the wrapped empty-MMR root. It may leave physical peak words stale, because a peak is read only when its bit in the newly written count is one; the first append at count zero therefore overwrites height zero before using it. LIVE-to-REFUND writes the REFUND tag and claim deadline and preserves only ID, owner and bond semantically. REFUND-to-FREE deletes the ID key and writes the FREE tag while its union words may remain physically stale. FREE and REFUND getters apply the masks above, and no append/root operation may read a stale word hidden by the tag or count. Consequently reuse of a formerly sealed, nonempty cell has exactly the same first-append root as a fresh cell.

Nonpayable maintenance alone owns `gcCursor`. An eligible call includes FREE and unmodified cells in this exact scan and update:

```
cell[j] = (oldCursor + j) mod 1024, for j = 0..7
gcCursor = (oldCursor + 8) mod 1024
```

In ACTIVE, an ordinary LIVE cell becomes REFUND iff `expiry<=block.timestamp` and its ID is not referenced by the stored normal best; its deadline is fixed as saturating `uint64` addition

of the claim window to the stored expiry, never to the keeper's scan time. A pre-existing REFUND becomes FREE only when `block.timestamp>claimDeadline`. One scan never both creates and forfeits the same REFUND. Migration and historical modes use the stricter rules below. No call scans all 1,024 cells.

The refundable bond is a LIVE liability and then a REFUND liability; conversion moves the same amount between the two checked scalars without transferring ETH. The owner calls `claimSessionBond(bytes32sessionId,addressrecipient)` through the bounded ID-to-cell lookup. It accepts either that exact REFUND before its deadline, or in current ACTIVE mode an expired LIVE cell not referenced by the normal best before `satAdd64(expiry,claimWindow)`. ARMED LIVE claims are suppressed until its normal/recovery boundary closes, then use that generation's frozen migration deadline. It always requires the exact owner, nonzero recipient and `block.timestamp<=claimDeadline`; checks-effects-interactions clears the mapping/cell/counters/liability before transfer. A failed recipient transfer reverts the whole claim. Every external Settlement mutation first requires the shared nonreentrancy guard to be NOT_ENTERED; session claims, reward claims and surplus sweep hold it across their external CALL. A nested canonical, session, reward or sweep mutation therefore rejects without changing state. A recipient may catch that nested rejection and return normally, in which case the outer claim succeeds exactly once. Claims remain unpausable on ACTIVE, ARMED, READY, FROZEN and historical components. After the deadline, maintenance clears the REFUND and its ETH becomes surplus; no owner or sink is called.

The Settlement account maintains checked `totalRewardFunding` equal to the sum of its three class buckets and enforces `balance>=liveBondLiability+refundBondLiability+totalRewardFunding`. Nonrefundable rent, forfeited bonds and forced ETH are surplus and never inflate a refund or reward bucket. A separate permissionless, non-gating `sweepSessionSurplus()` sends exactly the balance above all three accounted liabilities to the immutable `dataRent` sink under one shared reentrancy guard. Sink failure reverts only that sweep and cannot block GC, claims, READY, cutover or the forced lane. The calibrated rent/bond combination prices the full possible occupancy through TTL and the 86,400-second refund window. A Sybil can still buy all optional DA cells for that bounded period; the forced/escape lane is independent and this is not a protocol-liveness claim.

Each occupied cell owns a fixed `bytes32peaks[12]` and `uint16count<=2100`; it stores no record array. A peak word at height h is meaningful exactly when bit h of count is one; zero-bit words may be stale. Append uses the old count as the leaf index, combines the stored peak on the left while successive count bits are one, writes the carry at the first zero bit and increments count. The wrapped MMR root is recomputed from at most 12 meaningful peaks. The exact bag order and proof grammar are frozen in Appendix A.

Only the owner may append, and a candidate may reference only a session made immutable by `sealSession`. Sealing fixes `(root,count,expiry)`; there is no unseal or append-after-seal operation. A candidate supplies at most 16 sorted unique `(sessionId,count,root)` references. At submission each must equal a live sealed session. Normal acceptance requires `expiry>=normalDeadline+REORG_MARGIN_SECONDS`; recovery requires `expiry>=block.timestamp+REORG_MARGIN_SECONDS`. Opening requires its expiry to cover proving, settlement and the reorg margin:

$$\begin{aligned} expiry &\geq now + P\_PROVE\_MAX + W\_SETTLE\_SECONDS \\ &\quad + REORG\_MARGIN\_SECONDS, \\ expiry &\leq now + DATA\_TTL. \end{aligned}$$

Here `P_PROVE_MAX=recovery.proofTimeMaxSeconds=900` and `W_SETTLE_SECONDS=recovery.settlementWindowSeconds=1200` in the exact execution profile; they are not independent implementation constants. Profile validation separately enforces `DATA_TTL>=P_PROVE_MAX+W_SETTLE_SECONDS+REORG_MARGIN_SECONDS`.

For every blob attached to `postData(session,manifests[])`, the contract:

1. obtains `versionedHash=BLOBHASH(i)` from the same transaction. The exact KZG relation is `versionedHash=bytes1(0x01)||SHA256(commitment48)[1:32]`; the point-evaluation precompile below checks this equality as part of acceptance, so no alternate version byte, digest function or truncation is legal;
2. computes legacy Keccak over this packed fixed-width sequence:

   ```
   "slot-chain-data-fs-v2" || u256(settlementChainId) ||
   u256(protocolVersion) ||
   sessionId || versionedHash || fullBodyRoot || u16(blockOrdinal) ||
   u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
   chunkRoot || address20(publisher) || u64(validUntil)
   ```

   It interprets the digest big-endian and reduces it modulo the BLS scalar-field modulus to obtain z;
3. makes a 50,000-gas `STATICCALL` to the EIP-4844 point-evaluation precompile at address `0x0a` with the exact 192 bytes `versionedHash32||z32||y32||commitment48||proof48`. It requires call success and exactly 64 return bytes equal to `u256(4096)||BLS_MODULUS`, where the exact modulus is

   ```
   hex32("73eda753299d7d483339d80809a1d805" ||
         "53bda402fffe5bfeffffffff00000001")
   ```

   short, long, reverting or wrong-constant returns reject; and
4. appends the exact leaf from §A to the session MMR.

The proof receives blob polynomials privately, recomputes $P(z) = y$, and decodes the canonical blob codec. A discretionary body byte string is `u32(txCount)||(u32(txLength)||signedRawTx)*`; its root is `H("slot-chain-body-v1"||u32(byteLength)||bodyBytes)`. Blob payload is `u32(chunkByteLength)||chunkBytes||zeroPadding`, split into 31-byte chunks; each blob field element is `0x00||chunk31`. Exactly 4,096 elements are used, unused payload bytes are zero, and noncanonical encodings reject. A manifest entry binds `(blockOrdinal,chunkIndex,chunkCount,chunkByteLength,fullBodyRoot,chunkRoot,sessionId,recordIndex)`. For each block entries are in strict chunk-index order, cover exactly `0..chunkCount-1`, have `chunkCount<=MAX_CHUNKS_PER_BODY=9`, and concatenate to at most `MAX_BODY_BYTES=1,142,748`. The circuit recomputes every chunk root, the full discretionary body root and then the full Ethereum transactions root. The signed `dataManifestRoot` is *per block*. Its leaves contain only that block's entries, use local positions `0..m-1`, repeat that block's candidate-wide `blockOrdinal`, and are sorted by chunk index. Candidate calldata concatenates these independently rooted lists in increasing block ordinal; no global manifest tree exists. Duplicate, extra, overlapping or uncovered data is invalid. `dataManifestRoot` is the exact per-block tree in Appendix A, not a free-form publisher value.

Data publication has no consensus-coupled reimbursement. A separate service may pay it, but empty payment state cannot revert canonical settlement.

6.1 Canonical reconstruction availability

Blob sessions make an in-flight candidate provable; their TTL and garbage collection are not permanent state archival. On canonical commit, full nodes must retain the canonical raw bodies, receipts, Ethereum trie nodes and every runtime-bytecode preimage needed to execute from `CanonicalCore.stateRoot`. Bytecode is mandatory because an Ethereum account leaf authenticates only `codeHash`, not the corresponding bytes. A reconstruction package is a content-addressed set of those objects plus the canonical block headers; each code object is keyed by `H(runtimeBytecode)` and must cover every account reachable during replay, including implementation code reached through proxy, beacon or delegation indirection. A consumer rejects any object whose transaction/body/header hash, trie path, bytecode hash or final state root disagrees with the L1-canonical commitments, so an archive can be anonymous and malicious without gaining safety authority.

Renewing a recovery round refreshes its L1 anchor, queue cutoff and proof target; it cannot invert a state root or recreate destroyed bytes. Therefore the 2,164-second protocol bound begins only once a prover has the canonical reconstruction package and any required unexpired kind-0 bytes. A warm full node satisfies this condition; a cold-start prover is bounded by data download plus proof time while at least one archive or Ethereum's applicable DA history still serves the package. If no complete copy exists, safety remains intact but progress stops. No governance action may replace the state root, skip the missing prestate, or call that condition a builder failure.

7 Settlement and recovery state machine

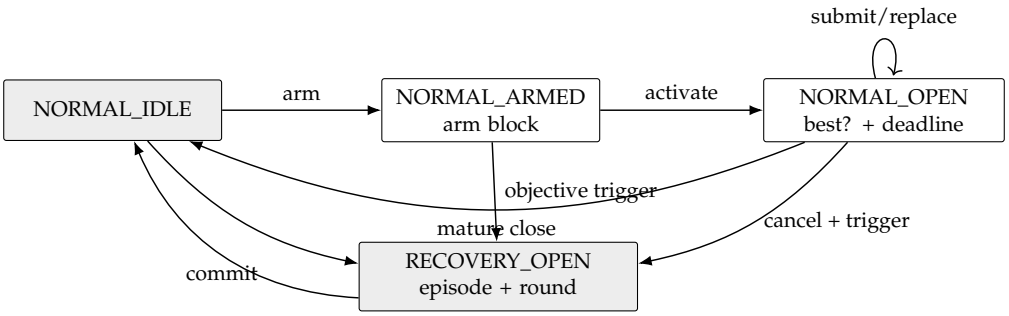


Figure 1: The complete mode machine. Payments are outside every arrow.

7.1 Normal context and candidates

Normal context creation is two-phase. Anyone calls `armNormalContext()` in `NORMAL_IDLE`; the call begins with `sync()`, stores only `armBlockNumber=block.number`, and emits `NormalContextArmed`. It accepts no caller-supplied anchor or hash. In a later L1 block anyone calls `activateNormalContext()`. It again begins with `sync()` and requires the arm's block-number age to be from 1 through `MAX_ARM_AGE_BLOCKS` (255), reads `armBlockHash=blockhash(armBlockNumber)`, and hashes a supplied RLP header to that value. It then freezes `baseCanonical`, `deadline=block.timestamp+W_SETTLE_SECONDS`, `minAdmissibleSlot=max(0,currentL2Slot-DELTA_TIP)`, the current stable `(admissionVersion,admissionRoot)`, the arm header fields, and `requiredThrough=deadline+T_INCLUDE_MAX_SECONDS`. It derives the sole normal `contextId` and emits `NormalContextActivated`. No candidate is accepted

while armed. If age exceeds 255 blocks, any caller may invoke `armNormalContext()` again; after its leading `sync()` it atomically replaces only the stale arm number with the current block, emits `NormalContextRearmed`, and returns `REARMED`. A nonstale arm cannot be replaced. Thus an abandoned arm cannot suppress normal operation.

Builders collect and issue tier-1 signatures only after the activation event. The execution anchor for every candidate is the exact arm block, whose hash and parsed header are already frozen; replacements hash their supplied RLP header to that stored hash and never repeat an EIP-2935 recency lookup. A reorg that removes the arm block necessarily changes the context. If the arm block survives but the activation transaction is reorged, reactivation intentionally recreates the same context: the builder must retain and honor its prior same-context signature. Evidence compares the exact common anchor/force tuple and treats this derived context ID as the signed promise namespace; it does not need activation storage or an EIP-2935 lookup after the builder has produced both signatures. Thus a shallow reorg cannot erase culpability, and an attacker cannot choose an anchor without the builder signing both conflicting registered-domain headers.

Activation is permissionless, reserves no builder, and does not block submissions. An empty activated window merely cancels at deadline, so an opener gains no censorship right. Every candidate first proves its frozen-prefix boundary in the circuit. It also supplies a canonical single-leaf Merkle proof at `endCursor` under the queue's current root/count, or proves `endCursor==currentCount`. Settlement verifies the canonical depth-64 singleton proof, which contains exactly 64 sibling hashes, and requires the resulting `currentBoundaryDueAt` to exceed `requiredThrough`. This prevents an old but valid anchor from hiding a post-anchor append. Because due times are monotone, one current boundary leaf proves that every item due through the landing horizon is consumed. Later candidates use the same base, deadline, minimum, admission pair and horizon. Their total order is lexicographic:

$$(\text{block count descending},\ \text{tip slot descending},\ \text{tip hash ascending}).$$

Only a strictly better candidate replaces `normalBest`; the stored best contains `endCursor` and the minimum expiry of every referenced data session. Provisional replacement changes no canonical state and pays nobody. At or after the deadline, `sync()` reads `ForcedQueue.dueAt(best.endCursor)` and commits only if that live boundary is greater than `block.timestamp` and `block.timestamp+REORG_MARGIN_SECONDS<=best.minDataExpiry`. Thus a late close cancels rather than committing a block that omits an already-due head, even beyond the assumed inclusion bound, or whose authenticated data has passed its resubmission margin. Data-session GC cannot evict a session referenced by the stored best; its own leading `sync()` clears an expired best first. Because `FORCE_DELAY>=W_SETTLE_SECONDS+T_INCLUDE_MAX_SECONDS`, an append after open cannot violate the frozen acceptance horizon. Settlement then recomputes recovery causes against the new core.

7.2 Activation

While normal, `sync()` opens one persistent recovery episode if either:

- `currentL2Slot>canonical.tipSlot+DELTA_FINAL_LAG`; or
- the forced queue's stored `dueAt[canonical.messageCursor]` is no greater than `block.timestamp`.

A due forced head has precedence over an immature normal window, which is canceled. At a mature deadline, a best that consumes the required due prefix commits before causes are recomputed; an omitting best is canceled. No best is promoted across activation. The contract increments `episode`, records causes and the base-canonical hash, then runs the bounded seat-duty pass in §9: an objectively missed primary duty is allocated, failed over or marked breached as applicable, while a full duty ring fails open to economic vacancy. This path performs no Market call, bond movement or token transfer. Breach enforcement and bond terminalization are later permissionless noncanonical operations against the retained exact receipt; force-only recovery creates no aggregator penalty.

Every round freezes `roundStartSlot`, the preceding L1 block number/hash, current `forceRoot/forceCutoff`, current `(admissionVersion,admissionRoot)`, and `escapeSlot= max(roundStartSlot+ESCAPE_OFFSET,canonical.tipSlot+1)`. Its `expiresAt` is `GENESIS_TIMESTAMP+escapeSlot+DELTA_TIP`. The exact ID is:

```
recoveryId = H("slot-chain-recovery-v2",
               u256(settlementChainId), address20(contract),
               u64(episode), u64(revision), bytes32(baseCanonicalHash),
               u64(roundStartSlot), u64(anchorNumber), bytes32(anchorHash),
               bytes32(forceRoot), u64(forceCutoff), u64(admissionVersion),
               bytes32(admissionRoot), u64(escapeSlot), u8(causes))
```

At `block.timestamp>expiresAt`, and never before, `sync()` replaces the expired target with revision $r + 1$ and fresh round fields, then returns `SYNCED`. It does not change episode/base/cause, create another duty, promote, terminalize a bond or pay. The old proof is already inadmissible, so permissionless refresh cannot grief a still-live proof. New queue appends enter the refreshed cutoff. An envelope admitted during recovery receives `dueAt=max(enqueuedAt+FORCE_DELAY,lastDueAt,currentRound.expiresAt+1)`; therefore it cannot become due while a frozen round that omits it remains admissible.

7.3 Recovery acceptance

Recovery is first-valid, not heaviest. A tier-2 or tier-3 candidate MUST:

1. extend the exact current stored canonical state;
2. bind current `episode`, `revision`, `recoveryId`, frozen admission pair, anchor and queue target;
3. land at least `F_L1` L1 blocks after its anchor, have first slot at least `roundStartSlot`, tip no older than `DELTA_TIP`, and no slot later than wall clock plus skew;
4. consume the exact maximum forced-message prefix from the current cursor, bounded by the current round's frozen cutoff; and
5. satisfy its tier-specific signature/data restrictions and validity proof.

The first valid L1 transaction atomically commits the tuple, records the current L1 block *number only*, clears the recovery episode and returns to normal. Later proofs are stale. Candidate rejection cannot undo activation because of the early-return wrapper in §3.

8 Forced queue and unsigned escape

`ForcedEnvelope` is a tagged union. Kind 0 contains `(sender,nonce,l2ChainId,rawTxHash, byteLength,gasLimit,maxFee,validUntil,refundAddress,deposit)`. Admission canonically decodes the EIP-2718 transaction, requires its recovered sender and all duplicated fields to match, and requires `msg.sender==sender`. Launch exposes no relayed kind-0 entry; any future EIP-712 relay requires a new function, profile and codec. The sole launch entry is

```
enqueueForcedTransactionV2(bytes rawTransaction, uint64 validUntil,
                           address refundAddress)
  payable returns (uint8 result, uint64 queueIndex)
```

Its selector is the first four Keccak bytes of `"enqueueForcedTransactionV2(bytes,uint64, address)"`. Word 0 is the canonical bytes offset `0x60`, word 1 the canonically padded validity deadline and word 2 the canonically padded nonzero refund address. The tail is one length word, the nonempty raw EIP-2718 bytes and zero right-padding; total calldata length is exactly `132+ceil32(rawTransaction.length)` bytes. The adapter re-encodes and byte-compares the complete call before state. It recovers the sender and derives nonce, chain ID, raw transaction hash, raw byte length, intrinsic gas, gas limit and maximum fee from those exact bytes; `msg.sender` must equal the recovered sender, the transaction chain must be the configured L2, and `msg.value` must equal the checked shared-schedule deposit. No duplicated caller field or caller hash is authoritative. Kind 1 admission receives one `BridgeAdmissionEnvelope` containing one `uint64liquidityFee` and the complete raw eleven-field `IBridge.Message`:

```
(uint64 id, uint64 fee, uint32 gasLimit, address from,
 uint64 srcChainId, address srcOwner, uint64 destChainId,
 address destOwner, address to, uint256 value, bytes data)
```

Its Solidity name and the envelope field order are exact:

```
struct MessageV1 {
  uint64 id; uint64 fee; uint32 gasLimit; address from;
  uint64 srcChainId; address srcOwner; uint64 destChainId;
  address destOwner; address to; uint256 value; bytes data;
}
struct BridgeAdmissionEnvelopeV2 {
  MessageV1 message; uint64 liquidityFee;
}
sendMessageV2(MessageV1 message, uint64 liquidityFee)
  payable returns (bytes32 msgHash, MessageV1 normalizedMessage)
enqueueBridgeCreditV2(MessageV1 message, uint64 liquidityFee)
  payable returns (uint8 result, uint64 queueIndex)
```

Their selectors are the first four Keccak bytes of respectively `"sendMessageV2((uint64, uint64,uint32,address,uint64,address,uint64,address,address,uint256,bytes) ,uint64)"` and the same expanded arguments under `enqueueBridgeCreditV2`. After either selector, word 0 is the Message offset `0x40` and word 1 is the canonically padded liquidity fee. The Message begins at argument byte `0x40`, has eleven head words in the shown order, and its data offset word is exactly `0x160`; that tail is one length word, exact data and zero

right-padding. Thus canonical calldata length is `452+ceil32(data.length)` bytes including selector. Each entry re-encodes the decoded arguments and requires byte-for-byte equality and exact length, so aliases, gaps, overlap, trailing bytes, high integer/address padding or nonzero right-padding reject before state.

The SourceBridge first applies the normalized overwrites, then freezes the legacy V1 hash codec exactly:

```
abi.encode("TAIKO_MESSAGE", normalizedMessage) =
  [0x40, 0x80] || [u256(13), "TAIKO_MESSAGE", zero-pad-to-32] ||
  [the eleven normalized Message ABI head words, dataOffset=0x160] ||
  [u256(data.length), data, zero-pad-to-32]
msgHash = keccak256(that complete 512+ceil32(data.length)-byte preimage)
```

The return from `sendMessageV2` has `msgHash` in word 0, Message offset `0x40` in word 1 and the same canonical Message tuple/tail. Both adapter returns are exactly 64 bytes: result QUEUED=1 carries the new index, SYNCED=2 carries `UINT64_MAX`, and ALREADY_QUEUED=3 is permitted only for a zero-value kind-1 query and carries the stored index. Kind 0 never returns ALREADY_QUEUED. Unknown result, bad padding, or any other sentinel/index combination rejects at its caller. The SourceBridge overwrites `id`, `from` and `srcChainId` with its next ID, the actual caller and immutable chain context before deriving `keccak256(abi.encode("TAIKO_MESSAGE",message))`, data length and `calldataHash=keccak256(message.data)` over exactly the unpadded dynamic bytes. Empty data therefore uses Ethereum's canonical `keccak256("")=c5d2460186f7233c927e7db2dcc703c0e500b653ca82273b7bfad8045d85a470`. For kind 1, `byteLength=uint32(message.data.length)` exactly after canonical ABI decoding; for kind 0 it is the exact EIP-2718 raw-transaction byte length. No caller-supplied length, ABI-padded length, descriptor length or event length is accepted. It requires a nonzero `liquidityFee` and a checked `message.value+message.fee>0`; a zero settlement amount has no ticket, Pool authorization or DONE encoding and is rejected before custody or ID mutation. No hash, length or funding ticket supplied by the caller is authoritative. Only the configured `BridgeInboxAdapter` may enqueue the resulting fixed V11 projection and it has no queue expiry. The source Bridge retains message value plus execution and liquidity fees while recording per-credit pending, user-pull and LP-pull liabilities; every outflow must leave `address(this).balance>=pendingEscrow+userPullLiability+lpPullLiability`. The adapter separately temporarily retains its caller-funded processing fee only across the sync decision; a successful append forwards it entirely to ForcedQueue. Its only forced effect is an idempotent inbox/signal credit keyed by the immutable identifier

```
sourceDomainId = H("slot-chain-source-domain-v4" || u64(srcChainId) ||
  bytes32(sourceGenesisHash) || address20(bridgeCreditRegistry) ||
  address20(terminalSignalVerifier) || address20(srcBridge) ||
  bridgeExecutionHash || bytes32(registryNamespace))
destinationDomainId = H("slot-chain-destination-domain-v7" ||
  u64(destChainId) || bytes32(destinationGenesisHash) ||
  address20(bridgeInboxAdapter) || address20(activeSettlementRouter) ||
  address20(terminalSignalVerifier) || address20(inboxApplyRouterV2) ||
  address20(inboxCreditStoreV2) || address20(protocolReleaseAuthorityV2) ||
  address20(terminalDomainRegistrarV2) ||
  address20(terminalAccumulatorV2) || address20(nativeLiquidityPoolV2) ||
  address20(destinationBridge) ||
```

```
  bytes32(destinationBridgeExecutionHash) ||
  bytes32(destinationInfrastructureHash) ||
  bytes32(destinationNamespace))
creditId = H("slot-chain-bridge-credit-id-v6" || u64(srcChainId) ||
  sourceDomainId || u64(srcEpoch) || address20(srcBridge) ||
  destinationDomainId || msgHash || u64(liquidityFee))
escrowId = H("slot-chain-bridge-escrow-v2" || creditId)
```

Both domain descriptors are append-only and source-approved; no message sender supplies an arbitrary destination trust root. The destination descriptor binds the full stable recovery path: L2 genesis, non-proxy `BridgeInboxAdapter`, immutable `ActiveSettlementRouter`, `TerminalSignalVerifier`, `InboxApplyRouterV2`, `InboxCreditStoreV2`, protocol-lifetime `ProtocolReleaseAuthorityV2`, immutable `TerminalDomainRegistrarV2`, protocol-lifetime `TerminalAccumulatorV2`, immutable `NativeLiquidityPoolV2` and frozen destination Bridge facade. The destination infrastructure hash is the Appendix-defined commitment to the deployed runtimes and constructor immutables of every one of those components; address-only binding is insufficient. The source and destination Bridge addresses are fresh immutable non-proxy lifetime constants for that domain; neither Resolver rotation nor a replacement proxy may change their callback, custody, status or terminal identity. An incompatible implementation requires a separately named endpoint and new domain, while every old endpoint remains callable for its old credits. Arbitrary destination invocation and RETRIABLE processing remain later ordinary bridge operations. The two forced-envelope variants have separate byte-exact leaves.

Launch accepts kind 1 only for the L1-to-slot-chain direction, where:

```
srcChainId = settlementChainId = block.chainid
```

Consequently the adapter omits an EIP-2935/MPT proof of its own L1. It performs a bounded `staticcall` directly to the descriptor's non-proxy `BridgeCreditRegistry`, checks its runtime hash and exact fixed-size return, and compares the immutable `CreditRecord`. A reorg that removes the source record necessarily removes every later enqueue transaction; anyone may retry from the still-durable record on the surviving chain. Cross-L1 sources require a separately reviewed protocol version and are not silently routed through this path.

The kind-0 v2 and kind-1 v11 leaf, descriptor and disposition decoders are protocol-lifetime queue ABIs. Every future execution profile must retain their exact semantics for all entries below `queueCount`; a migration may add a new tagged kind but cannot reinterpret an existing byte. The permanent queue keeps every fixed-width descriptor and kind-1 credit record. For kind 0, a candidate supplies raw EIP-2718 bytes matching `rawTxHash` while unexpired; if no party retains those bytes, the head becomes permissionlessly `EXPIRED_NO_TX` after the bounded seven-day validity horizon, using only the stored descriptor. Thus migration neither assumes an expiring blob for a durable bridge credit nor turns missing user bytes into permanent head-of-line blocking.

The deployed V1 `sendMessage(Message)` selector, counter, event, signal, status and recall semantics remain byte-for-byte V1. No Resolver update or interface probe redirects a V1 caller. Launch V2 is a separate, fresh immutable non-proxy SourceBridge with one additive `sendMessageV2(Message,uint64)` selector, where the second argument is the liquidity fee. It normalizes caller calldata by overwriting `id`, `from` and `srcChainId` with its checked next ID, the actual external caller and the immutable chain context before hashing. It

resolves the final destination Bridge and rejects that address as the effective caller. It requires exact `msg.value=message.value+message.fee+liquidityFee`, derives the complete Message hash and exact `keccak256(message.data)` hash/raw length itself, sets checked `enqueueBy=block.timestamp+MAX_BRIDGE_ENQUEUE_DELAY`, and creates only a DIRECT NEW credit. There is no V2 Vault selector, DRAFT state, capsule, restorable token, reverse asset outflow or `getOrDeploy` path. Before incrementing the ID or writing escrow/Registry state it also preserves the V1 safety invariants exactly: `srcOwner`, `destOwner` and `to` are each nonzero; `destChainId` differs from the immutable source chain and equals the selected destination-domain record; the checked sum `message.value+message.fee` is nonzero. If `gasLimit=0`, `fee` must be zero. Otherwise the checked `uint256(gasLimit)-V2_POST_CALL_GAS_RESERVE` must be nonzero; this is the exact destination invocation budget rule. Registry insertion, adapter join, the durable V11 decoder and the validity circuit repeat all those predicates. The corpus sets each owner/target address to zero separately, exercises both gas/fee branches, rejects value and fee both zero, exercises one-below/exactly-at/one-above the reserve, and requires byte-identical nonce, balance, liability, Registry and Queue rollback.

The immutable `BridgeCreditRegistryV2` recomputes `creditId`, accepts writes only from its exact configured SourceBridge capability, and rejects duplicates. Authorization creation, Bridge custody and liability creation are one non-reentrant rollback domain. The Registry stores only this fixed-size authorization and never stores or returns raw Message bytes:

```
(bytes32 sourceDomainId, uint64 srcEpoch, address srcBridge,
 bytes32 bridgeExecutionHash, uint64 emittedAtBlock, bytes32 msgHash,
 bytes32 destinationDomainId, uint64 destChainId,
 address destinationBridge,
 uint64 enqueueBy, address sender, address srcOwner, address destOwner,
 uint256 value, uint64 fee, uint64 liquidityFee,
 bytes32 calldataHash, uint32 calldataLength,
 bytes32 escrowId, uint64 protocolVersion,
 bytes32 releaseManifestHash, bytes32 executionProfileHash)
```

The SourceBridge alone owns mutable custody:

```
SourceLiability(value, executionFee, liquidityFee, status, queueIndex,
                pullClass, pullBeneficiary, pullAmount)
```

The adapter's source read boundary is exactly two calls:

```
creditAuthorizationV2(bytes32 creditId) view returns
 (bytes32 sourceDomainId, uint64 srcEpoch, address srcBridge,
  bytes32 bridgeExecutionHash, uint64 emittedAtBlock, bytes32 msgHash,
  bytes32 destinationDomainId, uint64 destChainId,
  address destinationBridge, uint64 enqueueBy, address sender,
  address srcOwner, address destOwner, uint256 value, uint64 fee,
  uint64 liquidityFee, bytes32 calldataHash, uint32 calldataLength,
  uint64 protocolVersion, bytes32 releaseManifestHash,
  bytes32 executionProfileHash)

creditLiabilityV2(bytes32 creditId) view returns
 (uint256 value, uint64 executionFee, uint64 liquidityFee, uint8 status,
```

```
uint64 queueIndex, uint8 pullClass, address pullBeneficiary,
uint256 pullAmount, uint256 totalLiveLiability)
```

Their selectors are respectively `0x05ecb6c2` and `0xc978978a`; calldata is exactly 36 bytes, gas is capped at 200,000 for each call, and return length is exactly 704 (22 words) and 288 bytes. The escrow ID is independently derived from the queried credit ID; it is not a duplicated CAU2 word. The stored `CreditAuthorizationV2` tuple order is normative: its `escrowId` word immediately precedes the three version-metadata words `protocolVersion`, `releaseManifestHash` and `executionProfileHash`, exactly as displayed above. The getter preserves that stored-field order after intentionally omitting only `escrowId`, which the adapter derives as `H("slot-chain-bridge-escrow-v2"||creditId)` from the queried key; the omission is not permission to reorder or omit any other stored field. All narrow words have canonical zero high padding. Before either read the adapter checks the descriptor-pinned runtime and exact `componentConfigHashV2()` result. It joins every duplicated amount/identity, requires `status=NEW=1`, `queueIndex=UINT64_MAX`, `pullClass=0`, zero beneficiary and zero pull amount, and checks `totalLiveLiability>=value+fee+liquidityFee` and `BALANCE(sourceBridge)>=totalLiveLiability`. Short/trailing return, malformed padding, substituted field, insolvency or call fault rejects before Queue, Registry, Bridge or adapter mutation. In particular, an unqueued NEW credit must match the active route's protocol version, destination chain, release-manifest hash and execution-profile hash. A credit minted before cutover cannot enter the successor Queue; it remains cancellable after its immutable `enqueueBy` deadline. Already-queued credits remain processable through their append-only historical destination domain and components and do not block a later migration. Only the liability moves `NEW->QUEUED->DELIVERED/RECALLED` or `NEW->CANCELLED`. Those two bounded read-only views let the adapter cross-check every immutable field, exact NEW status and aggregate solvency. Queue append, `markQueuedV2`, Registry state, Bridge liability/balance and adapter record are one transaction. Liability remains exactly value plus fee plus liquidity fee through QUEUED. Strict post-`enqueueBy` cancellation and proved FAILED recall reclassify the full amount from pending escrow to the source owner's USER_PULL. Proved DONE authenticates the consumed L2 funding ticket and reclassifies the full amount to its immutable L1 recipient's LP_PULL; it never drops liability or creates unowned SourceBridge surplus. Liability decreases only when the corresponding pull succeeds. Withdrawal is unpausable and CEI-safe, derives the claimant from the transaction frame, accepts only a nonzero claimant-selected recipient, and restores the claim if the recipient call fails or reenters.

The L1 `BridgeDomainRegistry` retains two append-only namespaces. The first is an exact source package keyed by `(destinationChainId,protocolVersion)`; the second is historical source-domain, destination-domain and frozen-endpoint support used by terminal proofs. Successful `REGISTER_RELEASE` authorizes a permissionless preparation call which deploys the target SourceBridge, credit Registry, quota manager and adapter package, then stages its exact manifest, target-registration hash, source-descriptor ID, authorization ID, component addresses, package root and count. Preparation never overwrites a current alias, active ingress binding or active route. A conflicting duplicate fails atomically. The historical proof namespace remains unusable until permissionless `confirmDestinationRegistration(protocolVersion,canonicalSequence,proof)` atomically reads the exact sequence-tagged full canonical tuple through `ActiveSettlementRouter`, requires its L1 canonicalization block to be at least `F_L1` deep, and authenticates the exact domain-registration record under that tuple's `stateRoot`. It never accepts a root or tuple from the caller. The active profile pins one immutable, acyclic `RegistrationMptVerifierDescriptorV2`;

no default verifier exists. Its exact statement is

```
struct RegistrationStorageStatementV2 {
  uint256 settlementChainId; address activeSettlementRouter;
  address bridgeDomainRegistry; bytes32 routeKey;
  uint256 destinationChainId; uint64 protocolVersion;
  uint64 canonicalSequence; bytes32 stateRoot;
  address terminalDomainRegistrar; bytes32 registrarCodeHash;
  bytes32 storageTrieKey; bytes32 expectedValue;
}
```

`routeKey=keccak256(abi.encode(ROUTE_KEY_TYPEHASH,sourceDomainId,bridgeExecutionHash, destinationDomainId))`. The exact type literal is

```
"RegistrationStorageStatementV2(uint256 settlementChainId,"
"address activeSettlementRouter,address bridgeDomainRegistry,bytes32 routeKey,"
"uint256 destinationChainId,uint64 protocolVersion,uint64 canonicalSequence,"
"bytes32 stateRoot,address terminalDomainRegistrar,bytes32 registrarCodeHash,"
"bytes32 storageTrieKey,bytes32 expectedValue)"
```

with the displayed fragments concatenated without whitespace or newline. `REGISTRATION_STATEMENT_TYPEHASH` and `publicInputSchemaHash` are both Keccak of those bytes, and `statementHash=keccak256(abi.encode(TYPEHASH,fields...))`. Registry derives all twelve fields from its staged route, immutable chain/contract identities and the Router-returned canonical history; proof calldata supplies none of them.

The packed proof schema is committed by

```
REGISTRATION_MPT_PROOF_SCHEMA_LITERAL = concat(
  "MptProofV1=be16(accountNodeCount)||be16(storageNodeCount)||",
  "(be16(nodeLength)||canonicalRlpNode)*accountNodeCount||",
  "(be16(nodeLength)||canonicalRlpNode)*storageNodeCount;",
  "rootToLeaf;EthereumKeccak;canonicalHexPrefix;canonicalRlp;",
  "selectedNodesValidated;unselectedInlineOpaque;emptyBranchValue;",
  "absenceRejected;valueRequired")
proofSchemaHash = keccak256(REGISTRATION_MPT_PROOF_SCHEMA_LITERAL)
```

The two hash layers are

```
REGISTRATION_MPT_VERIFIER_CONFIG_TYPE_LITERAL = concat(
  "RegistrationMptVerifierConfigV2(bytes32 publicInputSchemaHash,",
  "bytes32 proofSchemaHash,bytes4 selector,uint16 maximumNodesPerPath,",
  "uint16 maximumTotalNodes,uint16 maximumNodeBytes,",
  "uint32 maximumProofBytes,uint64 verificationGasLimit)")
registrationMptVerifierConfigurationHash = keccak256(abi.encode(
  keccak256(REGISTRATION_MPT_VERIFIER_CONFIG_TYPE_LITERAL),
  publicInputSchemaHash, proofSchemaHash, selector, 65, 130, 600, 78264, 11000000))

REGISTRATION_MPT_VERIFIER_DESCRIPTOR_TYPE_LITERAL = concat(
  "RegistrationMptVerifierDescriptorV2(address verifier,bytes32 runtimeHash,",
```

```
  "bytes32 configurationHash,bytes32 publicInputSchemaHash,",
  "bytes32 proofSchemaHash,bytes4 selector,uint16 maximumNodesPerPath,",
  "uint16 maximumTotalNodes,uint16 maximumNodeBytes,",
  "uint32 maximumProofBytes,uint64 verificationGasLimit)")
registrationMptVerifierDescriptorHash = keccak256(abi.encode(
  keccak256(REGISTRATION_MPT_VERIFIER_DESCRIPTOR_TYPE_LITERAL),
  verifier, runtimeHash, registrationMptVerifierConfigurationHash,
  publicInputSchemaHash, proofSchemaHash, selector, 65, 130, 600, 78264, 11000000))
```

The configuration layer deliberately excludes verifier, runtime hash and itself; the descriptor repeats every configuration field and rejects unless its `configurationHash` equals the independently recomputed first layer. The immutable verifier exposes only

```
registrationMptVerifierConfigHashV2() external view returns (bytes32)
REGISTRATION_MPT_VERIFIER_CONFIG_READ_GAS = 50000
```

Registry first checks `EXTCODEHASH`, then makes a zero-value exact-four-byte `STATICCALL` with that gas stipend and accepts exactly one 32-byte word equal to the recomputed configuration hash. Empty, short, trailing, faulting or mismatched returns reject before proof dispatch.

The verifier interface is exactly

```
verifyRegistration(
  RegistrationStorageStatementV2 calldata statement, bytes calldata proof)
    external view returns (bytes32 statementHash)
```

with selector from `"verifyRegistration((uint256,address,address,bytes32,uint256,uint64,uint64,bytes32,address,bytes32,bytes32,bytes32),bytes)"`. Its ABI head is the twelve statement words followed by the proof offset word `0x1a0`; the tail is canonical bytes length/data/zero-padding with no trailing calldata. Its total calldata length is exactly `452+ceil32(proof.length)` bytes; the 78,264-byte cap measures the packed inner `proof.length`, not the outer ABI envelope. BridgeDomainRegistry makes one zero-value `STATICCALL` forwarding exactly 11,000,000 gas and accepts exactly 32 return bytes equal to its locally recomputed statement hash. Revert, OOG, malformed return, substituted statement or a Boolean result rejects before support state changes.

`proof` is `MptProofV1`: big-endian `u16(accountNodeCount)||u16(storageNodeCount)` followed by each root-to-leaf canonical-RLP node as `u16(nodeLength)||nodeBytes`. Each path has at most 65 nodes, the two paths have at most 130 nodes, each node is at most 600 bytes, and the whole proof is at most `4+130*(2+600)=78,264` bytes. Any noncanonical framed path node, wrong inline/hash reference, surplus node/byte or gas use above 11,000,000 rejects. The account proof fixes the registrar address and code hash; the storage proof fixes one nonzero `registrationCommitment[protocolVersion]` word. No Solidity struct or adjacent slot is inferred from that one path. The registrar exposes `registrationCommitmentV2(uint64)viewreturns(bytes32)`. Its selector is `0xd972f446`; callers use a zero-value `STATICCALL` with exactly 75,000 gas and exactly 36 calldata bytes (selector plus one canonically padded `uint64` word), and accept exactly 32 return bytes. Dirty high bytes, missing, short, trailing, reverting or caller-dependent records reject. The registrar stores only:

```
registrationCommitment =
```

```
  H("slot-chain-destination-registration-v1" ||
    u64(protocolVersion) || releaseManifestHash ||
    u256(destinationChainId) || destinationNamespace ||
    destinationDomainId || address20(destinationBridge) ||
    destinationInfrastructureHash || executionProfileHash)

baseSlot = H("slot-chain-terminal-domain-registrar.registrationCommitment.v1")
elementSlot = H(bytes32(uint256(protocolVersion)) || baseSlot)
storageTrieKey = H(elementSlot)
```

The account trie key is `keccak256(address20(terminalDomainRegistrar))`. The root reference is exactly `stateRoot`; a child reference shorter than 32 bytes must equal the complete inline child RLP and a 32-byte reference must equal Keccak of the next complete node. Other reference lengths reject. Canonical RLP uses the shortest length form, forbids leading length zeros and uses long form only for payloads of at least 56 bytes. A branch has exactly 17 items; an extension/leaf has exactly two. Hex-prefix flags are only 0–3, the even form has a zero low nibble, an extension path is nonempty, and a leaf is accepted only at exact key exhaustion. The account leaf is exactly `[nonce,balance,storageRoot,codeHash]`; nonce and balance are minimal unsigned integers of at most 32 bytes and both roots are exact 32-byte strings. The storage proof starts at that authenticated `storageRoot`, consumes the exact `storageTrieKey`, and returns the canonical minimal-RLP unsigned value equal to `expectedValue`. Missing, surplus or out-of-order nodes, trailing container bytes and an alternative yet value-equivalent RLP all reject. The storage-trie leaf value is canonical RLP of the minimal big-endian unsigned integer represented by the nonzero 32-byte commitment (and therefore preserves leading-zero semantics through the expected value). L1 recomputes the full commitment from its staged release manifest and accepts only exact equality; substitution of any field, mapping key, code hash, account or root rejects. That destination proof does not authorize a source cutover. Each release adds at most 64 entries; there is no global cap, deletion, redirect or reinterpretation.

An inline child on the selected path must byte-equal the next framed node, which is then fully decoded and semantically validated. An unselected branch child is authenticated by the already verified parent-node hash but cannot affect membership; it is therefore checked only as one canonically framed RLP item of an allowed empty/hash/inline reference type. The verifier must not recursively walk an unselected inline subtree. Doing so would make verification gas depend on unrelated state and permit gas amplification without strengthening the selected-key proof. The seventeenth branch item must be the canonical empty byte string: all keys proved here are fixed 32-byte hashes, so no such key can terminate as the prefix of another at a branch value.

The release gas certificate uses the exact production compiler, optimizer and EVM revision and includes full 65-node account and storage paths with maximum eligible branch references, selected inline nodes, and deeply nested unselected inline items near key exhaustion. Deployment is permitted only while `ceil(1.30*measuredWorstAcceptedGas)<=11,000,000`. The frozen reference build's worst accepted fixture consumes 7,940,171 gas, so the gate requires 10,322,223 gas. Any compiler, RLP traversal or EVM-gas-schedule change invalidates the certificate and requires remeasurement before release.

The source package becomes `ARM_READY` only at `stagedAtBlock+214`. This is an explicit L1 code/configuration/package review delay, not MPT finality and not evidence that target tx0 executed. `PUBLISH_MIGRATION_ARM` exact-reads the target's version-keyed package and rejects unless manifest, target-registration hash, source descriptor, bundle, adapter, package root and

count all still match and the review delay has elapsed. No newest/global pointer exists. Thus merely staged or ARM_READY target $v+1$ cannot affect the active route for v.

Both SourceBridge credit preview and payable send perform the same O(1) active route read. In one fail-closed revert domain BridgeDomainRegistry first makes a zero-value exact-selector STATICCALL to Router migrationActivationContextV1() (selector 0x7cf70319) with exactly 100,000 gas and accepts exactly 320 bytes. It requires MACT||bytes28(0), canonical padding, lifecycle IDLE=0, and every remaining context word zero. Only after that read succeeds does Registry make the zero-value exact-selector STATICCALL to activeSettlementStateV1() with exactly 50,000 gas and accept exactly 256 bytes: eight ABI words, word 0 equal to ASR1||bytes28(0), canonical padding and gate phase ACTIVE. MACT or ASR1 empty, short, trailing, malformed-magic, noncanonical, reverting or OOG return data fail the same route read before preview returns or send/enqueue writes. No private Router field or caller-supplied lifecycle value may replace the MACT read. The returned activeProtocolVersion selects exactly (destinationChainId,activeProtocolVersion); Registry then requires the calling bundle's exact (sourceDomainId,bridgeExecutionHash) and never falls back to another version. In Router ARMING, READYING, ABORTING, ACTIVATING, REGISTERING or PUBLISHING, or gate ARMED/READY, both preview and send reject without writes.

For external conformance, the package getter is bridgeRoutePackageV1(uint256,uint64) (selector 0x52e2562f) with a 100,000-gas static-read cap and exactly 480 return bytes (15 ABI words): BRP1||bytes28(0), state (STAGED=1, ARM_READY=2, CONSUMED=3), destination chain ID, protocol version, staged block, ready block, target-registration hash, source-descriptor ID, authorization ID, package root, package count, then the left-padded SourceBridge, credit Registry, quota manager and adapter addresses. Integers use canonical high-zero padding. Bad magic, length, padding, state, zero field, count, component code/configuration or root rejects the arm before any Router, gate or source write.

The cross-contract route read is not an abstract service view. SourceBridge preview/send and the active Bridge adapter use the identical zero-value STATICCALL

```
activeBridgeRouteV2(
  uint256 destinationChainId, address sourceBridge,
  bytes32 sourceDomainId, bytes32 bridgeExecutionHash,
  address checkedTarget)
selector = 0x9266ca8d
ACTIVE_BRIDGE_ROUTE_READ_GAS = 1200000
```

with exactly 164 calldata bytes and exactly 384 return bytes (twelve ABI words): ABR2||bytes28(0), active protocol version, source chain ID, source registration epoch, source domain ID, bridge execution hash, destination domain ID, left-padded destination Bridge, release-manifest hash, execution-profile hash, invocation-policy hash and the left-padded checked target echoed from the calldata. The first three integers have canonical high-zero padding and the checked target is nonzero. SourceBridge must pass sourceBridge=address(this); an adapter must be the Router-active exact binding for that source Bridge. Registry independently executes the ordered MACT–ASR1–RTR2–BRX1 reads, binds both raw row hashes to the consumed primitive package, then makes the exact BID1 decision for the checked target. The caller must compare the returned target to its own Message preimage; a caller-supplied separate check target is not authority. Any caller, value, calldata length, return length, magic, padding, version, source tuple, binding, code/configuration, revert or OOG mismatch rejects before preview returns or send/enqueue writes. Neither caller retries

another version.

The versioned primitive getters used by REGISTER, stage, consume and ABR2 are:

```
bridgeRouteExpansionV1(uint64 version) selector 0x118464c5
  calldata 36; gas 200000; return 1952 bytes / 61 words / BRX1
bridgeInvocationPolicyV1(uint64 version) selector 0xc1b06888
  calldata 36; gas 250000; return 2176 bytes / 68 words / BIP1
bridgeInvocationDecisionV1(uint64 version,address target) selector 0x6bfa2ef0
  calldata 68; gas 100000; return 128 bytes / BID1
profileIngressMembershipV2(uint64 version,bytes32 authorizationId)
  selector 0x4c4455b8; calldata 68; gas 100000;
  return 160 bytes / PIM2
```

BRX1 is magic, version, Bridge authorization ID, source chain ID, source genesis hash, source namespace, destination namespace, `destinationRegistrationCommitment`, versioned invocation-policy hash and count, then the 28 canonical source-descriptor words and the following 23 canonical Source-bundle constructor words:

```
settlementChainId, sourceChainId, sourceGenesisHash, registryNamespace,
sourceBridgeRuntimeHash, sourceBridgeStorageLayoutHash, bridgeKernelHash,
sourceTerminalVerifier, officialVault[0], officialVault[1], officialVault[2],
sourceOnlyRole=1, nativeQuotaManagerRuntimeHash, nativeQuotaPeriod,
nativeEthQuota, sourceDomainRegistrar, sourceRegistrationEpoch,
supportRegistry, supportRegistryRuntimeHash, supportRegistryConfigurationHash,
sourceTerminalVerifierRuntimeHash,
sourceTerminalVerifierConfigurationHash, legacyV1Bridge
```

The 23-word suffix is registration storage, not BRD1 calldata: REGISTER derives it from the strict profile, commits it through `H(BRX1)` in RTR2 and stores the exact bytes. BRD1 can therefore reconstruct SBD1 after process restart without a caller witness or an in-memory `SettlementRegistration` object. It rejects unless re-encoding the descriptor and constructor tuple reproduces the stored 61 words and all derived Source identities. BIP1 is magic, version, the versioned policy hash, count, then 64 sorted unique left-padded address slots; the active prefix includes address zero and the suffix is all zero. The decoder recomputes the policy hash from that list and version. BID1 uses a bounded binary search over this same canonical BIP tuple; it has no duplicated boolean/index storage. PIM2 performs a bounded lookup in the same sorted authorization-ID tuple committed by PIR2 and has no second membership mapping.

Preparation and consumption likewise use fixed Router-to-Registry calls, never a caller-supplied package tuple:

```
stageBridgeRoutePackageV1(uint64 protocolVersion) selector = 0x9dad437b
  gas = 12000000; return = BRS1 || u64(version) || packageRoot || u64(readyBlock)
consumeBridgeRouteArmReadyV1(uint64 protocolVersion,
                             bytes32 targetRegistrationHash)
  selector = 0x68034634; gas = 2500000
  return = BRC1 || u64(version) || targetRegistrationHash || packageRoot
```

Each return is exactly 128 bytes (four ABI words) with right-zero magic and canonical high-zero integer padding. Both calls require `msg.sender=ActiveSettlementRouter`, `msg.value=0`,

exact calldata length (36 and 68 bytes respectively), pinned Registry code/configuration and exact returndata. Stage is reachable only inside Router's non-reentrant permissionless `prepareBridgeRoutePackageV1(uint64)` (selector `0x1ccbc24a`), after exact public REGISTER_RELEASE stores have been re-read. BRD1 uses exactly 36 calldata bytes, canonical uint64 high-zero padding, zero value, a fixed 25,000,000-gas conformance budget and a nonzero permissionless caller; wrong selector, length, padding, value or budget rejects before staging. It returns exactly `BRD1||u64(version)||packageRoot||u64(readyBlock)` (128 bytes). Stage first observes MACT in `BRIDGE_PREPARING`; consume is reachable only while Router lifecycle is `ACTIVATING`. Each path completes all external reads before its first write. Stage reads MACT, RTR2, BRX1, PIR2, PIM2, PIA2, BIP1; consume reads MACT, ASR1, RTR2, BRX1, PIR2, PIM2, PIA2, BIP1. They share one fail-closed revert domain:

```
migrationActivationContextV1() selector = 0x7cf70319
  gas = 100000; return = 320-byte MACT
activeSettlementStateV1() selector = 0x4a95c306
  gas = 50000; return = 256-byte ASR1
targetReleaseRegistrationV2(uint64) selector = 0xf588fec3
  gas = 100000; calldata = 36 bytes; return = 512-byte RTR2
```

MACT must name ACTIVATING and the source/target tuple. ASR1 must name gate `READY`; its generation, active Settlement/version and complete target tuple must equal MACT, while its active Settlement/version remain the current source. RTR2 must bind that target Settlement, predecessor version, manifest and registration hash. The MACT `activationContextHash` is the opaque correlation commitment to Router's complete callback frame defined by the migration-activation commitment below; because compact MACT does not disclose that complete frame, Registry requires it nonzero but does not treat an independently unrecomputable hash as an additional authorization. Authorization is the Router-only caller plus the exact lifecycle and all decoded primitive joins; no caller may supply any of those return values. Registry then checks the unconsumed `ARM_READY` row and all live package code/configuration before setting CONSUMED. Direct calls, wrong Router frame/version/hash, premature block, already-consumed row, bad return or downstream fault revert the whole Router/PVM/factory/bundle/Registry journal. REGISTER_RELEASE is store-only and never sets `stagedAtBlock`. A failed first BRD1 attempt leaves no package; the later successful BRD1 starts review at its own block. Once staging succeeds, idempotent exact BRD1 reuse preserves the original `stagedAtBlock` and `readyBlock`; no replay can reset or accelerate the 214-block review. The primitive package root is exactly `H("slot-chain-source-route-package-v2"||H(RTR2)||H(BRX1)||H(PIR2)||H(PIM2)||H(PIA2)||H(BIP1))`. RTR2 uses target-registration domain v3 and includes `bridgeRouteExpansionHash=H(BRX1)`. A retained route row carries an explicit `PRIMITIVE` or `LEGACY` kind. Only a consumed PRIMITIVE row can serve ABR2; an unconsumed LEGACY row alone may use the older proved-registration finality rule. A null model object is never a row-kind discriminator.

V1 and V2 never share a Bridge account. The existing V1 proxy, Resolver, balance, nonce, status, pause/lock, QuotaManager and every Vault flow remain unchanged. V2-aware applications explicitly accept a fresh Bridge address and ContextV2; applications hard-coding V1 remain V1-only.

The SourceBridgeV2 bundle is deployed permissionlessly through a manifest-pinned immutable CREATE2 factory. Its descriptor binds factory address/runtime/config, bundle

salt/init-code hash and the derived bundle-deployer runtime, the legacy V1 Bridge address that must not be reused, resulting SourceBridge address/runtime/config/layout, the Bridge-unique Registry and V2-only QuotaManager, support Registry/registrar/ epoch, immutable terminal verifier and Router binding, pauser and SignalService. The bundle-deployer address must equal the final 20 bytes of

```
keccak256(0xff || factory || salt || keccak256(init_code)).
```

The bundle init-code hash commits every acyclic child code/configuration primitive and excludes only the three addresses that its constructor derives from its own address. The exact Appendix formulas derive the bundle via CREATE2 and the Bridge, Registry and QuotaManager via its constructor's CREATE nonces 1–3 in one atomic transaction. A front-run is accepted only when the complete current bundle, code and configuration are exact; otherwise activation fails closed. No child address is embedded in the bundle CREATE2 preimage, so the graph is acyclic; the outer descriptor separately binds every derived address and current runtime/configuration.

The profile does not select an arbitrary factory. It fixes one protocol-root Source factory address, runtime hash and configuration hash; that configuration commits the compiled Source-bundle and Bridge-adapter creation artifacts. Missing or mismatched live factory code/configuration makes REGISTER/BRD fail closed. The exact factory hash and view are

```
sourceBundleFactoryConfigurationHash = H(
 "slot-chain-source-bundle-factory-config-v1" ||
 u256(settlementChainId) || address20(protocolVersionManager) ||
 bundleCreationHash || bundleRuntimeHash ||
 adapterCreationHash || adapterRuntimeHash ||
 sourceBridgeCreationHash || sourceBridgeRuntimeHash ||
 creditRegistryCreationHash || creditRegistryRuntimeHash ||
 quotaManagerCreationHash || quotaManagerRuntimeHash ||
 bytes4(0xf3514e73) || bytes4(0x8fafca22) ||
 u64(15000000) || u64(2000000) || u64(50000) || u64(500000))
sourceBundleFactoryConfigV1() // selector 0x74bab161; 640 bytes / SBF1
 returns (bytes4 magic,uint256 settlementChainId,address protocolVersionManager,
  bytes32 bundleCreationHash,bytes32 bundleRuntimeHash,
  bytes32 adapterCreationHash,bytes32 adapterRuntimeHash,
  bytes32 sourceBridgeCreationHash,bytes32 sourceBridgeRuntimeHash,
  bytes32 creditRegistryCreationHash,bytes32 creditRegistryRuntimeHash,
  bytes32 quotaManagerCreationHash,bytes32 quotaManagerRuntimeHash,
  bytes4 bundleDeploySelector,bytes4 adapterDeploySelector,
  uint64 bundleDeploymentGas,uint64 adapterDeploymentGas,
  uint64 componentConfigReadGas,uint64 postcheckReserveGas,
  bytes32 configurationHash)
```

Every bytes4 is left-aligned with 28 trailing zero bytes; every address and narrow integer is canonically padded. SBF1 is callable while root-inactive for Factory validation, but both deployment entries first exact-read their own 128-byte PRA1 and require ACTIVE. This check is repeated on every call, including after process restart; a cache or prior success is not authority.

The bundle deployment entry is exactly

```
deploySourceBundleV1(bytes32 salt,bytes descriptor,bytes constructorArgs)
  selector 0xf3514e73; value=0; gas=15000000; calldata=1668 bytes
```

The head is `(salt,0x60,0x380)`; the first tail is length 752 followed by the exact packed 28-field Source descriptor and 16 zero padding bytes; the second tail is length 736 followed by the 23 static constructor words listed for BRX1. There is no gap, alias or suffix. Factory decodes both tails, re-derives the descriptor ID, bundle init code and CREATE2 deployer, derives SourceBridge, Registry and QuotaManager from CREATE nonces 1–3, and exact-reads the root BridgeDomainRegistry and root-lifetime SourceTerminalVerifier before creation. The four-code set (deployer plus three children) must be wholly absent/free or wholly present; a partial set rejects. Balance-only prefunds are preserved. Any nonce/code collision on a CREATE target rejects; exact reuse requires every code/configuration and constructor-derived identity to match. The return is exactly 288 bytes:

```
SBD1 = (bytes4 magic,address bundleDeployer,address sourceBridge,
        address creditRegistry,address quotaManager,address terminalVerifier,
        uint8 created,bytes32 bundleInitCodeHash,bytes32 sourceDescriptorId)
```

The per-release adapter is independently deployed by that same factory through

```
deployBridgeAdapterV1(uint64 version,bytes32 sourceDescriptorId,
 bytes32 configurationHash,address sourceBridge,address creditRegistry,
 address router,address queue,address sealAuthority)
  selector 0x8fafca22; value=0; gas=2000000; calldata=260 bytes
```

All eight arguments are static canonical words. Factory exact-reads PVM1 from `sealAuthority`, requires it to name this ACTIVE SourceBundleFactory, Router and Queue, then exact-reads the SourceBridge, credit Registry, Router and Queue code/configuration and independently recomputes the adapter configuration. The return is exactly 224 bytes:

```
SAD1 = (bytes4 magic,address adapter,bytes32 salt,bytes32 initCodeHash,
        bytes32 runtimeHash,bytes32 configurationHash,uint8 created)
```

Both entries reserve 500,000 gas for complete postchecks. Short/trailing/dirty return, wrong deterministic address, partial code, wrong live dependency, constructor mismatch or any late fault reverts every CREATE, cache-independent publication and retained row in the call. Exact existing deployments are idempotent and return `created=0`; first creation returns 1.

The adapter address is

```
saltA = H("slot-chain-bridge-adapter-salt-v1" || u64(version) ||
          sourceDescriptorId)
initHashA = H(adapterCreationCode || abi.encode(
  adapterConfigHash,sourceBridge,creditRegistry,router,queue,sealAuthority))
adapter = last20(H(0xff || factory || saltA || initHashA)).
```

Here `sealAuthority=ProtocolVersionManager`. Its exact 100-byte component configuration is

```
address20(creditRegistry) || address20(sourceBridge) ||
address20(router) || address20(queue) || address20(sealAuthority)
```

and binds every constructor-fixed authority address. The real compiled creation bytecode hash must replace the model artifact and be sealed into the root deployment receipt before this design can be implemented; a process-local behavioral object is never deployment authority.

Router retains one protocol-lifetime BridgeDomainRegistry, immutable factories, source bundles by descriptor ID and an append-only version-to-descriptor map. Current pointers are aliases only and preparation cannot change them. Every successor release has a fresh domain and fresh SourceBridge/Registry/Quota accounts: the strict profile derives the bundle salt from the protocol-root manifest namespace and protocol version, so even a byte-identical implementation cannot alias a predecessor's private accounting state. Historical adapters keep their original bundle only for cancellation, terminal release and refunds; both the public credit-ID preview and payable send path use the same exact Router-active version lookup and therefore cannot mint from a historical bundle. Each bundle owns its private nonce, status, pull and aggregate-liability mappings. It has no owner, upgrade, reinitializer, delegatecall or public mutation helper. The source generation is a fixed nonzero registration number and every credit binds its exact Bridge execution descriptor. At launch all source components share settlement Ethereum; cross-L1 support requires a new reviewed kind/domain. Admission also requires `0<srcChainId<=UINT64_MAX`, nonzero `sourceDomainId`, `0<srcEpoch<=UINT64_MAX`, `0<emittedAtBlock<=UINT64_MAX`, nonzero `msgHash`, `destChainId=l2ChainId<=UINT64_MAX`, a registered nonzero `destinationDomainId` and a nonzero `srcBridge`; no narrowing cast or destination-chain inference is permitted.

`BridgeInboxAdapter` is a non-proxy immutable contract. Its profile-bound runtime and constructor values fix the credit registry, source Bridge, immutable settlement router, singleton ForcedQueue and ProtocolVersionManager. Its sole mutable configuration is a zero-to-nonzero `destinationDomainId` slot sealed once by that manager during manifest activation; the setter deletes its authority bit and cannot run twice. Enqueue rejects while the slot is zero. It has no upgrade, delegatecall, mutable target, arbitrary-call or caller-supplied authenticated value path. A different destination uses a distinct adapter/domain; an old adapter remains executable for its existing records.

The adapter is the exactly-once machine `NONE->QUEUED(index)`, keyed by `creditId`. Any caller supplies the separate queue processing fee and the full message preimage. The payable adapter initially retains `msg.value`, derives both domains and the ID, direct-reads the immutable record and live source liability, and validates all fixed bounds including `block.timestamp<=enqueueBy`. It first calls the router's nonpayable

```
syncIngressV2() returns
  (uint8 result, uint64 activeProtocolVersion, uint64 routerGeneration)
```

as a registered immutable adapter. Its selector is the first four Keccak bytes of `"syncIngressV2()"`; calldata is exactly four bytes and returndata is exactly 96 bytes. STAMP=1 requires both narrow words nonzero; SYNCED=2 requires both zero. Unknown result, noncanonical high padding, short/trailing return or fault rejects and rolls the adapter back. If that call changes mode, the adapter records no credit or queue leaf, adds the still-local full value to `claimable[msg.sender]`, and returns `SYNCED`; a CEI pull withdrawal is callable independently of protocol mode and the caller retries in a later transaction. If unchanged, the same adapter immediately calls the router's payable

```
appendFromAdapterV2(uint64 activeProtocolVersion, uint64 routerGeneration,
                    uint8 kind, bytes descriptorBytes)
```

```
  payable returns (uint8 result, uint64 queueIndex)
```

with the exact STAMP words, its immutable kind and canonical fixed descriptor. Its selector is the first four Keccak bytes of `"appendFromAdapterV2(uint64,uint64,uint8,bytes)"`. The four-word head is version, generation, canonically padded kind and offset `0x80`; the tail is the exact length, 220-byte kind-0 or 541-byte kind-1 descriptor, and zero right-padding. Total calldata is respectively 388 or 708 bytes, with no other length valid. The exact 64-byte return is QUEUED=1 plus the new non-sentinel index; every other result, padding, length or index rejects. This second entry point does not call `sync()` again: it requires the registered caller, ACTIVE phase, and exact current version and generation stamp, then forwards all value with the exact descriptor to the queue. The immutable adapter has no entry point that can split, defer or redirect this pair and makes the second call before any other external call in the same top-level transaction. EVM call execution supplies no interleaving between them. A stale stamp after any migration transition rejects and returns the value by call-level rollback; it cannot preserve a second state transition inside a reverting payable call. A successful router retains zero wei. The queue checks `queueCount<UINT64_MAX`, appends at the current count and timestamps the deadline; then the source Bridge changes NEW to QUEUED at that exact index and the adapter records the returned index, all in the same revert domain. No reservation, source proof, PENDING state or finalization call exists. A repeated zero-value query returns the stored index; a duplicate with funds reverts. The source event alone never means queued.

Kind 0 requires a nonexpired `validUntil`, `validUntil<=enqueuedAt+MAX_FORCE_VALIDITY_SECONDS`, and `accountedGas=max(gasLimit,intrinsicGas,MIN_FORCE_ACCOUNTED_GAS)`. The initial validity horizon is seven days. Kind 1 uses the permanent conservative bound `accountedGas=MAX_FORCE_MESSAGE_GAS=5,000,000` for its encoded credit and routed pin. Kind 0 requires `0<accountedGas<=5,000,000` and `0<byteLength<=131,072`; kind 1 permits an empty Message payload and requires `byteLength<=131,072`. The active profile fixes one shared five-word fee schedule

```
(fixedIngressWei, executionWeiPerAccountedGas,
 proofWeiPerAccountedGas, permanentWeiPerByte,
 maximumAcceptedFeeWei).
requiredDeposit = fixedIngressWei
  + accountedGas * (executionWeiPerAccountedGas
                    + proofWeiPerAccountedGas)
  + byteLength * permanentWeiPerByte.
```

All additions and multiplications are checked `uint256`; every coefficient and the cap is nonzero, `requiredDeposit<=maximumAcceptedFeeWei`, and the release is invalid unless the maximum gas/byte geometry and `UINT64_MAX*maximumAcceptedFeeWei` are representable. Each payable adapter requires `msg.value=descriptor.deposit=requiredDeposit` exactly; overpayment and underpayment both revert before retained state. `fixedIngressWei` is the release-calibrated upper bound for all admission costs independent of `accountedGas` and `byteLength`, including the fixed descriptor, event, cold-account and fixed storage-write geometry. `executionWeiPerAccountedGas`, `proofWeiPerAccountedGas` and `permanentWeiPerByte` are respectively upper bounds, with the release's explicit safety margin, for execution gas, proving cost and permanent publication/storage of one raw input byte. The profile records the L1 base-fee, blob-fee and L2/prover price caps and margin used to derive all four amounts; a release whose benchmark corpus exceeds any bound rejects. The five words are identical in every authorization of one profile and are componentwise nondecreasing across successor

profiles, so an old exact adapter never receives a cheaper active schedule after migration. This deposit covers execution, proof credit and permanent calldata/state costs. It is distinct from the SourceBridge's already escrowed `value+fee+liquidityFee`. The adapter verifies kind-1 source escrow and all static bounds before calling the router; failure leaves neither deposit, queue leaf nor partial credit. The mandatory codec corpus pins kind-0 raw-transaction lengths and kind-1 `Message.data.length` at 0, 31, 32, 33 and 131,072 bytes (with kind 0 rejecting zero), and rejects a one-byte mismatch between the normalized Message, descriptor, fee and per-block byte accounting.

Only `ActiveSettlementRouter` may append, and it accepts these two entry points only from an exact typed `authorizedIngress[caller]` entry. The target release/profile precommits the complete authorization root and deployed adapter objects. Preparation stages the exact inactive package and begins its 214-block L1 review. The proof-authenticated activation journal consumes that exact `ARM_READY` entry, activates the SourceBridge, seals and binds its adapter, then publishes the new active version. Any fault restores every package, binding, consumed bit and alias. The next transaction may both create a credit and enqueue it; there is no post-cutover support gap and no governance action. Address, object, domain or kind-role conflicts reject, while old exact adapters remain usable for status, cancellation and refund of their historical records. Only the exact adapter deployment named by the active authorization may append; after cutover an old adapter cannot enqueue a still-NEW predecessor credit. That source credit remains permissionlessly cancelable after its strict `enqueueBy`. Every adapter fixes this router and queue in constructor/configuration. The router calls the active Settlement's `sync()` only in nonpayable `syncIngress` under a non-reentrant guard and reports `SYNCED` without accepting value if any transition occurred. Its adapter caller preserves that transition, creates no admission record and keeps the entire fee locally claimable by the original caller. Otherwise it computes `enqueuedAt=block.timestamp` and `dueAt=max(enqueuedAt+FORCE_DELAY,lastDueAt,recoveryExpiry+1)` (the last term is zero outside recovery), and appends atomically. Neither a sync-result Boolean, enqueue timestamp nor due time is accepted in router calldata; payable append rechecks the shared gate is ACTIVE and the exact stamp before forwarding value, without a second sync decision. Router, queue and adapter are nonreentrant, and failure of any later append/Bridge/adapter write reverts the value transfer and all state. No public caller can invoke the forwarding path. `ForcedQueue` stores one authoritative `QueueDescriptor` per index. It is the complete fixed-width kind-0 or kind-1 leaf preimage in Appendix A, excluding only kind-0 `rawTx` bytes themselves but including their hash. Thus sender, nonce, chain IDs, byte/gas fields, fee, validity, refund address, enqueue/due times, deposit and every bridge-credit field remain durable. Kind 1 requires DIRECT and zero Vault/capsule fields. The Merkle leaf is always recomputed from this stored descriptor; admission atomically writes the descriptor, deposit/escrow accounting, append frontier and root or writes nothing. The fee is deliberately nonrefundable after append: it buys FIFO storage, proving and processing even when an expired or invalid kind-0 item has no Ethereum transaction. The queue stores `depositPrefix[i+1]=depositPrefix[i]+deposit[i]` with checked addition. It is also the sole owner of `cursor`; neither Settlement, Protocol nor either ingress adapter has a second descriptor array, root, count, cursor or due-time cache. Settlement reads the queue's `dueAt` directly for the canonical head and a stored best's live boundary; there is no callback, cached deadline or duplicated scalar. Missing indices return `UINT64_MAX`. Because that value is also a valid deadline, every force-due or live-boundary decision first requires `cursor<count`; only then may it compare `dueAt(cursor)` to the timestamp. A consumed/absent boundary is never due, including at timestamp `UINT64_MAX`; a present final boundary whose stored deadline is exactly `UINT64_MAX` becomes due at equality. Thus `dueAt`

is nondecreasing and force-due/coverage checks are one authoritative O(1) read with one O(1) existence check. The queue exposes exactly two cursor-authority transitions:

```
advanceCursor(uint64 expectedStart, uint64 end,
              address proofBeneficiary)              // active Settlement only
setActiveSettlement(address expectedOld, address new) // router only
```

`advanceCursor` requires `cursor=expectedStart<=end<=count` and writes the cursor and, in O(1), moves `depositPrefix[end]-depositPrefix[expectedStart]` from unconsumed escrow to `claimable[proofBeneficiary]`. The beneficiary is the exact nonzero public input authenticated by the accepted proof; it is not chosen by the queue caller. The queue remains the ETH custodian and enforces the solvency lower bound `address(this).balance>=unconsumedEscrow+totalClaimable`; every claim-ledger mutation updates `totalClaimable` by the same checked amount, so no global sum or scan is performed. ETH forced into the contract without an append is surplus, never enters either liability and cannot make an append, advance or withdrawal revert. It exposes an unpausable CEI pull withdrawal. There is no per-item callback and no "unused" refund. `setActiveSettlement` requires the stored old address and is called only inside bootstrap or an atomic version switch. The ingress router never advances consumption, and a frozen Settlement cannot do so.

The queue is a protocol-lifetime, depth-64 append-only Merkle vector with fixed capacity `UINT64_MAX` items: the final leaf is deliberately unused. It stores root, `uint64` count/cursor, cumulative deposit prefix and a 64-node append frontier. Admission rejects only when `count=UINT64_MAX`; no version migration rotates or resizes the queue. A canonical DFS range multiproof authenticates the consumed leaves and, when present, the first excluded boundary leaf under the frozen `(forceRoot,forceCutoff)`. It supplies no fully covered subtree and no unused hash; at most 257 sibling hashes authenticate a 65-leaf block range. Skipping, reordering, changing the start index or claiming an early boundary changes the reconstructed root. Kind-0 raw bytes are enqueue calldata and their hash is in the leaf; blobs are forbidden on this availability path. The deliberately unused final tree leaf avoids a nonexistent height-64 frontier slot when the old index has all 64 bits set. At one billion appends per second the usable space lasts more than five centuries, so exhaustion is outside any credible Ethereum lifetime while remaining an explicit hard stop.

Opening or refreshing recovery freezes only the current stored `(root,count)`. Settlement requires the requested cutoff to equal the live queue count and may validate the wrapped root by folding the fixed 64-word append frontier; it never recomputes the root from retained descriptors. Full descriptor-history root construction exists only as an off-chain differential test oracle.

Each proved block consumes the unique maximum FIFO prefix satisfying all three bounds simultaneously. `forceGasBudget` is 13,000,000 exactly for the first release-activation block and 20,000,000 for every steady block:

$$count \le 64, \qquad \sum byteLength \le 1{,}048{,}576, \qquad \sum accountedGas \le forceGasBudget.$$

Every admitted item individually fits all three. The circuit exposes `(start,end,count,totalBytes,totalAccountedGas,forceRoot,forceCutoff,nextDueAt,forceGasBudget)` and verifies the range multiproof. Maximality means either `end=forceCutoff`, or the authenticated next leaf would exceed at least one remaining per-block cap; an early stop with a fitting leaf rejects. Candidate totals are independently capped as stated in §5. Launch enforces the

separate activation and steady Anchor/force/margin inequalities in §12, plus the analogous witness-byte bounds. Because an item is at most 5,000,000 accounted gas, even the activation cap always admits the due FIFO head; a 20m backlog is split maximally under 13m in that one block and 20m thereafter rather than making activation invalid.

For the candidate's exact consumed interval, Settlement reads at most 256 descriptors plus the first excluded boundary descriptor when one exists under the frozen cutoff. Internal per-block boundaries are already the next consumed descriptor. It computes

```
H("slot-chain-force-descriptor-list-v2" || u64(start) ||
  u16(consumedCount) || u8(hasBoundary) ||
  (u64(index)||u8(kind)||u16(descriptorByteLength)||descriptorBytes)*)
```

as `forcedDescriptorCommitment`. The number of rows is the consumed count plus the zero-or-one boundary and is at most 257. `descriptorBytes` is the exact kind-specific packed sequence used by the leaf after its ASCII domain and index; its length is the unique constant for that kind. The circuit reconstructs every leaf, range proof, per-block maximum, candidate total and processing-fee transfer from these L1-authenticated descriptors. It additionally requires raw bytes for an unexpired kind 0 and the durable credit record for kind 1. An expired kind 0 needs neither historical calldata nor any unstored preimage.

Forced envelopes are auxiliary protocol inputs, not blindly copied into the Ethereum transaction list. Each block's transaction order is: (1) the exact profile-defined AnchorV4 transaction, (2) one deterministic `InboxApplyRouterV2` system transaction, (3) upfront-valid kind-0 raw transactions in FIFO order, then (4) discretionary transactions. Sequential pre-state classification puts expired, wrong-nonce, underfunded or underpriced kind-0 items only in `InboxApplyRouterV2`'s disposition vector; they have no Ethereum transaction, nonce or receipt. Upfront-valid kind 0 appears as an ordinary Ethereum transaction, whose success or revert has ordinary nonce/gas/receipt semantics. Kind 1 always produces its idempotent inbox credit inside `InboxApplyRouterV2` and never calls the eventual destination. The system calldata commits `(forceStart,forceEnd,(queueIndex,disposition,txIndex,resultHash)*)`; the circuit derives the transaction and receipt roots from this exact order. The full consumed fee interval is pull-credited to the proof beneficiary by the same L1 canonical commit. `refundAddress` remains part of the signed kind-0 descriptor for transaction-level compatibility but creates no queue-fee entitlement in this version; changing consumption or payment requires changing the descriptor and proof statement.

Disposition bytes are fixed: `0=EXPIRED_NO_TX`, `1=NONCE_NO_TX`, `2=FUNDS_NO_TX`, `3=FEE_NO_TX`, `4=INCLUDED_TX` and `5=BRIDGE_CREDIT`. Classification tests in that order; the first applicable no-transaction reason wins. Codes 0–3 require `txIndex=UINT32_MAX` and `resultHash=0`. Code 4 requires the exact Ethereum transaction index; `resultHash` is legacy Keccak-256 of the raw signed EIP-2718 transaction bytes. This ordinary Ethereum transaction hash is known before execution. It is deliberately not a receipt hash: putting a later receipt hash in the earlier InboxApply calldata would create a fixed point through cumulative gas. Code 5 requires `txIndex=UINT32_MAX` and this exact durable credit hash:

```
H("slot-chain-bridge-credit-result-v11" || u64(queueIndex) || creditId ||
  msgHash || u256(srcChainId) || sourceDomainId ||
  u64(srcEpoch) || address20(srcBridge) ||
  bridgeExecutionHash || u64(emittedAtBlock) || destinationDomainId ||
  u256(destChainId) || u64(enqueueBy) || address20(sender) ||
```

```
  address20(srcOwner) || address20(destOwner) || u256(value) ||
  u64(fee) || u64(liquidityFee) || calldataHash ||
  u8(refundMode) || address20(refundVault) ||
  refundCapsuleHash || escrowId)
```

InboxApplyRouterV2 recomputes it from the authenticated descriptor before routing the exact tuple and `resultHash` to `InboxCreditStoreV2`. That immutable store, neither Bridge nor legacy SignalService, owns the persistent destination pin. Resolver rotation and SignalService upgrades cannot retarget, fabricate or orphan an existing credit. Unknown codes or noncanonical auxiliary fields reject.

AnchorV4 and InboxApplyRouterV2 use a profile-defined, unsigned custom L2 system transaction type whose sender is injected by the post-cutover fork, not recovered from ECDSA. That type is invalid under every legacy fork, and the two reserved sender addresses are chosen with no known signing key. The circuit requires their exact profile transactions once each at indices 0 and 1 and rejects any ordinary, forced or discretionary transaction recovered from either sender. These reserved senders and the router/store pin ABI are permanent across every V2 profile; a later fork must still dispatch old routes through this router. InboxApplyRouterV2 stores the last applied L2 block number and rejects a second application in that block. An ordinary caller therefore cannot invoke the privileged path before or after cutover.

The launch fork deploys one protocol-lifetime non-proxy immutable `InboxApplyRouterV2` before every endpoint-specific immutable `InboxCreditStoreV2`. Their permanent operations are:

```
struct InboxCreditV2 {
  uint64 queueIndex; uint64 srcChainId; bytes32 sourceDomainId;
  uint64 srcEpoch; address srcBridge; bytes32 destinationDomainId;
  bytes32 msgHash; bytes32 resultHash; bytes32 sourceContextHash; uint256 value;
  uint64 executionFee; uint64 liquidityFee;
}
struct InboxRowV2 {
  uint64 queueIndex; uint8 disposition; uint32 txIndex;
  bytes32 resultHash; bytes kind1Descriptor;
}
InboxApplyRouterV2.apply(uint64 forceStart, InboxRowV2[] calldata rows)
markReceivedFromInboxBatch(InboxCreditV2[] calldata rows)
    returns (bytes4 magic)
routeConfigHashV2() view returns (bytes32)
verifyInboxCreditV2(bytes32 creditId) view returns
  (bytes4 magic, bytes32 returnedCreditId, bytes32 resultHash,
   uint64 processBy, uint256 value, uint64 executionFee,
   uint64 liquidityFee, bytes32 sourceContextHash)
liquidityQuoteV2(bytes32 creditId) view returns
  (bytes32 resultHash, uint256 value, uint64 executionFee,
   uint64 liquidityFee, uint64 processBy)
DestinationBridge.liquidityFundingStateV2(bytes32 creditId) view returns
  (bytes32 destinationDomainId, address destBridge, address inboxCreditStore,
   uint8 status, uint64 terminalIndexPlusOne)
```

Every endpoint store constructor authorizes the same InboxApplyRouterV2; only that router

may call the store batch operation, and only after that endpoint's one-shot activation. Router state includes checked `nextQueueIndex`, which the circuit requires to equal `CanonicalCore.messageCursor`. Every block calls the router once with its complete contiguous zero-to-64 forced-row interval, including kind-0 dispositions, in global queue order. It first validates the whole vector, exact result hashes, no duplicate `(domain,creditId)`, and each one-shot route `destinationDomainId->(store,Bridge,routeConfigHash,codeHash)`. Only the registrar may append a route; domain, store and Bridge are each unique and no route can be deleted or redirected. The router rechecks code/config/domain and Bridge on every use, partitions only pin side effects into maximal contiguous same-domain runs, then makes at most 64 fixed-selector, zero-value calls and requires the exact success magic. It advances `nextQueueIndex` only after all calls succeed. A missing route, conflict or later call failure reverts the cursor and every earlier store write. There is no sorting and no loop over historical routes. The batch selector is the canonical Solidity selector for the signature above. Precisely, it is the first four Keccak bytes of `"apply(uint64,(uint64,uint8,uint32,bytes32,bytes)[])"`. Calldata after the selector has `forceStart` in word 0 and the canonical rows offset `0x40` in word 1. At byte `0x40` is the row count $n \leq 64$, followed by n element offsets relative to the beginning of the element-head area (byte `0x60` of the arguments). Each element encoding is five head words in the displayed field order; its bytes offset is exactly `0xa0`, followed by the bytes length, data and zero right-padding. A kind-0 row requires zero descriptor length. A kind-1 row requires exactly the Appendix 541-byte durable descriptor body, padded to 544 bytes. Element offsets must be contiguous and minimal; aliases, gaps, overlap, trailing calldata, nonzero padding or a length inconsistent with the row kind rejects. Consequently the largest 64-row calldata is 49,252 bytes including selector. The bound statement's `inboxSystemCalldataHash` is Keccak of this complete byte string, while `inboxSystemTxHash` is Keccak of the complete type-`0x7f` envelope that contains it. The canonical Inbox rows in the L1 activation call are the data-availability preimage for both hashes; a proof-only hash is insufficient. Success returndata is exactly one 32-byte ABI word equal to `0x49425632` (ASCII "IBV2") followed by 28 zero bytes; empty, short, long, malformed or different returndata rejects even if the call itself succeeds. The getter likewise must return exactly 32 bytes, and its value is

```
H("slot-chain-inbox-route-config-v1" ||
  address20(inboxApplyRouterV2) || address20(destinationBridge) ||
  address20(terminalDomainRegistrarV2) ||
  destinationDomainId)
```

The router pins that hash and the store runtime hash at registration and rechecks both by bounded `STATICCALL/EXTCODEHASH` before any batch call. Runtime hash, not a caller assertion, pins the getter semantics. A successful no-op implementation cannot advance the cursor because the same transaction's post-state circuit must find every fresh pin at the Appendix-defined slot; the conformance corpus includes a magic-returning no-op. That pin stores the exact authenticated result hash, source-context hash, value, execution fee, liquidity fee and `processBy`; none is inferred from a later Message. `liquidityQuoteV2` returns exactly five canonical ABI words and is callable only through a registrar-routed Store identity. The Pool checks the Store runtime/configuration and route before reading it; short/trailing/malformed return or a zero result hash rejects without opening an authorization. The validity circuit proves those stored words equal the consumed V11 descriptor/result hash, so an under- or over-debit cannot be created by substituting a quote. `liquidityFundingStateV2` likewise returns exactly five canonical ABI words under the manifest-pinned Bridge code/configuration. It preserves the existing IBridge enum

encoding NEW=0, RETRIABLE=1, DONE=2, FAILED=3 and RECALLED=4. NEW and RETRIABLE are the only fundable statuses and require `terminalIndexPlusOne=0`; every terminal status has the checked terminal index plus one. Padding, short/trailing data or an inconsistent Store/domain/ Bridge tuple rejects. The canonical empty vector requires `forceStart=nextQueueIndex`, makes no store call and leaves the cursor unchanged, but still records the exact L2 block number; a second apply in that block rejects. The Store lookup selector is the first four Keccak bytes of `"verifyInboxCreditV2(bytes32)"`. Bridge calls it by zero-value `STATICCALL` with exactly 100,000 gas and accepts only the canonical 256-byte return with magic `0x49435632` (`ICV2`), identical returned credit ID, nonzero result hash, canonical integer padding and no trailing data. The constructor-fixed destination Bridge is the only caller that receives a nonempty row. Bridge first recomputes the v6 credit ID and typed source context from the supplied Message/contexts, so the Store performs one direct mapping lookup and never searches pins by a partial tuple. `kind1Descriptor` is empty for dispositions 0–4 and exactly the Appendix-defined 541-byte kind-1 descriptor for disposition 5; all other lengths reject. The router decodes that descriptor, recomputes `creditId` and the complete `bridge-credit-result-v11` hash, and derives the destination route. The circuit separately proves the same row/descriptor against the consumed L1 queue leaf. No absent owner, value, execution-hash, capsule or escrow field is treated as authoritative calldata.

Every registered store is nonproxy, unpausable, callback-free and permanently implements this exact all-or-revert pin ABI. Each run is completely validated before its first write. Activation is blocked unless the profile's system gas margin covers the maximum *reachable* cold path under the same count and accounted-gas caps proved by the circuit. In a steady block the route-call maximum is `min(64,floor(20,000,000/5,000,000))=4`, hence at most four route calls and sixteen fresh pin writes. In the first activation block the analogous 13,000,000-gas cap admits at most two kind-1 rows and eight fresh pin writes. The mandatory benchmark set includes all-cold four-kind-1 steady execution, the 64-row steady mix of 61 minimum-gas kind-0 rows plus three kind-1 rows, and the 64-row activation mix of 62 minimum-gas kind-0 rows plus two kind-1 rows. It also proves that the next non-fitting FIFO leaf is excluded. An unreachable 64-kind-1/256-write trace is not an activation requirement. Each kind-1 row conservatively carries `accountedGas=MAX_FORCE_MESSAGE_GAS=5,000,000`; opcode repricing may reduce throughput but cannot leave an old head underfunded. The 20,000,000 per-block forced-gas budget therefore admits at most four kind-1 rows per block. The store validates widths and nonzero fields and performs all writes atomically. The release-owned Store's logical key is exactly

```
H("slot-chain-inbox-credit-slot-v5" || creditId)
```

under the profile-pinned storage layout; Store/domain/Bridge are immutable constructor values and therefore are not repeated in the key. Each pin occupies exactly four mapping values: `inboxResultHash[slot]`, `inboxSourceContextHash[slot]`, `inboxValue[slot]` and `inboxPackedTerms[slot]`. Packed-term bits 0–63 are `processBy`, 64–127 `executionFee`, 128–191 `liquidityFee`, and 192–255 are zero. A nonzero result hash is the received bit. An absent result stores those four words, with `processBy=block.timestamp+BRIDGE_PROCESS_TTL_SECONDS`. The initial TTL is 2,592,000 seconds. The contract exposes no proxy, upgrade, delegatecall, mutable writer/reader/domain target, arbitrary storage primitive or pause gate. A repeated row succeeds without writes only when the stored result, context, amounts and deadline match and never extends `processBy`; any conflict, ordering error or excess row reverts the whole batch before a write. `verifyInboxCreditV2` indexes the supplied `creditId`, requires

the received bit and nonzero result, and requires `msg.sender` to be the constructor-fixed destination Bridge for that exact local domain before returning all eight exact words. Runtime hash, constructor router/Bridge/gate values, the one-shot sealed local-domain slot and initial empty mappings are migration-proved; the noncircular construction descriptor deliberately excludes the derived domain value. This is a new bytes32-domain ABI; it never overloads legacy `getSignalSlot(uint64,address,bytes32)`.

BridgeV2 introduces these exact static context tuples:

```
struct SourceContextV2 {
  uint64 protocolVersion; uint8 kind; bytes32 creditId; bytes32 msgHash;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  address sourceBridge; bytes32 sourceBridgeExecutionHash;
  uint64 emittedAtBlock; uint64 queueIndex;
}
struct DestinationContextV2 {
  uint256 destinationChainId; bytes32 destinationDomainId;
  address destinationBridge; bytes32 releaseManifestHash;
  bytes32 executionProfileHash;
}
```

Here `protocolVersion`, manifest and profile are the immutable destination release registered for that domain; `kind=1`. The remaining words are recomputed from the durable pin and V11 descriptor. Address, `uint8` and `uint64` ABI padding is canonical. The Bridge rejects any caller-supplied word mismatch before status, Pool authorization, value or external effects. The typed context commitment is

```
SOURCE_CONTEXT_TYPE_LITERAL =
  "SourceContextV2(uint64 protocolVersion,uint8 kind,bytes32 creditId,"
  "bytes32 msgHash,bytes32 sourceDomainId,uint64 sourceRegistrationEpoch,"
  "address sourceBridge,bytes32 sourceBridgeExecutionHash,"
  "uint64 emittedAtBlock,uint64 queueIndex)"
SOURCE_CONTEXT_TYPEHASH = keccak256(concat(the four quoted fragments))
sourceContextHash = keccak256(abi.encode(SOURCE_CONTEXT_TYPEHASH,
  all SourceContextV2 fields in the displayed order))
DESTINATION_CONTEXT_TYPE_LITERAL =
  "DestinationContextV2(uint256 destinationChainId,bytes32 destinationDomainId,"
  "address destinationBridge,bytes32 releaseManifestHash,"
  "bytes32 executionProfileHash)"
DESTINATION_CONTEXT_TYPEHASH = keccak256(concat(the three quoted fragments))
destinationContextHash = keccak256(abi.encode(DESTINATION_CONTEXT_TYPEHASH,
  all DestinationContextV2 fields in the displayed order))
```

Each literal is the byte concatenation of its displayed quoted fragments with no inserted whitespace or newline. InboxApplyRouterV2 derives it only from the V11 descriptor, queue index and registrar-bound release version, and the Store pins it with the credit. `verifyInboxCreditV2` returns exactly eight canonical ABI words including that hash, the stored result and both fee terms. Destination Bridge recomputes the credit ID and source-context hash from the supplied tuple and requires equality; queue index, emission block and

source execution hash are therefore not caller assertions hidden behind `creditId`. The existing send ABI remains V1; the complete additive V2/recovery ABI is:

```
struct LiquiditySettlementV1 {
  bytes32 ticketId; address l1Recipient; uint256 settlementAmount;
}
sendMessageV2(Message, uint64 liquidityFee) -> (bytes32 msgHash, Message)
messageContextV2(bytes32 msgHash)
  -> (bytes32 creditId, SourceContextV2, DestinationContextV2)
markQueuedV2(bytes32 creditId, uint64 queueIndex)
cancelUnqueuedMessageV2(bytes32 creditId)
finalizeDoneV2(bytes32 creditId, uint64 protocolVersion,
               uint64 canonicalSequence, uint64 leafIndex,
               LiquiditySettlementV1 settlement,
               bytes32[64] siblings)
processMessageV2(Message, SourceContextV2, DestinationContextV2)
  -> (Status, StatusReason)
retryMessageV2(Message, SourceContextV2, DestinationContextV2,
               bool isLastAttempt)
failMessageV2(Message, SourceContextV2, DestinationContextV2)
expireMessageV2(bytes32 creditId)
recallMessageV2(bytes32 creditId, uint64 protocolVersion,
                uint64 canonicalSequence, uint64 leafIndex,
                bytes32[64] siblings)
withdrawSourceCreditV2(address payable recipient)
withdrawLiquidityClaimV2(address payable recipient)
messageStatusV2(bytes32 creditId) -> Status
invocationContextV2() view returns (SourceContextV2, DestinationContextV2)
ActiveSettlementRouter.canonicalAt(uint64 protocolVersion,
                                   uint64 canonicalSequence)
  -> CanonicalHistoryV2
TerminalSignalVerifier.verifyTerminalSignal(
  bytes32 destinationDomainId, address destBridge,
  bytes32 creditId, uint8 terminal, bytes32 liquiditySettlementHash,
  uint64 protocolVersion,
  uint64 canonicalSequence, uint64 leafIndex,
bytes32[64] siblings) -> bytes32
```

Immediately around an allowed recipient CALL, Bridge installs those complete tuples in transient storage. `invocationContextV2` succeeds only while that exact outer call frame is live; it reverts before installation, after the finally-style clear, and during any failed/reverted attempt. The shared non-reentrant guard forbids a nested Bridge attempt from replacing the context. An implementation that lacks EIP-1153 support for the launch fork is invalid; persistent scratch storage is not an alternative.

Recipient invocation is a frozen release policy, not an application allowlist. The profile publishes a list of at most 64 addresses, sorted by unsigned 160-bit value and free of duplicates, denied addresses and

```
INVOCATION_POLICY_TYPEHASH = keccak256(
```

```
  "InvocationPolicyV2(uint16 count,bytes32 addressesHash,bytes4 hookSelector)")
addressesHash = keccak256(address20(denied[0]) || ... || address20(denied[n-1]))
invocationPolicyHash = keccak256(abi.encode(
  INVOCATION_POLICY_TYPEHASH, uint16(n), addressesHash,
  IMessageInvocable.onMessageInvocation.selector))
```

The destination Bridge configuration, source support record and release profile all bind that hash and exact list; historical lists cannot be changed. The list must contain the zero address, destination Bridge, both SignalServices, native QuotaManager, manifest pauser, every V1 Vault named by the release, any DelegateController, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, NativeLiquidityPoolV2, TerminalAccumulatorV2, TerminalSignalVerifier, every context/helper endpoint and every other system or Bridge-trusting endpoint enumerated by the profile. SourceBridge rejects a denied `message.to` before any ID, escrow or Registry write. Destination Bridge repeats the check; a historical or corrupted omission transitions to FAILED with zero target, quota, fee and Pool consumption, enabling the ordinary source refund proof. When discovered inside a Pool-funded call, the authorization remains OPEN and the ticket remains byte-identical; the exact FAILED wrapper result is joined by the Pool before it clears the authorization. The corpus includes a target denied only at destination after source authorization, proving no LP principal is consumed. ProtocolVersionManager materializes the source-side list once at finalized domain-support confirmation. BridgeDomainRegistry exposes only

```
invocationPolicyV2(bytes32 destinationDomainId, address target) view returns
  (bytes32 invocationPolicyHash, bool denied, bytes4 magic)
```

with exactly 96 bytes, canonical Boolean/padding and magic `0x49505632` (`IPV2`). SourceBridge pins Registry code/configuration and a gas limit, requires the returned hash to equal the supported domain and rejects on any call/length/value fault. Destination Bridge uses its deployment-proved local mapping with the same hash/list; neither side accepts a Merkle proof, caller Boolean or mutable application registry.

All other addresses are permissionless and may acquire code after send through CREATE or CREATE2; code presence is decided only at execution. No-code accounts are treated as EOAs. Empty data with zero value succeeds without CALL. Empty data with positive value and data of length one to three use an ordinary CALL and therefore the recipient fallback/receive path. For data length at least four, a contract CALL is permitted only when the first four bytes equal `IMessageInvocable.onMessageInvocation(bytes).selector` and the complete calldata is the canonical ABI encoding of that one bytes argument; malformed or another selector is invocation-prohibited. Invocation-prohibited processing makes no target CALL: it creates the destination owner's value pull and the successful processor's execution-fee pull, consumes quota and atomic Pool funding, and appends DONE atomically. EOA delivery uses ordinary CALL semantics and never interprets a selector. The profile corpus covers zero/1/3/4-byte boundaries, unknown and post-CREATE2 accounts, each denyset member, malformed hook ABI and a target attempting context/reentrancy substitution. This permission does not let a later release turn a previously unprivileged counterfactual address into an old-Bridge-trusting system endpoint. Every future privileged endpoint that accepts a Bridge/context call must constructor-pin exactly one fresh `destinationDomainId` and its fresh DestinationBridge, and must revalidate the complete transient `DestinationContextV2` inside its hook. It must reject every historical Bridge and any registry-wide "recognized Bridge" predicate. Each successor

profile, deployment commitment and verifier enforce that rule for every newly privileged endpoint before publishing its address. The conformance corpus pins the counterfactual sequence send-to-X, deploy-new-privileged-X, call-from-old- Bridge and requires X to reject without effect. An application that chooses to trust historical Bridges is ordinary application risk and cannot be named a protocol/Vault/system endpoint by a release. `CanonicalHistoryV2` is the fixed tuple

```
(uint64 protocolVersion, bytes32 executionProfileHash,
 uint64 canonicalSequence, uint48 l2BlockNumber, bytes32 blockHash,
 uint64 tipSlot, bytes32 stateRoot, uint64 messageCursor,
 bytes32 winningDataCommitment, uint256 nextBaseFee,
 uint64 nextExcessBlobGas, bytes32 terminalRoot,
 uint64 terminalCount, uint64 canonicalizedAtBlock)
```

and the only caller-supplied proof payload is

```
(uint64 protocolVersion, uint64 canonicalSequence,
 uint64 leafIndex, bytes32[64] siblings)
```

The verifier reads the exact sequence-tagged ring entry from Router, requires `leafIndex<terminalCount`, derives the exact domain-separated leaf from index, domain, Bridge, credit ID, terminal and liquidity-settlement hash, and folds exactly 64 siblings to the stored root. There is no caller-supplied Canonical tuple, dynamic array, RLP node, current EVM state proof or trailing element. `terminal` is exactly 1 for DONE or 2 for FAILED; all other values reject. FAILED requires `liquiditySettlementHash=bytes32(0)`. DONE requires

```
liquiditySettlementHash = H("slot-chain-liquidity-settlement-v1" ||
  ticketId || address20(l1Recipient) || u256(settlementAmount))
settlementAmount = message.value + message.fee.
```

For DONE, `ticketId`, `l1Recipient` and the resulting `liquiditySettlementHash` are each nonzero; checked addition rejects overflow. A Pool deposit whose derived ticket ID is zero reverts before creating a row or accepting value. SourceBridge recomputes that hash from the caller-supplied fixed settlement tuple and the immutable credit, so a proof copier cannot redirect the LP pull. The verifier constructor fixes only `ActiveSettlementRouter` and depth 64; it is generic across every append-only destination endpoint. Its explicit domain and Bridge arguments are low-level leaf inputs, never source authority. Each historical source Bridge terminal transition loads `destinationDomainId` from its immutable `CreditAuthorization`, bounded-static-reads the corresponding permanent `destBridge` from `BridgeDomainRegistry`, and passes those values. It rejects if the descriptor is absent or differs from the proof; no caller or proof chooses either value. A D1 source endpoint can therefore finalize D1 and a later registered D2 without changing its verifier. Here `Message` is the existing IBridge tuple and `msgHash=hashMessage(Message)`. Destination processing first requires the context's destination Bridge to equal its own address and the context's domain to equal its one-shot sealed local destination domain. It then recomputes that domain from the pinned chain ID, genesis, BridgeInboxAdapter, active settlement router, terminal verifier, InboxApplyRouterV2, InboxCreditStoreV2, protocol-release authority, terminal-domain registrar, terminal accumulator, Bridge address/frozen execution hash, infrastructure hash and namespace. Only then do they recompute `creditId` from the Message hash and context before any status, value or external effect. No caller-supplied foreign domain may query a pin through a different funded

Bridge. Source terminal functions load the immutable authorization and current permanent liability by credit ID, while destination expiration loads the permanent inbox pin; neither needs the Message preimage. Queue marking is authorized only to the exact adapter in the recorded destination domain. The read-only status view accepts the derived key. The SourceBridge records both contexts. Its profile-pinned V2 event carries the credit ID, message hash, both contexts and the normalized Message; destination changes emit the credit ID and new status.

Native delivery uses one protocol-lifetime immutable `NativeLiquidityPoolV2`, not a finite reserve, native minting or a long-lived reservation. Any address may create or top up its own ticket and bind the L1 reimbursement recipient:

```
depositLiquidityV2(address l1Recipient, bytes32 salt) payable
  returns (bytes32 ticketId, uint256 availableWei)
withdrawLiquidityV2(bytes32 ticketId, address payable recipient,
                    uint256 amount)
  returns (uint256 remainingAvailableWei)
processWithLiquidityV2(bytes32 ticketId, address destinationBridge,
  MessageV1 message, SourceContextV2 sourceContext,
  DestinationContextV2 destinationContext)
  returns (uint8 resultingStatus, uint8 statusReason)
retryWithLiquidityV2(bytes32 ticketId, address destinationBridge,
  MessageV1 message, SourceContextV2 sourceContext,
  DestinationContextV2 destinationContext, bool isLastAttempt)
  returns (uint8 resultingStatus, uint8 statusReason)
ticketAccountingV2(bytes32 ticketId) view returns
  (address depositor, address l1Recipient, uint256 availableWei)
poolAccountingV2() view returns
  (bytes4 magic, bytes32 configurationHash, bool active, bool entered,
   bytes32 activeAuthorizationHash, uint256 totalAvailableWei,
   uint256 balanceWei, uint256 surplusWei)
```

The exact identifier is

```
ticketId = H("slot-chain-liquidity-ticket-v2" || u256(chainId) ||
  address20(pool) || address20(depositor) || address20(l1Recipient) || salt).
```

Deposit requires nonzero value and recipient and rejects the cryptographically exceptional zero ID. An absent row creates the committed tuple; a present row may only checked-add when both stored identities equal the same preimage. Withdrawing or consuming the final wei captures the tuple before deleting the row. Recreating the exact tuple and salt intentionally recreates the same ID; the same salt under another depositor or recipient derives another ID. There is no global nonce, approval, operator, transfer, salt registry, tombstone, lifetime counter or mapping enumeration. Old DONE claims never consult the current Pool row: each terminal leaf independently binds its credit, ticket, captured L1 recipient and exact amount.

Every withdrawal and funded attempt requires `msg.sender=ticket.depositor`. Withdrawal is CEI-first under the Pool guard, calls only the nonzero chosen recipient, and restores the exact row and aggregate on revert or reentrancy. A Pool-funded attempt is also initiated at the Pool, not at the Bridge: this prevents any current or future registered Bridge from claiming to be an LP and debiting an arbitrary lifetime ticket. The Pool bounded-static-reads the registrar-routed Store quote and Bridge state with exactly `POOL_EXTERNAL_READ_GAS=50000` each. It

requires exact five-word canonical returns, the pinned result hash, domain, Store and Bridge, status NEW or RETRIABLE, zero terminal index, nonexpired `processBy`, and checked `amount=value+executionFee>0`. It never trusts caller-supplied amount or fee fields.

The Pool then opens one transient storage authorization:

```
authorizationHash = H("slot-chain-native-liquidity-attempt-v1" ||
  u256(destinationChainId) || destinationDomainId || address20(pool) ||
  address20(destinationBridge) || address20(inboxCreditStore) || creditId ||
  ticketId || address20(depositor) || resultHash || u256(amount) ||
  sourceContextHash || destinationContextHash || u8(operation) ||
  u8(isLastAttempt ? 1 : 0))
```

where PROCESS=1 and RETRY=2; PROCESS requires `isLastAttempt=false`, while a last attempt is valid only for RETRY. The Pool stores the same exact tuple plus state OPEN under its entered guard. It calls only the registrar-bound nonretired Bridge with no value and canonical calldata. For positive Message gas the requested amount is exactly `V2_POOL_BRIDGE_PRE_CALL_GAS_BOUND+V2_ATTEMPT_PRE_CALL_GAS_BOUND+message.gasLimit`; owner-only zero-gas mode requests only the gas remaining above `POOL_AUTH_CLEANUP_GAS=50000`. Before CALL, after fixed memory and warm account costs, Pool enforces both the exact EIP-150 forwarding inequality and the 50,000-gas worst-case cleanup reserve, and copies at most 192 return bytes. That Bridge accepts the call only from the immutable Pool, recomputes the credit, contexts, quote, authorization and effective `processor=depositor`, then invokes its external only-self child attempt. Its exact entry and return are:

```
attemptFromLiquidityPoolV2(bytes32 ticketId, address depositor,
  MessageV1 message, SourceContextV2 sourceContext,
  DestinationContextV2 destinationContext, uint8 operation,
  bool isLastAttempt, bytes32 authorizationHash)
  returns (bytes4 magic, bytes32 creditId, uint8 resultingStatus,
           uint8 statusReason, uint64 terminalIndexPlusOne,
           bytes32 liquiditySettlementHash)
```

Its selector is the first four Keccak bytes of the exact fully expanded tuple signature in the publication corpus. Calldata is the canonical `1124`-byte vector for the 33-byte Message fixture, rejects every dynamic offset/padding/trailing-byte alias, and binds the authorization as its final head word. The Pool-to-Bridge wrapper returns exactly six canonical words (192 bytes):

```
(bytes4 magic, bytes32 creditId, uint8 resultingStatus, uint8 statusReason,
 uint64 terminalIndexPlusOne, bytes32 liquiditySettlementHash)
POOL_BRIDGE_RESULT_MAGIC = 0x4c415632 // ASCII LAV2
```

It never normalizes a low-level revert, OOG or malformed return into an accepted status. No permit or delegated-funder mode exists in this revision.

Inside that child frame only, the Bridge calls

```
consumeAuthorizedLiquidityV2(bytes32 ticketId, bytes32 creditId,
  address depositor, uint256 amount, bytes32 resultHash)
  returns (bytes32 returnedTicketId, address l1Recipient,
           uint256 consumedAmount)
```

The Bridge requests exactly the profile-pinned `V2_POOL_CONSUME_CALL_GAS_BOUND` for this call after enforcing the same EIP-150 relation and retaining its measured child suffix; that bound includes the Pool's value-CALL overhead and 100,000-gas callback. The Pool permits this one reentry only from the authorization's exact Bridge, for the OPEN authorization, matching active child attempt, caller, tuple and amount. It CEI-debits or deletes the ticket and aggregate, marks the authorization CONSUMED, transfers exactly the captured amount through the narrow callback below, and returns exactly 96 canonical bytes. Every other Pool entry, duplicate consume, direct Bridge call, target reentry and tuple substitution rejects while `entered=true`.

The native-value callback is:

```
DestinationBridge.acceptLiquidityValueV2(
  bytes32 creditId, bytes32 ticketId, uint256 amount)
  payable returns (bytes4 magic)
POOL_VALUE_CALLBACK_GAS = 100000
POOL_VALUE_MAGIC = 0x4e4c5632 // ASCII NLV2
```

Before consume, the child writes the exact active frame and

```
H("slot-chain-native-liquidity-acceptance-v3" ||
  u256(destinationChainId) || destinationDomainId ||
  address20(destinationBridge) || address20(nativeLiquidityPool) || creditId ||
  ticketId || address20(depositor) || resultHash || u256(amount) || attemptDigest)
```

as its sole expected callback. The Pool uses canonical 100-byte calldata, exact value and the 100,000-gas stipend and accepts only the exact 32-byte `NLV2` word. The callback requires the immutable Pool, same active child frame, exact tuple/value/commitment and no prior receipt; it only clears the expectation, stores the receipt and returns magic. Plain receive/fallback, duplicate or direct calls, wrong tuple/value, revert/OOG and short/trailing/bad magic fail. Bridge next requires the exact 96-byte Pool tuple, cleared expectation, exact receipt and raw-balance delta before any target call.

Target execution, quota debit, pull creation, DONE status and terminal append all remain in that same child frame. A target, quota, liability, callback, return-data or terminal fault reverts the child and therefore restores the ticket, Pool aggregate/value transfer, target, Bridge and authorization to OPEN byte-for-byte. The Bridge's outer catch may then write only the already-defined NEW, RETRIABLE or owner-authorized FAILED result. `isLastAttempt=true` still requires the effective processor to equal `destinationOwner`; an unrelated LP may only make a non-last retry. Manual failure and strict expiry never call the Pool. The executable model represents that EVM child-revert boundary with an exact single-ticket journal (ticket row, Pool balance and aggregate), the touched Bridge/target/quota/accumulator keys and the OPEN transient authorization. It restores those bounded fields before authenticating the 36-byte typed failure; it never snapshots or enumerates the Pool ticket mapping on this hot path.

On return to the Pool, it requires the exact magic and credit and joins the wrapper result to its authorization state. DONE requires CONSUMED, nonzero `terminalIndexPlusOne`, and a settlement hash recomputed from the captured ticket/recipient/amount. FAILED requires OPEN, nonzero terminal index and zero settlement hash. NEW or RETRIABLE requires OPEN and both terminal fields zero. RECALLED and unknown statuses are impossible, and every status accepts only its normative reason-code set. Any other status, low-level failure, mal-

formed return or state combination reverts the complete top-level call. The Pool clears the authorization before returning only the status/reason pair to its caller. Thus LP principal moves iff the same journal commits DONE; a failing target has no option on LP capital and cannot leave a lease or orphaned row.

The hard Pool invariant is

```
address(pool).balance >= totalAvailableWei
sum(live ticket.availableWei) = totalAvailableWei
surplusWei = address(pool).balance - totalAvailableWei.
```

The sum is a model invariant, never a production scan. Forced ETH is surplus and creates no ticket. Mapping size is bounded by currently funded rows; dust rows pay their own storage and no protocol path enumerates them. Deposit is disabled until the first registrar activation, while later releases preserve all tickets and the same Pool. Ticket and Pool views return exactly 96 and 256 bytes; Pool magic is `0x504c4132` (`PLA2`), Boolean/high padding is canonical, and an absent ticket is all-zero. Invalid mutating calls revert.

The exact event surface is:

```
event LiquidityDepositedV2(bytes32 indexed ticketId,
  address indexed depositor, address indexed l1Recipient,
  uint256 amount, uint256 availableWei)
event LiquidityConsumedV2(bytes32 indexed creditId, bytes32 indexed ticketId,
  bytes32 indexed destinationDomainId, address destinationBridge,
  address l1Recipient, address processor, uint256 amount)
event LiquidityWithdrawnV2(bytes32 indexed ticketId,
  address indexed depositor, address indexed recipient,
  uint256 amount, uint256 remainingAvailableWei)
```

The byte-exact corpus fixes all selectors, topics, calldata/returndata lengths, padding, authorization and callback commitments and negative substitutions.

There is deliberately no claim of backlog-independent delivery. One signed Pool transaction can be put through the forced path without locking principal while it waits, but it succeeds only if the credit and forced envelope remain within their deadlines and the owner has not withdrawn or spent the ticket. Paid FIFO saturation can outlast those deadlines. Permissionlessness means anyone may deposit and compete with their own capital; the first successful canonical transaction wins the source reimbursement, while a front-runner cannot spend the victim's ticket. The success-only liquidity fee compensates only voluntary successful execution, not failed gas or an objective delivery guarantee. Because delegation is deliberately absent, third-party LP processing rejects owner-only zero-gas and last-retry modes; those succeed only when the destination owner funds and calls through its own ticket. Direct manual fail and permissionless expiry remain outside the Pool. The protocol does not claim that passive third-party liquidity can exercise destination-owner authority.

The child rollback boundary is one exact external self-call, never an ordinary internal call and never a second public reentrancy frame:

```
executeAttemptV2(
  MessageV1 message, SourceContextV2 sourceContext,
  DestinationContextV2 destinationContext, address processor,
```

```
  uint8 expectedEntryStatus, bool isLastAttempt,
  bytes32 ticketId, bytes32 authorizationHash)
  external returns (uint8 resultingStatus, uint8 statusReason)
finalizeFailedAttemptV2(bytes32 creditId)
  external returns (uint8 resultingStatus, uint8 statusReason)
error TargetCallFailedV2(bytes32 attemptDigest)
```

Its selector is the first four Keccak bytes of the exact expanded signature

```
executeAttemptV2((uint64,uint64,uint32,address,uint64,address,uint64,address,
address,uint256,bytes),(uint64,uint8,bytes32,bytes32,bytes32,uint64,address,
bytes32,uint64,uint64),(uint256,bytes32,address,bytes32,bytes32),address,uint8,
bool,bytes32,bytes32)
```

The failure-finalizer selector is the first four Keccak bytes of `"finalizeFailedAttemptV2(bytes32)"`; the custom-error selector is the first four Keccak bytes of `"TargetCallFailedV2(bytes32)"`. The immutable implementation records `SELF=address(this)` at construction; both external entries require `msg.sender==SELF` and `address(this)==SELF`, so a direct caller, proxy or delegatecall context rejects before a read or write. The Pool-authenticated Bridge wrapper already holds the shared guard, authenticates the Message/pin/contexts/authorization, derives the effective processor from the Pool tuple and checks entry status, owner and last-attempt policy before making the self-call with zero value. The child defensively recomputes every field, consumes only that authorization and owns the complete Pool–target–quota–pull–terminal journal. The failure finalizer rechecks RETRIABLE, the same pin and absent terminal and owns only FAILED plus terminal; it never calls the Pool. No other entry may invoke one of those steps.

The direct, non-Pool `process/retry` entries never invoke `executeAttemptV2`: without an authenticated Pool authorization they return `UNFUNDED` before constructing a child-failure digest or touching the target. Thus the typed failure boundary cannot convert an unfunded direct attempt into a generic rejection.

The wrapper ABI-encodes only those derived values, reserves the profile-pinned `V2_OUTER_CATCH_GAS_RESERVE`, and makes a bounded external call to `SELF`; its success return must be exactly two canonical ABI words (64 bytes). A failed low-level target CALL never returns normally and never bubbles or hashes target returndata. Instead the Bridge computes

```
attemptDigest = H("slot-chain-destination-attempt-v2" || creditId ||
  sourceContextHash || destinationContextHash || address20(processor) ||
  u8(expectedEntryStatus) || u8(isLastAttempt ? 1 : 0) ||
  ticketId || authorizationHash)
```

and reverts the complete inner frame with the exact 36-byte canonical custom error `TargetCallFailedV2(attemptDigest)`. This restores the consumed ticket, Pool transfer, target effects and Bridge state to their entry values. The outer wrapper locally recomputes the digest and recognizes only that selector, exact length and exact word. Target returndata is discarded before the Bridge creates this error, so a target-crafted identical error cannot spoof the catch.

After an authenticated target-failure catch, non-owner initial failure and non-last retry return the unchanged NEW or RETRIABLE state without writes; an owner initial failure writes RETRIABLE in the outer frame; and an owner-last retry makes the zero-value

`finalizeFailedAttemptV2(creditId)` self-call. That second entry writes FAILED and appends its zero-settlement terminal leaf atomically after the first child has restored the ticket. If it reverts or returns anything but the exact canonical 64-byte FAILED result, the outer call reverts, leaving the entry status and ticket unchanged. Any first-frame revert other than the authenticated target-failure error, and any short/long return, noncanonical enum or unexpected status, discards revert bytes and maps only to `(expectedEntryStatus,ATTEMPT_ROLLED_BACK)` without an outer write. The release profile freezes both selectors, tuple grammars, custom-error selector/domain, valid status/reason matrix, self-call gas caps and catch reserve. Direct-inner, delegatecall, callback reentry, nested process/retry, target-crafted error, target failure with ticket then retry, owner-last rollback-before-FAILED, malformed return, and post-target quota/pull/terminal fault vectors prove the separation and byte-identical rollback.

The destination quota is exact: its only V2 key is

```
nativeQuotaKey = H("slot-chain-native-quota-v2" ||
                   destinationDomainId || "NATIVE_ETH")
destinationQuotaDebit = checked(message.value + message.fee).
```

Both a successful real CALL and invocation-prohibited DONE debit that amount in the same nested terminal journal. Target failure, UNFUNDED, NEW, RETRIABLE, manual failure and expiry debit zero. A missing key, overflow, insufficient quota or quota-call fault rolls back the target call, Pool transfer/ticket debit, Bridge status/pulls and terminal append together. Refund, source recall, withdrawal and terminal proof paths never read or debit quota. The fixed corpus covers value zero/fee positive, value positive/fee zero, both-zero rejection, addition overflow, quota exactly equal/one wei short and a post-target quota miss with byte-identical rollback.

On a proved DONE leaf, SourceBridge reclassifies exactly `value+fee+liquidityFee` from pending escrow to the authenticated `l1Recipient`'s LP pull. On FAILED or pre-enqueue cancellation, it reclassifies the same total to the source owner's user pull. Its exact conservation invariant is

```
totalLiveLiability = pendingEscrow + userPullLiability + lpPullLiability
address(sourceBridge).balance >= totalLiveLiability.
```

DONE never turns locked L1 ETH into ownerless surplus. Source liability falls only after the matching pull transfer succeeds. This removes deterministic lifetime-reserve exhaustion while leaving existing V1 native exits untouched. It does not make liquidity free: delivery liveness is explicitly economic and requires at least one LP willing to atomically fund execution from pre-existing L2 ETH.

The destination domain's fresh immutable non-proxy Bridge owns V2 status, retry state, ETH custody and terminal index for the domain's entire lifetime. Its address, runtime and storage never rotate and it never delegates execution. This prevents another implementation or address from seeing an empty status and processing a pinned credit twice, and keeps RETRIABLE liquidity in the same account. The sole V2 admission is `verifyInboxCreditV2`; there is no caller-supplied nonempty source-proof path on L2. `processMessageV2` is NEW-only. A non-owner invocation failure remains NEW; only a destination-owner failed initial invocation becomes RETRIABLE. `retryMessageV2` is RETRIABLE-only. Invocation-prohibited handling precedes owner checks and permissionlessly creates the destination owner's value pull. For a

real invocation, zero gas or `isLastAttempt=true` is owner-only; a non-last failure is side-effect free, while an owner last failure becomes FAILED. Manual failure is owner-only and pausable. Strictly after `processBy`, `expireMessageV2(creditId)` is raw-preimage-free, permissionless and unpausable from NEW or RETRIABLE. DONE, FAILED and RECALLED are terminal.

The V2 fee is a success-only bounty. No NEW, RETRIABLE, FAILED, manual-fail or expiry path pays it. A non-owner successful real invocation receives all fee; owner self-processing also receives all fee. For a real CALL, an insufficient caller-owned ticket returns UNFUNDED before the attempt. The nested attempt then consumes only the exact OPEN Pool authorization; a transfer or raw-balance failure reverts that frame rather than authorizing from a free-balance view. A successful target is followed by exact quota consumption, execution-fee pull-credit creation, terminal append and the hard `balance>=aggregateV2Liability` assertion in the same journal. Any quota, liability-floor, authorization or terminal failure rolls back Pool, target and Bridge, including an owner-last success. Deterministic no-CALL success paths may precheck their exact capacity. Every entry and withdrawal shares one top-level non-reentrant frame. Send/process/retry/manual-fail may be paused; cancellation, expiry, terminal append, DONE finalization, FAILED recall and pull withdrawal are outside every pause and ordinary quota gate. On transition to DONE or FAILED, the immutable destination Bridge makes exactly one call to the protocol-lifetime `TerminalAccumulatorV2`. Every process, retry, fail and expire entry shares one non-reentrant guard, and arbitrary recipient execution finishes before a terminal status is installed. The Bridge then writes DONE or FAILED and `terminalIndex=UINT64_MAX`, calls only `appendTerminalV2(creditId)`, and replaces the sentinel with the returned index before returning. The accumulator accepts no terminal or domain argument. The permanent ABI is:

```
TerminalAccumulatorV2.appendTerminalV2(bytes32 creditId)
    returns (uint64 terminalIndex)
DestinationBridge.terminalCommitmentV2(bytes32 creditId)
    view returns (bytes32 destinationDomainId, address destBridge,
                  uint8 terminal, uint64 terminalIndex,
                  bytes32 liquiditySettlementHash)
TerminalAccumulatorV2.terminalStateV2()
    view returns (uint64 count, bytes32 root)
event TerminalAppended(uint64 indexed index,
  bytes32 indexed destinationDomainId, address indexed destBridge,
  bytes32 creditId, uint8 terminal, bytes32 liquiditySettlementHash,
  bytes32 leaf,
  uint64 count, bytes32 root)
```

All selectors are canonical Solidity selectors of those signatures. Append returndata is exactly 32 bytes with a zero-padded `uint64`; commitment returndata is exactly five ABI words (160 bytes), with canonical high padding, `terminal=1` for DONE or 2 for FAILED and `terminalIndex=UINT64_MAX`. State returndata is exactly two ABI words. Short, trailing, noncanonical, overflow, revert or out-of-gas returndata rejects atomically. The accumulator uses exactly 50,000 gas for the commitment `STATICCALL`; the release gas benchmark must show 30% margin or revise this normative constant and the profile before launch. With that fixed gas stipend and exact return length it `STATICCALL`s the registered caller's `terminalCommitmentV2(creditId)`, requires the sealed domain, caller address, DONE/FAILED status, sentinel and the terminal-specific settlement-hash rule, then appends:

```
leaf = H("slot-chain-terminal-leaf-v2" || u64(terminalIndex) ||
         destinationDomainId || address20(destBridge) ||
         creditId || u8(terminal) || liquiditySettlementHash)
```

Any call or return failure reverts status, sentinel, leaf and count together. During a V1 recipient callback there is no concurrent sentinel, so a message targeting the accumulator cannot forge a leaf even though `msg.sender` is the Bridge. V2 additionally rejects the complete manifest denyset, including the accumulator, before any CALL.

The accumulator stores no per-credit "already appended" set. The exact immutable registered Bridge is the sole writer for its domain, and its permanent terminal status/index plus non-reentrant APPENDING sentinel make a second or concurrent append for one credit unreachable. The append and Bridge transition share one EVM revert domain, so a failed suffix restores frontier/count/root and the Bridge sentinel/status/index together.

The accumulator stores exactly a checked `uint64count`, wrapped root and 64-word frontier. At old index i it starts with the new leaf, combines `frontier[h]` on the left while bit h of i is one, writes the carry only to the first zero-bit frontier slot, increments count, then folds all 64 heights with canonical empty nodes to obtain the tree root and wrapped count/root. Stale frontier values at zero bits are ignored. Append rejects old `count=UINT64_MAX`; the final leaf is unused.

The complete leaf preimage, leaf, new count and root are emitted in `TerminalAppended`. Proofs are reconstructed from canonical L2 receipts by an open-source deterministic prover and at least two independently administered, replaceable log archives. Anyone verifies a supplied 64-sibling proof on L1; archives have no signing key or correctness authority. Loss of every leaf log copy can stop future proof generation and therefore terminal release, just as loss of every canonical state preimage stops recovery. Archive-independent proof generation is not claimed; it would require the rejected unbounded completed-subtree store. Production must benchmark carry depths 0–63 and the complete cold post-call suffix before fixing `V2_POST_CALL_GAS_RESERVE`. The same harness must measure and freeze nonzero profile fields `V2_ATTEMPT_PRE_CALL_GAS_BOUND` and `V2_OUTER_CATCH_GAS_RESERVE`, plus `V2_POOL_BRIDGE_PRE_CALL_GAS_BOUND` and `V2_POOL_CONSUME_CALL_GAS_BOUND`. For positive Message gas, the wrapper requests exactly `V2_ATTEMPT_PRE_CALL_GAS_BOUND+message.gasLimit` for the self-call and first applies the exact EIP-150 sufficiency relation while retaining the catch reserve; the inner target receives exactly `message.gasLimit-V2_POST_CALL_GAS_RESERVE`. For owner-only zero-gas execution, it forwards all gas above the outer reserve and the inner retains the post-call reserve. Production vectors cover first/cold Pool and quota slots, target OOG, custom-error construction/decoding, the second failure-finalizer self-call and outer guard cleanup. Until all five measured values, the fixed Pool read/callback paths and a 30-percent margin are published, no concrete execution profile is production authorised; the architecture and measurement procedure remain implementable.

Registration is append-only. Two protocol-lifetime non-proxy contracts are:

```
ProtocolReleaseAuthorityV2
TerminalDomainRegistrarV2
```

Only the registrar may register an accumulator writer. Release authority is bound to the fork-reserved system sender and manifest namespace, not a particular Anchor version or endpoint. Activation passes the value, never merely its hash:

```
struct ComponentDescriptorV2 {
  address component; bytes32 runtimeHash; bytes32 configHash;
}
struct DestinationBridgeDescriptorV2 {
  address bridge; bytes32 runtimeHash; bytes32 configurationHash;
  bytes32 storageLayoutHash;
  bytes32 bridgeKernelProfileHash; address inboxCreditStore;
  address terminalAccumulator; address terminalDomainRegistrar;
  address quotaManager; address nativeLiquidityPool;
  address bridgeSurplusSink;
}
struct ReleaseManifestV2 {
  uint64 protocolVersion;
  uint256 settlementChainId; uint256 destinationChainId;
  bytes32 destinationGenesisHash; bytes32 executionProfileHash;
  bytes32 manifestNamespace; bytes32 destinationNamespace;
  address anchorV4; bytes32 anchorRuntimeHash;
  bytes32 destinationDomainId; address destinationBridge;
  bytes32 destinationBridgeExecutionHash;
  DestinationBridgeDescriptorV2 destinationBridgeDescriptor;
  bytes32 destinationInfrastructureHash;
  bytes32 migrationVerifierDescriptorHash;
  bytes32 ingressAuthorizationRoot;
  address nativeLiquidityPool; bytes32 poolRuntimeHash;
  bytes32 poolConfigurationHash;
  ComponentDescriptorV2[10] components;
}
AnchorV4.activateRelease(ReleaseManifestV2 calldata manifest,
                         uint64 retirementQueueCount)
ProtocolReleaseAuthorityV2.activateRelease(
    ReleaseManifestV2 calldata manifest, uint64 retirementQueueCount)
TerminalDomainRegistrarV2.activateDomain(
    ReleaseManifestV2 calldata manifest, uint64 retirementQueueCount)
```

The release manifest is a fully static Solidity tuple: no string, bytes or dynamic array occurs in it. Every address/integer has canonical ABI padding and the ten components have fixed order. Its ABI tuple contains exactly 59 words (1,888 bytes). In the hash preimage, word 0 is `RELEASE_MANIFEST_TYPEHASH`, words 1–12 are the outer fields through `destinationBridgeExecutionHash`, words 13–23 are the eleven destination descriptor fields, words 24–29 are the remaining outer fields through `poolConfigurationHash`, and words 30–59 are the ten three-word components. The typehash is Keccak of this exact single ASCII string (shown wrapped only for typesetting):

```
ReleaseManifestV2(uint64 protocolVersion,uint256 settlementChainId,
uint256 destinationChainId,bytes32 destinationGenesisHash,
bytes32 executionProfileHash,bytes32 manifestNamespace,
bytes32 destinationNamespace,address anchorV4,bytes32 anchorRuntimeHash,
bytes32 destinationDomainId,address destinationBridge,
bytes32 destinationBridgeExecutionHash,
```

```
DestinationBridgeDescriptorV2 destinationBridgeDescriptor,
bytes32 destinationInfrastructureHash,bytes32 migrationVerifierDescriptorHash,
bytes32 ingressAuthorizationRoot,address nativeLiquidityPool,
bytes32 poolRuntimeHash,bytes32 poolConfigurationHash,
ComponentDescriptorV2[10] components)
ComponentDescriptorV2(address component,bytes32 runtimeHash,bytes32 configHash)
DestinationBridgeDescriptorV2(address bridge,bytes32 runtimeHash,
bytes32 configurationHash,bytes32 storageLayoutHash,
bytes32 bridgeKernelProfileHash,address inboxCreditStore,
address terminalAccumulator,address terminalDomainRegistrar,
address quotaManager,address nativeLiquidityPool,address bridgeSurplusSink)
```

The byte string contains no whitespace or newline between the three displayed type declarations. Dependencies are in ascending type-name order. Its commitment is the type-hashed `keccak256(abi.encode(...))` below. The function selector is Keccak's first four bytes over the canonical expanded ABI signature

```
activateRelease((uint64,uint256,uint256,bytes32,bytes32,bytes32,bytes32,
address,bytes32,bytes32,address,bytes32,
(address,bytes32,bytes32,bytes32,bytes32,address,address,address,address,address,address),
bytes32,bytes32,bytes32,address,bytes32,bytes32,
(address,bytes32,bytes32)[10]),uint64)
```

again with the displayed newlines removed. Tx0 calldata is that four-byte selector, `0x33f5ca80`, followed by the 59 manifest words and one canonical padded `uint64retirementQueueCount` word: exactly 1,924 bytes. Appendix golden vectors pin every boundary word and the final selector/hash. The dependency order is profile and primitive descriptors, Bridge bundle init/config, addresses, release manifest, then registration commitment; neither Bridge init/config nor the profile may depend on the manifest or derived domain. It is

```
releaseManifestHash = H(abi.encode(RELEASE_MANIFEST_TYPEHASH,
                                   canonicalReleaseManifestV2))
```

Zero values, duplicate component addresses, a Bridge different from component 10, a component count/order mismatch, or any named/hash field inconsistent with the Appendix grammars rejects. The ordinary tier-1/2/3 Settlement validity-verifier descriptor is committed transitively and unambiguously by `executionProfileHash`; the separately named `migrationVerifierDescriptorHash` is only the proof-first genesis/version-transition verifier. Neither field may substitute for the other, and registration recomputes both descriptor commitments from their distinct profile ranges before accepting the manifest.

The first AnchorV4 custom system transaction calls exactly `activateRelease(manifest, retirementQueueCount)`. Its canonical uint64 word must equal the migration statement's authenticated L1 ForcedQueue count; there is no optional/default watermark or Inbox-cursor substitute. The raw tx0 bytes contain the fixed manifest words immediately followed by that count and contain no MPT proof, statement hash, candidate digest or tx0 hash. This avoids a hash cycle. The validity circuit authenticates the already published L1 release manifest and exact L1 origin as separate statement inputs, requires this tx0 at index zero, and constrains its raw EIP-2718 hash and decoded values. Anchor recomputes the manifest commitment, stores

the active release once, calls the release authority and registrar with the identical value/count, and rolls all three back on any failure.

The authority recomputes the commitment, requires `msg.sender` and its `EXTCODEHASH` to equal the manifest Anchor, requires `tx.origin` to be the reserved Anchor-system sender, and requires Anchor's one-shot active manifest/count to match. It permanently records the release and exact retirement watermark. A direct call by that reserved sender, an EOA, Bridge or an Anchor from another manifest rejects. Its append-only read boundary is

```
releaseActivationV2(uint64 protocolVersion) view returns
  (bytes4 magic, bytes32 manifestHash, uint64 retirementQueueCount)
```

The selector is the first four Keccak bytes of that canonical signature. Registrar uses a zero-value `STATICCALL` with exactly 75,000 gas and accepts exactly 96 bytes with magic `0x52415632` (`RAV2`) and canonical padding. Missing, short, trailing, caller-dependent or reverting records reject. Replaying the same version/hash with a different count rejects; any Registrar failure rolls the Authority write back in the same tx0 journal. Before either side hashes or stores the manifest, L1 requires `settlementChainId=block.chainid<=UINT64_MAX`; L2 independently requires `destinationChainId=block.chainid<=UINT64_MAX`. The registrar constructor fixes only the protocol-lifetime release authority, InboxApplyRouterV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2. Its non-reentrant activation recomputes the same manifest hash, requires the authority's exact stored version/watermark, and compares every protocol-lifetime row plus the fresh release-owned Store, QuotaManager and DestinationBridge against the manifest. Components on L1 were already checked by ProtocolVersionManager; the L2 registrar never pretends to inspect cross-chain code.

Every destination release is fresh-only. Tx0 authenticates exact non-proxy runtime/configuration/layout for Store, QuotaManager and Bridge; arbitrary pre-activation Bridge surplus; zero nonce, liability and private status/pull/ terminal mappings; full initial nonzero V2-only quota; and `v2Active=false`. It authenticates the lifetime Pool code/configuration and its unchanged aggregate solvency, then one-shot registers the fresh domain/Bridge/Store as an append-only atomic-funding authority, installs the inbox route and accumulator writer, and seals Store plus `v2Active=true` in one rollback domain. Tx0 moves no native value, creates no LP ticket and mints nothing. There is no activation-gate account, legacy proxy branch, exact-reuse branch, owner, implementation slot or delegate target. A historical domain or Bridge can never be reactivated even if code is identical. Any partial state, duplicate identity, Pool insolvency, conflicting route or late failure reverts tx0 entirely.

InboxApplyRouterV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2 are protocol-lifetime objects repeated byte-exactly by every successor manifest. Their cursor, old routes/releases/registrations, Pool tickets and accumulator frontier/root/count remain intact; only the absent release, route, Pool funding route and one-to-one writer are appended. The Pool persistently stores the two O(1) reverse indexes `bridgeByDomain[domainId]=Bridge` and `domainByBridge[Bridge]=domainId`. Tx0 writes both in the same rollback journal as the Registrar route; a duplicate domain/Bridge rejects. A restarted implementation reconstructs only the ephemeral call harness from these rows, never from a post-activation object binding. The registrar stores the single Appendix-defined `registrationCommitment[protocolVersion]` only after the full seal. The accumulator has no public registrar or root setter and calls only the registered Bridge's bounded terminal getter during append. Status, sentinel, frontier/root/count update and terminal index are atomic. The application-level leaf sequence, not a versioned EVM

storage proof, remains the source release statement across later Settlement profiles.

Release rotation and old-endpoint reclamation are bounded and explicit. Successful tx0 replaces the registrar's one `activeDestinationBridge` with the fresh Bridge and records for the predecessor `retirementQueueWatermark[oldBridge]=retirementQueueCount`; neither may be rewritten. The count is the migration statement's exact L1 Queue count. Thus the retired adapter cannot append after the watermark, while `InboxApplyRouterV2.nextQueueIndex>=watermark` proves that every earlier queued predecessor descriptor has been applied.

The same L2 tx0 journal appends a local successor receipt; an L2 Bridge never reads the L1 Router receipt. Its ID and permanent views are:

```
destinationReceiptId = H("slot-chain-destination-activation-receipt-v2" ||
  u256(destinationChainId) || address20(terminalDomainRegistrarV2) ||
  u64(successorIndex) || u64(oldProtocolVersion) ||
  u64(newProtocolVersion) || oldManifestHash || newManifestHash ||
  oldDestinationDomainId || newDestinationDomainId ||
  address20(oldDestinationBridge) || address20(newDestinationBridge) ||
  u64(retirementQueueCount) || u64(activatedAtBlock))
destinationActivationReceiptV2(bytes32 destinationReceiptId) view returns
  (bytes4 magic, bytes32 returnedReceiptId, uint64 successorIndex,
   uint64 oldProtocolVersion, uint64 newProtocolVersion,
   bytes32 oldManifestHash, bytes32 newManifestHash,
   bytes32 oldDestinationDomainId, bytes32 newDestinationDomainId,
   address oldDestinationBridge, address newDestinationBridge,
   uint64 retirementQueueCount, uint64 activatedAtBlock, uint8 sealed)
destinationSuccessorReceiptV2(bytes32 oldDestinationDomainId,
                              address oldDestinationBridge) view returns
  (bytes32 destinationReceiptId, uint64 successorIndex, bytes4 magic)
```

The full return is exactly 448 bytes with magic `0x44525632` (`DRV2`), `sealed=1` and canonical padding; the index return is 96 bytes with magic `0x44535632` (`DSV2`). Registrar derives all fields from its current active release and the proof-bound identical manifest/count, increments a checked `uint64successorIndex`, and writes the immutable receipt, predecessor index, new active tuple and all route/Pool/writer seals in one rollback domain. Genesis uses canonical-zero predecessor fields and writes no predecessor index. A successor requires nonzero old fields, distinct new domain/Bridge, exact stored old manifest, absent predecessor index and exact watermark; partial or duplicate sealing rejects. The DRV2 and DSV2 reads are zero-value `STATICCALL`s with exactly 150,000 and 75,000 gas respectively. `activatedAtBlock` is the proof-authenticated L2 block executing tx0, not a later read-time block. The checked successor index may advance through `UINT64_MAX`; no subsequent activation is then representable and must reject without mutation.

Each release-owned `InboxCreditStoreV2` maintains a checked `uint64pinnedCount`, incremented only when its authorized InboxApply route writes an absent credit pin. The accumulator maintains a checked per-domain `uint64terminalizedPinnedCount`. It increments only in the same journal as a terminal append when the caller is that registrar-bound domain Bridge, the route and Store identities match, the pin exists, status is DONE or FAILED, and the `(domainId,creditId)` terminal guard was previously absent. A duplicate, unpinned credit, foreign Store/Bridge or failed append changes no counter, leaf, guard or status.

Historical reclamation has three additional fixed-width, address-keyed views:

```
routeStateV2(address destinationBridge) view returns
  (bytes4 magic, address returnedDestinationBridge,
   bytes32 destinationDomainId, address inboxCreditStore,
   bytes32 storeRuntimeHash, bytes32 storeConfigurationHash,
   uint64 protocolVersion, bytes32 releaseManifestHash,
   bytes32 executionProfileHash, uint256 destinationChainId,
   uint64 nextQueueIndex)
pinStateV2(address destinationBridge) view returns
  (bytes4 magic, address returnedDestinationBridge,
   bytes32 destinationDomainId, address authorizedInboxApply,
   address terminalDomainRegistrar, uint64 pinnedCount)
domainStateV2(address destinationBridge) view returns
  (bytes4 magic, address returnedDestinationBridge,
   bytes32 destinationDomainId, uint64 terminalizedPinnedCount)
```

Their selectors are respectively `0x610ee813`, `0x92aea0d5` and `0xa16ec5a1`. Every input is exactly 36 bytes: selector plus one canonically left-zero-padded address word. Every call is a zero-value `STATICCALL` with exactly 100,000, 75,000 and 75,000 gas respectively. The returns are exactly 352, 192 and 128 bytes, with first-word magic `RTS2`, `PNS2` and `DMS2`; all address and narrow-integer padding is canonical. Missing address reverse-index entries, zero identity fields, short/trailing data, dirty padding, revert or OOG reject. Router and Accumulator maintain permanent one-to-one `address20(Bridge)` reverse indexes written and rolled back atomically with route/domain registration. Store answers only for its immutable destination Bridge.

Anyone may call the old Bridge's bounded `reclaimSurplus()` only after a distinct successor has the Registrar-local sealed destination activation receipt above, InboxApply has reached the immutable watermark, `terminalizedPinnedCount[oldDomain]=oldStore.pinnedCount`, and the old Bridge's aggregate pull liability is zero. The call reauthenticates the old Bridge's aggregate execution-fee pull liability is zero. Atomic funding leaves no Pool state attached to a domain across transactions; unrelated ticket funds remain withdrawable by their LP. Activation derives and seals a fixed-width, primitive `ReclamationConfigV2` from the already authenticated manifest. It stores the old version/chain/domain/Bridge, manifest/profile/registration commitments, the six Router/Store/Authority/Registrar/Accumulator/Pool address/runtime/config triples, and the ForceSend constants below. The sink address must equal exact ExecutionProfileV2 word 70 and is also committed by the destination Bridge descriptor, component configuration, execution hash and activation-state commitment. From this retained config alone reclaim selects the exact Registrar and Authority addresses, checks

```
struct ReclamationComponentV2 {
  address account; bytes32 runtimeHash; bytes32 configurationHash;
}
struct ReclamationConfigV2 {
  uint64 protocolVersion; uint64 destinationChainId;
  bytes32 releaseManifestHash; bytes32 registrationCommitment;
  bytes32 executionProfileHash; bytes32 destinationDomainId;
  address destinationBridge; address bridgeSurplusSink;
  ReclamationComponentV2[6] components; // rows 4,5,6,7,8,9
  address forceSendHelper; bytes32 forceSendCreate2Salt;
  bytes32 forceSendInitcodeHash; bytes32 forceSendCompilerBuildHash;
```

```
  bytes32 forceSendEvmRulesHash;
  uint64 forceSendCreate2FixedGas; uint64 forceSendChildGas;
  uint64 forceSendPostcheckReserve; uint64 forceSendPrecreateGas;
}
```

All narrow fields are re-width-checked on decode. The six entries are exact manifest component positions 3–8 in zero-based order (human rows 4–9); no caller supplies or reorders them. Its canonical static projection is exactly 35 ABI words (1,120 bytes). `forceSendPrecreateGas` is rederived from the other gas fields and equality-checked, rather than trusted as a second authority. The reclaim path then checks `EXTCODEHASH` and the exact four-byte/50,000-gas `componentConfigHashV2()` read (selector `0xf6c0f7d2`, zero value, exactly 32 return bytes) for each, then reads DSV2, DRV2, `registrationCommitmentV2` and RAV2. It strictly joins the direct receipt to the old version/manifest/domain/Bridge, successor index, nonzero distinct successor, sealed bit and watermark; RAV2 must repeat the old manifest and watermark, while the registration view must equal the locally recomputed `registrationCommitment`. No latest-tip lookup or private Registrar map read is permitted.

Only after those roots authenticate the old manifest commitment may the Bridge select and code/config-authenticate the retained Router, Store, Accumulator and Pool. It then makes RTS2 to that already authenticated Router, PNS2 to the manifest-pinned Store and DMS2 to the manifest-pinned Accumulator. RTS2 must repeat the Bridge/domain/version/manifest/profile/chain, the exact manifest-pinned Store address/runtime/config triple and a cursor at least the receipt watermark. PNS2 must repeat the Bridge/domain, manifest-pinned Router and Registrar and the Store count. DMS2 must repeat the Bridge/domain and the same count. In particular, an RTS2-returned Store triple cannot authenticate the Router that returned it: Router code/config was already rooted in the manifest. The Pool has no reclamation state view; only its manifest-pinned code/config is checked. Target dispatch is by the exact 20-byte account address, so an equal-looking object installed at an alias cannot answer.

These are the complete fixed reclamation reads: exactly six `EXTCODEHASH` operations and thirteen `STATICCALL`s (six config reads plus DSV2, DRV2, registrationCommitmentV2, RAV2, RTS2, PNS2 and DMS2). Reclaim never follows an in-memory object reference, enumerates a map or reads another component's private state. A restart rehydrates only the primitive `ReclamationConfigV2`, the six address-keyed call records, address-keyed native balances and a new Bridge instance. The executable model exercises A→B→C reclamation and an old SourceBridge claim through that target-free state.

If the remaining Bridge balance is zero, the Bridge sets `retired=true` and returns `RECLAIMED_ZERO` without CREATE2; low gas or an occupied helper cannot block empty cleanup. For a positive surplus it executes exact 22-byte initcode

```
0x73 || bridgeSurplusSink[20] || 0xff  // PUSH20 sink; SELFDESTRUCT
salt = keccak256("slot-chain-force-send-create2-salt-v1" || oldBridge[20])
helper = last20(keccak256(0xff || oldBridge || salt || keccak256(initcode)))
```

with `CREATE2(value=surplus)`. No CALL reaches the sink and recipient code cannot reject, reenter or consume gas. Sink, helper and the six component accounts, Bridge, quota manager, pauser, signal service and Pool must be pairwise address-disjoint. Sink and helper must additionally be disjoint from every fixed privileged destination target in the release denyset, including both Bridge-context endpoints, the delegate controller and every official V1 vault. The exact initcode hash, helper derivation, handwritten Osaka build hash, EIP-6780 execution-rules

hash and gas certificate are component configuration inputs. A balance-only counterfactual helper is legal under EIP-684: its prefunded balance is included with the Bridge value and the constructor self-destructs the total to the sink. A helper with nonzero nonce or nonempty code makes CREATE2 fail and leaves all state unchanged. The handwritten-build commitment is `0xa81af0253ed5abd3752b67c07e0c84760813a7294979ff57beefe43b0b61c551`; the EIP-6780/Osaka rules commitment is `0x81a66caeeb505a605f1fb2619833286272e08d37f7589155125455efad4c17`

Let `C_create2` be the release-certificate bound fixed caller/opcode/hash/ memory cost, `G_child` the measured worst-case constructor consumption including a cold new beneficiary, and `R_post` the fixed post-check reserve. Immediately before CREATE2 the exact sufficient guard is

```
gasleft >= C_create2 + max(G_child + R_post, ceil(64*G_child/63))
```

with checked arithmetic; the release benchmark succeeds at that threshold and fails at one less. This model fixes the certificate values at 33,000, 75,000 and 20,000 gas respectively, hence a 128,000-gas guard. CREATE2 must return the exact nonzero derived helper and the Bridge postbalance must be zero, otherwise the whole call rolls back. The helper has no surviving runtime. Reclaim never asserts an exact sink balance delta because unrelated forced ETH is legal.

The mechanism relies on same-creation-transaction SELFDESTRUCT value transfer under the profile-pinned Osaka/EIP-6780 semantics. A future fork that removes that transfer must reject incompatible activation profiles; if semantics change after deployment, only already-unowned raw surplus cleanup can become stranded. Reclaim is reachable only after all pull liabilities are zero, so this dependency can never block a user claim, protocol processing or chain liveness. No Pool ticket is enumerated or consumed. Its result is

```
enum ReclaimResult { REJECTED, RECLAIMED_ZERO, RECLAIMED_VALUE }
returns (ReclaimResult result, uint256 amount)
```

so a failed CREATE2 cannot masquerade as a valid zero-balance reclaim. Only after the exact zero-postbalance check does it set the one-shot `retired` bit; partial transfer, later processing, double reclaim and any state mismatch revert or return REJECTED without mutation. The old Bridge authenticates its immutable *direct* successor receipt; it never requires that successor to remain the Registrar's latest active tip. Therefore A may first reclaim from its sealed A→B receipt after B→C or any later rotation, while B independently uses its B→C receipt. DRV2 was written only after the complete atomic successor seal, so reclaim neither reads a Python-only activation object nor requires a third unbounded historical lookup. Historical status, terminal index/events and proof verification remain readable forever. Historical SourceBridges remain callable until every USER_PULL and LP_PULL is withdrawn; a DONE reimbursement is never reclaimable as surplus. Repeated release rotation changes no ticket owner, source liability or terminal proof.

Every execution proof additionally authenticates the accumulator account and proves that the candidate's `terminalRoot` and `terminalCount` equal its exact post-state root/count slots. These fields are part of `CanonicalCore` and `executionOutputsCommitment`; tier-2 and tier-3 use the same public-output constraint. A caller cannot supply an unconstrained root; no checkpoint publisher exists in this revision. The same proof authenticates the protocol-lifetime InboxApplyRouterV2 account and, for each block, requires its pre-state `nextQueueIndex` to equal that block's `messageStart` and its post-state value to equal `messageEnd`. Before proof

acceptance, the L1 base invariant is `baseCanonical.messageCursor=ForcedQueue.cursor`, while the proof's first L2 pre-state cursor equals the same start. The final proved L2 post-state cursor and new `CanonicalCore.messageCursor` both equal the candidate end. After proof verification and before its history write, the active Settlement calls `ForcedQueue.advanceCursor(start,end,proofBeneficiary)`. That call moves the exact cumulative processing-fee interval to the beneficiary's pull claim. A failed call reverts the entire canonical commit. L1 never writes the L2 router; the end value is authenticated L2 poststate adopted by the proof. Launch bootstrap proves all three base values equal in the imported L2 state. A later migration occurs only at a canonical boundary, checks the two L1 values directly, and inherits L2 equality inductively from the last accepted proof; the target profile must preserve that same cursor slot and transition. No activation transaction may repair, overwrite or select among divergent cursors.

The vector root is

```
H("slot-chain-terminal-root-v2" || u64(terminalCount) || treeRoot)
```

where the Appendix fixes empty nodes, node hashing and bit order.

Every versioned Settlement is a permanent non-proxy contract with no `SELFDESTRUCT`, delegate target, history pause or code upgrade. Its internally written history cell contains the exact sequence plus the full `CanonicalCore` and `canonicalizedAtBlock`; a read requires exact tag equality. The protocol-lifetime non-proxy `ActiveSettlementRouter` keeps an append-only mapping from protocol version to exact Settlement address, `EXTCODEHASH` and execution-profile hash. Historical entries cannot be deleted, redirected or reactivated. One constructor-bound, tagged `LEGACY_BOOTSTRAP` descriptor is the sole exception: it binds the deployed legacy Inbox proxy, expected proxy/runtime/configuration identity and the `legacyDeploymentHash`, but is never inserted into the version mapping or canonical history. While `genesisConsumed=false`, the active-state getter returns that immutable proxy as the source. A failed or orphaned genesis activation preserves this branch; successful GENESIS_IMPORT sets `genesisConsumed=true` in the same atomic switch, after which the branch is permanently tombstoned and cannot be reused. Registration, `canonicalAt` and every later migration reject a legacy-tagged row. Its only switch is:

```
struct CanonicalCoreV2 {
  uint48 l2BlockNumber; bytes32 tipHash; uint64 tipSlot; bytes32 stateRoot;
  uint64 messageCursor; bytes32 winningDataCommitment; uint256 nextBaseFee;
  uint64 nextExcessBlobGas; bytes32 terminalRoot; uint64 terminalCount;
}
struct MigrationActivationFixedV2 {
  uint8 transitionKind; uint64 migrationGeneration;
  uint64 seatGeneration; uint64 sourceProtocolVersion;
  uint64 targetProtocolVersion;
  uint64 sourceCanonicalSequence; bytes32 candidateDigest;
  CanonicalCoreV2 outputCore; address proofBeneficiary;
  uint256 anchorNumber; bytes32 anchorHash; uint64 forceCutoff;
  uint64 preInboxLastAppliedPlusOne;
}
activateVersionWithMigration(
  MigrationActivationFixedV2 calldata fixedInput,
```

```
  ReleaseManifestV2 calldata manifest,
  InboxRowV2[] calldata rows,
  bytes calldata importedHeaderRlp,
  bytes calldata proof)
```

The selector is the first four Keccak bytes of this exact canonical expanded signature, with no whitespace:

```
activateVersionWithMigration((uint8,uint64,uint64,uint64,uint64,uint64,bytes32,
(uint48,bytes32,uint64,bytes32,uint64,bytes32,uint256,uint64,bytes32,uint64),
address,uint256,bytes32,uint64,uint64),
(uint64,uint256,uint256,bytes32,bytes32,bytes32,bytes32,address,bytes32,
bytes32,address,bytes32,
(address,bytes32,bytes32,bytes32,bytes32,address,address,address,address,address,address),
bytes32,bytes32,bytes32,address,bytes32,bytes32,
(address,bytes32,bytes32)[10]),
(uint64,uint8,uint32,bytes32,bytes)[],bytes,bytes)
```

The selector is `0x17a548ed`. The first two tuples are static and occupy 22 and 59 words. Therefore the top-level ABI head is exactly 84 words: words 0–21 are `fixedInput`, 22–80 are `manifest`, and words 81–83 are the rows, imported-header and proof offsets. The rows offset is exactly `0x0a80`. The other offsets are the immediately following canonical tails: rows encode as one length word, n element offsets and n contiguous five-word element heads plus their bytes tails; then the header and proof each encode one length word, bytes and zero right-padding. Aliases, gaps, overlap, nonminimal offsets, nonzero padding or trailing calldata reject. There are at most 64 rows; kind-0 descriptor bytes are empty and kind-1 bytes are exactly 541. GENESIS_IMPORT requires a nonempty canonical RLP header of at most 2,048 bytes; VERSION_MIGRATION requires zero header bytes. Proof length is nonzero, no greater than the descriptor's `maximumProofBytes`, and that bound is at most 131,072 bytes. Later VERSION_MIGRATION activation is permissionless only while the Router's exact public migration gate is READY. GENESIS_IMPORT instead uses one separately delayed, finite campaign while that gate remains ACTIVE on the constructor-bound legacy branch. Its exact target and review commitment are final before the admission-fence cutoff; during the inclusive campaign window any caller may land the proof-carrying atomic cutover. No caller, including governance, has a privileged landing path or post-cutoff approval. A separate Router execution lifecycle is the global reentrancy journal:

```
RouterLifecycle = IDLE=0 | ARMING=1 | READYING=2 |
                  ABORTING=3 | ACTIVATING=4 | REGISTERING=5 |
                  PUBLISHING=6
```

Every Router and ProtocolVersionManager mutator begins only in IDLE. The later-migration arm, readiness and abort set their distinct lifecycle before the first external interaction. GENESIS_IMPORT enters ACTIVATING before its first external read or bounded STATICCALL; a failing proof or callback reverts that write with the whole transaction. Every successful control-plane call restores IDLE only as its final write. During ACTIVATING, and only then, the Router stores a nonzero `activeActivationContextHash`; outside that lifecycle every activation-only context field is canonical zero. The public ACTIVE/ARMED/READY gate does not change during a callback. Genesis moves directly from legacy ACTIVE to target ACTIVE inside the

same transaction; local READY exists only between LGAR and LGFN in that transaction and is never a canonical stopping state. This lifecycle is not a governance pause and has no public setter.

8.1 Immutable delayed protocol authority

No proxy, owner-controlled executor or generic-call timelock is a protocol authority. The release uses exactly one non-proxy, ownerless `ProtocolChangeTimelockV1` and one non-proxy, ownerless `ProtocolVersionManagerV2`. Their runtime hashes and constructor-fixed configuration are release/profile fields. Neither runtime contains `DELEGATECALL`, an upgrade path, a target setter, an arbitrary `call(address,bytes)` path or a self-call that can reach one. The timelock's only proposer is the immutable DAO proposer; execution is permissionless. This deliberately retains governance as the validity/release safety root, but makes every protocol change publicly visible for at least `PROTOCOL_CHANGE_DELAY_SECONDS=604800` seconds and removes every bypass around that delay.

The timelock accepts only operation kinds

```
REGISTER_RELEASE=1 | REGISTER_FORK_VERIFIER=2 |
PUBLISH_GENESIS_CAMPAIGN=3 | PUBLISH_MIGRATION_ARM=4 |
REPLACE_PENDING_FORK_VERIFIER=5 | SPLIT_LATEST_FORK_VERIFIER=6
```

and rejects every other value. Its exact identity is

```
protocolChangeOperationId = H(
 "slot-chain-protocol-change-operation-v1" ||
 u256(settlementChainId) || address20(protocolChangeTimelock) ||
 address20(protocolVersionManager) || u64(operationNonce) ||
 u8(operationKind) || u32(payloadBytes) || H(payload))
```

where `payloadBytes` is nonzero and at most 149,088. This exact maximum is the 2,240-byte REGISTER_RELEASE envelope plus the maximum 146,848-byte canonical execution profile; no other operation kind may exploit trailing capacity because its decoder requires its own exact static length. The sole proposer calls `queueProtocolChangeV1(uint8,bytes)`; the timelock assigns the next checked nonzero `uint64` nonce, records the exact hash/length and `executeAfter=checked(uint64(block.timestamp)+604800)`, and emits the complete payload. Any caller later supplies the same nonce, kind and bytes to `executeProtocolChangeV1(uint64,uint8,bytes)`. Execution requires `block.timestamp>=executeAfter`, moves QUEUED to EXECUTING before the only external call, invokes only the immutable manager's `applyProtocolChangeV1(uint64,uint8,bytes)`, requires the sole exact `0x50415031` (PAP1) return word, and writes EXECUTED last. Failure reverts the whole operation to QUEUED. Only the DAO proposer may call `cancelProtocolChangeV1(uint64,uint8,bytes)`, and only while QUEUED; it authenticates the complete preimage and writes CANCELED without an external call. NONE=0, QUEUED=1, EXECUTING=2, CANCELED=3, EXECUTED=4 and EXPIRED=5 are the only states. EXPIRED is reachable only for the three fork-change kinds by the permissionless function below. CANCELED, EXECUTED and EXPIRED retain the tuple forever, nonces are never reused, and the EXECUTING guard makes same-block execute/reenter/cancel order atomic.

PUBLISH_MIGRATION_ARM has the additional fixed execution bound `MIGRATION_ARM_EXECUTION_WINDOW_SECONDS=604800`: both Timelock and PVM independently require

`executeAfter<=block.timestamp<=checked(executeAfter+604800)`. Both endpoints are inclusive. The preceding seven-day delay is unchanged; the second seven-day interval only bounds how long a mature arm can remain dormant. Other operation kinds retain their separately specified timing rules except the three fork-change kinds. REGISTER_FORK_VERIFIER, REPLACE_PENDING_FORK_VERIFIER and SPLIT_LATEST_FORK_VERIFIER have the fixed `FORK_CHANGE_EXECUTION_WINDOW_SECONDS=86400`. Both Timelock and PVM independently require `executeAfter<=block.timestamp<=checked(executeAfter+86400)`. After the upper endpoint any caller may terminalize the still-QUEUED row as EXPIRED; execution can never race successfully after that timestamp. This prevents a reviewed route payload from remaining dormant and executable after its review evidence has become stale. The generic PCO1 row and operation identity are unchanged.

The five mutating ABIs are exact:

```
queueProtocolChangeV1(uint8 kind,bytes payload)
 returns (bytes4 magic,bytes32 operationId,uint64 nonce,uint64 executeAfter)
cancelProtocolChangeV1(uint64 nonce,uint8 kind,bytes payload)
 returns (bytes4 magic)
executeProtocolChangeV1(uint64 nonce,uint8 kind,bytes payload)
 returns (bytes4 magic,bytes32 operationId)
applyProtocolChangeV1(uint64 nonce,uint8 kind,bytes payload)
 returns (bytes4 magic)
expireForkChangeV1(bytes32 operationId)
 returns (bytes4 magic,bytes32 operationId)
```

Their selectors are `0xbd5c80a8`, `0x5701c308`, `0x31d81ea2`, `0xaf3927f6` and `0x379c9ecc`; return lengths/magics are respectively 128/PCQ1 `0x50435131`, 32/PCC1 `0x50434331`, 64/PCE1 `0x50434531`, 32/PAP1 `0x50415031` and 64/PFX1 `0x50465831`. Queue calldata has a two-word head and exact bytes offset `0x40`; the other three dynamic calls have a three-word head and exact offset `0x60`. Expiry calldata is the selector plus one exact operation-ID word. Each dynamic call has one length word, contiguous bytes, zero right-padding and no suffix. Narrow arguments use canonical padding. Apply is Timelock-only; the other four have the caller rules above.

The payload is canonical ABI for exactly one fixed schema:

1. REGISTER_RELEASE is exactly `abi.encode(uint64expectedPredecessorProtocolVersion, ReleaseManifestV2,SettlementDeploymentDescriptorV1,bytesprofileBytes)`. Its head is the canonically padded expected-predecessor word, 59 manifest words, then the eight descriptor words `(addressfactory,bytes32factoryRuntimeHash,bytes32factoryConfiguration bytes32salt,bytes32initCodeHash,addresstargetSettlement,bytes32targetRuntimeHash, bytes32targetConfigurationHash)` followed by the sole dynamic offset, exactly `0x08a0`. Its tail is one length word, 1–146,848 canonical profile bytes and zero right-padding, making total payload length `2240+ceil32(profileBytes.length)`. Alias, gap, nonminimal offset, nonzero padding or trailing bytes reject. The manager strictly decodes and byte-for-byte re-encodes the strict canonical ABI profile grammar below, requires `executionProfileHash=H("slot-chain-execution-profile-v2"||u32(profileBytes. length)||profileBytes)`, and checks every manifest profile-derived field, ingress row, verifier, component and Market read tuple against that decoded profile. It checks CREATE2 derivation, the live root factory identity and target code/configuration, requires the expected predecessor to equal Router's current active protocol version

and `0<=expectedPredecessorProtocolVersion<protocolVersion` (zero is the GENESIS_IMPORT sentinel), recomputes the descriptor's 232-byte packed hash, then derives

```
targetRegistrationHash = H(
 "slot-chain-target-registration-v3" || u64(protocolVersion) ||
 address20(targetSettlement) || targetRuntimeHash ||
 targetConfigurationHash || settlementDeploymentDescriptorHash ||
 executionProfileHash || migrationActivationProfileRecordHash ||
 releaseManifestHash ||
 u64(expectedPredecessorProtocolVersion) || u64(settlementChainId) ||
 ingressAuthorizationRoot || migrationVerifierDescriptorHash ||
 bridgeRouteExpansionHash)
```

from the same manifest/profile and current Router predecessor, where `bridgeRouteExpansionHash` is the nonzero Keccak hash of the exact canonical 1,952-byte BRX1 return. The `targetReleaseRegistratio` wire ABI retains its V2 name; the target-registration *hash grammar* is V3 and no V2 hash is accepted. It may only append that exact version/deployment/registration, its profile-derived ingress rows and its Market authorization to the named immutable registries. A different predecessor, target, manifest or registration requires a new delayed operation and cannot reuse this record.

2. REGISTER_FORK_VERIFIER is the exact 576-byte concatenation of the canonical 480-byte row `abi.encode(bytes4forkDigest,uint64firstParentSlot,uint64lastParentSlotExclusive, addressverifier,bytes32runtimeHash,uint64beaconSlotGindex,uint64executionPayloadGindex, uint64stateRootGindex,uint64prevRandaoGindex,uint64timestampGindex,uint64blockHashGindex bytes32witnessSchemaHash,bytes32configurationHash,bytes4selector,uint64gasLimit)` and the 96-byte review envelope defined in the ScheduleOracle subsection. The row is exactly 15 static words. Every descriptor field is nonzero, `firstParentSlot` may be zero only for the constructor-installed initial row, the interval is nonempty, the selector is exactly `0x7e981e0b`, and `configurationHash` is recomputed from the displayed constants under the ScheduleOracle grammar below. The runtime/configuration getter is checked, and `forkDigest` occupies the high four word bytes with 28 zero trailing bytes; it is the permanently tombstoned ScheduleOracle registry key used by `sealScheduleWindowV1`.
3. REPLACE_PENDING_FORK_VERIFIER is the exact 1,536-byte concatenation of the canonical predecessor, old-latest and replacement 15-word rows and the 96-byte review envelope. The three row parts use the preceding field order and canonical ABI word padding; no offset, length word or suffix is present. The review envelope is the exact grammar in the ScheduleOracle subsection. The expected predecessor/old registration hashes, unreachable-boundary checks, used-digest tombstone and atomic state transition are exactly the rules in the ScheduleOracle subsection.
4. SPLIT_LATEST_FORK_VERIFIER is the exact 1,056-byte concatenation of the canonical current-latest row, canonical successor row and review envelope. It is the forward-only latest-row split specified in the ScheduleOracle subsection; both rows and their boundary are committed before the delayed operation is queued.
5. PUBLISH_GENESIS_CAMPAIGN is `abi.encode(uint64forceCutoffBlock,uint64proposalCutoffBloc uint64quiesceNotBeforeBlock,uint64resumeByBlock,uint64resumeByTimestamp, uint64reviewFinalizedByBlock,addresstargetSettlement,uint64targetProtocolVersion, bytes32targetManifestHash,bytes32targetRegistrationHash)`. Nonce, generation, review commitment and campaign ID are derived, never supplied.

6. PUBLISH_MIGRATION_ARM is `abi.encode(uint64expectedSourceProtocolVersion, uint64targetProtocolVersion,bytes32targetManifestHash,bytes32targetRegistrationHash)`. Generation, arm time and hard abort time are derived.

Canonical ABI means the displayed number of 32-byte words, zero high padding for every unsigned/address value, fixed bytes left-aligned with zero trailing padding, and no suffix. The manager decodes by kind, re-encodes byte-for-byte, exact-reads the EXECUTING timelock row, records the operation ID consumed before its first downstream call, and dispatches only the kind's named contracts and selectors. It has no raw target or calldata branch. Every manager mutation requires its private lifecycle IDLE. Applying an operation writes APPLYING and the consumed ID before the first external call; lease expiry writes ABORTING before its first Router call; each path restores IDLE only as its final write, and any failure reverts the entire frame. A downstream callback therefore cannot execute another operation, register another release or interleave a lease abort. REGISTER_RELEASE appends Router registration, profile-derived registry rows and Market authorization in this one Timelock–PVM rollback domain and exact-checks every fixed return before PAP1; partial publication is impossible. REGISTER_RELEASE is also the only way to install AggregatorSeatMarket's append-only Settlement authorization: the manager derives it solely from the same manifest and target registration checked by Router. There is no separate Release Manager.

REGISTER_RELEASE uses this one fixed cross-contract journal:

```
ActiveSettlementRouter.registerTargetReleaseV2(
 uint64 expectedPredecessorProtocolVersion,
 ReleaseManifestV2 manifest,
 SettlementDeploymentDescriptorV1 deployment,
 bytes profileBytes)
 returns (bytes4 magic,bytes32 manifestHash,bytes32 targetRegistrationHash)
ActiveSettlementRouter.targetReleaseRegistrationV2(uint64 protocolVersion)
 returns (bytes4 magic,uint64 returnedProtocolVersion,
          uint64 expectedPredecessorProtocolVersion,address targetSettlement,
          bytes32 targetRuntimeHash,bytes32 targetConfigurationHash,
          bytes32 settlementDeploymentDescriptorHash,
          bytes32 executionProfileHash,
          uint64 componentConfigurationReadGas,
          bytes32 migrationActivationProfileRecordHash,
          bytes32 dataSessionConfigurationHash,
          bytes32 releaseManifestHash,
          bytes32 bridgeRouteExpansionHash,
          bytes32 settlementValidityVerifierDescriptorHash,
          bytes32 kind0IngressAuthorizationId,
          bytes32 targetRegistrationHash)
ActiveSettlementRouter.migrationActivationProfileV2(uint64 protocolVersion)
 returns (bytes4 magic,uint64 returnedProtocolVersion,
          bytes32 executionProfileHash,bytes32 activationProfileRecordHash,
          address verifier,bytes32 verifierRuntimeHash,
          bytes32 verifierConfigurationHash,bytes32 verifyingKeyHash,
          bytes32 proofSystemId,bytes32 publicInputSchemaHash,
          bytes4 verifierSelector,uint32 maximumProofBytes,
          uint64 verificationGasLimit,uint64 supportedL1BlockGasLimit,
```

```
          uint64 worstCaseActivationAdoptionGas,uint64 sourceFreezeGasLimit,
          uint64 targetAdoptionGasLimit,uint64 queueMigrationGasLimit,
          uint64 activationContextReadGasLimit,uint64 postStateReadGasLimit,
          uint64 legacyStateReadGasLimit,uint64 legacyArmGasLimit,
          uint64 legacyFinalizeGasLimit,uint64 postCallbackReserveGas)
ProtocolVersionManagerV2.profileIngressRootV2(uint64 protocolVersion)
 returns (bytes4 magic,uint64 returnedProtocolVersion,
          uint16 count,bytes32 ingressAuthorizationRoot)
ProtocolVersionManagerV2.profileIngressAuthorizationV2(bytes32 authorizationId)
 returns (bytes4 magic,bytes32 returnedAuthorizationId,
          ProfileIngressAuthorizationV2 row)
```

The selectors are respectively `0x9aa71eff`, `0xf588fec3`, `0xc65ff64e`, `0x2d2bbe23` and `0x2181b974`. Router registration calldata is the four-byte selector followed by the exact REGISTER_RELEASE payload; its selector is the Keccak selector of the fully expanded 59-word manifest tuple printed in the activation ABI and the eight-word deployment tuple above. Returns are 96 bytes for registration, 512/RTR2, 768/MPR2, 128/PIR2 and 832/PIA2 bytes; magics are `0x52545232`, `0x4d505232`, `0x50495232` and `0x50494132`. The ingress row return is magic, returned ID and the 24 static struct words in the published field order. Unknown version/ID, short/trailing data or noncanonical padding rejects rather than returning a zero row. RTR2's added gas, descriptor-hash and kind-0 authorization-ID words are derived from profile word 111, words 271–280 and the exact kind-0 PIA2 row during registration and are immutable. Because `targetRegistrationHash` already commits `executionProfileHash` and `releaseManifestHash`, this compact copy is a recoverable activation witness for the same read budget, descriptor and authorization, not a second authority. Any mismatch with the transient profile rejects before the row is written. Activation requires that stored ID to pass PIM2 membership under the exact PIR2 root before reading PIA2; it then joins kind, adapter, runtime, configuration and constructor commitment.

The fixed MPR2 words are decoded from canonical profile bytes and committed as

```
migrationActivationProfileRecordHash = H(
 "slot-chain-migration-activation-profile-v2" || u64(protocolVersion) ||
 executionProfileHash || address20(verifier) || verifierRuntimeHash ||
 verifierConfigurationHash || verifyingKeyHash || proofSystemId ||
 publicInputSchemaHash || bytes4(verifierSelector) ||
 u32(maximumProofBytes) || u64(verificationGasLimit) ||
 u64(supportedL1BlockGasLimit) || u64(worstCaseActivationAdoptionGas) ||
 u64(sourceFreezeGasLimit) || u64(targetAdoptionGasLimit) ||
 u64(queueMigrationGasLimit) || u64(activationContextReadGasLimit) ||
 u64(postStateReadGasLimit) || u64(legacyStateReadGasLimit) ||
 u64(legacyArmGasLimit) || u64(legacyFinalizeGasLimit) ||
 u64(postCallbackReserveGas))
```

Router stores that immutable record with the registration; PVM verifies its verifier descriptor hash against the manifest. Genesis and later activation recover every verifier/gas word only from MPR2, recheck its record/profile/ registration hashes and target code/configuration, and reject a caller-supplied profile byte, descriptor or gas value. The transient full ABI preimage is therefore needed only to validate and register the compact activation record and ingress rows; activation never attempts to invert `executionProfileHash`.

Before any protocol-change write, PVM checks EXTCODEHASH of the actual Timelock caller and makes a 100,000-gas exact PCT1 STATICCALL back to that caller. It recomputes the governance-delay descriptor from the actual chain, caller address/code, returned DAO/PVM/delay/domain and requires equality to its immutable descriptor. REGISTER_RELEASE additionally requires profile words 16–19 to equal those live observations. A Timelock with a substituted DAO or caller-dependent PCT1 response therefore cannot use an otherwise authentic queued payload.

PVM then derives all expected values from the raw strict profile and joins the complete control tuple to the actual root deployment. It checks its own EXTCODEHASH against word 21 and exact-reads its own 1,184-byte PVM1 view; Router independently checks `msg.sender` EXTCODEHASH and makes the same 100,000-gas, four-byte PVM1 STATICCALL before its first write. Each caller strictly decodes PVM1, recomputes the complete PVM configuration hash against word 22, checks the six non-Market control accounts in this order: Router, ForcedQueue, BuilderRegistry, ScheduleOracle, BridgeDomainRegistry and SourceBundleFactory; it then exact-reads SBF1, checks the root-lifetime SourceTerminalVerifier at words 166–168, and checks Market EXTCODEHASH and its fixed `componentConfigHashV2()` against the PVM1 root and independently recomputes the Market configuration from the actual five constructor values. It next checks the protocol-root ERC-2470 address and EXTCODEHASH; ERC-2470 has no protocol-specific configuration getter. It directly checks the ordinary Settlement verifier's EXTCODEHASH and exact configuration getter from profile words 271–280. It then derives the kind-0 init code and address from the profile artifact and checks that adapter's EXTCODEHASH, `componentConfigHashV2()` and KCS1 in that order. Finally it checks target EXTCODEHASH, `componentConfigHashV2()` (selector `0xf6c0f7d2`), DSV1, TCS2 and SVD2 in that order. PVM's postread repeats the verifier, three adapter and six target observations. Short/trailing, bad padding, revert/OOG or caller-dependent PCT1/PVM1/control getter behavior rejects in the common outer revert domain. This exact order is part of the cold-state gas certificate and MUST NOT be reordered. The ten destination component rows remain profile/manifest commitments at this L1 step; their live identities are proved by the L2 migration/activation gate, never by impossible cross-chain STATICCALLs. After complete validation but before any registration write, PVM writes APPLYING/consumed. It then calls Router registration with exactly `releaseRouterRegistrationGas`. The historical model value is not a release authority: the production PVM configuration instead contains nonzero immutable `releaseRouterRegistrationGas`, `releaseMarketInstallationGas`, `releasePostreadGas` and `releasePostCallbackReserveGas`. Each is bound by the PVM configuration hash and the release gas certificate; the 3,000,000/500,000/100,000 figures in model fixtures are lower-bound examples only. Before each nested call PVM checks EIP-150-forwardable gas plus all remaining stipends, bounded copy cost and the post-callback reserve. The maximum 146,848-byte cold-state registration benchmark must succeed at the published values and fail at one less. Router requires the manager caller in PVM APPLYING while Router is IDLE, independently re-derives expected values from the same raw profile and repeats the PVM1, control, Market, factory and target live reads as its own caller before writing REGISTERING. It then appends the version row and restores Router IDLE last. Before publishing that row, Router derives one fixed compatibility projection from the 59 manifest words: the four protocol-lifetime component descriptors at zero-based component positions 3, 5, 6 and 7, plus the release's fresh `destinationDomainId` and destination Bridge. The bootstrap release seeds the immutable lifetime projection. Router stores an exact `protocolVersion->projection` row and one-shot reverse maps `destinationDomainId->protocolVersion` and `destinationBridge->protocolVersion`. Ev-

ery later registration requires exact equality with the lifetime projection and absence from both fresh-key maps before inserting all three rows. Profile-deployment installation then exact-reads only its version row, the lifetime projection and the two reverse keys; it never scans prior release registrations. Missing, colliding or disagreeing indexes fail closed, and the common REGISTER_RELEASE journal rolls every compatibility/index write back on any later fault. Caller-dependent getter behavior, short/trailing returns, revert/OOG or any late mismatch reverts the common outer journal. Because these are STATICCALLs, the target is nonproxy/selfdestruct-free and Ethereum code cannot change during the transaction; after Router returns PVM nevertheless repeats target EXTCODEHASH/config/DSV1/TCS2 and exact-checks all Router stores, closing mutable return substitution rather than relying on a second read by the same caller. PVM exact-checks RTR2/hash pair, stores the sorted ingress rows and root internally, calls Market installation with exactly `releaseMarketInstallationGas`, and then makes `releasePostreadGas`-bounded exact STATICCALL postreads of the Router registration, MPR2 activation profile, PIR2 root, every PIA2 row and Market SAT1 row. Market installation itself direct-reads the same Router registration with exactly 100,000 gas before its write. The call target is the Market constructor's immutable `activeSettlementRouter`; it is never an install argument or a caller-supplied expected row. Only after every postread agrees does PVM return PAP1; Timelock then writes EXECUTED. This order gives Market an already existing authenticated Router row, while the outer revert domain removes every Router/PVM/Market write if any later call or decode fails. A registration may be armed only when its stored predecessor still equals Router's active version; registration does not itself close admissions.

The exact authority descriptors are

```
marketAuthorityConfigurationHash = H(
 "slot-chain-pvm-derived-market-authority-config-v1" || u16(124) ||
 u256(marketChainId) || address20(aggregatorSeatMarket) ||
 u256(settlementChainId) || address20(protocolVersionManager) ||
 address20(activeSettlementRouter))
aggregatorSeatMarketConfigurationHash = H(
 "slot-chain-aggregator-seat-market-config-v2" || u16(1376) ||
 marketAuthorityConfigurationHash || profileWord[24] || profileWord[25] ||
 profileWord[68] || profileWord[73] ||
 profileWord[84] || profileWord[85] ||
 concat(profileWord[86],...,profileWord[100]) ||
 profileWord[120] || profileWord[121] ||
 concat(profileWord[107],...,profileWord[110]) ||
 concat(profileWord[252],...,profileWord[266]))
governanceDelayAuthorityDescriptorHash = H(
 "slot-chain-protocol-change-timelock-v1" ||
 u256(settlementChainId) || address20(protocolChangeTimelock) ||
 protocolChangeTimelockRuntimeHash || address20(daoProposer) ||
 address20(protocolVersionManager) || u64(604800) ||
 H("slot-chain-protocol-change-operation-v1"))
protocolVersionManagerConfigurationHash = H(
 "slot-chain-protocol-version-manager-config-v1" ||
 u256(settlementChainId) || address20(protocolChangeTimelock) ||
 governanceDelayAuthorityDescriptorHash ||
```

```
 address20(activeSettlementRouter) || activeSettlementRouterRuntimeHash ||
 activeSettlementRouterConfigurationHash ||
 address20(forcedQueue) || forcedQueueRuntimeHash ||
 forcedQueueConfigurationHash ||
 address20(builderRegistry) || builderRegistryRuntimeHash ||
 builderRegistryConfigurationHash ||
 address20(scheduleOracle) || scheduleOracleRuntimeHash ||
 scheduleOracleConfigurationHash ||
 address20(aggregatorSeatMarket) || aggregatorSeatMarketRuntimeHash ||
 aggregatorSeatMarketConfigurationHash ||
 address20(bridgeDomainRegistry) || bridgeDomainRegistryRuntimeHash ||
 bridgeDomainRegistryConfigurationHash ||
 address20(sourceBundleFactory) || sourceBundleFactoryRuntimeHash ||
 sourceBundleFactoryConfigurationHash ||
 address20(sourceTerminalVerifier) || sourceTerminalVerifierRuntimeHash ||
 sourceTerminalVerifierConfigurationHash || manifestNamespace ||
 u64(604800) || u64(604800) || u64(604800) || u16(64) ||
 u64(releaseRouterRegistrationGas) ||
 u64(releaseMarketInstallationGas) || u64(releasePostreadGas) ||
 u64(releasePostCallbackReserveGas))
protocolVersionManagerDescriptorHash = H(
 "slot-chain-protocol-version-manager-v1" ||
 address20(protocolVersionManager) || protocolVersionManagerRuntimeHash ||
 protocolVersionManagerConfigurationHash)
```

The Timelock's immutable `componentConfigHashV2()` is exactly `governanceDelayAuthorityDescriptorHash`; PVM1 word 12 is the same value. Thus protocol-root finalization joins the live role-3 runtime/configuration to the PVM authority row rather than accepting two independently valid Timelock identities. For this protocol root `marketChainId=settlementChainId`. The inner authority hash binds the five topology values, while the outer 1,376-byte projection additionally binds the penalty sink, inclusion/evidence/reorg bounds, every seat economic value, seat/duty capacity, all historical-read gas bounds, both quote-maturity clocks, all nine mutation/rotation gas bounds, standby-lease and reverse-auction improvement bounds, the canonical economic profile hash, and the live Router runtime/configuration identity. Market stores those exact constructor values; its fixed `componentConfigHashV2()` return and profile word 37 are both `aggregatorSeatMarketConfigurationHash`. The hash is recomputed from the listed profile words and cannot be publisher-selected. All three topology addresses are distinct. PVM and Router additionally compare profile words 20/23/26/29/32/35/38/41 to the actual PVM identity and complete PVM1 root. PVM1 binds the runtime/configuration pairs in words 24–25, 27–28, 30–31, 33–34, 36–37, 39–40 and 42–43. Each caller independently checks EXTCODEHASH and the fixed-gas live configuration getter for Router, ForcedQueue, BuilderRegistry, ScheduleOracle, Market, BridgeDomainRegistry and SourceBundleFactory, exact-reads SBF1, and checks the terminal identity at profile words 166–168. The Market configuration is also independently recomputed from the actual constructor inventory. Thus a publisher cannot make a self-consistent alternate control graph. Substituting any lookalike control component makes registration fail before a Router, PVM or Market write. Words 18 and 37 are fully recomputable from other profile words and the strict profile decoder rejects either mismatch. Word 22 is deliberately checked

at the live-root join, not by the context-free ABI decoder: its preimage includes the four measured protocol-root release gas caps, which are immutable PVM1 values but are not duplicated as per-release profile fields. PVM and Router therefore exact-read PVM1 and recompute word 22 before their respective first writes; a nonzero publisher value alone never authorizes registration. The final eight configuration values are minimum governance delay, maximum live VERSION_MIGRATION duration, migration-arm execution-window seconds, review-finality blocks, `releaseRouterRegistrationGas`, `releaseMarketInstallationGas`, `releasePostreadGas` and `releasePostCallbackReserveGas`. The last four are protocol-root immutable MAX_PROFILE caps measured before the PVM is deployed against the 146,848-byte/281-field cold worst case, not per-release mutable values or guessed constants. Every later release certificate must measure at or below these caps; exceeding one requires the independent new-root migration described above. The executable model's 15,000,000/1,000,000/500,000/2,000,000 values are fixtures only and MUST NOT populate a production deployment without the cold max-profile benchmark. All addresses, runtime hashes, namespace and descriptor hashes are nonzero; the two contracts, DAO proposer and every downstream component are pairwise distinct where their roles differ.

Configuration reads use exactly 100,000 gas and these canonical ABIs:

```
protocolChangeTimelockConfigV1() // selector 0xb80095ca; 160 bytes
 returns (bytes4 magic,address daoProposer,address protocolVersionManager,
          uint64 minimumDelaySeconds,bytes32 operationDomain) // PCT1
protocolVersionManagerConfigV1() // selector 0x4deb7821; 1184 bytes
 returns (bytes4 magic,uint256 settlementChainId,address timelock,
          address activeSettlementRouter,address forcedQueue,
          address builderRegistry,address scheduleOracle,
          address aggregatorSeatMarket,bytes32 aggregatorSeatMarketRuntimeHash,
          bytes32 aggregatorSeatMarketConfigurationHash,address bridgeDomainRegistry,
          address sourceBundleFactory,bytes32 timelockDescriptorHash,
          bytes32 manifestNamespace,uint64 minimumDelaySeconds,
          uint64 maximumLiveMigrationSeconds,
          uint64 migrationArmExecutionWindowSeconds,
          uint16 reviewFinalityBlocks,
          uint64 releaseRouterRegistrationGas,
          uint64 releaseMarketInstallationGas,uint64 releasePostreadGas,
          uint64 releasePostCallbackReserveGas,
          bytes32 activeSettlementRouterRuntimeHash,
          bytes32 activeSettlementRouterConfigurationHash,
          bytes32 forcedQueueRuntimeHash,bytes32 forcedQueueConfigurationHash,
          bytes32 builderRegistryRuntimeHash,
          bytes32 builderRegistryConfigurationHash,
          bytes32 scheduleOracleRuntimeHash,
          bytes32 scheduleOracleConfigurationHash,
          bytes32 bridgeDomainRegistryRuntimeHash,
          bytes32 bridgeDomainRegistryConfigurationHash,
          bytes32 sourceBundleFactoryRuntimeHash,
          bytes32 sourceBundleFactoryConfigurationHash,
          address sourceTerminalVerifier,
          bytes32 sourceTerminalVerifierRuntimeHash,
```

```
          bytes32 sourceTerminalVerifierConfigurationHash) // PVM1
protocolChangeOperationV1(bytes32) // selector 0x4b80fe68; 256 bytes
 returns (bytes4 magic,uint8 state,uint64 nonce,uint8 operationKind,
          uint32 payloadBytes,bytes32 payloadHash,
          uint64 queuedAt,uint64 executeAfter) // PCO1
migrationArmFreshAfterV1() // selector 0xbc4707fb; 64 bytes
 returns (bytes4 magic,uint64 armFreshAfter) // MAF1
```

The magics are PCT1 `0x50435431`, PVM1 `0x50564d31`, PCO1 `0x50434f31` and MAF1 `0x4d414631`; `operationDomain` is exactly `H("slot-chain-protocol-change-operation-v1")`. Revert, OOG, short/trailing return, bad magic, bad padding, wrong code hash, wrong constant or descriptor mismatch rejects publication before a protocol component changes.

A later migration arm appends one non-extendable manager arm row:

```
abortAfterTimestamp = checked(uint64(block.timestamp) + 604800)
versionMigrationArmId = H(
 "slot-chain-version-migration-arm-v2" || u256(settlementChainId) ||
 address20(protocolVersionManager) || u64(generation) ||
 u64(sourceProtocolVersion) || u64(targetProtocolVersion) ||
 targetManifestHash || targetRegistrationHash || u64(block.timestamp) ||
 u64(abortAfterTimestamp) || protocolChangeOperationId)
liveVersionMigrationLeaseV1() // selector 0xaaac4c97; 320 bytes
 returns (bytes4 magic,uint8 state,uint64 generation,
          uint64 sourceProtocolVersion,uint64 targetProtocolVersion,
          bytes32 targetManifestHash,bytes32 targetRegistrationHash,
          bytes32 armId,uint64 armedAtTimestamp,uint64 abortAfterTimestamp) // VML1
```

The magic is `0x564d4c31`; NONE=0 and LIVE=1 are the only live-view states, and NONE requires every other word zero. Arm rows are append-only and keyed by `armId`; they are historical evidence, never a singleton liveness lock. A permanent `generation->armId` reverse index makes the live lookup O(1); the indexed row must reproduce the complete gate tuple. On every call the getter exact-reads Router's active settlement state and migration gate. It returns that exact row only while the gate is ARMED/READY and otherwise returns canonical NONE, including immediately after successful activation. REGISTER_RELEASE and PUBLISH_MIGRATION_ARM likewise derive the current predecessor only from Router's exact ASR1 ACTIVE read; PVM has no shadow active-version authority. A new arm requires Router IDLE/ACTIVE, the current Router version and the registered target predecessor, so a CONSUMED/ABORTED historical row cannot block the next generation. Router activation exact-reads this row, requires the complete live tuple and `block.timestamp<abortAfterTimestamp` before verifier dispatch, and never accepts a caller-supplied lease. At or after the bound, `permissionlessAbortExpiredMigrationV1()` (selector `0xea3e96c2`) is callable by anyone and executes the Router ABORTING journal. If activation already won an earlier transaction, the derived live getter is already NONE and the call is a no-op; otherwise it aborts ARMED or READY and restores source ACTIVE. At the exact expiry timestamp activation rejects, so abort wins independent of ordering. No operation can refresh, replace or extend a LIVE lease. A proving outage, lost governance key or abandoned target therefore cannot leave admissions closed forever.

PVM also stores one monotone `armFreshAfter` watermark, initialized to zero. A PUBLISH_MIGRATION_ARM row is eligible only when `queuedAt>armFreshAfter` strictly. A

successful permissionless expiry abort atomically writes `armFreshAfter=block.timestamp` before IDLE; any abort fault rolls that write back with the Router and source state. Therefore every arm queued before that abort, including a mature sibling for the same source/target and a row queued at the exact abort timestamp, is permanently unusable. Only a row queued strictly later, which then serves the full new seven-day delay and lands within its finite execution window, can arm again. Freshness advances on abort, not on successful activation. Consequently the DAO may visibly prequeue a future-source migration and, if its source becomes ACTIVE while that row is in its finite execution window, arm a consecutive upgrade. This is an explicit, advance-noticed governance upgrade capability; it is not dormant authority and does not bypass either delay or source-version check.

The remaining PVM–Router mutation boundaries are

```
publishLegacyGenesisCampaignV1(
 bytes32 operationId,uint64 executeAfter,
 uint64 forceCutoffBlock,uint64 proposalCutoffBlock,
 uint64 quiesceNotBeforeBlock,uint64 resumeByBlock,
 uint64 resumeByTimestamp,uint64 reviewFinalizedByBlock,
 address targetSettlement,uint64 targetProtocolVersion,
 bytes32 targetManifestHash,bytes32 targetRegistrationHash)
 returns (bytes4 magic,uint64 nonce,uint64 generation,
          bytes32 reviewCommitment,bytes32 campaignId)
armVersionMigrationV1(
 bytes32 operationId,bytes32 armId,uint64 armedAtTimestamp,
 uint64 abortAfterTimestamp,uint64 expectedSourceProtocolVersion,
 uint64 targetProtocolVersion,bytes32 targetManifestHash,
 bytes32 targetRegistrationHash)
 returns (bytes4 magic,uint64 generation,bytes32 returnedArmId)
abortExpiredVersionMigrationV1(bytes32 armId)
 returns (bytes4 magic,bytes32 returnedArmId,uint64 generation)
```

Their selectors are `0x5f0ed7f5`, `0xe3bcfcb4` and `0xc4eee12d`; exact returns are 160/LGP1 `0x4c475031`, 96/VMA1 `0x564d4131` and 96/VMB1 `0x564d4231`. Each call receives exactly 8,000,000 gas, is nonpayable and PVM-only, checks canonical calldata and starts from Router lifecycle IDLE. Publication writes PUBLISHING before its first external profile/legacy read, independently recomputes the complete EXECUTING operation ID/timing/registration/review/campaign and restores IDLE last. PVM requires the LGP1 fields and then exact-postreads LGC1.

For arm, PVM first writes APPLYING/consumed and the derived LIVE VML1 lease, then calls Router. Router writes ARMING, exact-reads that lease and the stored RTR2/MPR2 target, performs the source callback, writes ARMED and restores IDLE last. PVM checks VMA1 and exact-postreads ASR1 plus VML1 before PAP1. Expiry abort writes PVM ABORTING, calls the Router abort selector while VML1 is still LIVE and expired, exact-checks VMB1/ASR1, then clears VML1 to canonical NONE, sets `armFreshAfter` to the abort timestamp, and restores manager IDLE. If a sealed successor receipt proves activation already won, PVM skips the Router mutation, terminalizes only the matching stale lease and still exact-checks the receipt/ASR1. Any callback, return, postread or final write failure reverts both contracts and the source callback. No raw Router selector or second order is legal.

Successful migration uses a separate Router-to-PVM terminal callback:

```
consumeVersionMigrationLeaseV1(
  bytes32 armId, bytes32 oldAuthorizationId,
  bytes32 activationReceiptId)
 returns (bytes4 magic,bytes32 returnedArmId,uint64 generation) // VMC1
```

Its selector is `0xebb624e2`; the Router supplies exactly 200,000 gas, uses exact 100-byte calldata (selector plus three `bytes32` words), zero value and the constructor-bound PVM, and accepts exactly 96 bytes with magic `0x564d4331`. Only after ARV1/ASV1 and ASR1 prove the same generation, source/target, manifests, registration, old authorization and sealed successor index does PVM clear the singleton LIVE lease and restore IDLE. The append-only arm row and the abort freshness watermark remain unchanged. Revert, OOG, bad magic, short/trailing data or a post-clear fault rolls Router, PVM, source, target, Queue and receipts back atomically.

The Router derives the target Settlement, manifest, registration, execution profile and immutable verifier solely from the stored delayed gate or genesis campaign and rejects a caller-supplied address or authority. For a later migration it first performs every source, target, Queue, receipt and index preflight, reconstructs the complete typed statement and invokes the profile-pinned verifier by bounded STATICCALL while lifecycle is IDLE and the public gate is READY. For genesis it first validates the exact live campaign, enters ACTIVATING, exact-reads the campaign-bound QUIESCENT LGS1 boundary while the public Router gate remains ACTIVE on legacy, reconstructs the statement and invokes the verifier by bounded STATICCALL before LGAR. Launch header/MPT/SignalService authentication is part of that same verifier and statement, not an opaque Boolean. A later switch checks on L1 that the old `CanonicalCore.messageCursor=ForcedQueue.cursor`; it does not pretend to read the live L2 Inbox cursor. The preflight reads the old canonical core, sequence and last-commit block; Queue root/count/frontier/cursor/liabilities, balance and last due time; the exact fresh target predicate; and all code, configuration and object identities. In particular, before any activation write it loads RTR2's kind-0 authorization ID, exact-reads PIR2/PIM2/PIA2, joins that member to the registered version/root and repeats its adapter's exact EXTCODEHASH, configuration and KCS1 constructor-state checks from registration. It also loads the compact ordinary-verifier descriptor hash from RTR2, strictly decodes the target's complete SVD2 getter under RTR2's stored configuration-read gas, recomputes that hash and repeats the returned verifier's exact EXTCODEHASH/configuration checks. The target proof executes the release Anchor and validates L2 component rows 4–10. A missing component, collision, bad verifier or proving outage therefore leaves the source and Queue unchanged.

After verifier success, Router computes the one shared join key:

```
activationContextHash =
 H("slot-chain-l1-activation-context-v1" ||
   u256(settlementChainId) || address20(router) || u8(transitionKind) ||
   u64(generation) || u64(seatGeneration) || u64(sourceProtocolVersion) ||
   u64(targetProtocolVersion) || sourceManifestHash || targetManifestHash ||
   targetRegistrationHash || address20(sourceSettlement) ||
   address20(targetSettlement) || u64(sourceCanonicalSequence) ||
   baseCanonicalHash || statementHash || candidateDigest ||
   canonicalCoreV2Hash(outputCore) || u64(targetCanonicalSequence) ||
   address20(forcedQueue) || queueRoot || u64(queueCount) ||
   u64(startCursor) || u64(endCursor) || address20(proofBeneficiary) ||
```

```
   u64(block.number))
```

Here `sourceCanonicalSequence=0` and `targetCanonicalSequence=0` are mandatory sentinels for GENESIS_IMPORT; the Router rejects any other genesis source sequence before calling the verifier. The target sequence is checked `sourceCanonicalSequence+1` for VERSION_MIGRATION. For a later migration, Router writes ACTIVATING and this complete context only after verifier success. For genesis it writes ACTIVATING first as the reentrancy journal, then derives the context, campaign-bound arm ID and launch ID from the exact QUIESCENT LGS1 boundary and stored campaign before the proof STATICCALL. The compatibility implementation's campaign fence prevents that boundary from moving once clean. Each component authenticates it through the bounded STATICCALL

```
migrationActivationContextV1() returns
 (bytes4 magic,uint8 lifecycle,bytes32 activationContextHash,
  address sourceSettlement,address targetSettlement,uint64 generation,
  uint64 sourceProtocolVersion,uint64 targetProtocolVersion,
  bytes32 targetManifestHash,bytes32 targetRegistrationHash)
```

whose selector is `0x7cf70319`, magic is `0x4d414354` (MACT), and returndata is exactly 320 bytes. Wrong lifecycle, target, tuple, length, magic, padding, revert or OOG rejects. Outside ACTIVATING, MACT remains callable as the sole lifecycle view: it returns the current ARMING/READYING/ABORTING/IDLE value and canonical zero for every activation-context field. Later-migration arm, readiness and abort callbacks authenticate that lifecycle and obtain their target tuple from `activeSettlementStateV1()`; genesis LGAR authenticates ACTIVATING and the exact live campaign ID, generation, target and activation context. At every transaction and external-call boundary the context is nonzero exactly in ACTIVATING; the adjacent lifecycle/context stores have no intervening external interaction.

For VERSION_MIGRATION, the exact source callback is

```
freezeMigrationSourceV1(bytes32 activationContextHash) returns
 (bytes4 magic,bytes32 returnedActivationContextHash,
  bytes32 sourcePostStateCommitment)
```

with selector `0x45a80913`, 36-byte calldata, 96-byte return and magic `0x4d46525a` (MFRZ). It is nonpayable and accepts only the immutable Router, exact MACT tuple and local MIGRATION_READY state. It rechecks its canonical core/sequence, arm receipt, closed boundary and zero-live readiness, then writes only MIGRATION_READY to FROZEN plus the context and frozen block. All canonical history, claims, refunds, rewards, reservations, liabilities and balances remain byte-for-byte live on the permanent source. It makes no external interaction after the MACT read and returns

```
H("slot-chain-source-freeze-poststate-v1" || activationContextHash ||
  address20(sourceSettlement) || u8(FROZEN=4) || u64(generation) ||
  u64(sourceProtocolVersion) || u64(sourceCanonicalSequence) ||
  baseCanonicalHash || u64(frozenAtBlock)) .
```

The target adoption callback is

```
adoptMigrationCanonicalV1(
```

```
 uint8 transitionKind,uint64 generation,uint64 seatGeneration,
 uint64 sourceProtocolVersion,
 uint64 targetProtocolVersion,uint64 sourceCanonicalSequence,
 bytes32 targetManifestHash,bytes32 candidateDigest,
 CanonicalCoreV2 outputCore)
returns (bytes4 magic,uint64 targetCanonicalSequence,
         bytes32 targetPostStateCommitment)
```

Its exact expanded selector is `0x3286443c`; calldata is 580 bytes, returndata is 96 bytes and magic is `0x4d43414e` (MCAN). It is the only PREACTIVE mutation entry and accepts only Router plus the exact MACT target, version, manifest and registration. It requires untouched canonical/history, mutation and DataSession regions, then writes the output core, target sequence, history tag/cell, both L1 block fields, the ordinary canonical event, stored context and locally prepared ACTIVE state. For a later migration Router's public gate remains READY; for genesis the legacy branch remains unconsumed and the Router remains ACTIVATING. In both cases the target remains PREACTIVE to ordinary callers. After its initial MACT read it makes no external call. Its returned commitment is

```
H("slot-chain-l1-adoption-v1" || u256(settlementChainId) ||
  address20(router) || address20(targetSettlement) || activationContextHash ||
  u8(transitionKind) || u64(generation) || u64(sourceProtocolVersion) ||
  u64(targetProtocolVersion) || u64(sourceCanonicalSequence) ||
  targetManifestHash || candidateDigest || canonicalCoreV2Hash(outputCore) ||
  u64(targetCanonicalSequence) || u64(canonicalizedAtBlock)) .
```

Ordinary commits continue to use `advanceCursor`. Activation instead uses the sole Router-only combined Queue transition:

```
migrateAuthorityForActivationV1(
 bytes32 activationContextHash,address sourceSettlement,
 address targetSettlement,bytes32 expectedRoot,uint64 expectedCount,
 uint64 startCursor,uint64 endCursor,address proofBeneficiary)
returns (bytes4 magic,bytes32 returnedActivationContextHash,
         uint256 creditedWei,bytes32 queuePostStateCommitment)
```

with selector `0x9461f698`, 260-byte calldata, 128-byte return and magic `0x514d4947` (QMIG). It authenticates Router and MACT, old authority, root/count/current cursor and `start<=end<=count`; checked-adds the exact prefix fee interval to the beneficiary's pull claim, advances the cursor and changes authority to the target atomically. It calls no source, target, beneficiary or arbitrary sink, and changes no leaf, deadline, root or count. Its poststate commitment is

```
H("slot-chain-queue-migration-poststate-v1" || activationContextHash ||
  address20(forcedQueue) || address20(targetSettlement) || queueRoot ||
  u64(queueCount) || u64(endCursor) || address20(proofBeneficiary) ||
  u256(creditedWei) || u256(postAccountedLiabilityWei) ||
  u256(postTotalPullClaimableWei)) .
```

Source, target and Queue each implement the common view

```
migrationActivationPostStateV1(bytes32 activationContextHash) returns
 (bytes4 magic,uint8 role,bytes32 returnedActivationContextHash,
  bytes32 postStateCommitment)
```

with selector `0x66e664cb`, 36-byte calldata, 128-byte return, magic `0x4d415053` (MAPS), and roles SOURCE=1, TARGET=2, QUEUE=3. Router performs exactly these three bounded STATICCALLs after QMIG and independently recomputes all commitments. The source reproduces MFRZ, target reproduces MCAN, and Queue reproduces QMIG. No mutating external call is legal after QMIG except the terminal VMC1 call below. Router makes that sole exception only after sealing the activation receipt and successor row, publishing the new ACTIVE state, clearing its transient context and restoring its own lifecycle to IDLE. VMC1 may mutate only the constructor-bound PVM lease/lifecycle. Its failure or noncanonical return still reverts the same outer Ethereum transaction and thus restores QMIG, all three MAPS states, receipts, the public Router state and the old LIVE lease. No target, beneficiary, arbitrary sink or other mutating call is legal in this suffix.

The exact VERSION_MIGRATION journal is verifier STATICCALL; write ACTIVATING/context; MFRZ; MCAN; publish the exact kind-0 ingress binding; execute the target SourceBridge inactive-to-active call; install its source authority indexes; execute and post-read exact BRC1; one-shot seal the destination adapter; publish its Bridge ingress binding; QMIG; the three MAPS reads; internal target- registration consumption, receipt and successor-index writes; the final current- target/public-ACTIVE word; clear context; write Router IDLE; exact VMC1 CALL; check its 96-byte return and exact-postread canonical NONE lease; and restore PVM IDLE. ACTIVATING makes every newly published ingress binding inert until the current-target/public-ACTIVE write. VMC1 is the last external call and the sole post-QMIG mutating call. Every callback, post-read, return decoder and later write is in the outer transaction's single revert domain. A fault at any boundary restores READY/IDLE, the old source and Queue authority, cursor/payment state, empty target, every prepared adapter's prior seal, the inactive target SourceBridge, both ingress/profile maps, unused registration and absent receipt, so permissionless retry or mature abort remains possible.

"Empty PREACTIVE target" is an exact predicate, not a claimed constructor default. The target Settlement's internal DataSession/reward region requires all 1,024 cell tags FREE; empty ID-to-cell and live-owner maps; `liveCount=refundCount=occupiedCount=0`; `nextSessionSequence=0`, `gcCursor=0`; `migrationRefundGeneration=0`, `migrationRefundClaimDeadline=0`, all three 256-cell reward receipt rings empty, all three reward buckets and `totalRewardFunding` zero, the shared Settlement guard in its canonical NOT_ENTERED state; and zero live- and refund-bond liability. The actual target Settlement ETH balance is deliberately not required to be zero: CREATE2 prefunding or forced ETH is surplus and cannot veto activation. Deployment may occur while the source has live sessions, because the protocol-lifetime gate is shared; cutover separately requires that gate to be READY after authenticating the source-local live-session count zero. Exact fresh creation code plus the absence of every PREACTIVE mutation surface proves the fixed cells and hidden maps empty. Logs are not storage and are not part of this predicate. Before cutover the Router uses the exact `dataSessionAccountingV1()` ABI to recheck every observable scalar, guard word and configuration hash; it does not pretend to enumerate a mapping. It also repeats the source-readiness handshake defined below. Both reads must report the same generation/version/manifest, closed boundary, zero LIVE count, NOT_ENTERED guard and registered DataSession configuration immediately before any authority, queue or canonical write.

Settlement constructor provenance is authenticated by the delayed registration's exact 232-

byte `SettlementDeploymentDescriptorV1`:

```
(address20(factory), bytes32(factoryRuntimeHash),
 bytes32(factoryConfigurationHash), bytes32(salt), bytes32(initCodeHash),
 address20(targetSettlement), bytes32(targetRuntimeHash),
 bytes32(targetConfigurationHash))
```

Its hash is `H("slot-chain-settlement-deployment-v1"||u32(232)||packedDescriptor)`. Every field is nonzero and `targetSettlement=low20(keccak256(0xff||address20(factory)||salt||initCodeHash))`. Registration must publish the complete init-code artifact and immutably bind this descriptor inside `targetRegistrationHash`; arm, activation, cancellation and the activation receipt must bind that same hash. The Router checks the root factory and target `EXTCODEHASH`, then applies the exact four-byte `componentConfigHashV2()` call with 50,000 gas and exactly 32 return bytes to the target, and rechecks the derivation at arm and activation. Hidden mapping emptiness follows only from that authenticated constructor and the lack of PREACTIVE mutation entry points; it is never inferred from runtime/config getters alone. The same authenticated Settlement init code and its target configuration commitment bind the internal DataSession/reward storage layout, shared gate, `dataRent` sink and the rule that only session open/post and `fundRewardClassV1` are payable while fallback and receive reject; there is no second custody contract or cross-contract session authority. The delayed target registration, arm/cancel tuple, statement, deployment commitment, MACT context and sealed receipt all carry the same nonzero `targetRegistrationHash`; every mismatch rejects before ACTIVATING.

The source callback freezes only new work; its permanent history and all mature reward, refund, bond and Queue claims remain claim-only and do not gate cutover. The imported base remains only in the old version's history; MCAN writes the target's proof-authenticated output cell. QMIG is the only activation operation that moves Queue cursor/payment/authority, and it is followed only by the fixed poststate STATICCALLs. Failure of any step reverts freeze, adoption, payment, authority, receipt, indexes and the final switch together.

The router's stable `syncIngress` status/refund preflight remains callable by every historically registered immutable BridgeInboxAdapter. The payable `appendFromAdapter` path is narrower: it requires the caller to equal the current active authorization's exact adapter deployment and authorization ID. Its nonpayable half resolves the active version, calls that Settlement's bounded `sync()`, preserves a SYNCED transition without accepting value, and otherwise returns the exact version/generation stamp. Its payable half makes no second sync decision and forwards the exact value and descriptor only after that stamp still matches the singleton ForcedQueue's ACTIVE authority and the same active adapter/authorization binding; a predecessor adapter cannot append after its retirement watermark. Historical adapter status, cancellation and pull-refund functions remain callable without an append capability. No Settlement calls the ingress/version router during an ordinary canonical commit; the activation-only MACT read and Router's MAPS postreads are the fixed exceptions. Ordinary commits call only the queue's non-callback `advanceCursor`; activation uses QMIG. Router `canonicalAt(protocolVersion,canonicalSequence)` performs a bounded fixed-selector `STATICCALL` to the permanently registered Settlement, requires the exact return length, code hash, version/profile and cell tag, and returns no caller fields. This gives target replacement the same delayed trust root as the validity verifier without adding a canonical-path dependency.

The non-proxy, unpausable `TerminalSignalVerifier` accepts only the fixed 64-sibling proof

against one exact routed tuple and the exact domain/Bridge/credit/terminal leaf. Source finalize/recall verifies and mutates the source credit in one transaction; there is no publication stage to front-run. Sparse 4,096-block L2 jumps cannot choose a cell, and at most one canonical write per L1 block gives a live version's 256 cells at least 3,072 seconds of retention. Frozen versions retain their last 256 cells forever. The Bridge stores `terminalIndex[creditId]`. Any archive reconstructs the siblings from the canonical complete `TerminalAppended` event sequence, and L1 verifies them against the selected history cell. No archive signature or privileged indexer is accepted. Source terminal functions never call legacy SignalService. No V2 failure, recall, delivery finalization or view falls back to `msgHash`. The release profile pins every selector, event, status, liability and accumulator slot plus positive and negative lifecycle vectors.

V2 launch contains no asset codec or Vault flow. ERC20, ERC721, ERC1155 and all official Vault calls remain V1, with their existing V1 status, quota and recall semantics. A future asset protocol must define bidirectional round-trip and delivered-backing accounting, immutable asset policy and canonical/deployment identity, mapping-rotation history, destination release-versus-mint supply conservation and an unpausable exit. It requires a new kind, domain, release, profile, codec and audit and cannot reinterpret DIRECT.

The fresh L2 Bridge's sealed `v2Active` state enables processing only after the proof-authenticated tx0 atomically seals its Store and Bridge. Source V2 send becomes eligible only in the same successful activation journal after its pre-activation 214-block L1 package review; it does not wait for a later MPT proof. Every old `sendMessage` call and all L2-to-L1 sends retain V1 process/retry/fail/recall. InboxApplyRouterV2 makes only the bounded manifest-routed store calls above, and neither router nor store invokes a message recipient. Legacy proof-based `proveSignalReceived` remains unchanged for V1 and is never a V2 terminal dependency. AnchorV4 and InboxApplyRouterV2 are profile-defined system transactions: their custom type, reserved senders, system nonces, zero beneficiary tip and fee treatment are consensus fork rules, not ordinary user accounts. Their gas and the batch loop still count against the block limit. Deployment and migration require vector-tested InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2 and TerminalAccumulatorV2 bytecode. Legacy SignalService remains V1-only and, after a completed launch or outside an active genesis campaign, may be upgraded without entering a V2 trust path. From campaign publication through local resume/freeze, however, its exact checkpoint implementation and upgrade authority are covered by the genesis campaign fence above. Authorizing an unreviewed caller is invalid.

An unexpired kind-0 item must reveal raw bytes hashing to its stored `rawTxHash`; the circuit decodes them and matches every duplicated descriptor field before execution. Once `validUntil` is strictly before the L2 block timestamp, the circuit emits the unique `EXPIRED_NO_TX` disposition from the authenticated durable descriptor without needing those bytes. Consequently a disappeared sender or pruned history can delay its own item only until the bounded validity horizon, never wedge the FIFO permanently. A user that wants execution remains an available source for its own signed bytes. The 2,164-second recovery target is conditional on unexpired head bytes being retrievable; the unconditional protocol bound adds at most `MAX_FORCE_VALIDITY_SECONDS`.

For a normal signed block, `forceRoot/forceCutoff` are authenticated by its L1 anchor. For recovery they equal the current round's directly stored pair. The circuit consumes only up to that cutoff, so a later append cannot invalidate an already signed block or in-flight proof. All blocks in one candidate use the same cutoff. A due current head cannot be bypassed: `sync()` cancels an immature window, or closes a mature best only when its live boundary is not due,

before opening recovery.

The 1,500-second force delay applies to the queue head, not to an arbitrary position behind existing prepaid work. An admitted sender can compute its worst-case number of escape blocks from the gas ahead at admission. As with L1 itself, the protocol does not promise backlog-independent latency against an adversary willing to buy all available capacity; it promises that builders cannot selectively veto the FIFO head.

An escape block is deterministic under a governance-frozen `executionProfileHash`. That profile commits the L2 fork rules; complete header-field derivation (gas limit, base fee, extra data, withdrawals/requests roots and randomness); the deployed AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2 and TerminalAccumulatorV2 bytecode/constructor hashes; reserved senders, nonces and gas; AnchorV4 checkpoint calldata and `ancestorsHash` transition; and the transaction/receipt derivation above. There is no protocol signature or account nonce. The permanent unsigned typed system envelope is exactly

```
0x7f || rlp([chainId, systemKind, systemNonce, gasLimit, to, value, data])
```

with minimal RLP integers, a 20-byte `to`, zero value, no signature fields and `systemNonce=block.number-firstV2BlockNumber` independently for each kind. `firstV2BlockNumber` is the protocol-lifetime value `launchImportedCanonical.l2BlockNumber+1`; launch installs it in the cutover commitment and execution profile, every later profile must reproduce it exactly, and checked subtraction rejects a pre-cutover block. Kind 0 is Anchor at transaction index 0 and kind 1 is InboxApply at index 1; the fork injects their distinct reserved senders. Ordinary accounts cannot submit type `0x7f`. System gas counts against the block limit, but the envelope has zero tip and is exempt from account balance/base-fee charging. Its receipt is exactly `0x7f||rlp([status,cumulativeGasUsed,bloom,logs])`; both transactions contribute normally to transaction/receipt roots and cumulative gas.

The type's state transition is fully separate from EIP-1559 signing. Decoding requires `chainId=block.chainid=destinationChainId<=UINT64_MAX`; a foreign, zero or nonminimal chain ID rejects before execution. The profile maps each kind to exactly one reserved sender and target. Both `CALLER` and `ORIGIN` are that reserved sender. Sender and target start warm in addition to the fork's ordinary precompile warm set. No sender account nonce is read, compared or incremented; no sender balance is read, debited or credited; `GASPRICE` returns zero while `BASEFEE` retains the block's ordinary value. There is no access list, creation target, blob field or value transfer. Intrinsic gas is exactly `21000+4*zeroCalldataBytes+16*nonzeroCalldataBytes`; execution starts with `gasLimit-intrinsicGas`. A malformed envelope or intrinsic gas above the exact kind gas limit makes the block invalid.

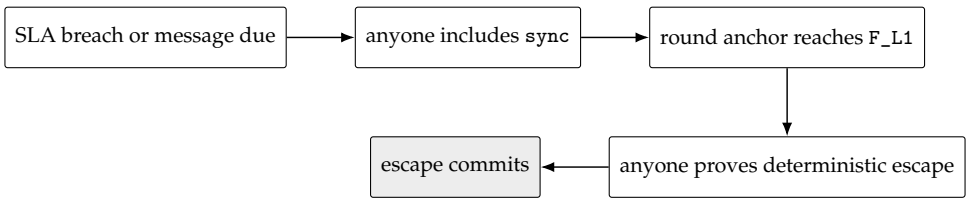
The transaction hash is legacy Keccak-256 of the complete typed envelope above, and that same byte string is the transaction-trie value. After execution let `spentBeforeRefund=gasLimit-gasRemaining` (therefore including intrinsic gas). Under the launch fork's London refund quotient 5, `refundGas=min(refundCounter,floor(spentBeforeRefund/5))` and `gasUsed=spentBeforeRefund-refundGas`; unused or refunded gas creates no account credit. REVERT or exceptional halt produces status zero and reverts state/logs under ordinary EVM rules while retaining that gas accounting; success produces status one. The typed receipt bytes above are the receipt-trie value, and `cumulativeGasUsed` is the ordinary running sum in exact transaction order. The block rejects any mismatch in envelope bytes, injected environment, warm set, state change, transaction hash/root, receipt bytes/root, gas used or cumulative gas.

The steady kind-0 call has gas limit 1,000,000, selector `0x523e6854` and ABI tuple

```
(uint48(anchorNumber), anchorHash, anchorStateRoot)
```

from the candidate's one batch anchor. The first pending-release activation kind-0 call has gas limit 12,000,000 and the activation selector/calldata defined above. Its kind-1 forced budget is 13,000,000 and that block permits no discretionary gas. Kind 1 always targets the protocol-lifetime InboxApplyRouterV2, has gas limit 5,000,000, selector `0x6b326168` for `apply(uint64,(uint64,uint8,uint32,bytes32,bytes)[])`, and uses the exact ABI rows derived in §8; the empty vector is still a canonical call. Activation uses 12m Anchor + 13m forced + 5m margin, exactly the 30m block limit. Steady blocks use 1m Anchor + 20m forced + 5m margin, at most 30m. Its timestamp is `GENESIS_TIMESTAMP+escapeSlot`, coinbase is the protocol sink and there is no discretionary body. A repeated batch anchor still performs the exact per-block ancestors transition. Changing the execution profile requires a delayed protocol-version upgrade and new full-block golden vectors; runtime guessing is forbidden.

For a cartel with no usable builder block, the recovery sequence is:



With the initial bounds, activation inclusion (120 s), the frozen `ESCAPE_OFFSET=1,900s` (covering `T_DEPTH_MAX=900s`, proving at 900 s and 100 s margin), submission inclusion (120 s), and clock margin (24 s) total 2,164 seconds, below `DELTA_FINAL_LAG=3,600s`. Tip admissibility is separate: the escape slot is at the end of the offset, so only submission and skew—144 seconds—must fit `DELTA_TIP=1,200s` when proof generation meets its bound.

9 Perpetual aggregator seat market

The aggregator market is an optional service and payment layer around the permissionless proof path. It has four installed ranks—one primary and three standbys—and a continuous reverse auction with four shared waiting cells. It has no auction epoch or fixed close window. A healthy funded primary persists until competitive handover, voluntary exit, its one objective duty, funding expiry or migration. No seat, operator action, premium balance or Market call is a precondition for proof submission, canonical commit, forced recovery, recovery renewal or migration readiness.

9.1 Authority, identity and bounded call graph

Settlement alone owns the installed roster, immutable seat terms and service intervals, one staged-handover commitment, duties, successor selection, failover, breach receipts, migration state and retained duty history. Every canonical transition is bounded, scans at most four seat/duty cells and performs no external call. If optional seat state is absent, stale, economically unmaintained or locally full, Settlement closes the affected economics to vacancy and continues canonical recovery.

The protocol-lifetime non-proxy `AggregatorSeatMarket` alone owns pending offers, native-ETH SLA bonds, premium custody, reserves and pull credits. It has no canonical writer, migration coordinator, `armMigration` or `abortMigration` entry point. It stores append-only exact Settlement authorizations installed only by the immutable `ProtocolVersionManagerV2`'s typed REGISTER_RELEASE operation. The manager derives the authorization ID, target, runtime/configuration, selectors, gas bounds, version and chain from the exact release manifest and target registration; neither the DAO proposer nor an execute caller supplies an authorization row. Duplicate or conflicting installation rejects, and no delete or redirect exists. A Market call may read Settlement only through a separately specified bounded `STATICCALL`; it checks exact chain, version, address, runtime/configuration hash, selector, return length and magic. No caller-supplied authority, phase, generation, operator, close time or validity Boolean is accepted.

The sole installation and audit views are

```
installSettlementAuthorizationV1(
 uint64 protocolVersion,address target,bytes32 runtimeHash,
 bytes32 configurationHash,bytes4 expectedMagic,
 bytes32 targetManifestHash,
 bytes32 targetRegistrationHash)
 returns (bytes4 magic,bytes32 authorizationId)
settlementAuthorizationV1(bytes32 authorizationId) returns
 (bytes4 magic,uint64 protocolVersion,address target,bytes32 runtimeHash,
 bytes32 configurationHash,bytes4 expectedMagic,
 bytes32 targetManifestHash,
  bytes32 targetRegistrationHash)
```

Their selectors are `0xb1a3fef9` and `0x1693ae01`; returns are exactly 64 and 256 bytes; install magic is `0x53414931` (SAI1) and getter magic is `0x53415431` (SAT1). Install is nonpayable, ProtocolVersionManagerV2-only and legal only inside its consumed REGISTER_RELEASE frame. It recomputes the V1 authorization ID below from the Market's immutable chain/address plus the supplied manifest-derived fields, then makes exactly one 100,000-gas STATICCALL to Router's `targetReleaseRegistrationV2(protocolVersion)`. It requires the exact 512-byte RTR2 return and compares its version, target, runtime hash, configuration hash and target-registration hash with the supplied fields before writing an unused ID once. Market does not call the target during installation: PVM and Router already performed independent target live reads, and PVM repeats them after Router returns and before invoking Market. After the Market callback PVM exact-reads Router and Market stores in the same common rollback domain. The manager exact-checks the full return; a callback cannot install a second row because its operation/frame is already consumed. The three target read selectors are fixed to `0xcf52185b`, `0x76d5ecd4` and `0x9a649489` for state, term and duty respectively, each with exactly 100,000 gas. They are protocol constants, not installer fields; every configuration hash commits the implementations of those exact ABIs.

The sole offer-book state read is the no-argument selector `seatMarketTargetStateV1()` with canonical ABI return

```
(address target, uint256 settlementChainId, uint64 protocolVersion,
 bytes32 runtimeHash, bytes32 configurationHash, bytes4 magic,
 uint8 phase, uint64 seatGeneration)
```

where `target=address(this)`, `runtimeHash=address(this).codehash`, `magic=0x53454154` (ASCII `SEAT`), and phase is exactly ACTIVE=1, ARMED=2, READY=3 or FROZEN=4. Return length is exactly 256 bytes; address, `uint64`, `bytes4` and `uint8` padding must be canonical. Market checks the target's pinned `EXTCODEHASH` before a descriptor-gas-bounded `STATICCALL` and compares every word with the append-only authorization. Revert, OOG, short/trailing data or any mismatch fails closed. `AggregatorSeatMarket.syncSeatGenerationV1()` returns exactly `(bytes4MSG1,uint8result,uint8purgedCount,uint8targetPhase,uint64generation)` as five ABI words, where result is UNCHANGED=0 or SYNCED=1 and `purgedCount<=4`; it reads the current target, terminalizes every stale waiting tranche in one rollback domain and commits the generation cache last. A new offer also performs that same exact target read and accepts only ACTIVE with the cached generation; it does so before incrementing a sequence, retaining ETH or displacing another offer. Requote, pending exit/refund and pull withdrawal perform no target read and use only already-owned records. Stage/apply authenticate the same generation only inside their enclosing atomic Settlement call. Premium, installed-release, breach and duty-reclamation paths may make only the two separately bounded historical reads below:

```
seatMarketRecordV1(bytes32 seatTermId) view returns
  (bytes4 magic, bytes32 authorizationId, uint64 seatGeneration,
   bytes32 returnedSeatTermId, bytes32 trancheId, address operator,
   uint64 responsibilityStart, uint64 premiumCap, uint64 serviceCloseAt,
   uint64 termRemovedAt, uint64 lastLiabilityAt, uint16 liveDutyCount,
   bytes32 latestDutyId, uint8 latestDutyDisposition,
   uint64 latestDutyDispositionAt, bytes32 breachReceiptId,
   uint64 breachRecordedAt)
seatDutyRecordV1(bytes32 dutyId) view returns
  (bytes4 magic, bytes32 authorizationId, uint64 seatGeneration,
   bytes32 returnedDutyId, bytes32 seatTermId, bytes32 trancheId,
   uint8 disposition, uint64 dispositionAt, uint64 lastLiabilityAt,
   bytes32 breachReceiptId, uint64 breachRecordedAt)
```

Both magic words are `0x53485231` (ASCII `SHR1`); exact return lengths are 544 and 352 bytes. Every narrow word and address has canonical padding. The append-only target authorization pins both selectors, separate gas limits, runtime/configuration, version and chain. Market checks code hash, makes one bounded STATICCALL, and compares authorization, generation and returned ID before using a word. Revert, OOG, short/trailing/noncanonical data, unknown record, a live-duty count above four or an invalid disposition rejects. Historical records are permanent after close and cannot be redirected by a successor. These are the only post-install Settlement reads made by Market; the activation-only MACT/MFRZ/MCAN/MAPS protocol is separate. No caller-supplied close cap, breach receipt, duty binding, magic or Boolean is authoritative.

9.2 Executable Settlement–Market mutation wire

The optional seat system has exactly three call families. Roster staging, installation and stage reconciliation use the single Settlement-to-Market wire below. Premium, reserve, release and breach operations are permissionless Market entry points which read permanent historical Settlement records directly. Authorization rotation is a permissionless Market operation which reads sealed Router receipts directly. Historical economics and rotation never enter a Settle-

ment mutation journal, and no caller supplies a `ServiceView`, manager object, receipt tuple, generation, clock, close cap or validity Boolean.

`installSettlementAuthorizationV1` additionally writes the append-only reverse index `authorizationIdByTarget[target]`. A nonzero target may have exactly one authorization for the protocol lifetime; a second ID for the same target reverts even when its other fields are well formed. Thus `executeSeatMutationV1()` obtains the authorization from `msg.sender` before reading caller state. The CREATE2/non-proxy release rules already give every version a distinct target, so this uniqueness rule removes ambiguity without restricting a valid upgrade.

Fixed views. Market exposes selector `0x04420e26`,

```
seatMarketWireStateV1() returns
 (bytes4 magic,uint8 journal,uint256 marketStateVersion,
  uint256 crossWireNonce,bytes32 lastReceiptHash,
  bytes32 currentAuthorizationId,uint8 generationInitialized,
  uint64 cachedGeneration,uint8 stagePresent,bytes32 stageId,
  bytes32 stageAuthorizationId,uint64 stageGeneration,bytes32 offerId,
  bytes32 trancheId,address operator,address payout,uint256 askWeiPerSecond,
  uint8 selectedRank,bytes32 outgoingTermId,bytes32 lineupCommitment,
  uint64 handoverAt,uint64 expiresAt,bytes32 reserveId,uint256 reserveWei)
```

as exactly 768 bytes / 24 words with magic `MWV1=0x4d575631`. The journal is IDLE=0 or EXECUTING=1. Address and narrow-word padding is canonical. When `stagePresent=0`, words 9–23 are zero; when it is one, stage, authorization, offer, tranche, operator, payout, lineup commitment, handover and expiry are nonzero; `selectedRank` is exactly 0–3 and may be zero; the outgoing term may be zero; and ask may be zero. Exactly `askWeiPerSecond=0` iff `reserveId=0` and `reserveWei=0`; a positive ask has a nonzero reserve ID and amount. V1 has no funding-mode, outgoing-reserve reuse or contingent-premium field. `marketStateVersion` increments exactly once for each successful state-changing top-level Market operation; `crossWireNonce` increments only for a successful state-changing roster wire operation. Both are `uint256`. Market stores only one `lastReceiptHash`; it never appends receipt history.

Settlement exposes these three exact views:

```
seatMutationIntentV1() returns
 (bytes4 magic,uint8 status,uint8 operation,uint256 intentSequence,
  bytes32 authorizationId,uint64 generation,
  uint256 expectedMarketStateVersion,uint256 expectedCrossWireNonce,
  bytes32 expectedLastReceiptHash,bytes32 stageId,
  bytes32 preLineupCommitment,bytes32 postLineupCommitment,
  bytes32 incomingTermId,bytes32 outgoingTermId,uint64 installRevision,
  bytes32 intentHash)

seatLineupWireV1() returns
 (bytes4 magic,bytes32 authorizationId,uint64 generation,
  uint64 lineupRevision,bytes32 lineupCommitment,uint8 termCount,
  uint8 primaryState,bytes32[4] termIds,uint256[4] asksWeiPerSecond,
  uint64 primaryMinimumTenureUntil,uint64 primaryServiceEligibleUntil)
```

```
seatInstallRecordV1(bytes32 termId) returns
 (bytes4 magic,bytes32 authorizationId,uint64 generation,
  bytes32 returnedTermId,bytes32 trancheId,bytes32 offerId,
  address operator,address payout,uint256 askWeiPerSecond,
  uint64 installedAt,uint64 installRevision)
```

Their selectors are `0x8ccb8cfb`, `0x39a3e69a` and `0xaacea017`; returns are exactly 512/SMI1, 544/SLV1 and 352/SIR1 bytes, with magics `0x534d4931`, `0x534c5631` and `0x53495231`. SMI status is NONE=0, PENDING=1 or EXECUTING=2 and operation is STAGE=1, APPLY=2, EXPIRE=3, INVALIDATE=4 or MIGRATION_CANCEL=5. Every operation has a published used/zero word mask. SLV1 has VACANT=0, HEALTHY=1 or NONSERVING=2; unused array cells are zero, the first `termCount` IDs are nonzero and unique, standby asks are in canonical order, and only HEALTHY may carry nonzero primary tenure/eligibility. SIR1 echoes its nonzero key and derives the same term ID from its remaining fields. All three reject short/trailing data, unknown enums and dirty padding.

The intent hash is exactly

```
H("slot-chain-seat-mutation-intent-v1" || u256(chainId) ||
  address20(settlement) || address20(market) || u8(status) || u8(operation) ||
  u256(intentSequence) || authorizationId || u64(generation) ||
  u256(expectedMarketStateVersion) || u256(expectedCrossWireNonce) ||
  expectedLastReceiptHash || stageId || preLineupCommitment ||
  postLineupCommitment || incomingTermId || outgoingTermId ||
  u64(installRevision)).
```

An ordinary wrapper derives `intentSequence=committedSequence+1` but commits the counter only after a state-changing receipt. Thus arbitrarily many NO_FEASIBLE or UNDERFUNDED calls restore the entire Settlement and Market prestate, including both sequence counters and the last receipt.

The one roster mutation. The sole Market entry is nonpayable selector `0xc97d49ee` with exactly four calldata bytes:

```
executeSeatMutationV1() returns
 (bytes4 magic,uint8 result,uint8 operation,uint256 intentSequence,
  bytes32 intentHash,uint256 preStateVersion,uint256 postStateVersion,
  uint256 preWireNonce,uint256 postWireNonce,
  bytes32 preLastReceiptHash,bytes32 receiptHash,bytes32 stageId,
  bytes32 offerId,bytes32 trancheId,address operator,address payout,
  uint256 askWeiPerSecond,uint8 selectedRank,bytes32 outgoingTermId,
  uint64 handoverAt,uint64 expiresAt,bytes32 reserveId,uint256 reserveWei,
  bytes32 creditId,uint256 amount)
```

The return is exactly 800 bytes / 25 words with `SMR1=0x534d5231`. Results are NO_FEASIBLE=0, UNDERFUNDED=1, STAGED=2, APPLIED=3, RESTORED=4 and OWNER_TERMINALIZED=5. The first two are strict Market no-ops: pre/post version and nonce are equal, the pre-last-receipt is echoed and `receiptHash=0`. NO_FEASIBLE zeros words 11–24. UNDERFUNDED zeros stage, rank, outgoing term, clocks, reserve ID and credit ID; it reports only offer, tranche,

operator, payout, ask, required reserve in word 22 and the strictly smaller gross available free premium in word 24. Market writes no counter, journal, event-derived record or receipt for either no-op. Every other result has checked `postStateVersion=preStateVersion+1`, `postWireNonce=preWireNonce+1`, and writes `lastReceiptHash` last. The receipt hash is

```
H("slot-chain-seat-mutation-receipt-v1" || u16(800) ||
  exactSMR1BytesWithWord10Zero).
```

The embedded intent hash already binds chain, Settlement, Market and every request field; the receipt preimage retains the magic and every other response word and zeroes only its own word. Every result has a fixed used/zero union mask; zero ask always has zero reserve ID/amount and cannot create or rekey a zero-valued reserve. STAGED uses the complete stage identity and clocks, has zero credit ID, and echoes its reserve amount in word 24. APPLIED uses that identity, rekeys only the incoming reserve, and requires both credit ID and word-24 amount to be zero even when an outgoing term exists. RESTORED has zero reserve/credit IDs and word 24 is the returned stage reserve; OWNER_TERMINALIZED has a zero reserve pair and a nonzero owner-credit/amount pair. No result may borrow a used word from another union arm.

Settlement first enters PREPARING, exact-reads MWV1, validates its immutable Market binding and writes EXECUTING SMI1. STAGE exposes the unchanged current lineup. APPLY first writes the complete incoming term, outgoing close/removal and post-lineup under the lock, then exposes SMI1, SLV1 and the incoming SIR1; otherwise Market could not authenticate the install. EXPIRE first clears the local stage under the same rollback frame. Market writes its EXECUTING lock, maps `msg.sender` through the unique reverse authorization, checks the pinned code/configuration and SEAT target state, then exact-reads SMI1 and the operation's required SLV1/SIR1 rows. SHR1 is reserved for later asynchronous economics and is not read by APPLY. Except for the lock, every external read and validation finishes before the first offer, tranche, reserve, credit or accounting write. APPLY validates the outgoing term's absence from the post-lineup and installs/rekeys only the incoming stage reserve. It neither closes nor reuses an outgoing reserve; later permissionless reconciliation reads permanent Settlement history. Market writes IDLE last and returns SMR1. Settlement strictly decodes it, exact-reads MWV1 again, recomputes the receipt and compares every touched stage/counter/hash field, performs its remaining local write (STAGE records the returned stage), commits its sequence and clears its journal last. Any revert, OOG, malformed return or postread mismatch bubbles through the one outer EVM transaction and rolls back both contracts; neither implementation uses a manual snapshot, catch-and-continue or compensating write.

Canonical sync, recovery and migration arm never read or call Market. If they invalidate a stage they clear the local stage and write the sole durable PENDING INVALIDATE intent; arm monotonically upgrades it to MIGRATION_CANCEL. Nonpayable selector `0xc96a3543`, `executePendingSeatMutationV1()`, is permissionless in ACTIVE, ARMED, READY and FROZEN, performs no leading sync, exact-reads MWV1, fills the three expected Market words, changes PENDING to EXECUTING and invokes the same Market entry. STAGE, APPLY and ordinary EXPIRE require the exact ACTIVE target and the current cached generation. INVALIDATE restores only while its authorization is still current/enabled and its generation equals that cache; otherwise it owner-terminalizes. MIGRATION_CANCEL instead authenticates the stage's pinned authorization and accepts its canonical target in ACTIVE, ARMED, READY or FROZEN, ignores current-generation equality, and always owner-terminalizes. Returning an aborted target to ACTIVE never resurrects or restores a canceled stage. The

durable single slot, not a Market call, is therefore the only canonical-path output; failure can stall the optional auction but cannot stall canonical chain or migration progress.

Historical economics and rotation. Permissionless payable `sponsorPremiumV1()` has selector `0x7004fb96`, exactly four calldata bytes and zero returndata. It requires nonzero `msg.value`, performs no external call or refund, and atomically adds that value to both the Market balance and the accounted `freePremium` bucket with checked `uint256` arithmetic. It advances `marketStateVersion` exactly once because the accounting projection changed, but leaves Settlement `lineupRevision`, Market `crossWireNonce` and `lastReceiptHash` unchanged. It rejects while the roster wire or credit-claim lock is active. ETH arriving without this selector, including forced ETH, remains unaccounted surplus and cannot fund an ask. Thus any account may permissionlessly fund positive reverse-auction asks without gaining rank, refund or withdrawal authority.

The direct Market selectors are

```
accrueSeatPremiumV1(bytes32)          0x8a719803
reconcileSeatReserveV1(bytes32)       0x0b29a6a9
requestSeatBondReleaseV1(bytes32)     0xae9f7df2
finalizeSeatBondReleaseV1(bytes32)    0xc1b2deb2
enforceSeatBreachV1(bytes32)          0xf3817632
```

with exactly 36 calldata bytes and a common exact 192-byte return `(bytes4MEC1,uint8result, bytes32termId,bytes32creditId,uint256amount,uint64deadline)`, where `MEC1=0x4d454331`. Market derives the tranche and stored historical authorization from `termId`, then reads the 544-byte term row and, when nonzero, its 352-byte duty row. It does not require the authorization to be current, ACTIVE or equal to the cached generation. Every read and check precedes economic writes. A well-formed but immature call may return NOOP with byte-identical state; authority/encoding mismatch reverts. The exact history predicate is selector `0xdc312dc8`, `isDutyHistorySafeV1(bytes32,bytes32,bytes32)`, with 100 calldata bytes and exact 64-byte return `(bytes4MHS1,uint8safe)`, magic `0x4d485331`. Invalid structure or authority reverts; only a valid but unmatured history returns zero.

Permissionless nonpayable `rotateSettlementAuthorizationV1()` has selector `0x817be013`, four calldata bytes and exact 256-byte return

```
(bytes4 MRO1,uint8 result,uint8 purgedCount,bytes32 oldAuthorizationId,
 bytes32 newAuthorizationId,bytes32 activationReceiptId,
 uint64 successorIndex,bytes32 blockingStageId)
```

with magic `0x4d524f31`. It reads, in that order, Router selectors `seatSuccessorReceiptV1(bytes32)=0x2540efff` (96/ASV1), `activationReceiptV1(bytes32)=0x0a4434d0` (1024/ARV1), then old/new SEAT state and the already installed SAT1 authorization. It recomputes every receipt field before any write. The successor SAT1 row and reverse-index row must already exist and be disabled from REGISTER_RELEASE; rotation neither creates nor rewrites either row. If an old stage remains it returns RECONCILIATION_REQUIRED with only old authorization and blocking stage ID nonzero and changes no state. Anyone first executes the old Settlement's pending wrapper and retries. ADVANCED purges at most four old pending offers, disables the old insertion authorization, advances exactly one hop and increments only `marketStateVersion`; it leaves cross-wire nonce and last wire receipt unchanged.

Rotation never calls Settlement mutably or accepts a manager object, supplied receipt or migration-stage Boolean.

Locks and gas certificate. While Settlement is PREPARING/EXECUTING, every mutating public entry rejects; only SEAT/SMI1/SLV1/SIR1/SHR1 getters remain callable by the immutable Market where specified. While Market is EXECUTING, every mutation rejects and only MWV1 remains callable. Market uses a separate CLAIMING effects-before-transfer lock for pull credits; no roster/economic operation performs an external value callback. A STATICCALL callback executes its complete subtree in static mode, so it cannot reenter a state-changing Market path.

Profile words 254–262 bind, respectively, `seatMarketMutationCallGas`, `seatMutationIntentReadGas`, `seatLineupWireReadGas`, `seatInstallRecordReadGas`, `seatMarketPostReadReserveGas`, `seatWirePostCallReserveGas`, `seatDutyHistorySafeReadGas`, `seatSuccessorReceiptReadGas` and `activationReceiptReadGas`; existing words 108–110 bind MWV1, term and duty reads. Before every CALL/STATICCALL, the implementation prepares memory, warms and verifies the target by EXTCODEHASH, then checks at the instruction boundary both (i) enough gas to forward the full named stipend under EIP-150's all-but-one-64th rule and (ii) the named post-call reserve after a callee burns the entire stipend. It issues zero value and zero output area, then bounded RETURNDATACOPY only after exact RETURNDATASIZE. The release certificate measures the cold four-offer/four-seat APPLY, every historical operation and rotation, proves the configured outer stipend and inner reserves under the profile's EVM gas schedule, and proves that one gas unit less fails before the external call with byte-identical state.

The non-proxy `ActiveSettlementRouter` is the protocol-lifetime root of the Settlement graph and immutably owns the shared migration gate, `ForcedQueue` and InboxApply deployment descriptor. There is no replaceable `L1HeaderOracle`: every Settlement authenticates historical L1 hashes by direct EIP-2935 system-contract reads plus strict canonical RLP-header decoding under the profile-pinned fork rules, and fresh hashes by native `BLOCKHASH`. A deployment without those primitives requires a different reviewed execution profile rather than an authority object. Every live, PREACTIVE and historical Settlement must identity-alias those exact objects. The Router neither reads nor writes a live L2 object. ProtocolVersionManagerV2 keeps the append-only target authorizations under its manifest-derived Market-only branch, while Router is the sole *L1* activation-receipt and successor-index store. The separately domained L2 Registrar receipt exists only for destination-Bridge reclamation; its counter, ID and magic are never accepted on L1, and the L1 receipt is never accepted on L2. Historical targets expose only premium accrual/claim, reserve reconciliation, breach enforcement, installed release, bond-credit claim and duty-history reclamation; they cannot accept new offers, stages, sessions or ingress.

Every successful switch appends one exact sealed receipt. Its identity is

```
receiptId = H("TAIKO_ACTIVATION_RECEIPT_V1" ||
  u256(settlementChainId) || address20(router) || u64(routerGeneration) ||
  u64(successorIndex) || u8(transitionKind) ||
  u64(sourceProtocolVersion) || u64(targetProtocolVersion) ||
  sourceManifestHash || targetManifestHash ||
  sourceAuthorizationId || targetAuthorizationId ||
  targetRegistrationHash ||
```

```
  address20(sourceSettlement) || address20(targetSettlement) ||
  oldDestinationDomainId || newDestinationDomainId ||
  address20(oldDestinationBridge) || address20(newDestinationBridge) ||
  u64(queueWatermark) || candidateDigest || outputCanonicalHash ||
  u64(outputCanonicalSequence) || activationContextHash ||
  transitionAuxiliaryHash || sourcePostStateCommitment || adoptionCommitment ||
  queuePostStateCommitment || u64(seatGeneration) || u64(activatedAtBlock))
```

The Router exposes only these fixed views:

```
activationReceiptV1(bytes32 receiptId) view returns
  (bytes4 magic, bytes32 returnedReceiptId, uint256 settlementChainId,
   address router, uint64 routerGeneration, uint64 successorIndex,
   uint8 transitionKind, uint64 sourceProtocolVersion,
   uint64 targetProtocolVersion, bytes32 sourceManifestHash,
   bytes32 targetManifestHash, bytes32 sourceAuthorizationId,
   bytes32 targetAuthorizationId, bytes32 targetRegistrationHash,
   address sourceSettlement,
   address targetSettlement, bytes32 oldDestinationDomainId,
   bytes32 newDestinationDomainId, address oldDestinationBridge,
   address newDestinationBridge, uint64 queueWatermark,
   bytes32 candidateDigest, bytes32 outputCanonicalHash,
   uint64 outputCanonicalSequence, bytes32 activationContextHash,
   bytes32 transitionAuxiliaryHash, bytes32 sourcePostStateCommitment,
   bytes32 adoptionCommitment,
   bytes32 queuePostStateCommitment, uint64 seatGeneration,
   uint64 activatedAtBlock, uint8 sealed)
seatSuccessorReceiptV1(bytes32 oldAuthorizationId) view returns
  (bytes32 receiptId, uint64 successorIndex, bytes4 magic)
bridgeSuccessorReceiptV1(bytes32 oldDestinationDomainId,
                         address oldDestinationBridge) view returns
  (bytes32 receiptId, uint64 successorIndex, bytes4 magic)
```

The receipt return is exactly 1,024 bytes with magic `0x41525631` (`ARV1`), `sealed=1` and canonical padding; each index return is exactly 96 bytes with magic `0x41535631` (`ASV1`). The Router checks and increments one global `uint64successorIndex`, derives every field from the verified activation/current registrations, and writes receipt plus both predecessor indexes in the same journal before ACTIVE. A GENESIS_IMPORT has canonical-zero predecessor authorization/domain/Bridge, must be the first global successor at index one, and writes exactly one permanent `ASV1[bytes32(0)]` bootstrap index; there is no Bridge predecessor index. Before either row is written, Router exact-reads the newly ACTIVE target and requires the registered target address, settlement chain, protocol version, runtime hash, configuration hash, `SEAT` magic, ACTIVE phase and constructor generation zero. Any mismatch reverts the target adopt, Queue migration, registration, ARV1/ASV1 writes and legacy freeze in their single outer transaction. Its `sourceManifestHash` is the authenticated `legacyDeploymentHash`, not zero, and its `transitionAuxiliaryHash` is the exact nonzero legacy-abandonment receipt hash defined below. VERSION_MIGRATION requires every predecessor field nonzero, a distinct target, absent indexes, exact queue watermark, `targetRegistrationHash`, `outputCanonicalSequence`, shared context and all three independently recomputed poststate commitments, and requires

`transitionAuxiliaryHash=0`; no key can be consumed twice. The full receipt is immutable and a zero, partial, mismatched or unsealed write cannot survive rollback.

Market rotation accepts a successor only after bounded code/config/length checks on both Router views and exact old/new authorization, versions, manifests, target registration, Settlements, generation, candidate/output, activation context, the canonical-zero transition auxiliary, all three poststate commitments and seal. An unreconciled Market stage yields the strict no-write RECONCILIATION_REQUIRED result; it is never canceled inside rotation. OldBridge reclamation does not use these L1 views; it uses the proof-bound Registrar-local destination receipt specified in the release lifecycle below. A caller-supplied receipt, index, successor address or Boolean is never authority on either layer.

9.3 Orthogonal lifecycle and conservation

Lifecycle dimensions are deliberately independent:

```
OfferLocation  = NONE | PENDING | STAGED
TrancheUsage   = OFFER | STAGED | INSTALLED | CLOSED_UNINSTALLED
BondDisposition = NONE | OWNER_CREDITED | PENALTY_CREDITED
ReserveLifecycle = ABSENT | UNSTARTED | OPEN | CLOSED_TAIL
```

Only PENDING occupies a waiting cell and only STAGED occupies the single stage. Installation sets location to NONE but preserves the immutable `SeatTerm.offerId`. The stage retains the capacity of its selected waiting cell, so at all times

```
pendingCount + stagedCount <= 4
stagedCount <= 1.
```

Staging changes the pair by $(-1, +1)$; ordinary or lineup expiry restores the same offer by $(+1, -1)$ and can never overwrite or refund a fifth entry.

Every cross-contract record uses one fixed-width legacy-Keccak identity; no ABI dynamic encoding, string enum or caller-selected ID is accepted:

```
authorizationId = H("TAIKO_SEAT_TARGET_AUTHORIZATION_V1" ||
  u256(marketChainId) || address20(market) || u256(settlementChainId) ||
  u64(protocolVersion) || address20(target) || runtimeHash ||
  configurationHash || bytes4(expectedMagic) || targetManifestHash ||
  targetRegistrationHash)
trancheId = H("TAIKO_SEAT_TRANCHE_V1" || authorizationId ||
  u64(seatGeneration) || address20(operator) || u256(bondWei) ||
  u256(creationSequence))
offerId = H("TAIKO_SEAT_OFFER_V1" || authorizationId ||
  u64(seatGeneration) || trancheId || address20(payout) ||
  u256(askWeiPerSecond) || u256(eligibleAtTimestamp) ||
  u256(eligibleAtBlock) || u256(quoteSequence))
stageId = H("TAIKO_SEAT_STAGE_V1" || authorizationId ||
  u64(seatGeneration) || lineupCommitment || offerId ||
  u256(selectedRank) || u256(handoverAt) || u256(expiresAt))
seatTermId = H("TAIKO_SEAT_TERM_V1" || authorizationId ||
  u64(seatGeneration) || offerId || trancheId ||
```

```
  u64(installedAt) || u64(installRevision))
lineupCommitment = H("TAIKO_SEAT_LINEUP_V1" || u64(lineupRevision) ||
  primaryTermId || standby0TermId || standby1TermId || standby2TermId)
dutyId = H("TAIKO_SEAT_DUTY_V1" || seatTermId || u64(dutySequence) ||
  u64(baseCanonicalSequence) || u64(baseTipSlot))
selectionId = H("TAIKO_SEAT_SELECTION_V1" || seatTermId || trancheId ||
  offerId || u64(selectedCanonicalSequence) || u64(selectedAt) ||
  u64(targetTip) || u8(selectionSource) || predecessorDutyIdOrZero)
breachReceiptId = H("TAIKO_SEAT_BREACH_V1" || dutyId || seatTermId ||
  trancheId || u64(recordedAt))
```

Absent lineup terms and predecessor duty are `bytes32(0)`; selection source is DUTY_FAILOVER=1 or HEALTHY_EXPIRY=2. In the two legacy-u256 offer-clock fields and two legacy-u256 stage-clock fields, the value is first required to fit its public uint64 wire and that exact uint64 value is then widened to u256 for hashing; no wider hidden clock is accepted. All widths are checked before hashing, all counters are monotone, and a hash collision reverts rather than overwriting an existing record. Market and Settlement recompute the same ID at every handoff; the release bundle contains cross-language golden and one-field- substitution vectors for each domain.

Each tranche stores immutable operator, bond, creation sequence, offer and term bindings, one-shot release timestamp and terminal disposition. The exact terminal credit is

```
creditId = H("TAIKO_SEAT_BOND_CREDIT_V1" || u256(marketChainId) ||
             address20(market) || trancheId || u8(OWNER|PENALTY))
```

and stores immutable beneficiary and amount plus one claimed bit. A terminal transition requires `disposition=NONE`, moves the exact bond from escrow to exactly one owner or penalty credit, and writes the disposition atomically. Claim marks the exact credit before transfer and never erases history.

Event	**Offer location**	**Tranche usage**	**Bond disposition**
Accept	NONE → PENDING	create OFFER	NONE
Displace/purge	PENDING → NONE	OFFER → CLOSED_UNINSTALLED	OWNER_CREDITED
Request pending exit	PENDING → NONE	OFFER → CLOSED_UNINSTALLED	NONE until delay
Finalize pending exit	unchanged	unchanged	OWNER_CREDITED
Stage	PENDING → STAGED	OFFER → STAGED	unchanged
Ordinary/lineup expiry	STAGED → PENDING	STAGED → OFFER	unchanged
Migration cancellation	STAGED → NONE	STAGED → CLOSED_UNINSTALLED	OWNER_CREDITED
Install	STAGED → NONE	STAGED → INSTALLED	unchanged
Healthy release	unchanged	remains INSTALLED	OWNER_CREDITED
Breach enforcement	unchanged	remains INSTALLED	PENALTY_CREDITED

A pending exit immediately leaves ranking/capacity and sets `pendingRefundAt=exitRequestedAt+`

EXIT_DELAY_SECONDS; anyone finalizes at or after equality. A staged offer cannot exit. Never-installed tranches never use installed release, and installed tranches never use pending refund. An earlier installed release request cannot defeat later valid breach enforcement.

Market accounting obeys

```
accountedMarketBalance = bondEscrow + outstandingOwnerBondCredits
  + outstandingPenaltyCredits + freePremium + reservedPremium
  + outstandingPremiumClaims
actualMarketBalance >= accountedMarketBalance
reservedPremium = sum(live PremiumReserve.reservedWei).
```

Forced ETH above the accounted amount is surplus and creates no claim.

Staging moves exactly askWeiPerSecond*SEAT_RUNWAY_SECONDS from freePremium into the exact stage reserve. Apply rekeys it to the exact seat term with no balance delta. An installed standby is UNSTARTED and earns nothing; after canonical promotion, the first noncanonical economic call derives the true start from immutable Settlement service state and may not trust a Market sentinel.

For a started service,

```
claimMaturedThrough = max(0, now - PREMIUM_CLAIM_DELAY_SECONDS)
accrueTo = max(lastAccruedAt,
  min(claimMaturedThrough, Settlement.previewPremiumCap(seatTermId),
      premiumFundedUntil))
earnedWei = askWeiPerSecond * (accrueTo - lastAccruedAt).
```

Accrual moves the checked earned value from that reserve to the immutable payout pull credit before withdrawal. The Settlement.previewPremiumCap notation means the authenticated premiumCap word returned by seatMarketRecordV1; there is no third selector. That word is the earliest applicable immutable close, satisfaction, objective failover, unfunded eligibility end, or ring-full recovery cap; omitted maintenance can never increase it.

For an atomic healthy close let $a = lastAccruedAt$, $f = premiumFundedUntil$, $c = \min(authenticatedCloseAt, f)$ and $m = \max(a, \min(c, now - PREMIUM\_CLAIM\_DELAY\_SECONDS))$, with saturating subtraction and $a \le c \le f$. It allocates

```
maturedWei  = ask * (m-a)
tailWei     = ask * (c-m)
unearnedWei = ask * (f-c).
```

The transaction credits matured value, returns only unearned value and retains the tail as CLOSED_TAIL until c+PREMIUM_CLAIM_DELAY_SECONDS. Canonical/asynchronous closes make no Market call; later permissionless reconciliation authenticates the permanent Settlement cap, credits earned value and returns only proven unearned value. Bond terminalization and duty-cell reclamation are forbidden while a reserve remains.

9.4 Continuous reverse auction and service clocks

Pending offers sort by ascending askWeiPerSecond, eligibleAtTimestamp, eligibleAtBlock, quoteSequence, then operator address. A fifth offer must strictly beat the worst under this full order or revert without retaining funds; no absolute/BPS threshold applies

to the four-cell waiting book. Displacement creates the exact owner bond credit atomically. Stale target/generation offers cannot rank or displace. Acceptance derives, with checked arithmetic, `eligibleAtTimestamp=block.timestamp+quoteMaturitySeconds` and `eligibleAtBlock=block.number+quoteMaturityBlocks` from profile words 252 and 253; neither clock nor maturity is caller-supplied. Both equalities must be reached before selection. The two-clock delay is committed by the complete Market configuration hash and cannot differ across release tooling, runtime and model.

Submission also proves at admission time that `askWeiPerSecond*SEAT_RUNWAY_SECONDS` fits `uint256`; an offer that would overflow only later at staging is never accepted. Operator is always `msg.sender`, the supplied nonzero payout is only a beneficiary, and exact `msg.value= SLA_BOND_WEI`. No user supplies target, authorization, generation, maturity clock, rank, outgoing term, service clock or refund owner.

The primary and standby comparisons deliberately use different rules. A healthy primary is competitively replaceable by every strictly cheaper ask; there is no minimum decrement because a perpetual reverse auction must continue to discover price. Only a full-lineup replacement of the worst standby uses the anti-churn threshold. For incumbent ask x define, with full-precision ceil-mul-div,

```
relative = ceil(x * minimumAskImprovementBps / 10000)
required = min(x, max(minimumAskImprovementWeiPerSecond, relative))
improves = candidateAsk < x && x-candidateAsk >= required.
```

The BPS value is at most 10,000 and the absolute/BPS pair is not both zero. The outer `min(x,..)` is consensus critical: for every positive x, a zero ask can replace it even when the absolute floor exceeds x. A zero incumbent cannot be competitively displaced and instead leaves only through its bounded standby lease or ordinary lifecycle.

Requote is pending-only and requires the immutable operator, exact live offer/tranche, no exit, exact active authorization/generation, nonzero payout and an ask within the immutable maximum. It may lower the ask, or retain the ask while changing payout; increases and no-ops revert. It keeps the tranche and bond, derives a checked fresh quote sequence and offer ID, and copies both original `eligibleAtTimestamp` and `eligibleAtBlock` exactly. Quote maturity is the one-time admission delay of the bonded tranche, not a delay which a current holder may push forward: no successor quote may postpone either dimension. Every revision voluntarily takes the fresh sequence's tie-break position. Before inserting the successor, Market deletes the superseded pending quote; the old ID is immediately stale and repeated payout changes cannot create unbounded live or historical contract storage. Requote performs no ETH movement or Settlement call. Consequently four zero-ask offers cannot freeze all four cells by changing payout before maturity: at the original timestamp/block equality every current successor is stageable by any caller.

Direct and promoted primary service share:

```
SEAT_RUNWAY_SECONDS >= MIN_PRIMARY_TENURE_SECONDS
  + HANDOVER_EXECUTION_BUFFER_SECONDS + SLA_TAIL_SECONDS
HANDOVER_EXECUTION_BUFFER_SECONDS >= HANDOVER_DELAY_SECONDS
  + STAGE_GRACE_SECONDS + T_INCLUDE_MAX_SECONDS
SLA_TAIL_SECONDS = DELTA_SLASH_LAG - DELTA_RECOVERY_LAG
currentL2Slot(t) = t - GENESIS_TIMESTAMP
responsibilityBaseTip = max(canonical.tipSlot,currentL2Slot(t))
```

```
responsibilityBaseCanonicalSequence = canonical.canonicalSequence
responsibilityBaseTime = GENESIS_TIMESTAMP + responsibilityBaseTip
targetTip = responsibilityBaseTip + DELTA_RECOVERY_LAG
recoveryAt = responsibilityBaseTime + DELTA_RECOVERY_LAG
failoverAt = responsibilityBaseTime + DELTA_FINAL_LAG
slashAt = responsibilityBaseTime + DELTA_SLASH_LAG
minimumTenureUntil = responsibilityStart + MIN_PRIMARY_TENURE_SECONDS
premiumFundedUntil = responsibilityStart + SEAT_RUNWAY_SECONDS
serviceEligibleUntil = premiumFundedUntil - SLA_TAIL_SECONDS.
```

State mutation before genesis rejects; equality is slot zero. All operations are checked-width and require `recoveryAt<failoverAt<slashAt`. The same responsibility-base formula is used for direct vacancy fill, competitive primary replacement, duty failover, healthy-expiry promotion and voluntary primary exit. Thus a newly responsible primary never inherits an already-aged recovery, failover or slash clock from a stale canonical tip; its first prospective thresholds are measured no earlier than its actual service start. Standby tenure remains separately bounded by `MIN_STANDBY_TENURE_SECONDS`. Primary tenure gates only competitive replacement and voluntary primary exit; duty completion, failover, funding expiry, ring-full vacancy and migration override it.

A qualifying canonical commit refreshes a prospective, not-yet-allocated duty only when it has a greater canonical sequence, reaches the old target tip and lands no later than the old recovery equality. The refresh recomputes the full base from the new canonical tip and current L2 slot; it never preserves an already-aged time. Once a duty is allocated, target and all three thresholds are immutable. Allocation, failover and breach require strict `now>recoveryAt`, `now>failoverAt` and `now>slashAt`; equality remains curable. A promoted standby receives fresh clocks and never inherits a predecessor duty, tenure or base. Duty and selection IDs bind canonical sequence, not L2 height.

Anyone may call `stageBest()`. Settlement first performs its bounded leading sync. If sync changes canonical/recovery state it returns SYNCED with no Market call. Otherwise Market scans at most four offers in total order and selects the first mature, current and structurally feasible transition: direct primary into vacancy, strictly cheaper primary replacement preserving all standbys, deterministic standby insertion when fewer than four terms exist, or replacement of the exact worst standby in a full lineup under the threshold above. The first structurally feasible candidate must be fully funded from the same gross `freePremium` snapshot; staging cannot anticipate, close or reuse the outgoing primary or standby reserve. An underfunded feasible candidate returns UNDERFUNDED byte-identically and does not skip to a more expensive offer. Candidate selection, reserve debit, PENDING-to-STAGED and Settlement stage recording are one revert domain.

Replacement handover is `max(active.minimumTenureUntil,now+HANDOVER_DELAY_SECONDS)`; vacancy fill or standby insertion uses `now+HANDOVER_DELAY_SECONDS`. Expiry is handover plus `STAGE_GRACE_SECONDS`. APPLY owns the closed interval `handoverAt<=now<=stageExpiresAt`; ordinary permissionless expiry is valid only for `now>stageExpiresAt`. At exact equality an expiry call rejects without mutation, so transaction ordering cannot preempt the last valid installation instant. The stage binds target, generation, selected rank, outgoing primary or zero, offer, reserve and

```
lineupCommitment = H("TAIKO_SEAT_LINEUP_V1" || u64(lineupRevision) ||
  primarySeatTermId || standbySeatTermIds[0] ||
```

```
  standbySeatTermIds[1] || standbySeatTermIds[2]).
```

Absent ranks are zero. It also requires `stageExpiresAt+T_INCLUDE_MAX_SECONDS<=serviceEligibleUntil` for a live primary. No later quote replaces a stage or moves its deadlines.

Permissionless apply repeats leading sync and requires exact unchanged stage, lineup, authorization, generation, health, maturity and headroom. Opening recovery tombstones the stage before returning SYNCED. Apply derives the term only at its actual timestamp:

```
installRevision = checked(lineupRevision + 1)
seatTermId = H("TAIKO_SEAT_TERM_V1" || authorizationId ||
  u64(seatGeneration) || offerId || trancheId ||
  u64(installedAt) || u64(installRevision)).
```

Market rekeys the reserve and binds the tranche in the same transaction in which Settlement changes the roster and revision. Settlement also writes the permanent reverse index `seatTermByTranche[trancheId]=seatTermId`; a nonzero entry rejects before any roster write. Term-ID and tranche-ID collision checks are therefore two constant-time mapping reads. APPLY never scans retained term history, regardless of how long the perpetual auction has operated.

The lineup is always compact and unique: rank 0 is the serving primary or one selected-but-unstarted successor, ranks 1–3 are unstarted standbys, there is no standby without rank 0 and standby asks are nondecreasing. Equal-ask incumbents remain ahead of a newly installed equal-ask standby. For each pending offer in total order, structural lanes are tried in this order: fill vacancy; challenge a healthy primary; insert after all standby asks not greater than the candidate when fewer than four terms exist; otherwise challenge only rank 3 and stably reinsert among the surviving standbys. If a primary challenge lacks tenure or headroom, the same offer may still use a feasible standby lane. No caller chooses a rank or outgoing term. A full-lineup standby lane exact-reads `seatInstallRecordV1(rank3TermId)` and authenticates authorization, generation, term, tranche/offer binding, operator and ask against SLV1 before deriving tenure and lease from its `installedAt`. Removing reserve reuse does not remove this SIR1 authority read; malformed, substituted or reverting SIR1 leaves both contracts byte-identical.

Every installed term stores `standbyLeaseExpiresAt=installedAt+MAXIMUM_STANDBY_LEASE_SECONDS`. The lease applies while `responsibilityStart=0`, including after selection as an unstarted rank-0 successor. At exact `now>=standbyLeaseExpiresAt`, leading sync removes every due unstarted term in one four-cell pass, records `serviceCloseAt=standbyLeaseExpiresAt`, preserves survivor order and advances lineup revision once. Any stage over the old lineup is durably invalidated without calling Market; Market later returns the untouched reserve from permanent history. If a selected successor expires, the first surviving standby is selected under the same immutable predecessor semantics, or the lineup becomes vacant. Starting primary responsibility permanently disables the former standby lease. The profile enforces

```
MAXIMUM_STANDBY_LEASE_SECONDS >= MIN_STANDBY_TENURE_SECONDS
  + HANDOVER_DELAY_SECONDS + STAGE_GRACE_SECONDS + T_INCLUDE_MAX_SECONDS.
```

A full-lineup standby stage additionally requires the outgoing rank-3 tenure at equality and `stageExpiresAt+T_INCLUDE_MAX_SECONDS<=standbyLeaseExpiresAt`; therefore leading sync cannot remove it during its valid APPLY interval.

Installed exit request is one-shot and operator-only. Maturity is derived from the term's role at finalization, not frozen at request. A serving primary uses `max(exitRequestedAt+EXIT_`

`DELAY_SECONDS,minimumTenureUntil)`; an unstarted standby uses `max(exitRequestedAt+ EXIT_DELAY_SECONDS,installedAt+MIN_STANDBY_TENURE_SECONDS)`. Therefore a standby that requests and is later promoted must also satisfy its fresh primary minimum tenure. The immutable one-shot `exitRequestedAt` can be written only from zero, and a request requires `seatTermId!=selectedSuccessorTermId`; the latter is the exact term ID in the current immutable selection record, or zero when no successor is selected. Finalization is permissionless, begins with sync and removes the term only on a fresh retry if sync changed state. Until removal the bond and every attached duty remain live. Funding expiry may close service earlier; later sponsorship cannot extend an installed term or postpone its exit.

9.5 Public seat ABI freeze

The Market's user-facing mutations are exactly

```
submitSeatOfferV1(address,uint256)
 returns (bytes4,uint8,bytes32,bytes32,bytes32,bytes32,
          uint64,uint64,uint256)                    0xae6526dd MOF1/288
requoteSeatOfferV1(bytes32,address,uint256)
 returns (bytes4,uint8,bytes32,bytes32,bytes32,
          uint64,uint64,uint256)                    0x25920dbe MRQ1/256
requestPendingSeatExitV1(bytes32)
 returns (bytes4,uint8,bytes32,bytes32,uint64)       0x37a00d91 MPR1/160
finalizePendingSeatExitV1(bytes32)
 returns (bytes4,uint8,bytes32,bytes32,uint256)      0xfed8975c MPF1/160
claimSeatBondCreditV1(bytes32)
 returns (bytes4,uint8,bytes32,address,uint256)      0xa4da9607 MBC1/160
claimSeatPremiumCreditV1(bytes32)
 returns (bytes4,uint8,bytes32,address,uint256)      0x20b34d18 MPC1/160
syncSeatGenerationV1()
 returns (bytes4,uint8,uint8,uint8,uint64)           0x66ea370e MSG1/160
```

Submit results are ACCEPTED=1 and ACCEPTED_AND_DISPLACED=2; requote and the two pending-exit transitions use result 1. Claim result 1 means claimed and is never returned twice. Submit's fields after the result are offer, tranche, displaced offer, owner-credit, eligibility timestamp/block and quote sequence; absent displacement fields are zero. Requote returns old offer, new offer, tranche, the unchanged eligibility clocks and new sequence. Pending request returns offer, tranche and refund time; finalization returns tranche, credit and amount. Claims echo credit, immutable beneficiary and amount. Sync returns UNCHANGED=0 or SYNCED=1, purged count at most four, exact target phase and cached generation. All result unions have published used/zero masks, every narrow word has canonical padding, and state is written before a claim transfer.

The fixed Market views are

```
pendingSeatOffersV1()                                       0xe7631a19
 returns (bytes4,uint8,bytes32[4])                          MPO1/192
seatOfferV1(bytes32)                                        0xc552344a
 returns (bytes4,uint8,bytes32,bytes32,bytes32,uint64,
          address,address,uint256,uint64,uint64,uint256,uint8) MOV1/416
seatTrancheV1(bytes32)                                      0x6407311c
```

```
 returns (bytes4,uint8,bytes32,bytes32,uint64,address,
          uint256,bytes32,bytes32,uint64,uint64,uint8,bytes32) MTV1/416
seatBondCreditV1(bytes32)                                      0xd25ae0f4
seatPremiumCreditV1(bytes32)                                   0x554075d2
 returns (bytes4,uint8,bytes32,address,uint256,uint8)           MCV1/192
seatMarketAccountingV1()                                      0x6abe143e
 returns (bytes4,uint256,uint256,uint256,uint256,uint256,
          uint256,uint256,uint256,uint256,uint256,uint256) MAC1/384
```

An `exists=0` keyed view echoes only its queried ID and zeros all other payload words. Live-offer status is PENDING=1 or STAGED=2. Tranche disposition is NONE=0, OWNER=1 or PENALTY=2. Accounting fields after magic are state version, wire nonce, free premium, reserved premium, bond escrow, outstanding owner credits, outstanding penalty credits, outstanding premium claims, accounted balance, actual balance and forced surplus. The last two must satisfy `actual>=accounted` and `surplus=actual-accounted`.

Settlement's public auction and exit wrappers are

```
stageBestSeatV1()
 returns (bytes4,uint8,bytes32,bytes32,bytes32,uint8,bytes32,
          uint64,uint64,uint256,uint256)             0x0c9abfb4 SSB1/352
applySeatStageV1()
 returns (bytes4,uint8,bytes32,bytes32,bytes32,uint8,
          uint64,uint64,bytes32)                     0xd7ce3714 SSA1/288
expireSeatStageV1()
 returns (bytes4,uint8,bytes32,bytes32,bytes32,uint256)
                                                      0xc323777c SSE1/192
requestInstalledSeatExitV1(bytes32)
 returns (bytes4,uint8,bytes32,uint64,uint64)         0x07e884eb SXQ1/160
finalizeInstalledSeatExitV1(bytes32)
 returns (bytes4,uint8,bytes32,bytes32,uint64,uint64,bytes32)
                                                      0xd210e9d0 SXF1/224
```

Stage results are SYNCED=0, NO_FEASIBLE=1, UNDERFUNDED=2 and STAGED=3; their payload is stage, offer, tranche, selected rank, outgoing term, handover, expiry, required reserve and available free premium. NO_FEASIBLE zeros the payload. UNDERFUNDED reports only the structurally selected offer/tranche, required and available values and changes no state, version, nonce or receipt. Apply is SYNCED=0 or APPLIED=1 and returns stage, installed term, outgoing term, rank, installation time, lineup revision and commitment. Expire is SYNCED=0, RESTORED=1 or OWNER_TERMINALIZED=2. Exit request is SYNCED=0, REQUESTED=1 or ALREADY_REQUESTED=2; finalization is SYNCED=0 or REMOVED=1. Every SYNCED return zeros every field after result, commits only the leading sync and requires a fresh retry. Exit paths are Settlement-local: they never call Market, do not require at least one remaining seat, preserve unresolved duty history, and later reserve/bond reconciliation is permissionless.

Both contracts are non-proxy, ownerless and unpausable. Their constructors pin the Router, PVM, counterpart, chain IDs, code/configuration hashes, sinks, economic profile and every gas/timing bound. They reject wrong topology, missing code, BPS above 10,000, a zero improvement pair, overflowable reserve products, inconsistent runway/standby-lease relations

or an UNCALIBRATED profile. There is no generic call, setter, pause, admin withdrawal or caller-installable authority. Forced ETH is excluded from accounting and no V1 surplus withdrawal exists.

9.6 Duty, failover, release and reclamation

Service start allocates no duty. It stores the exact responsibility-base tip above, the current canonical sequence and objective recovery/failover/slash thresholds. A qualifying ordinary commit at or before recovery refreshes that prospective base. After strict `now>recoveryAt`, Settlement's one four-cell pass allocates at most one duty; equality remains curable. If no duty cell is locally reusable, or the checked sequence is exhausted, the same sync closes optional seat economics to vacancy at objective `recoveryAt` and immediately opens SLA recovery. It creates no duty, calls no Market and never waits for `DELTA_FINAL_LAG`.

The pass applies objective failover/slash to existing duties before cure and then performs prospective/selection work. A commit after strict failover may still transition the failed-over predecessor to SATISFIED through slash equality; it never restores the removed former primary and starts a selected successor only when its exact selection record names that duty. This cure is valid regardless of whether a permissionless sync executed the failover in an earlier transaction or the committing transaction observes and applies it first. A commit after strict slash may advance canonical state but cannot cure BREACHED. Recovery candidates use one O(1) preflight followed by that same pass. Failed history adoption restores all state.

Selection records are immutable and bind unique selection ID, term/tranche/ offer, revision/time and source DUTY_FAILOVER or HEALTHY_EXPIRY. A healthy expiry allocates no fake duty. Every roster removal records immutable `termRemovedAt`. A selected-but-unstarted successor counts as seatless for the existing `DELTA_FINAL_LAG/G_MAX` recovery floor; a pointer can never suppress recovery or backdate liability. A usable later commit/recovery starts it with fresh thresholds, while an unusable revision vacates the lineup.

Installed release is permitted only for exact closed Settlement history, an INSTALLED tranche with no disposition, a refundable terminal duty and no live reserve. Anyone may create its one-shot request. Define

```
dispositionStableAt = dispositionAt + REORG_STABILITY_SECONDS (or 0)
evidenceSafeAt = lastLiabilityAt + EVIDENCE_DELAY_SECONDS
                 + REORG_STABILITY_SECONDS
finalizeReleaseAt = max(releaseRequestedAt + RELEASE_CHALLENGE_SECONDS,
                        dispositionStableAt, evidenceSafeAt)
reserveMatureAt = settlementCap + PREMIUM_CLAIM_DELAY_SECONDS (or 0)
ownerTerminalAt = max(finalizeReleaseAt, reserveMatureAt).
```

Permissionless finalization uses the exact authenticated `seatMarketRecordV1`, requires zero live duties and the matching term/ tranche/operator/times, reconciles and deletes the reserve, then credits only the immutable operator. For breach, `penaltyTerminalAt=max(recordedAt+REORG_STABILITY_SECONDS,reserveMatureAt);` permissionless enforcement authenticates the exact receipt and duty binding from both historical getters, reconciles the reserve and credits only the immutable penalty sink. Neither path can overwrite a terminal disposition.

Market's monotone `isDutyHistorySafe(dutyId,seatTermId,trancheId)` requires the exact three-way binding, terminal bond disposition, no live offer/stage/roster/reserve, all timing horizons, and an identity that can never be reused. Immutable closed term and binding history

remain queryable forever. Settlement's `reclaimDutyCell` begins with leading sync; if sync changes state it returns SYNCED with no Market call. Otherwise it authenticates the exact predicate and caches only a local reusable bit. Credit withdrawal is irrelevant, so a beneficiary cannot pin history.

9.7 Migration interaction

ProtocolVersionManager alone consumes the delayed typed arm operation, which creates the non-extendable LIVE lease. In one revert domain it requires Router lifecycle IDLE, writes ARMING before any external interaction, writes Router ACTIVE-to-ARMED, then calls the old Settlement's manager-only completion callback. The callback rereads the complete router tuple, runs leading sync internally without a generic SYNCED early return and with READY transition explicitly disabled, recomputes its inputs, increments `seatGeneration`, closes services non-fault, excuses unresolved duties and selected successors, tombstones the stage and returns exact fixed-length `SEAT_ARMED_MAGIC` binding generation, versions, manifest, target registration and resulting seat generation. Any mismatch, short/trailing return or local fault reverts the global arm. After all local and Router writes succeed, IDLE is the transaction's final store.

Settlement stores an exact `unresolvedDutyCount`, incremented only when a duty is attached and decremented exactly once when an OPEN or FAILED_OVER duty becomes SATISFIED, BREACHED, EXCUSED or EXCUSED_MIGRATION. OPEN to FAILED_OVER leaves the count unchanged. The count is bounded by the four-cell duty ring, and migration READY requires it to be zero. Retained append-only duty history is never scanned to decide readiness; an exhaustive recount is a debug/test invariant only.

The two manager-only selectors are the canonical ABI selectors for

```
completeSeatMigrationArmV1(uint64,uint64,uint64,bytes32,bytes32)
    returns (bytes response)
completeSeatMigrationAbortV1(uint64,uint64,uint64,bytes32,bytes32)
    returns (bytes response)
```

Their selectors are respectively `0x91d657cf` and `0xc69d5579`. Their five calldata words are, in order, Router generation, active version, target version, target manifest hash and target registration hash; every `uint64` word has zero high padding and trailing calldata rejects. The returned `bytes` payload is exactly 100 bytes:

```
bytes4 magic || u64(routerGeneration) || u64(activeProtocolVersion) ||
u64(targetProtocolVersion) || bytes32(targetManifestHash) ||
bytes32(targetRegistrationHash) ||
u64(seatGeneration)
```

with `magic=0x5341524d` (ASCII `SARM`) for arm and `0x53414254` (ASCII `SABT`) for abort. Full ABI returndata is exactly 192 bytes: offset `0x20`, length `0x64`, the payload and 28 zero pad bytes. PVM validates the complete encoding and every field; raw 100-byte, short, long, nonzero-padding or substituted responses revert the Router word, Settlement callback and all local cleanup together.

Abort is symmetric: the manager authenticates the expired immutable LIVE lease, writes lifecycle ABORTING while the public gate remains ARMED or READY, calls the manager-only

abort callback, and requires `SEAT_ABORTED_MAGIC` bound to the canceled tuple, target registration and retained seat generation. It then records cancellation, clears the target tuple, restores the old-version ACTIVE word and writes IDLE last. It performs no post-abort callback or same-transaction sync and accepts no fresh deadline. The next ordinary entry may sync under ACTIVE. Abort never resurrects a term, quote, duty, successor or stage. Market is not called by either canonical callback. Later permissionless reconciliation executes any durable MIGRATION_CANCEL through the frozen old Settlement. Market rotation then authenticates Router's immutable successor receipt, requires no old stage, purges at most four pending offers, disables the old authorization and advances one receipt per call. Historical economic claim/reconciliation surfaces remain available; only the exact ACTIVE tip can resynchronize generation and accept new offers.

10 Economics, slashing and payments

Event	Objective evidence	Effect
Same-context equivocation	Two distinct protocol-domain signatures for one key/slot/context	Slash that schedule window's tranche once; tombstone key; emit reporter entitlement.
Aggregator SLA breach	Exact duty passes its strict slash threshold uncured	Record BREACHED in Settlement; later permissionless enforcement reconciles the reserve and terminalizes that tranche once to the penalty credit.
Force-only activation	Due queue head	No aggregator penalty; open recovery.
Skip or absence	None that distinguishes fault from non-receipt	Not slashable; promise may be revoked.
Invalid proof/data	Verifier failure	Reject; no canonical or payment effect.

The release commits one canonical `taiko.slot-chain.economic-profile.v2` JSON artifact and its measurement commit. Its identity is byte exact. Starting from the parsed top-level JSON object, make a deep copy, require the `profileId` key and replace only its value by JSON `null`; serialize UTF-8 with recursive lexicographic key order, comma and colon separators with no whitespace, Unicode characters unescaped, and non-finite numbers forbidden. If those bytes are `J`, then

```
economicProfileHash = SHA256(
  "slot-chain-economic-profile-v2" || u32(byteLength(J)) || J)
profileId = "0x" || lowercaseHex(economicProfileHash).
```

No alternate whitespace, key order, Unicode escaping or self-referential `profileId` spelling is accepted. ExecutionProfileV2 word 266 MUST equal this hash and every duplicated economic word MUST equal the value projected from this same calibrated object.

BuilderRegistry configuration is independently derived, never supplied as an unrelated label. Let `R` be the following exact packed bytes:

```
u256(builderLease.chainId) || address20(builderLease.address) ||
builderLease.runtimeHash || u8(builderLease.decimals) ||
```

```
u256(leasePerWindowAtomic) || u256(maximumBondAtomic) ||
u256(reporterRewardCapAtomic) || u64(evidenceDelaySeconds) ||
u64(reorgMarginSeconds) || u16(maximumBuilders) ||
u16(maximumAssignedSlots) || u16(entryDelayWindows) ||
u16(maximumLiveWindows) || u16(maximumTrancheAheadWindows) ||
u16(maximumLiabilityGenerations) ||
u8(maximumGenerationMovesPerWindow) ||
address20(builderPenaltySink) || u64(rewardClaimWindowSeconds) ||
u8(rewardClassCount) ||
for each reward class in JSON array order:
  u8(classId) || Keccak256(UTF8(name)) || u256(fixedWei) ||
  u256(perExecutionGasWei) || u256(perPublishedByteWei) || u256(capWei)
builderRegistryConfigurationHash = Keccak256(
  "slot-chain-builder-registry-economic-config-v2" ||
  u32(byteLength(R)) || R).
```

ExecutionProfileV2 word 31 MUST equal that derived hash. Every integer encoded as `u8`, `u16` or `u64` here or in ExecutionProfileV2 is a production blocker when it is at least 2^8, 2^{16} or 2^{64}, respectively; hashing a wider JSON number never makes it encodable. The reward claim window is not recoverable from that hash. Production therefore requires `rewards.claimWindowSeconds==dataSession.refundClaimWindowSeconds`; Settlement receives that exact value from ExecutionProfileV2 word 106. Its reward reuse margin is the exact `reorgMarginSeconds` in word 85. Registration derives word 31 and both constructor scalars from the same calibrated JSON object, so the registry hash, target timing and executable profile cannot select different schedules. The calibrated V2 schedule contains exactly reward class IDs 1, 2 and 3, mapped respectively to authenticated settlement tiers 1, 2 and 3. Missing or additional classes, duplicate IDs, or any remapping of a tier to another class are production blockers; neither a prover nor a caller supplies a class ID.

Units are not interchangeable: native amount is `wei`, native rate is `wei/second`, and builder amount is atomic units of the one profile-bound builder lease token. That token binds chain ID, nonzero address, `EXTCODEHASH`, decimals and the manifest proof that it is the immutable, non-proxy, no-hook/no-fee/no-rebase/no-confiscation artifact required by Section 4.2; a proxy runtime hash is not a valid asset identity. Its fields are exactly `leasePerWindowAtomic`, `maximumBondAtomic` and `reporterRewardCapAtomic`; no generic untyped "bond" scalar is decoded. For one builder slash,

```
reporterRewardAtomic = min(reporterRewardCapAtomic, builderSlashAtomic)
builderPenaltyAtomic = builderSlashAtomic - reporterRewardAtomic.
```

Both operations are checked in that token's atomic units. The production profile and Registry constructor both require `5*reporterRewardCapAtomic<=leasePerWindowAtomic`. Therefore even a builder that Sybil-self-reports or front-runs its own evidence loses at least 80 percent of the slashed tranche to the immutable penalty sink, in addition to tombstoning. Comparing reporter and builder addresses would not provide this property and is not a substitute for the inequality.

There is one top-level immutable sink table. Each entry is the typed pair `(assetId,address)`. `builderPenalty` must use BUILDER_LEASE; `dataRent`, `seatPenalty`, `forcedExpiry` and `bridgeSurplus` must use NATIVE_ETH. For `bridgeSurplus`, that typing authorizes only

the primitive beneficiary address used by the fixed ForceSend initcode; no receiver interface, callback, runtime hash or mutable sink code is trusted by reclamation. All sink addresses are nonzero and a nested or component-local sink is invalid; in particular DataSession cannot substitute a rent sink. Reporter reward goes only to the evidence submitter and the remainder only to the typed builder penalty sink. Seat SLA bond, premiums and ingress deposits are native ETH and never enter builder-token accounting.

A production profile must have `status=CALIBRATED`, contain every known key and no unknown key, and make every required identity/measurement nonnull. It rejects wrong unit strings, malformed decimal strings, asset-chain mismatch, zero identity, unchecked arithmetic or any failed relation, including:

```
dataSession geometry = (86400,86400,1024,2,2100,8,6)
1024 * refundableBondWei <= UINT256_MAX
refundableBondWei + baseRentWei <= UINT256_MAX
2100 * 126972 * rentPerPublishedByteWei <= UINT256_MAX
DATA_TTL >= P_PROVE_MAX + W_SETTLE_SECONDS + REORG_MARGIN_SECONDS
refundClaimWindowSeconds >= T_INCLUDE_MAX_SECONDS + REORG_MARGIN_SECONDS
rewardClaimWindowSeconds = refundClaimWindowSeconds

PREMIUM_CLAIM_DELAY_SECONDS >= REORG_STABILITY_SECONDS

SEAT_RUNWAY_SECONDS >= MIN_PRIMARY_TENURE_SECONDS
  + HANDOVER_EXECUTION_BUFFER_SECONDS + SLA_TAIL_SECONDS

HANDOVER_EXECUTION_BUFFER_SECONDS >= HANDOVER_DELAY_SECONDS
  + STAGE_GRACE_SECONDS + T_INCLUDE_MAX_SECONDS

SLA_BOND_WEI >= MAX_ASK_WEI_PER_SECOND
                  * PREMIUM_CLAIM_DELAY_SECONDS

SLA_BOND_WEI >= MAX_ASK_WEI_PER_SECOND
                  * MIN_PRIMARY_TENURE_SECONDS
  + MAX_AVOIDED_SERVICE_COST_WEI + COLLUSION_SAFETY_MARGIN_WEI

maximumBondAtomic <= 2^192-1
reporterRewardCapAtomic <= 2^192-1
5 * reporterRewardCapAtomic <= leasePerWindowAtomic
maximumBondAtomic * (Q_MAX+1) < 2^256.
```

The first bond inequality prevents cheap closed-tail capital grief; the second bounds the direct subsidy when one actor fails a predecessor to promote its own maximum-ask standby. Neither claims to insure unbounded external application loss. Every other geometry, retention, proof-size, queue-solvency and fee relation in the parameter table is validated by the same checked profile decoder. The committed example remains UNCALIBRATED and therefore cannot be used to deploy or activate a release.

Equivocation evidence is idempotent per `(builder,window)`; separate windows slash separate tranches. `L_LEASE` is calibrated against at most 76 assigned slots but is not insurance for unbounded signed value. Tombstoning is atomic with the first valid slash; future snapshots

exclude the key.

Evidence does not depend on retaining an obsolete `admissionRoot`. It contains two distinct low-*s* EIP-712 headers with the same recovered nonzero key, slot, tier, context ID, admission version, admission root, episode, revision and recovery ID, and also equal anchor and force-root tuples. The context ID is an opaque nonzero promise namespace inside the exact Router-registered Settlement domain; Registry does not pretend to reconstruct an overwritten normal base or recovery-round record from fields absent from the evidence. Signing two different complete `SlotChainBlock` EIP-712 struct digests for one such domain/context/slot is the complete culpable act; equal `blockHash` fields do not make two otherwise different signed structs non-conflicting. It supplies one proof under the signed admission root and one under the *current* admission root for the same address and `registrationIndex`; the latter must be that still-retained active/liability generation. Its current stored tranche root separately authenticates the window leaf, which is `RESERVED(W)` or `LIABLE(W)`. The two complete struct hashes must differ. Signing two conflicting protocol headers in one such context is slashable even if the key was not ultimately selected at that schedule position; builders must not sign off-ticket headers. The evidence transaction derives W from the slot, requires the tranche leaf's stored window to equal W, and lands by `evidenceReplayDeadline(W)=evidenceDeadline(W)+REORG_MARGIN_SECONDS`. Evidence available by `evidenceDeadline(W)` therefore has the full margin for reorg detection and re-inclusion; first publication delayed into the replay tail has no replay guarantee. Address reuse is forbidden while that retained generation exists, so the current proof cannot select a newer incarnation. This rule keeps evidence verification fixed-depth while the liability-capacity invariant retains every slashable generation through the replay deadline. An accepted `RESERVED(W)->SLASHED(W)` or `LIABLE(W)->SLASHED(W)` transition decrements that generation's `unreleasedTrancheCount` exactly once. For an ACTIVE generation whose exact RESERVED window is represented in the 17-bit reservation bitmap, it also clears that one derived bit; a LIABILITY generation has no reservation bit to clear. It preserves `reservationBaseWindow` and `maximumLiableUntil`, writes the zero-amount terminal tranche leaf, and atomically updates the tranche root, the active registry root when applicable, escrow accounting and both pull credits. If and only if the current admission leaf is not yet tombstoned, it also installs the first tombstone and updates the admission root/version; later per-window slashes preserve that already- tombstoned admission pair. Evidence against an already terminal leaf rejects without another counter decrement or credit. Consequently later close-count witnesses never have to close an already-SLASHED bitmap entry and generation release cannot be stranded by a stale unreleased counter.

Canonical settlement performs no reward calculation, token balance write or external call. The authenticated tier deterministically selects reward class 1, 2 or 3. For a calibrated class with immutable `(fixedWei,perExecutionGasWei,perPublishedByteWei,capWei)`, the later claim path computes

```
rewardWei = min(capWei,
  fixedWei + perExecutionGasWei * rewardExecutionGas +
  perPublishedByteWei * rewardPublishedBytes)
```

with cap-aware checked arithmetic; no intermediate overflow can turn into a larger payment or revert canonical settlement. The byte metric counts only the distinct sealed bytes consumed by the winning canonical candidate. It is not data-rent reimbursement, does not pay losing candidates or improvement levels, and does not change the rule that data publication has no consensus-coupled reimbursement.

Every successful ordinary normal or recovery commit governed by `SettlementStatementV2` emits

```
CandidateCommittedV2(bytes32 indexed candidateId,
  address indexed beneficiary, uint8 rewardClass,
  uint256 rewardExecutionGas, uint64 rewardPublishedBytes,
  bool receiptStored, uint8 receiptIndex, bytes32 receiptCommitment)
```

where `candidateId=statementHash`, `beneficiary=proofBeneficiary`, `rewardClass=tier` and `receiptIndex=uint8(uint256(candidateId))`. The fixed three class-local `MAX_REWARD_RECEIPTS_PER_CLASS=256` rings store

```
struct RewardReceiptV1 {
  bytes32 candidateId; address beneficiary; uint8 rewardClass;
  uint256 rewardExecutionGas; uint64 rewardPublishedBytes;
  bytes32 executionProfileHash; uint64 committedAtBlock;
  uint64 committedAtTimestamp; uint64 claimUntil; bool claimed;
}
rewardReceiptCommitment = H("slot-chain-reward-receipt-v1" ||
  candidateId || address20(beneficiary) || u8(rewardClass) ||
  u256(rewardExecutionGas) || u64(rewardPublishedBytes) ||
  executionProfileHash || u64(committedAtBlock) ||
  u64(committedAtTimestamp) || u64(claimUntil))
```

The candidate ID, beneficiary, profile hash and both timestamps are nonzero; the block number and timestamp must fit `uint64`, and `claimUntil=committedAtTimestamp+rewardClaimWindowSeconds` uses the immutable word-106 constructor value and checked addition. Receipt construction is a total internal decision tree over explicitly checked candidate, metric, timestamp, deadline and cell-reuse conditions; it contains no external call, unbounded work or catch-all exception path. A named conversion or deadline failure only skips the optional receipt and sets `receiptStored=false`; it never reverts the canonical transition. The commit preflight checks the immutable timing/registry pins and only the selected class/index cell before a canonical write; view and claim probe exactly three candidate-ID cells. No production path scans a ring. Global ring integrity follows inductively from constructor-empty state and the sole internal receipt writer. A corrupt selected cell or immutable pin rejects atomically rather than becoming an optional miss, while an unrelated cell cannot veto canonical progress. When all receipt fields are well typed, `receiptCommitment` is the exact commitment below even if a live collision makes `receiptStored=false`; it is zero only when conversion or deadline arithmetic prevents construction of a well-typed receipt. The event is never itself an entitlement. Each authenticated reward class owns an independent 256-cell ring, indexed by the candidate ID's low byte. An empty cell has `candidateId=0`. A claimed cell becomes reusable only at or after `committedAtTimestamp+rewardReorgMarginSeconds`; an unclaimed cell becomes reusable only strictly after `claimUntil` and at or after `claimUntil+rewardReorgMarginSeconds`. Both timing scalars are the exact word-106/word-85 constructor values described above. Failed equality, a live same-class collision or overflow only skips storage. A class-1 receipt can never occupy or evict a class-2 or class-3 cell. `claimed` is excluded from the commitment and is the sole mutable receipt field.

The one-time genesis-import and later migration-transition proofs do not use `SettlementStatementV2`, do not emit `CandidateCommittedV2` and cannot create or overwrite a reward receipt or fund-

ing word while the target is PREACTIVE/ACTIVATING. Their prover compensation, if any, is an external release-campaign expense. This keeps the authenticated empty-target and MCAN poststate independent of an otherwise undefined migration reward tier or metering output.

There is no separately deployed distributor and therefore no uncommitted address, code hash or constructor authority. Each immutable Settlement owns three separately accounted native-ETH funding buckets and exposes exactly

```
fundRewardClassV1(uint8 rewardClass) payable
claimRewardV1(bytes32 candidateId) returns (uint256 paidWei)
rewardReceiptV1(bytes32 candidateId) returns
 (bytes4 magic, bool present, address beneficiary, uint8 rewardClass,
  uint256 rewardExecutionGas, uint64 rewardPublishedBytes,
  bytes32 executionProfileHash, uint64 committedAtBlock,
  uint64 committedAtTimestamp, uint64 claimUntil, bool claimed,
  bytes32 receiptCommitment)
```

with selectors `0x15e08308`, `0xaa5498cb` and `0x3ed526c7`. Each mutating call has exactly 36 calldata bytes and no suffix. Successful funding returns zero bytes; a successful claim returns exactly one canonical 32-byte `uint256paidWei` word. The view has zero value, exactly 36 calldata bytes and exactly 384 return bytes; `magic=0x52525631` (RRV1). It returns `present=true` only when the low-byte cell in exactly one of the three class-local rings equals the query, and then returns that exact stored fields and recomputed commitment. It probes the three candidate-ID words in fixed class order, rejects an impossible duplicate exact hit as corrupt state, and loads a complete receipt only for the sole hit. A miss returns `present=false` and canonical zero for every remaining word. Funding requires class 1, 2 or 3, positive value and checked bucket addition. Only that entry point increases an accounted bucket. It rejects PREACTIVE but remains permissionless after first activation in ACTIVE, ARMED, READY, FROZEN and historical targets, and neither receipts nor funding participate in migration readiness. Direct donations and forced ETH are surplus, are never credited or spent by the reward path, and remain sweepable only by the existing permissionless `sweepSessionSurplus` route to `dataRent`. Reward buckets have no withdrawal or sweep path. At every exit `totalRewardFunding=sum(funded[1..3])` and the complete session/reward liability sum is at most the contract's native balance.

A permissionless claim supplies only `candidateId`; Settlement probes the same low-byte cell in exactly three class-local rings, requires one exact ID, unclaimed status, matching execution-profile hash and `block.timestamp<=claimUntil`, and recomputes the receipt commitment. It exact-STATICCALLs the profile-pinned, non-proxy BuilderRegistry at the stored address and checked runtime hash using

```
rewardClassV1(uint8 rewardClass) returns
 (bytes4 magic, bytes32 builderRegistryConfigurationHash,
  uint8 returnedClassId, uint256 fixedWei,
  uint256 perExecutionGasWei, uint256 perPublishedByteWei, uint256 capWei)
```

The selector is `0x3d273ee7`; calldata is exactly 36 bytes, value is zero, the stipend is the profile's fixed 50,000-gas `componentConfigurationReadGas`, returndata is exactly 224 bytes, `magic=0x52435631` (RCV1), the configuration hash equals stored profile word 31, the echoed class equals the receipt class and all padding is canonical. Before that read, the claim path repeats the exact four-byte `componentConfigHashV2()` STATICCALL with the same stipend and requires exactly 32 return bytes equal to word 31. BuilderRegistry stores its three reward rows in

constructor-written, writerless slots. It is a protocol-lifetime identity, so all V2 execution profiles must derive the same word-31 class table; varying reward rates requires a future independently reviewed versioned registry ABI, not reinterpretation of this getter. That configuration hash commits the exact ordered class table and the same claim/reorg timing whose executable values arrive as profile words 106 and 85. Settlement stores no duplicate class table or second reward-configuration hash; the exact profile-pinned RCV1 row is the sole payout-rate authority.

After the exact read and cap-aware calculation, Settlement enters the same reentrancy guard used by session claims and surplus sweep, marks `claimed=true` and debits the exact class bucket and `totalRewardFunding` before the sole beneficiary CALL. A zero reward succeeds without a CALL. Insufficient funding, getter failure, a reverting transfer, an uncaught nested guard rejection or any mismatch reverts the whole transaction and restores the receipt, bucket, total and balance. A beneficiary may catch a nested guarded rejection and return normally; then the outer claim succeeds exactly once. An old immutable Settlement retains its receipts, registry binding and funding after Router migration, so release retirement cannot strand a still-live claim. Receipt recording and its event create no payment liability beyond available class funding; ring occupancy or an empty funding bucket can forfeit a reward but can never revert or delay canonical state. Logs alone are not entitlements. Burns are never recycled into the same episode's bounty, and no reward path calls a seat Market or data session. Successful calls emit exactly

```
RewardClassFundedV1(uint8 indexed rewardClass,address indexed funder,
  uint256 amount,uint256 classFundingAfter,uint256 totalFundingAfter)
RewardClaimedV1(bytes32 indexed candidateId,address indexed beneficiary,
  uint8 rewardClass,uint256 paidWei)
```

including `RewardClaimedV1` for a successful zero-valued claim; a reverted call emits nothing.

Seat premium and bond accounting follows the conservation and maturity rules in §9. Canonical progress never pushes ETH, waits for Market or infers a penalty from mere non-receipt. A breach requires the objective retained duty receipt. Each seat bond has one terminal disposition, while every premium withdrawal is backed by an exact reserve debit and delayed through `PREMIUM_CLAIM_DELAY_SECONDS`. The example economic profile remains UNCALIBRATED and is invalid for deployment.

11 Security and liveness analysis

Adversary/failure	Protocol result	Bound or residual
Invalid execution or forged state	Proof rejects.	Finalized-state safety reduces to proof-system and L1 safety.
Aggregator offline/byzantine	Objective duty recovery/failover proceeds without Market; any prover uses tier 2 or 3.	At most the configured recovery trigger plus 2,164 s understated inclusion/proof and head-data bounds; economic enforcement is asynchronous.
No seat or no standby	State remains valid; escape has no seat dependency.	Same protocol bound; reward may be zero.

All builders collude/absent	Unsigned escape advances anchor and forced queue.	Meaningful discretionary throughput falls to forced envelopes; no builder censorship veto.
Capital-ranked registration churn	A healthy displaced generation remains a non-tombstoned liability key for its already-sealed tickets; removal from the active table prevents new reservations/snapshots, and the newcomer becomes effective only after eight windows.	Four moves per window and strictly increasing victim bond bound transition rate/capital, but a funded adversary may still capture the normal top-64 market; the unsigned escape lane is the independent liveness floor.
Builder equivocation	Competing signed branches; one may win normal order.	Direct per-window slash; preclose promises remain revocable.
Builder skips/withholds body	Honest tail may be unprovable or omitted.	Not objectively slashable; eventual escape restores state progress but may revoke promises.
Data-session spam	Exact rent/bond buys at most 1,024 bounded LIVE/REFUND cells.	A funded Sybil may censor the optional session market through TTL plus refund grace; escape uses no blob session.
Missing/pruned kind-0 bytes	Unexpired head waits; after validity it is consumed as EXPIRED_NO_TX from its durable descriptor.	Adds at most seven days; a kind-1 descriptor is durable, while successful destination invocation still needs Message bytes.
V1 or V2 endpoint upgrade exploit	V1 remains untouched; every V2 source/destination Bridge is a fresh immutable non-proxy account.	A new endpoint/domain cannot redirect an old writer; every old endpoint retains its terminal/refund path.
Unauthorized source-support fill	Version manager rejects entries absent from the active delayed manifest.	At most 64 audited additions per release; no attacker-consumable lifetime cap.
Message bytes disappear	Enqueue expires to source cancellation; a pinned credit expires to destination FAILED.	The full DIRECT ETH value, execution fee and liquidity fee refund remains permissionless by credit ID; successful destination invocation is lost.
Source escrow drain	The fresh SourceBridge checks every payout against pending, user-pull and LP-pull aggregate liabilities.	DONE reclassifies the full escrow to the proof-bound LP recipient; cancel/FAILED reclassifies it to the user, and liability falls only on a successful pull.

No willing destination LP	The credit remains UNFUNDED without consuming Pool capital or blocking unrelated credits, then may expire to FAILED.	Delivery liveness is economic: permissionless tickets remove deterministic finite-reserve exhaustion but cannot compel capital at an unattractive `liquidityFee`.
Target or post-call tail fails	The only-self attempt frame reverts Pool consume, transfer, target and Bridge effects; target-failure policy runs only after the authenticated catch.	The caller's ticket remains available; owner-last failure appends FAILED only after that rollback.
Guardian pause or zero quota	Risky sends/process/retries stop; expiry, terminal verification and reserved refunds continue.	Frozen V2 proof/refund paths have no transitive pause or ordinary quota dependency.
Foreign destination replay	Local domain and `destBridge==address(this)` checks reject before status or value effects.	The immutable pin store and accumulator both bind one exact domain/Bridge pair.
Terminal proof-format change	Canonical state carries a stable application-level depth-64 vector root/count.	Source verification does not parse the destination EVM state format; later roots preserve old leaves.
Retired destination reclaim	Reclaim waits for successor receipt, Queue watermark, pin/terminal equality and zero pull liability.	Atomic Pool funding leaves no domain-scoped capital; only unaccounted Bridge surplus reaches the pinned sink, and tickets are never enumerated or swept.
Legacy call to installed V2 code	The fresh release-owned Bridge and Store remain inactive and the custom system transaction type is invalid.	Proved tx0 atomically seals Store and Bridge once; legacy V1 addresses are never redirected.
All canonical prestate copies lost	No party can construct the next EVM witness from a root alone; proof generation stops without changing canonical state.	Explicit information-theoretic availability limit; restore any root-verifiable archive copy.
Transparent-mempool front-run	First valid recovery transaction wins.	Proof statement binds beneficiary, episode, revision and outputs; copying cannot redirect payment or state.
Payment-vault exhaustion	Claim is zero or partial.	Never blocks safety or canonical liveness; economic liveness is conditional.
L1 reorg within policy	Contract state, burns and claims roll back together.	Clients replay canonical logs and proofs against new version/hash.

L1 censorship/outage	Neither activation nor proof can land.	Outside model after `T_INCLUDE_MAX_SECONDS`; no L2 protocol can force its L1 contract to execute.
Genesis campaign reaches QUIESCENT but no valid target proof lands	Canonical state and Queue authority do not move; before either hard expiry a valid proof may retry, and at expiry anyone or the next ordinary call restores legacy ACTIVE.	Exact scan and resume-profile checks ensure the bounded pause does not expire recoverable data or change a validity/custody deadline. A corrected target needs a higher-nonce delayed campaign.

The escape lane establishes strong transaction-entry permissionlessness even when the capital-ranked builder set is captured. It does not make the normal builder market egalitarian: top-64 ranking and per-window capital remain a meaningful entry barrier. The product must distinguish these two claims.

12 Implementation blueprint

12.1 L1 modules and bounded storage

- `BuilderRegistry`: protocol-lifetime non-proxy, with 64 active cells, 1,072 liability generations, 512-leaf tranche roots, authoritative generation/address reverse locators, 17-bit per-generation traversal bitmaps, exact unreleased-tranche counters, stored aggregate pull-credit liability, tombstones, replacement counters and the versioned `admissionRoot`. Every production transition is fixed-depth or constant-size; exhaustive recounts exist only as model/off-chain audits.
- `ScheduleOracle`: protocol-lifetime non-proxy sealed window root/seed under the version-independent `slot-chain-seed-v2` and frozen allocation/ placement rules, plus the one-shot delayed fork-verifier registry, in a ring covering `MAX_LIVE_WINDOWS=268`. `MAX_EARLY_SEAL_WINDOWS=8` covers the earliest snapshot geometry. In window units the capacity is at least:

  ```
  MAX_EARLY_SEAL_WINDOWS + MAX_CANDIDATE_WINDOWS +
  ceil((W_SETTLE_SECONDS + DELTA_FINAL_LAG + DELTA_TIP +
        REORG_MARGIN_SECONDS + EVIDENCE_DELAY_SECONDS) / 384) + 2
  ```

  An entry is overwritten only after it is `EXPIRED`; the same transaction checks the release predicate and no stored normal best/round reference.
- Settlement's internal DataSession region: exclusive session ETH custody inside the immutable Settlement and a 1,024-cell FREE/LIVE/REFUND ring, at most two LIVE cells per owner, one global checked session sequence, exact live/refund/occupied counters and a live-owner-only mapping. Each LIVE cell has a derived owner/sequence ID, `bytes32peaks[12]`, checked count at most 2,100, sealed root/count, expiry and bond; REFUND reuses that cell for one bounded owner/bond/deadline claim. There is no on-chain leaf array, permanent per-owner nonce or claim mapping. Only nonpayable ordinary/migration/refund maintenance advances the cursor by exactly eight cells; payable open selects one caller-hinted FREE cell and never runs maintenance. Expiry and migration perform no callback, and surplus-sweep failure cannot gate any protocol transition.

- `ForcedQueue`: protocol-lifetime immutable non-proxy depth-64 append-only Merkle root/count/frontier, last due time, complete fixed-width per-index descriptors, deposit prefix, unconsumed escrow and pull claims. An expired kind-0 descriptor reconstructs its leaf and prefix accounting without historical calldata, while kind-1 credits remain durable. Only the router appends and only the active Settlement advances the cursor/payment interval.
- `BridgeDomainRegistry`: non-proxy append-only source, destination and frozen-endpoint-support mappings. Only bounded entries from the manifest being atomically activated by ProtocolVersionManager may be added, and destination support additionally requires a finalized canonical L2 proof of the exact registrar record; there is no delete, redirect or age-proportional iteration.
- `BridgeInboxAdapter`: exactly-once NONE/QUEUED records and caller-funded queue processing fees. It direct-reads same-L1 CreditRecord and Bridge liability state, rejects while PREACTIVE and, after router SYNCED, commits that transition, creates no record and pull-credits the still-local full caller fee before a nonreverting return. A successful append forwards the whole fee to ForcedQueue and atomically marks the permanent source record QUEUED. It is non-proxy and immutable; its runtime and constructor-fixed registry, Bridge, router, Queue and version manager plus its one-shot sealed destination-domain slot are profile/domain-bound, with no mutable target, delegatecall or arbitrary-call selector.
- `InboxApplyRouterV2` and `InboxCreditStoreV2`: the router is a protocol-lifetime non-proxy dispatcher with one global cursor and append-only registrar-authenticated domain routes; every endpoint store fixes that same router and its own Bridge/domain. Full-vector validation, at most 64 contiguous-run calls and EVM rollback make mixed-domain application atomic. Neither has an upgrade, resolver, callback or caller-selected target.
- `ProtocolReleaseAuthorityV2`: protocol-lifetime non-proxy release manifest authenticator for the manifest-derived Anchor caller, permanent reserved system-transaction origin and namespace. Its version records are one-shot and have no endpoint target.
- `TerminalDomainRegistrarV2`: non-proxy immutable version-activation coordinator bound only to the release authority and accumulator. It derives endpoint targets from the authenticated manifest and atomically seals the fresh Store, Bridge `v2Active` bit, inbox route and accumulator writer. Records are permanent, one-shot and one-to-one.
- `TerminalAccumulatorV2`: protocol-lifetime non-proxy depth-64 append-only vector with checked count/wrapped root, a 64-word frontier and the complete canonical leaf event. It has one-to-one domain/Bridge writer registration and no root setter. Its sentinel/static-read handshake derives the terminal from the Bridge; each transition stores the returned index atomically.
- Fresh immutable non-proxy SourceBridgeV2 and DestinationBridgeV2 bundles retain V2 custody, aggregate/per-credit liabilities, status, pull credits and ContextV2 identity. V1 accounts and Vaults remain untouched. Every successor uses fresh domain/Bridge/Registry/Quota accounts; none has an upgrade, delegatecall, owner or reinitializer.
- `Settlement`: permanent non-proxy state machine with `PREACTIVE`, `ACTIVE`, `MIGRATION_ARMED`, `MIGRATION_READY` and `FROZEN` states; canonical core, one normal best, recovery episode/round, seat pointer, 256-cell sequence-tagged full-core history and three class-isolated 256-cell best-effort receipt rings. A delayed cutover stops opening new work and

reaches MIGRATION_READY after the current transition settles, so an adversary cannot veto migration by continuously creating a best or recovery round. Rewards and burn disposal remain separate; no loop is proportional to chain age.

- `ActiveSettlementRouter`: non-proxy, no generic target setter. Its append-only version registry permanently binds Settlement address/code/profile; proof-carrying activation binds the exact old core, same ForcedQueue/registry/schedule objects and migration-ready state, then atomically installs the proved target poststate. Delayed exact-target abort restores the unchanged old authority before cutover. Its stable stamped two-call sync/append facade preserves old adapters, while `canonicalAt(version,sequence)` only reads exact historical cells.
- Terminal verifier: immutable, read-only and unpausable; it checks one fixed 64-sibling vector proof against an exact router-routed version and sequence cell. The leaf separately binds index, domain, Bridge, credit and terminal status.

The non-authoritative `CanonicalArchive` is independently mirrored off-chain canonical bodies, receipts, state-trie nodes and runtime bytecode indexed by canonical block, state root and code hash. Every object is commitment-verifiable; no archive address, signature or availability vote appears in Settlement or the verifier statement.

All setters enforce parameter inequalities atomically and are disabled after the launch freeze except through the existing delayed governance path. Every counter has an explicit width and checked increment; no sentinel shares a legal value.

12.2 Historical authentication

Native `blockhash` plus a supplied matching RLP header authenticates a fresh normal arm block; EIP-2935 authenticates schedule carrier headers. Equivocation evidence needs no historical context read because its culpability predicate is two builder signatures under one registered domain/context/slot. A recovery round captures its immediately preceding block hash with `blockhash(block.number-1)` and later hashes a supplied RLP header to that stored value; it never tries to read an unavailable parent timestamp, queries an expired history entry or mutates the meaning of a past header. Its queue pair is round storage, not an MPT claim. EIP-4788 plus SSZ branches authenticates the schedule's parent beacon/execution payload; its state root plus MPT proofs authenticates the registry. Deployments lacking either primitive are unsupported by this profile and require a new exact profile and separate review.

The EIP-2935 read is byte exact and has no Solidity selector. The execution profile pins `HISTORY_STORAGE_ADDRESS=0x0000F90827F1C53a10cb7A02335B175320002935`, its fork-specific runtime hash, activation block, `HISTORY_SERVE_WINDOW=8191` and `HISTORY_READ_GAS=50000`. Settlement first checks that code hash, then STATICCALLs with zero value and exactly `u256(requestedBlockNumber)` as 32-byte big-endian calldata. The call must return exactly one nonzero 32-byte hash and the requested number must be in `[block.number-8191, block.number-1]`; revert, OOG, short/trailing/zero return, a pre-activation empty cell or a supplied RLP header whose Keccak/block number differs rejects. No wrapper address, cached Boolean or caller-supplied history root is accepted. This is the EIP-2935 `get` wire format; the SYSTEM_ADDRESS-only `set` path is never called by the protocol.

12.3 Executable execution profile

`executionProfileHash` is not a free-form governance label. Each `protocolVersion` maps im-

mutably to exactly one strict canonical Solidity ABI value

```
profileBytes = abi.encode(ExecutionProfileV2)
executionProfileHash = H("slot-chain-execution-profile-v2" ||
                         u32(byteLength) || profileBytes)
```

where `1<=byteLength<=146848`. V2 is protocol-lifetime frozen: the immutable V2 manager accepts only this schema. An incompatible extension is outside every authority path specified here and requires an independently reviewed new L1 protocol root/genesis migration; this manager cannot install or reinterpret it. Stored V2 hashes are never reinterpreted. There is no CBOR or legacy fallback.

Exact ABI grammar. `ExecutionProfileV2` is one tuple of 281 fixed value fields followed by the sole dynamic `bytesdeploymentCodeArtifacts`. Its outer root word is exactly `u256(0x20)`. Value field i below is the ABI word at tuple offset $32i$; word 281 is the sole offset and equals `u256(9024)=u256(0x2340)`. The total fixed head is therefore 282 words. The tail is the ordinary ABI bytes encoding `u256(artifactLength)||artifact||zeroPadding`, with no gap, alias, trailing word or nonzero padding. The complete field order is:

```
  0.. 15 core:
    uint64 schemaVersion, uint64 protocolVersion,
    uint256 settlementChainId, uint256 l2ChainId,
    bytes32 settlementGenesisHash, bytes32 destinationGenesisHash,
    bytes4 forkDigest, uint64 firstV2BlockNumber, uint64 genesisTimestamp,
    bytes32 manifestNamespace, bytes32 destinationNamespace,
    bytes32 routerNamespace, bytes32 poolNamespace,
    address anchorV4, bytes32 anchorRuntimeHash,
    bytes32 anchorConfigurationHash
 16.. 43 control objects, in this order:
    (address protocolChangeTimelock, bytes32 runtimeHash, bytes32 configHash),
    address daoProposer,
    (address protocolVersionManager, bytes32 runtimeHash, bytes32 configHash),
    (address activeSettlementRouter, bytes32 runtimeHash, bytes32 configHash),
    (address forcedQueue, bytes32 runtimeHash, bytes32 configHash),
    (address builderRegistry, bytes32 runtimeHash, bytes32 configHash),
    (address scheduleOracle, bytes32 runtimeHash, bytes32 configHash),
    (address aggregatorSeatMarket, bytes32 runtimeHash, bytes32 configHash),
    (address bridgeDomainRegistry, bytes32 runtimeHash, bytes32 configHash),
    (address sourceBundleFactory, bytes32 runtimeHash, bytes32 configHash)
 44.. 56 target artifact:
    address settlementFactory, bytes32 settlementFactoryRuntimeHash,
    bytes32 settlementFactoryConfigurationHash, bytes32 settlementSalt,
    bytes32 targetRuntimeHash, bytes32 targetCreationCodeHash,
    bytes32 targetAbiHash, bytes32 targetStorageLayoutHash,
    bytes32 targetConstructorSchemaHash, bytes32 targetCompilerBuildHash,
    bytes32 targetCompileTimeRulesHash,
    bytes32 targetImmutableReferencesHash, bytes32 targetLinkReferencesHash
 57.. 62 target verification infrastructure:
```

```
    (address settlementL1HistoryStorageAddress, bytes32 runtimeHash,
     bytes32 readConfigurationHash),
    (address registrationMptVerifier, bytes32 runtimeHash, bytes32 configHash)
 63.. 71 sinks:
    address builderLeaseToken, bytes32 builderLeaseTokenRuntimeHash,
    uint8 builderLeaseTokenDecimals, address builderPenaltySink,
    address dataRentSink, address seatPenaltySink, address forcedExpirySink,
    address bridgeSurplusSink, address protocolCoinbaseSink
 72.. 85 recovery uint64 values:
    settlementWindowSeconds, includeMaxSeconds, finalLagSeconds,
    tipLagSeconds, proveMaxSeconds, l1FinalityBlocks, depthMaxSeconds,
    clockSkewSeconds, escapeOffsetSeconds, forceDelaySeconds,
    maximumParentGapSlots, maximumForceValiditySeconds,
    evidenceDelaySeconds, reorgMarginSeconds
 86..100 seat values:
    uint64 seatRunwaySeconds, minimumPrimaryTenureSeconds,
    minimumStandbyTenureSeconds, handoverDelaySeconds, stageGraceSeconds,
    exitDelaySeconds, recoveryLagSeconds, slashLagSeconds,
    premiumClaimDelaySeconds, reorgStabilitySeconds, releaseChallengeSeconds;
    uint256 maximumAskWeiPerSecond, seatSlaBondWei,
    maximumAvoidedServiceCostWei, collusionSafetyMarginWei
101..106 DataSession values:
    uint256 dataSessionBondWei, uint256 dataSessionBaseRentWei,
    uint256 dataSessionRentPerPublishedByteWei,
    uint16 dataSessionBlobBaseFeeMultiplierBps,
    uint64 dataSessionMaximumTtlSeconds,
    uint64 dataSessionRefundClaimWindowSeconds
107..117 target gas uint64 values:
    activeSettlementStateReadGas, seatMarketStateReadGas,
    seatTermRecordReadGas, seatDutyRecordReadGas,
    componentConfigurationReadGas, registrationVerifierConfigReadGas,
    registrationVerifierCallGas, migrationActivationContextReadGas,
    migrationPostStateReadGas, targetAdoptionCallGas,
    targetPostCallbackReserveGas
118..137 compile-time rules:
    uint8 slotSeconds; uint16 scheduleWindowSlots; uint8 seatCount;
    uint8 dutyRingCapacity; uint8 forceTreeDepth; uint8 dataMmrDepth;
    uint16 dataSessionCellCount; uint16 maximumDataSessionsPerOwner;
    uint16 maximumDataRecordsPerSession; uint8 maximumGcSteps;
    uint8 maximumBlobsPerPost; uint16 canonicalHistoryCapacity;
    uint8 destinationComponentCount; uint8 ingressAuthorizationCount;
    address pointEvaluationPrecompile; uint32 pointEvaluationGas;
    uint256 blsModulus; uint32 blobGasUsed;
    uint32 maximumBlobPayloadBytes; uint16 maximumBlobChunkCount
138..157 MPR2 source:
    address migrationVerifier; bytes32 migrationVerifierRuntimeHash;
    bytes32 migrationVerifierConfigurationHash;
    bytes32 migrationVerifyingKeyHash; bytes32 migrationProofSystemId;
```

```
    bytes32 migrationPublicInputSchemaHash; bytes4 migrationVerifierSelector;
    uint32 migrationMaximumProofBytes; then uint64
    migrationVerificationGas, supportedL1BlockGasLimit,
    worstCaseActivationAdoptionGas, sourceFreezeGas, targetAdoptionGas,
    queueMigrationGas, activationContextReadGas, postStateReadGas,
    legacyStateReadGas, legacyArmGas, legacyFinalizeGas,
    postCallbackReserveGas
158..201 destination:
    address inboxSystemSender, address anchorSystemSender;
    for i=1..10, expanded in increasing i:
      (address destinationComponent_i, bytes32 runtimeHash, bytes32 configHash);
    bytes32 destinationBridgeStorageLayoutHash,
    bytes32 bridgeKernelProfileHash, bytes32 bridgeKernelAbiHash,
    bytes32 bridgeStatusLayoutHash, bytes32 bridgeCustodyLayoutHash,
    address destinationQuotaManager, bytes32 destinationQuotaManagerRuntimeHash,
    bytes32 destinationQuotaManagerConfigurationHash,
    uint256 destinationInitialNativeQuota, uint64 poolReadGas,
    uint64 poolAuthorizationCleanupGas, uint64 poolValueCallbackGas
202..227 source:
    address releaseSourceBundleFactory,
    bytes32 releaseSourceBundleFactoryRuntimeHash,
    bytes32 releaseSourceBundleFactoryConfigurationHash,
    bytes32 sourceBundleSalt,
    bytes32 sourceBundleInitCodeHash, address sourceBundleDeployer,
    bytes32 sourceBundleDeployerRuntimeHash, address legacyV1SourceBridge,
    address sourceBridge, bytes32 sourceBridgeRuntimeHash,
    bytes32 sourceBridgeConfigurationHash, bytes32 sourceBridgeStorageLayoutHash,
    address sourceCreditRegistry, bytes32 sourceCreditRegistryRuntimeHash,
    bytes32 sourceCreditRegistryConfigurationHash, address sourceQuotaManager,
    bytes32 sourceQuotaManagerRuntimeHash,
    bytes32 sourceQuotaManagerConfigurationHash, address sourceSupportRegistry,
    bytes32 sourceSupportRegistryRuntimeHash,
    bytes32 sourceSupportRegistryConfigurationHash, address sourcePauser,
    address sourceSignalService, uint64 sourceRegistrationEpoch,
    bytes32 sourceNamespace, bytes32 sourceGenesisHash
228..234 kind-0 ingress:
    address kind0IngressAdapter, bytes32 kind0IngressAdapterRuntimeHash,
    uint256 fixedIngressWei, uint256 executionWeiPerAccountedGas,
    uint256 proofWeiPerAccountedGas, uint256 permanentWeiPerByte,
    uint256 maximumAcceptedFeeWei
235..251 execution:
    uint64 l2BlockGasLimit, uint64 baseFeeElasticity,
    uint64 baseFeeChangeDenominator, uint64 blobTarget,
    uint64 blobUpdateFraction, bytes32 emptyOmmersHash,
    bytes32 emptyTransactionsRoot, bytes32 emptyWithdrawalsRoot,
    bytes32 emptyRequestsHash, uint64 l1HistoryFirstSupportedBlock,
    bytes32 l2HistoryStorageRuntimeHash,
    uint64 l2HistoryStorageActivationBlock, bytes32 headerRulesHash,
```

```
    bytes32 systemTransactionRulesHash, bytes32 forcedInputRulesHash,
    bytes32 stateTransitionAbiHash, bytes32 legacyExecutionRulesHash
252..266 seat wire and calibrated economics:
    uint64 quoteMaturitySeconds, uint64 quoteMaturityBlocks,
    uint64 seatMarketMutationCallGas, uint64 seatMutationIntentReadGas,
    uint64 seatLineupWireReadGas, uint64 seatInstallRecordReadGas,
    uint64 seatMarketPostReadReserveGas,
    uint64 seatWirePostCallReserveGas,
    uint64 seatDutyHistorySafeReadGas,
    uint64 seatSuccessorReceiptReadGas, uint64 activationReceiptReadGas,
    uint64 maximumStandbyLeaseSeconds,
    uint256 minimumAskImprovementWeiPerSecond,
    uint16 minimumAskImprovementBps, bytes32 economicProfileHash
267..270 kind-0 deployment provenance:
    bytes32 kind0IngressAdapterCreationCodeHash,
    bytes32 kind0IngressAdapterSalt,
    bytes32 kind0IngressAdapterInitCodeHash,
    bytes32 kind0IngressAdapterConfigurationHash
271..280 Settlement validity verifier:
    address settlementValidityVerifier,
    bytes32 settlementValidityVerifierRuntimeHash,
    bytes32 settlementValidityVerifierConfigurationHash,
    bytes32 settlementValidityVerifyingKeyHash,
    bytes32 settlementValidityProofSystemId,
    bytes32 settlementValidityPublicInputSchemaHash,
    bytes4 settlementValidityVerifierSelector,
    uint32 settlementValidityMaximumProofBytes,
    uint64 settlementValidityVerificationGas,
    uint64 settlementValidityPostVerificationReserveGas
281 bytes deploymentCodeArtifacts offset (=9024)
```

Words 202–204 are the release Source-bundle projection and MUST equal control words 41–43 byte-for-byte. They are named separately only to keep the source-descriptor range contiguous; they do not create a second factory or authority. The canonical decoder, PVM and Router all reject a mismatch before their first registration write. Execution word 247 `headerRulesHash` is fixed to `keccak256("slot-chain-force-send-eip-6780-osaka-v1")`. The profile validator and destination Bridge configuration both require that commitment; an execution environment without same-creation-transaction SELFDESTRUCT value transfer cannot activate this Bridge/reclamation design. Every unused high byte of `uint8/16/32/64`, every unused high byte of an address, and every unused low byte of `bytes4` is zero. Every field is nonzero except the three explicitly zero-permitted economic words 102–104. `schemaVersion=2`; the fixed geometry is `(1,384,4,4,64,12,1024,2,2100,8,6,256,10,2)`, followed by the canonical point-evaluation address `0x0a`, gas 50000, BLS modulus, blob gas 131072, maximum payload 126972 and chunk cap 9. Words 114/115/116/117 must equal words 152/153/150/157 respectively; disagreement is rejected, so there is one gas authority. Word 54 is derived, not a publisher-selected label:

```
compileTimeRulesBytes = concat(profileWord[118],...,profileWord[137])
targetCompileTimeRulesHash = H(
```

```
  "slot-chain-target-compile-time-rules-v2" || u16(640) ||
  compileTimeRulesBytes)
```

The concatenation is exactly the 20 canonical ABI words in the displayed order. A bounded word-table/assembly decoder is permitted and preferred; compiler-generated decoding of a 281-field tuple is not an implementation assumption. Fixed tuple arity means unknown, missing, duplicated or reordered fields cannot be represented canonically.

The sole dynamic artifact is a canonical two-artifact bundle, not a generic list:

```
u32(settlementCreationLength) || settlementCreationCode ||
u32(settlementRuntimeLength)  || settlementRuntimeCode  ||
u32(kind0CreationLength)      || kind0CreationCode      ||
u32(kind0RuntimeLength)       || kind0RuntimeCode
```

All four lengths are nonzero; each runtime is at most 24,576 bytes, the Settlement creation code is at most 40,032 bytes, and the kind-0 creation code is at most 48,544 bytes. The raw bundle is at most 137,744 bytes. With the 32-byte outer root, 9,024-byte tuple head, 32-byte dynamic length and canonical padding, the complete profile is at most exactly 146,848 bytes. There is no artifact count, offset, inner padding, alias, suffix or omitted entry. Words 49/48 and 267/229 equal the respective creation/runtime Keccak hashes. The build-conformance report derives both artifact pairs from their named single compiler outputs and rejects mixed invocations or metadata.

Kind-0 deterministic deployment. The kind-0 adapter is a fresh, non-proxy, release-scoped ERC-2470 deployment, not a SourceBundleFactory child and not a nineteenth root artifact. Let `F=0xce0042B868300000d44A59004Da54A005ffdcf9f`. Its exact derived values are

```
kind0Salt = H("slot-chain-kind0-ingress-salt-v1" ||
  u256(settlementChainId) || u256(l2ChainId) || manifestNamespace ||
  u64(protocolVersion) || address20(activeSettlementRouter) ||
  address20(forcedQueue))
kind0ConfigBytes =
  u256(settlementChainId) || u64(protocolVersion) ||
  address20(activeSettlementRouter) || routerRuntimeHash || routerConfigHash ||
  address20(forcedQueue) || queueRuntimeHash || queueConfigHash ||
  address20(protocolVersionManager) || u256(l2ChainId) ||
  u64(maximumForceValiditySeconds) || u64(5000000) || u32(131072) ||
  u64(21000) || u256(fixedIngressWei) ||
  u256(executionWeiPerAccountedGas) || u256(proofWeiPerAccountedGas) ||
  u256(permanentWeiPerByte) || u256(maximumAcceptedFeeWei)
kind0ConfigurationHash =
  H("slot-chain-kind0-ingress-config-v1" ||
    u32(448) || kind0ConfigBytes)
kind0ConstructorTail = abi.encode(
  settlementChainId, protocolVersion,
  activeSettlementRouter, routerRuntimeHash, routerConfigHash,
  forcedQueue, queueRuntimeHash, queueConfigHash,
  protocolVersionManager, l2ChainId,
```

```
  maximumForceValiditySeconds, uint64(5000000), uint32(131072),
  uint64(21000), fixedIngressWei, executionWeiPerAccountedGas,
  proofWeiPerAccountedGas, permanentWeiPerByte, maximumAcceptedFeeWei)
kind0InitCodeHash = H(kind0CreationCode || kind0ConstructorTail)
kind0Adapter = last20(H(0xff || address20(F) || kind0Salt ||
                        kind0InitCodeHash))
```

The nineteen constructor words are exactly 608 bytes and canonically padded. Profile words 267–270 and 228–229 MUST equal these derived creation/salt/init/config and address/runtime values, respectively. The configuration and constructor exclude the adapter address, Settlement, execution-profile hash, target hashes, release manifest, registration, authorization ID and ingress root; the dependency direction is therefore root primitives/artifact, then kind-0 deployment, then profile, target, manifest and registration, with no fixed point.

Deployment is the canonical zero-value ERC-2470 `deploy(bytes,bytes32)` call (selector `0x4af63f02`) with head `[0x40,kind0Salt]` and one contiguous length/data/zero-padded init-code tail. The factory account/runtime/raw deployment transaction are the same protocol-root constants specified for Settlement. An exact-code front-run is accepted after all postreads; a nonce/code collision, empty or different runtime, wrong configuration, alternate salt/init code or partial deployment rejects registration. Balance-only prefunding is preserved and is not a freshness veto.

The pinned constructor/build proof is the same strength as Settlement's: both artifacts come from named outputs of one layer-1 compiler invocation; reachable kind-0 init code accepts only the nineteen-word suffix, observes only zero `CALLVALUE`, the fixed `CALLER=F`, and the required `CHAINID=settlementChainId`, performs the exact finite configuration-slot writes, performs no external call/create/transient write, and returns the published runtime. Every claim, record, aggregate-liability and reentrancy slot is canonical zero. The runtime has no mutation path before Router authorization: an unregistered or merely staged adapter's first Router call reverts the entire outer frame, and it cannot create a record or refund credit. Registration and activation independently recheck EXTCODEHASH, `componentConfigHashV2()` and the authenticated constructor commitment.

During ACTIVATING, both old and new adapters are append-inert. The atomic ACTIVE write installs exactly the new typed append authorization; rollback restores the old sole authority and the new empty state. A historical adapter may forever expose records and withdraw already-created CEI pull refunds, and may remain allowlisted for the nonpayable sync preflight, but can never append, regain append authority or redirect a refund. Global Queue rows are not adapter-owned and survive the switch.

Word 57 is fixed to the L1 EIP-2935 system address `0x0000F90827F1C53a10cb7A02335B175320002935`; word 58 and the independent L2 runtime word 245 are each fixed to the Keccak-256 hash of the normative 83-byte runtime, `0x6e49e66782037c0555897870e29fa5e552daf4719552131a0abce779daec0a5d`; mere equality between publisher-selected hashes is insufficient. Word 59 is exactly

```
H("slot-chain-eip2935-read-config-v1" ||
  address20(0x0000F90827F1C53a10cb7A02335B175320002935) ||
  0x6e49e66782037c0555897870e29fa5e552daf4719552131a0abce779daec0a5d ||
  u64(l1HistoryFirstSupportedBlock) || u64(8191) || u64(50000)) .
```

Word 244 is that nonzero canonically padded uint64 L1 first-supported block; it is not an address and changing it without recomputing word 59 rejects. The L2 history address is a

schema constant equal to the same EIP-2935 system address, not another profile authority. Word 246 is only the nonzero release-bound L2 activation block and need not equal the L1 boundary; word 7 MUST be at least word 246. The protocol-root Router has one immutable `l1HistoryFirstSupportedBlock`; its configuration commitment includes that uint64 and the complete word-59 read configuration. Before its first REGISTER_RELEASE write the Router compares that immutable uint64 with decoded word 244 and rejects disagreement. The target constructor stores word 244, and every target-side history consumer loads that stored value as its lower bound; it has no second default or replaceable adapter boundary. Thus a later release may name a different boundary only after an independent protocol-root migration installs a Router with the same value. Both Router and target reject a zero history cell or any request below that bound or outside the 8191-block window.

Authenticated code artifacts and Settlement constructor. The first creation/runtime pair in the canonical two-artifact dynamic bundle is the Settlement pair. Both lengths are positive, `H(creationCode)=targetCreationCodeHash`, `H(runtimeCode)=targetRuntimeHash`, `runtimeLength<=24576`, and the final init code below is at most 49152 bytes. The two metadata commitments are exactly

```
targetImmutableReferencesHash =
  H("slot-chain-solc-immutable-references-v1" || u32(0))
targetLinkReferencesHash =
  H("slot-chain-solc-link-references-v1" || u32(0))
```

and the pinned build manifest must independently report empty Solidity `immutableReferences`, empty `linkReferences`, no library placeholder and byte-for-byte deployed runtime equal to `runtimeCode`. Empty compiler metadata alone is insufficient: the release verifier proves for all accepted constructor arguments that the only constructor RETURN path emits those exact runtime bytes; handwritten assembly/Yul runtime patching is forbidden.

`TargetParametersV2` is exactly the 281 fixed value words above, in the same order and canonical padding, excluding only word 281 and the dynamic tail. The constructor tail is exactly

```
abi.encode(TargetParametersV2) || executionProfileHash ||
migrationActivationProfileRecordHash || targetRuntimeHash || targetArtifactHash
```

or 285 words/9120 bytes. Consequently `creationLength<=40032` and `initCode=creationCode||constructorTail`. The constructor reads this fixed suffix, requires zero value, `msg.sender==settlementFactory`, and `block.chainid==settlementChainId`; it recomputes every hash and stores the exact initial inventory below. Let `P=keccak256("slot-chain.target.parameters.storage.v2")`; it writes profile fixed word i to slot `P+i` for every $0 <= i < 281$. It then writes `executionProfileHash`, `migrationActivationProfileRecordHash`, `targetRuntimeHash`, `targetArtifactHash`, `targetParametersHash`, `dataSessionConfigurationHash` and `targetConfigurationHash` to the seven ERC-7201-style slots obtained by Keccak of those exact lower-camel names prefixed with `"slot-chain.target.v2."`. It performs no other SSTORE and no TSTORE; all mappings, guards, counters, queues, sessions and migration fields therefore remain canonical zero. The constructor-poststate commitment is

```
H("slot-chain-target-constructor-poststate-v2" || u16(281) ||
  abi.encode(TargetParametersV2) || executionProfileHash || MPR2Hash ||
```

```
  targetRuntimeHash || targetArtifactHash || targetParametersHash ||
  dataSessionConfigurationHash || targetConfigurationHash)
```

and the build/storage proof checks exactly that inventory and non-collision of all 288 slots against the published storage layout.

The constructor environment-opcode allowlist is exact: CALLER is used only for the factory equality, CALLVALUE only for zero, CHAINID only for chain equality, and ADDRESS only in the two committed configuration derivations. All other environment, foreign-account, transient-state or external-effect observations are forbidden, including ORIGIN, GASPRICE, BALANCE, SELFBALANCE, EXTCODESIZE, EXTCODECOPY, EXTCODEHASH, BLOCKHASH, COINBASE, TIMESTAMP, NUMBER, PREVRANDAO, GASLIMIT, CHAINID outside the equality, BASEFEE, BLOBHASH, BLOBBASEFEE, GAS, CALLCODE, CALL, STATICCALL, DELEGATECALL, CREATE, CREATE2, SELFDESTRUCT, TLOAD and TSTORE. The build verifier proves the reachable init-code opcode trace obeys this allowlist and that its sole RETURN emits `runtimeCode`; a blacklist claim or source-code inspection is insufficient. Its returned runtime is argument- and environment-independent.

The target exposes the exact no-argument view

```
targetConstructorStateV2() returns
 (bytes4 magic,bytes32 constructorPoststateCommitment,
  bytes32 targetConfigurationHash)
```

with selector `0x654f7fce`, exactly four calldata bytes, zero value, the MPR2 `migrationPostStateReadGas` stipend, exactly 96 return bytes and magic `0x54435332` (TCS2) in canonically padded word zero. It recomputes the 288-word constructor inventory commitment from its own stored fixed inventory; it accepts no profile, witness or slot list and returns no cached caller-supplied commitment. PVM and Router independently derive the expected inventory and configuration from the strict raw profile. The authenticated constructor trace proves exactly those 288 SSTORE operations, no other SSTORE/TSTORE, and the authenticated runtime has no PREACTIVE mutation surface; hidden-map emptiness therefore follows from creation provenance rather than an impossible mapping enumeration.

Immediately after CREATE2, at registration, and after MCAN but before Router writes ACTIVE, the caller checks EXTCODEHASH and the already normative fixed-gas `componentConfigHashV2()` exact 32-byte getter against the independently derived `targetConfigurationHash`. It also exact-reads the existing 512-byte `dataSessionAccountingV1()` and requires its configuration word to equal the profile derivation and every counter, liability, generation, deadline and guard word to be canonical zero. Word 111 `componentConfigurationReadGas` is required to equal the protocol constant 50,000, so activation need not recover the full profile; the accounting stipend is MPR2's `migrationPostStateReadGas`. Short/trailing/bad-padding, revert or mismatch rolls the whole journal back.

The artifact descriptor is nine 32-byte words in this order: creation-code, runtime, ABI, storage-layout, constructor-schema, compiler-build, compile-time-rules, immutable-references and link-references hashes, and

```
targetArtifactHash = H("slot-chain-settlement-artifact-v2" || u16(288) ||
                       those nine words)
targetParametersHash = H("slot-chain-target-parameters-v2" || u16(8992) ||
                         abi.encode(TargetParametersV2))
```

The four target build-metadata hashes are not opaque labels. The release bundle publishes their four preimage files and hashes exact bytes:

```
targetAbiHash = H("slot-chain-target-abi-v2" || u32(abiBytes.length) || abiBytes)
targetStorageLayoutHash = H("slot-chain-target-storage-layout-v2" ||
                            u32(layoutBytes.length) || layoutBytes)
targetConstructorSchemaHash = H("slot-chain-target-constructor-schema-v2" ||
                                 u32(schemaBytes.length) || schemaBytes)
targetCompilerBuildHash = H("slot-chain-target-compiler-build-v2" ||
                            u32(buildBytes.length) || buildBytes)
```

`abiBytes` is the RFC 8785 JCS UTF-8 encoding (no BOM/newline) of the complete solc ABI JSON array; `layoutBytes` is the same encoding of the complete solc `storageLayout` object. `schemaBytes` is UTF-8, with no BOM/newline, of the exact ASCII literal `TargetConstructorV2(bytes32[281],bytes32,bytes32,bytes32,bytes32)`. `buildBytes` is RFC 8785 JCS of exactly the object keys `solcVersion`, `solcLongVersion`, `evmVersion`, `optimizerEnabled`, `optimizerRuns`, `viaIR`, `metadataBytecodeHash`, `metadataAppendCBOR`, `solcBinaryHash`, `sourceTreeHash` and `standardJsonInputHash`, `targetContractSourcePath`, `targetContractName`, `kind0ContractSourcePath` and `kind0ContractName`; unknown or missing keys reject. JSON strings/numbers/booleans have RFC 8785 semantics and each file is at most 1,048,576 bytes. The pinned build verifier regenerates all target and kind-0 metadata byte strings from the standard-json input and compares every hash before registration. The version/EVM/metadata values, all named hashes, both source paths and both contract names are JSON strings; `optimizerEnabled`, `viaIR` and `metadataAppendCBOR` are JSON booleans; `optimizerRuns` is an integer in $[0, 2^{32} - 1]$. The three inner hashes are lowercase `0x`-prefixed 32-byte hex and are exactly

```
solcBinaryHash = H(solc executable bytes)
standardJsonInputHash = H("slot-chain-solc-standard-json-v1" ||
                          u32(inputBytes.length) || inputBytes)
sourceTreeHash = H("slot-chain-source-tree-v1" || u32(entryBytes.length) ||
                   entryBytes)
entryBytes = concat, in unsigned UTF-8 bytewise path order, of
             u16(path.length) || path || u32(content.length) || content
```

`inputBytes` is the no-BOM/no-newline RFC 8785 JCS encoding of the complete standard-json input. Each source path is a nonempty relative POSIX UTF-8 path in Unicode NFC with neither `.`, `..`, empty segment, backslash nor leading slash; duplicates reject. Content is the exact source bytes named by the standard-json input (no newline or Unicode normalization), each path is at most 65535 bytes and the concatenation at most 16,777,216 bytes. Every standard-json `sources[path]` object contains exactly one `content` string and no `urls`; every import resolves to another entry in that same source map. The verifier invokes the pinned solc binary in an empty directory with only `--standard-json`, no base/include/allow paths, no import callback, no filesystem fallback and no network. Both named source paths obey the same path grammar and exist in `sources`; both contract names are nonempty ASCII Solidity identifiers. Each creation code, runtime code, ABI, storage layout and immutable/link-reference table must come from its exact named output object: `contracts[targetContractSourcePath][targetContractName]` or `contracts[kind0ContractSourcePath][kind0ContractName]`. Mixing fields between them or any other source/contract entry, selecting by a suffix or using a different compiler invocation rejects. ERC-2470 is a protocol-root dependency, not

a publisher-built artifact and therefore does not appear in this target compiler manifest. The kind-0 build metadata uses the same RFC 8785 rules. It publishes `kind0AbiHash`, `kind0StorageLayoutHash` and `kind0ConstructorSchemaHash`; the last is the length-prefixed hash, under `"slot-chain-kind0-constructor-schema-v1"`, of this no-newline ASCII literal (adjacent fragments concatenate without separators):

```
kind0AbiHash = H("slot-chain-kind0-abi-v1" ||
  u32(kind0AbiBytes.length) || kind0AbiBytes)
kind0StorageLayoutHash = H("slot-chain-kind0-storage-layout-v1" ||
  u32(kind0LayoutBytes.length) || kind0LayoutBytes)
"Kind0IngressAdapterConstructorV1(uint256,uint64,address,bytes32,"
"bytes32,address,bytes32,bytes32,address,uint256,uint64,uint64,"
"uint32,uint64,uint256,uint256,uint256,uint256,uint256)"
kind0ConstructorSchemaHash = H(
 "slot-chain-kind0-constructor-schema-v1" ||
 u32(kind0SchemaBytes.length) || kind0SchemaBytes)
kind0ImmutableReferencesHash =
 H("slot-chain-solc-immutable-references-v1" || u32(0))
kind0LinkReferencesHash =
 H("slot-chain-solc-link-references-v1" || u32(0))
```

Both named compiler tables MUST be empty; any immutable patch range, link reference, placeholder, DELEGATECALL, CALLCODE or SELFDESTRUCT in reachable adapter runtime rejects conformance. Ordinary external calls remain limited to the exact Router/Queue protocol ABIs and the guarded CEI pull-refund path; no library or generic target is reachable. Its semantic constructor poststate commitment is

```
kind0ConstructorPoststateCommitment = H(
 "slot-chain-kind0-constructor-poststate-v1" || u16(19) ||
 kind0ConstructorTail || kind0ConfigurationHash || kind0RuntimeHash)
```

The trace/storage proof binds that semantic commitment to the exact published storage layout and proves every other production slot remains zero. The adapter exposes

```
kind0ConstructorStateV1() returns
 (bytes4 magic,bytes32 constructorPoststateCommitment,
  bytes32 configurationHash)
```

with selector `0xf33625b3`, magic `0x4b435331` (KCS1), no arguments and exactly 96 canonical return bytes. Registration and activation each call both this getter and `componentConfigHashV2()` with RTR2's registered `componentConfigurationReadGas`, zero value and zero output area, and reject unexpected returndata before copying. Registration derives the KCS1 commitment from the transient profile and stores it in the kind-0 PIA2 row; activation exact-reads that PIR2/PIM2-authenticated PIA2 row and compares KCS1 to its stored commitment while also joining adapter address, runtime and configuration. Activation never recovers or accepts profile bytes. The two getters are caller-independent and make no external call.

The mandatory off-chain conformance tool emits a canonical binary report in addition to its human-readable diagnostics:

```
buildConformanceReportHash = H(
 "slot-chain-build-conformance-report-v2" || u16(853) || u8(2) ||
 verifierExecutableHash || standardJsonInputHash || sourceTreeHash ||
 executionProfileHash || migrationActivationProfileRecordHash ||
 targetCreationCodeHash || targetRuntimeHash || targetArtifactHash ||
 targetParametersHash || targetConfigurationHash ||
 constructorPoststateCommitment || targetAbiHash ||
 targetStorageLayoutHash || targetConstructorSchemaHash ||
 targetCompilerBuildHash || address20(kind0IngressAdapter) ||
 kind0IngressAdapterCreationCodeHash || kind0IngressAdapterRuntimeHash ||
 kind0IngressAdapterSalt || kind0IngressAdapterInitCodeHash ||
 kind0IngressAdapterConfigurationHash ||
 kind0ConstructorPoststateCommitment || kind0AbiHash ||
 kind0StorageLayoutHash || kind0ConstructorSchemaHash ||
 kind0ImmutableReferencesHash || kind0LinkReferencesHash)
```

All 26 hashes and the adapter address are nonzero. The release repository publishes the verifier executable bytes, complete input bundle, binary report and diagnostics; the input bundle also contains the exact ordinary-validity-verifier runtime and the `proofSystemId`-named precompile policy. Before emitting the report, the verifier matches that runtime to profile word 272 and proves the state-independent/runtime restrictions in the verifier-statement subsection; failure produces no conformant report. Thus the existing `executionProfileHash` report field binds the checked runtime, descriptor and policy without introducing another on-chain label. The DAO proposal metadata names this report hash before queueing. Two independently built verifier binaries must reproduce the same report and all profile outputs. The report is an auditable governance-release artifact, not an on-chain proof: REGISTER_RELEASE remains secure against untrusted executors and bytecode/state substitution, while deliberate authorization of malicious bytecode is the governance-root failure stated in Section 2. The deployment/configuration graph is then exactly

```
initCodeHash = H(creationCode || constructorTail)
target = last20(H(0xff || address20(settlementFactory) ||
                  settlementSalt || initCodeHash))
dataSessionConfigurationHash = the 340-byte formula in Section 3
targetConfigurationHash = H("slot-chain-settlement-config-v2" || u16(212) ||
  address20(target) || targetRuntimeHash || executionProfileHash ||
  migrationActivationProfileRecordHash || targetParametersHash ||
  targetArtifactHash || dataSessionConfigurationHash)
```

The target, profile, MPR2, PIA2 rows/root, configuration, deployment descriptor, manifest, registration and predecessor are derived outputs and MUST NOT occur inside the profile. Registration recomputes this DAG from profile bytes and compares all eight deployment words, the manifest chain/version/core/component joins, exactly two independently reconstructed PIA2 rows and their root. It also recomputes every source CREATE2/config/domain and destination component, infrastructure/Bridge/domain relation; a getter or just-written Router row is not an independent expected value.

The Settlement deployer is the protocol-root ERC-2470 Singleton Factory, not a release-publisher contract. The root fixes this complete identity:

```
factoryAddress = 0xce0042B868300000d44A59004Da54A005ffdcf9f
singleUseDeployer = 0xBb6e024b9cFFACB947A71991E386681B1Cd1477D
factoryCreationCodeBytes = 340
factoryCreationCodeHash =
  0x122b6b28aeddfd05fa3ce4348e93d357b3ce50d9ab7dda4e8ee524a5b9a6ab3b
factoryRuntimeBytes = 308
factoryRuntimeHash =
  0xc4d5542b53a8b779595a20a8ddd60e58a6c49d3c3decc2df83ced1c69c8ca807
factoryDeploymentTransactionHash =
  0x803351deb6d745e91545a6a3e1c0ea3e9a6a02a1a4193b70edfcd2f40f71a01c
factoryCreationCode =
  hex("6080604052348015610010576000 80fd5b506101348061002060003960 00f3fe")
  || factoryRuntimeCode
factoryRuntimeCode = hex(concat(
 "6080604052348015600f57600080fd5b506004361060285760003560e01c8063",
 "4af63f0214602d575b600080fd5b60cf60048036036040811015604157600080",
 "fd5b810190602081018135640100000000811115605b57600080fd5b82018360",
 "2082011115606c57600080fd5b8035906020019184600183028401116401000000",
 "0000831117156 08d57600080fd5b91908080601f016020809104026020016040",
 "5190810160405280939291908181526020018383808284376000920191909152",
 "5092955050913592506 0eb915050565b60408051600160016 0a01b03909216825251",
 "9081900360200190f35b6000818351602085016000f5939250505056fea26469",
 "706673582212206b44f8a82cb6b156bfcc3dc6aadd6df4eefd204bc928a4397",
 "fd15dacf6d5320564736f6c63430006020033"))
```

The root release bundle also publishes the exact raw non-EIP-155 deployment transaction below; its bytes, recovered sender, creation code, resulting address and transaction hash MUST all match the tuple above before a chain is supported:

```
hex(concat(
 "f9016c8085174876e8008303c4d88080b90154608060405234801561001057 60",
 "0080fd5b506101348061002060003960 00f3fe6080604052348015600f576000",
 "80fd5b506004361060285760003560e01c80634af63f0214602d575b600080fd",
 "5b60cf600480360360408110156041576000 80fd5b810190602081018135640 1",
 "0000000811115605b57600080fd5b8201836020820 11115606c57600080fd5b",
 "8035906020019184600183028401116401000000008311171560 8d57600080fd",
 "5b91908080601f016020809104026020016040519081016040528093929190 81",
 "8152602001838380828437600092019190915250929550509135925060eb91505056",
 "5b60408051600160016 0a01b03909216825251908190036020019 0f35b600081",
 "8351602085016000f5939250505056fea2646970667358221 2206b44f8a82cb6",
 "b156bfcc3dc6aadd6df4eefd204bc928a4397fd15dacf6d5320564736f6c6343",
 "00060200331b83247000822470"))
```

The only deployment wire used by a release is

```
deploy(bytes initCode,bytes32 salt) returns (address target)
selector = 0x4af63f02
calldata = selector || u256(0x40) || salt || u256(initCode.length) ||
           initCode || zeroPaddingToWord
```

```
returndata = bytes12(0) || address20(target) // exactly 32 bytes
```

Calldata has no suffix and value is zero. The factory passes the supplied bytes unchanged to `CREATE2(0,initCode,salt)`. The permissionless release workflow first recomputes target from the root factory, salt and init-code hash. If target has no code, any account may call the factory and MUST require the exact 32-byte returned address, then exact-check target EXTCODEHASH, configuration, TCS2 and DSV1. If target already has code, the workflow skips the factory call and performs the same exact postreads. A CREATE2 collision returns zero under this runtime; zero/wrong returndata or any postread mismatch fails deployment. Idempotence therefore belongs to the external workflow, not to a fictitious factory replay branch. This external transaction is not a nested protocol call; the release publishes its measured whole-transaction gas ceiling.

Profile words 44 and 45 MUST equal the root address and runtime hash. Word 46 is independently recomputed as

```
settlementFactoryConfigurationHash = H(
 "slot-chain-erc2470-factory-config-v1" || u16(147) ||
 address20(factoryAddress) || factoryRuntimeHash ||
 factoryCreationCodeHash || factoryDeploymentTransactionHash ||
 address20(singleUseDeployer) || bytes4(0x4af63f02) ||
 u32(49152) || u16(32) || u8(1))
```

where the final three fields commit the maximum accepted init code, exact return width and zero-value-only rule. PVM and Router compile and enforce this triple; neither trusts a self-reporting factory getter or publisher-selected behavior cases. REGISTER_RELEASE cannot rotate it. Changing any root tuple member requires the independent protocol-root migration. Registration safety then depends on the independently derived CREATE2 address and exact target code, configuration, constructor-state and zero-accounting reads; an absent or wrong factory makes a release unavailable, never accepted.

Release conformance rejects the former `a1617601` and A4 `a4000101400241000380` CBOR fixtures even if a caller rehashes them. It also gates deployed PVM, Router and target runtime at no more than 24576 bytes, target creation at no more than 40032 bytes, final initcode at no more than 49152 bytes, and publishes cold-state maximum-profile deployment and nested Timelock–PVM–Router registration gas measurements. Every fixed call budget includes EIP-150 forwardability and a named post-call reserve; no unmeasured 146848-byte profile is implementation ready.

Every profile contains exactly one immutable `MigrationTransitionVerifierDescriptorV2`:

```
struct MigrationTransitionVerifierDescriptorV2 {
  address verifier; bytes32 runtimeHash; bytes32 configurationHash;
  bytes32 verifyingKeyHash; bytes32 proofSystemId;
  bytes32 publicInputSchemaHash; bytes4 selector;
  uint32 maximumProofBytes; uint64 verificationGasLimit;
}
```

All fields are nonzero, `selector` equals `bytes4(keccak256("verifyMigrationTransition(bytes,uint256[2])"))`, and the two hash layers are acyclic and byte exact. First, `configurationHash` is the canonical verifier-instance configuration:

```
MIGRATION_VERIFIER_CONFIG_TYPE_LITERAL =
  "MigrationTransitionVerifierConfigV2(bytes32 verifyingKeyHash,"
  "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
  "uint32 maximumProofBytes,uint64 verificationGasLimit)"
MIGRATION_VERIFIER_CONFIG_TYPEHASH = keccak256(
  concat(the three quoted fragments))
configurationHash = keccak256(abi.encode(
  MIGRATION_VERIFIER_CONFIG_TYPEHASH, verifyingKeyHash, proofSystemId,
  publicInputSchemaHash, selector, maximumProofBytes, verificationGasLimit))
```

It deliberately excludes `verifier`, `runtimeHash` and itself; the immutable verifier getter is exactly

```
migrationVerifierConfigHashV2() external view returns (bytes32)
MIGRATION_VERIFIER_CONFIG_READ_GAS = 50000
```

Its selector is the first four Keccak bytes of `"migrationVerifierConfigHashV2()"`. PVM and Router first require `EXTCODEHASH(verifier)==runtimeHash`, then issue a zero-value `STATICCALL` with exactly those four calldata bytes, no suffix and the fixed 50,000-gas stipend. Success returns exactly one 32-byte word equal to `configurationHash`; empty, short, trailing, faulting or mismatched returns reject before any state write. Second,

```
MIGRATION_VERIFIER_DESCRIPTOR_TYPE_LITERAL =
  "MigrationTransitionVerifierDescriptorV2(address verifier,"
  "bytes32 runtimeHash,bytes32 configurationHash,bytes32 verifyingKeyHash,"
  "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
  "uint32 maximumProofBytes,uint64 verificationGasLimit)"
MIGRATION_VERIFIER_DESCRIPTOR_TYPEHASH = keccak256(
  concat(the four quoted fragments))
migrationVerifierDescriptorHash = keccak256(abi.encode(
  MIGRATION_VERIFIER_DESCRIPTOR_TYPEHASH, verifier, runtimeHash,
  configurationHash, verifyingKeyHash, proofSystemId,
  publicInputSchemaHash, selector, maximumProofBytes,
  verificationGasLimit))
```

Each type literal is the byte concatenation of its displayed fragments with no inserted whitespace or newline. These are ordinary canonical ABI words: address occupies the low 20 bytes; `bytes4` occupies the high four bytes; unsigned integers occupy the low bytes; every unused byte is zero. The profile decoder recomputes both layers and rejects noncanonical padding or any mismatch. PVM independently recomputes them while validating the complete manifest and requires its `migrationVerifierDescriptorHash` to equal the second layer. The Router repeats those checks immediately before bounded verifier dispatch and uses only the descriptor recovered from the pinned profile. The profile hash, release manifest and append-only historical registration therefore commit one identical descriptor; no constructor default, caller-supplied descriptor or global verifier exists. The verifier's separate implementation configuration hash is not a synonym for the descriptor hash. A successor may rotate the descriptor only by activating a new profile and release. Golden/reject vectors replace each field in turn and cover zero values, short/long getter returns, caller-substituted configuration, nonzero narrow-field padding and the former self-referential fixed-point construction; all must reject except

the one exact canonical vector. The same profile records nonzero `supportedL1BlockGasLimit`, `worstCaseActivationAdoptionGas`, `sourceFreezeGasLimit`, `targetAdoptionGasLimit`, `queueMigrationGasLimit`, `activationContextReadGasLimit`, `postStateReadGasLimit`, `legacyStateReadGasLimit`, `legacyArmGasLimit`, `legacyFinalizeGasLimit` and `postCallbackReserveGas`. These values are part of the exact executable profile bytes/hash and cannot be supplied by a caller or changed in-place. The Router copies at most the one exact ABI return length for each call and discards arbitrary revert data; no callee can force unbounded memory expansion. `worstCaseActivationAdoptionGas` is the measured cold-state upper bound for the *complete* L1 Router execution after transaction intrinsic gas: canonical ABI decode/re-encode and allocation, every cold Router/Queue/history/profile/manifest read, statement and transaction hashing, `EXTCODEHASH`, the configuration getter and its call overhead, verifier dispatch consuming the full `verificationGasLimit` including EIP-150 and memory/call overhead, and every post-verifier callback/write in the atomic Router/Queue/old-Settlement/new-Settlement/seat/receipt suffix. It is not a post-verifier-only estimate or a governance number detached from the release bytecode. A benchmark that omits any prefix, call overhead or suffix cannot populate this field.

Before each fixed-gas call, Router requires enough gas for the requested stipend under EIP-150 plus the sum of all remaining call stipends, bounded return-copy and `postCallbackReserveGas`. In particular the call-local available gas after memory and CALL/STATICCALL charges is at least `ceil(64*stipend/63)`, so a syntactically requested full stipend is actually forwardable. The reserve covers all three MAPS reads, receipt/index and target- registration writes, final ACTIVE/context clearing and IDLE. The release harness pins cold/warm access and compiler/fork call overhead, succeeds at each exact threshold, fails at one gas less, and injects full-stipend consumption and late OOG at every MFRZ/LGFN, MCAN, QMIG, MACT and MAPS boundary. A compiler, fork or bytecode change invalidates the measurements until republished.

The public-input schema is the exact static Solidity tuple below. The two core hashes are the Appendix canonical hashes of the complete supplied base and output `CanonicalCore`; neither hides a caller-selected partial tuple.

```
struct MigrationTransitionStatementV2 {
  uint256 settlementChainId;
  address activeSettlementRouter;
  bytes32 routerRuntimeHash; bytes32 routerConfigurationHash;
  uint8 transitionKind; uint64 migrationGeneration; uint64 sourceProtocolVersion;
  uint64 targetProtocolVersion; uint64 sourceCanonicalSequence;
  bytes32 executionProfileHash; bytes32 targetManifestHash;
  bytes32 targetRegistrationHash;
  bytes32 candidateDigest; bytes32 baseCanonicalHash;
  bytes32 outputCanonicalHash;
  address forcedQueue; bytes32 queueRuntimeHash;
  bytes32 queueConfigurationHash; bytes32 queueRoot;
  uint64 queueCount; uint64 startCursor; uint64 endCursor;
  bytes32 forcedDescriptorCommitment; address proofBeneficiary;
  uint256 anchorNumber; bytes32 anchorHash;
  bytes32 forceRoot; uint64 forceCutoff;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  bytes32 sourceBridgeExecutionHash;
```

```
  bytes32 releaseSystemCalldataHash; bytes32 inboxSystemCalldataHash;
  bytes32 releaseSystemTxHash; bytes32 inboxSystemTxHash;
  uint8 releaseSystemTxPosition; uint8 inboxSystemTxPosition;
  bytes32 importedHeaderHash; bytes32 importedStateRoot;
  bytes32 legacySignalCheckpointHash;
  bytes32 legacyDeploymentHash; bytes32 legacyArmId;
  bytes32 legacyLaunchId;
  bytes32 deploymentCommitment;
  uint64 preInboxLastAppliedPlusOne;
  uint64 postInboxLastAppliedPlusOne;
}
```

The typehash is Keccak of the exact canonical Solidity signature with the field names and order shown, and `statementHash=keccak256(abi.encode(TYPEHASH,fields...))`. `publicInputSchemaHash` is Keccak of that same signature. The plus-one cursor words use zero only for an uninitialized lifetime InboxApply router and otherwise encode the checked block number plus one. L1 derives every L1-authoritative field from current Router/Queue/history state or exact bounded calldata. `transitionKind` is exactly GENESIS_IMPORT=1 or VERSION_MIGRATION=2. For GENESIS_IMPORT, the imported header hash must equal the legacy Inbox's stable finalized-block hash authenticated by the exact QUIESCENT LGS1 and transaction-local LGAR; its state root equals `importedStateRoot`. The checkpoint hash is exactly `H("slot-chain-legacy-checkpoint-v1"||u48(header.number)||importedHeaderHash||importedStateRoot)` and the circuit proves the legacy SignalService record at that height has those values. The same genesis statement binds the registered compatibility proxy's nonzero `legacyDeploymentHash`, campaign/scan-bound `legacyArmId` and target-derived `legacyLaunchId`. For VERSION_MIGRATION, `importedHeaderHash`, `importedStateRoot` and `legacySignalCheckpointHash`, `legacyDeploymentHash`, `legacyArmId` and `legacyLaunchId` are all canonical zero; the base core and state root are derived exclusively from Router's exact sequence-tagged current Canonical. Mixing a legacy witness into a later migration, or zeroing one for launch, rejects. The sole deployment public input is derived as follows:

```
componentsHash = keccak256(abi.encode(manifest.components))
prestatePolicyHash = H("slot-chain-deployment-prestate-policy-v2" ||
                       u8(transitionKind))
poststatePolicyHash = H("slot-chain-deployment-poststate-policy-v2" ||
                        u8(transitionKind))
DEPLOYMENT_COMMITMENT_TYPEHASH = keccak256(
  concat(
   "DeploymentCommitmentV2(uint8 transitionKind,",
   "uint64 targetProtocolVersion,bytes32 targetManifestHash,",
   "bytes32 targetRegistrationHash,",
   "bytes32 destinationInfrastructureHash,bytes32 componentsHash,",
   "bytes32 poolConfigurationHash,uint64 retirementQueueCount,",
   "bytes32 prestatePolicyHash,bytes32 poststatePolicyHash)"))
deploymentCommitment = keccak256(abi.encode(
  DEPLOYMENT_COMMITMENT_TYPEHASH, transitionKind, targetProtocolVersion,
  targetManifestHash, targetRegistrationHash,
  manifest.destinationInfrastructureHash,
```

```
  componentsHash, manifest.poolConfigurationHash, queueCount,
  prestatePolicyHash, poststatePolicyHash))
```

`componentsHash` encodes the ten three-word rows in manifest order with no dynamic offset. Router derives every field and supplies the same one hash as both statement input and verifier expectation; there is no caller-selected or "observed" hash. The type literal is the single concatenation shown, with no display newline or extra whitespace.

For GENESIS_IMPORT, the prestate-policy tag means exact lifetime code/config, empty Router routes/cursor, empty release/domain/writer records, accumulator empty root/count/frontier, Pool zero ticket/accounted total, fresh Store/Quota/Bridge private state and arbitrary forced ETH surplus. For VERSION_MIGRATION it means exact lifetime code/config and byte-for-byte preservation of every old route/release/domain/writer, Inbox cursor, accumulator frontier/root/count and Pool ticket/accounting word, with only the new release/domain/writer/route absent and the new Store/Quota/Bridge fresh. Both poststate tags mean the exact tx0 manifest/count is sealed, only those new append-only entries and endpoint activation are added, all old lifetime state is preserved, no Pool ticket is created and no ETH is moved or minted. The circuit proves each enumerated account/code/storage and balance relation directly under authenticated pre/post roots and proves the complete output core. It also proves the deployment policy encoded by the single `deploymentCommitment` holds, `queueCount=retirementQueueCount`, positions are zero and one, and the two transaction hashes are legacy Keccak of the exact type-`0x7f` RLP envelopes reconstructed from the separately bound calldata hashes, fixed targets, kind, nonce, gas, zero value and destination chain ID.

Verifier calldata is exactly

```
verifyMigrationTransition(bytes proof, uint256[2] digestLimbs)
    external view returns (bytes32 statementHash)
```

with ABI head `[0x60,high128(statementHash),low128(statementHash)]`, then the canonical bytes length/data/padding tail. Proof length is at most the descriptor bound. ActiveSettlementRouter makes one descriptor-gas-bounded `STATICCALL`, after checking `EXTCODEHASH` and configuration, and accepts exactly 32 return bytes equal to its reconstructed `statementHash`. The circuit recombines both unreduced 128-bit limbs and proves this exact typed preimage. Revert, OOG, short/trailing return, wrong hash, substituted code/key/schema or a Boolean-shaped result rejects before any state write. The target Settlement never receives a verifier address, result capability or callback; its authority begins only after Router validates and atomically adopts the exact result.

The commitment dependency graph is acyclic and normative:

```
kernel/component configs/runtime descriptors
  -> Bridge execution/infrastructure/domain
  -> canonical profile bytes -> executionProfileHash
  -> ReleaseManifestV2 -> releaseManifestHash
  -> destination registration commitment
```

The profile may contain the manifest schema version, manager/authority/registrar addresses, namespaces, ABI and activation/validation rules. It MUST NOT contain the release-manifest hash, registration commitment, destination-registration commitment, or any field derived

from them; otherwise decoding rejects rather than attempting a hash fixed point. In particular it contains one nonzero `poolNamespace`, distinct from the router, manifest and destination namespaces. That primitive profile value is an input to component-9 `poolConfigurationHash`; it is not derived from the Pool address, manifest or destination domain. The Pool address/configuration then participates in the infrastructure and domain hashes in the direction shown above, so either side can independently reconstruct the graph without a fixed point.

- settlement/L2 chain IDs, both genesis hashes, the protocol-local Schedule route ID carried in the historical `forkDigest` field, EIP-2935 address and protocol-lifetime `firstV2BlockNumber`; BuilderRegistry, ScheduleOracle, its protocol-lifetime schedule-rules hash and every fork-verifier address/code/config/gas tuple, domain-registry and version-manager addresses, runtime hashes and exact immutable/configuration commitments; the legacy resume root, non-proxy fixed-key RISC0/SP1 adapters, their four immutable key-policy values, recursively closed remote-verifier dispatch graphs, and the direct legacy SignalService proxy/implementation/layout/authority fence (with ForkRouter rejected); every append-only source, destination and support descriptor and manifest schema/validation binding; each nonzero domain namespace, fresh immutable SourceBridge/Registry/Quota and DestinationBridge/Store/Quota bundle runtime/config/layout, CREATE2 factory/salt/init-code derivation, and the ban on every V2 proxy, owner, upgrade and delegate target, terminal verifier, immutable settlement router, its append-only ingress authorization rules, immutable ForcedQueue runtime and exact configuration hash, the kind-0 ingress adapter and Bridge inbox adapter, InboxApply router, InboxCredit Store, release authority, terminal registrar and accumulator runtime plus every constructor immutable, infrastructure hash, ABI, CreditRecord/liability/status/index layouts and direct-call gas limit;
- gas limit, EIP-1559 elasticity/change denominator, blob target/update fraction, empty ommers/withdrawals/requests constants and exact derivation of base fee, blob fields, logs bloom, randomness, parent beacon root, requests hash and every post-fork header field;
- the Anchor, InboxApplyRouterV2, InboxCreditStoreV2, the release authority, terminal registrar and accumulator, and both legacy SignalService contracts, fresh source/destination Bridge bundles, Bridge-unique quota managers and NativeLiquidityPoolV2 address/runtime/configurati every DIRECT-only V2 selector and ABI tuple; the unsigned custom system transaction type and unforgeable reserved senders; system nonce rules, fee exception, chain-ID equality, caller/origin injection, warm set, intrinsic gas, refund rule, zero GASPRICE, account non-transition, exact typed trie values, one-shot Store/Bridge seal and gas ceilings;
- the exact AnchorV4 checkpoint/ancestors transition, forced disposition rules, InboxApplyRouterV2 global cursor, route and batch transition, immutable store inbox slot, result and deadline storage, Bridge V2 aggregate and per-credit liability, both endpoint statuses, terminal indices, vector count, root, 64-word terminal frontier and full leaf event, pull credits, cancellation, expiry, delivery and recall transitions and the sole inbox-pin path; and
- at least six complete vectors, each covering parent, prestate, input, transaction, receipt, header and poststate: empty escape, valid kind 0, expired kind 0 without bytes, kind 1 under two separately registered immutable source generations, source and destination domain replacement, direct-record absence and mismatch, enqueue and cancel ordering, router-SYNCED refund and retry, capacity race, delayed append, PREACTIVE kind 0 and kind 1 rejection, legacy attempts to invoke inbound V2, first post-cutover release activation, pin survival across arbitrary legacy SignalService upgrades, endpoint-support manifest authorization/finality and finalized L2 registration proof, squatting/unauthorized/out-

of-order registration, mixed D1/D2/D1 routing, old-domain enqueue after new-domain activation, missing-route atomic rollback, multi-credit batch ordering/conflict, Message loss before enqueue and after pin, deadline expiration, DONE/FAILED terminal replay and a cap-limited multi-block prefix, plus extra-reserved-sender, unauthorized clone, mismatched/local-foreign domains, pooled-balance drain attempts, zero ordinary quota during V2 refund, legacy SignalService pause during V2 terminal proof, forbidden upgrade/delegatecall, exact legacy-send compatibility, DIRECT-only V2 and rejected Vault callers, old-domain accumulator append after new-domain activation, canonical-sequence cell collisions, accumulator-target caller confusion, sentinel/wrong-credit/duplicate-append, terminal leaf/index/root substitution, historic-count proof reconstruction, unauthorized/fake/full-core-discontinuous router switches, same-block history writes, migration-ready anti-griefing, conflicting result/terminal hash and every V1/V2 boundary.

For escape, parent hash/height/state and next fee/blob scalars come from the stored canonical core; timestamp/slot, L1 origin and forced inputs come from the round. Every remaining header field and both system transactions are a pure profile function of those values and the execution result. The circuit loads the profile selected by the public protocol version and proves its hash equals Settlement's immutable mapping. Launch tooling rejects a version lacking the ABI artifact, verifier-key binding or complete vectors. Thus a profile change is a new protocol version and review, not runtime interpretation.

12.4 Reorg rule

The canonical L1 log is the journal. A reorg truncates and replays, in order: Router lifecycle/context and public gate; source freeze or legacy cutover phase; target canonical/history adoption; ForcedQueue cursor, payment claim and authority; all three activation poststate commitments; sealed receipt/indexes; then ordinary canonical tuple, mode/version, normal arm/activation/best/deadline/ frozen admission, episode/round, seat duties/services/selections, retained breach receipts, Market credits/reserves, data-session root/count and reward eligibility. These activation writes are one L1 transaction and therefore truncate together; no indexer treats an intermediate callback event as a committed switch. Off-chain workers index by `(settlementChainId,contract,blockHash,logIndex)` and discard orphaned jobs. Withdrawal/bridge execution waits for the configured L1 finality policy.

12.5 Genesis and migration

Deployment occurs at least `D_SNAP+F_FINAL+sealMargin` before activation, so first live sources are post-deployment and sealable. New Settlement begins `PREACTIVE`; registry and scheduling preparation plus off-chain data preparation may run, but target session open/post/seal, every candidate and every kind-0/kind-1 forced-ingress or bridge enqueue call rejects before creating target-local state, a deposit, adapter record or queue leaf.

The legacy Inbox does not share this design's queue or expose a safe sidecar freeze. A sidecar cannot intercept direct calls to the deployed proxy's old proposal and forced-inclusion selectors. GENESIS_IMPORT therefore requires the delayed legacy governance path to install one final `LegacyGenesisCutoverInboxV1` implementation in that same proxy. It has byte-identical legacy storage layout/configuration, consumes only a published collision-free gap slot, pins

the non-proxy Router immutable, preserves every legacy event and bond-withdrawal path, and exposes no delegate target. Direct implementation calls reject `onlyProxy`. Its upgrade authorization is legal only in local ACTIVE and exact-reads LGC1 before authorizing. Any SCHEDULED campaign before either expiry bound blocks every proxy implementation upgrade, even before the admission cutoffs. A published campaign is immutable and noncancelable; the campaign-wide 600-block/7,200-second bound ends this fence through expiry and local resume. EXPIRED and a time-expired retained SCHEDULED tuple permit the ordinary delayed legacy upgrade path only after all local scan/campaign words are cleared through the independent permissionless resume described below. DRAINING, QUIESCENT, READY and FROZEN can never be bypassed by installing another implementation. A Router read failure, wrong tuple or implementation/profile mismatch fails closed rather than authorizing the upgrade. Router authenticates proxy/runtime, implementation address/runtime and the exact 17-word configuration during activation preflight and again at LGAR, LGS1 and LGFN. If an existing deployment cannot install this exact compatibility implementation, sound in-place GENESIS_IMPORT for that deployment is impossible; a separately initialized chain/state migration is required.

ProtocolVersionManager publishes at most one live delayed genesis campaign. The Router exposes its exact fail-closed view:

```
legacyGenesisCampaignV1() returns
 (bytes4 magic,uint8 status,uint64 nonce,uint64 generation,
  uint64 forceCutoffBlock,uint64 proposalCutoffBlock,
  uint64 quiesceNotBeforeBlock,uint64 resumeByBlock,
  uint64 resumeByTimestamp,uint64 reviewFinalizedByBlock,
  address targetSettlement,uint64 targetProtocolVersion,
  bytes32 targetManifestHash,bytes32 targetRegistrationHash,
  bytes32 reviewCommitment,bytes32 campaignId)
```

Its selector is `0x718b2ac7`, magic is `0x4c474331` (LGC1), and returndata is exactly 512 bytes. Status is NONE=0, SCHEDULED=1, CONSUMED=3 or EXPIRED=4; value 2 is reserved and rejects if observed. All high padding is zero. NONE requires every other word zero; every non-NONE status retains the same complete nonzero immutable tuple and campaign ID so proxy resume and reorg replay never depend on a cleared target. No behavioral or caller-supplied substitute is accepted. Let `publicationBlock=block.number`, `publicationTimestamp=block.timestamp` and `executeAfter` be the retained PCO1 value in the one EXECUTING PUBLISH_GENESIS_CAMPAIGN operation. Publication atomically enforces all of these checked inequalities:

```
publicationBlock <= reviewFinalizedByBlock
reviewFinalizedByBlock + 64 <= forceCutoffBlock
forceCutoffBlock + 64 <= proposalCutoffBlock
proposalCutoffBlock + 128 + 64 <= quiesceNotBeforeBlock
quiesceNotBeforeBlock < resumeByBlock
resumeByBlock - publicationBlock <= 600
executeAfter <= publicationTimestamp < resumeByTimestamp
resumeByTimestamp - publicationTimestamp <= 7200
resumeByTimestamp - executeAfter <= 7200
```

The operation payload carries every right-hand campaign bound, and PVM derives the

two publication values from the executing transaction; no caller supplies or later refreshes them. Thus a queued operation that is executed too late rejects, and SCHEDULED cannot freeze upgrades or proposer/configuration authority for an arbitrarily distant cutoff. The `reviewFinalizedByBlock` is at least `F_FINAL_BLOCKS=64` blocks before `forceCutoffBlock`. Once the campaign publication transaction succeeds, neither governance nor a copied proof can change or cancel the target or campaign. Expiry tombstones that campaign ID; every later canonical attempt increments `nonce`, even when the target and generation are unchanged. A reorg instead removes the orphaned campaign and all of its scan/quiescence state together; proofs from it reject unless the identical campaign is canonically republished.

Before publishing any campaign, including a higher-nonce retry, ProtocolVersionManager additionally exact-staticreads LGS1 and LGSS from the compatibility proxy. LGS1 must be the canonical all-cutover-words-zero ACTIVE row; LGSS must be EMPTY with every campaign, endpoint, cursor, count, byte, root, value, expiry and profile word zero. Thus Router cannot replace the LGC1 tuple while an expired attempt still has a partial/complete scan or a QUIESCENT snapshot whose old campaign ID is needed for resume. Anyone first calls the old campaign's permissionless resume; a bad, short or stale local view blocks publication, not legacy operation.

The compatibility implementation exact-staticreads LGC1 before every selector that can change the five checkpoint boundary fields. At and after `forceCutoffBlock`, payable forced ingress rejects before retaining ETH or writing the tail. At and after `proposalCutoffBlock`, proposal admission rejects before consuming a forced record, transferring its fee, incrementing a proposal ID or storing a hash. Existing proof/finalization remains open until QUIESCENT, and historical bond paths never close. An LGC1 revert, OOG, wrong code, magic, length, padding, nonce, campaign ID or stage fails closed. Once either `block.number>=resumeByBlock` or `block.timestamp>=resumeByTimestamp`, checkpoint/activation rejects and admissions resume even if nobody first calls Router expiry. These comparisons make same-block ordering irrelevant: an ingress transaction in a cutoff block cannot precede the fence.

A SCHEDULED campaign also applies hard preparation caps immediately when its publication transaction completes. Publication rejects unless the unfinalized proposal count and pending forced count are each at most 1,024. Until their respective cutoffs, proposal/forced admission rejects before mutation if it would exceed either count or the codec-derived 4,194,304-byte total scan bound. A canonical row consumes at most its profile-pinned maximum encoded bytes, so this is checked without trusting a later caller. Scan batches contain exactly `min(16,end-cursor)` consecutive rows; a non-final short batch is invalid. Hence every successful call makes maximum progress and at most 128 successful scan calls cover both 1,024-row maximum ranges, even when an adversary front-runs a keeper. The immutable stage inequalities above provide `F_FINAL_BLOCKS` blocks from force cutoff to proposal cutoff and `128+F_FINAL_BLOCKS` blocks from proposal cutoff to `quiesceNotBeforeBlock`. Entering QUIESCENT additionally requires at least 128 L1 blocks and 1,800 seconds remaining; the campaign-wide interval is already hard-capped at 600 L1 blocks and 7,200 seconds. Thus one included scan batch per block completes before quiescence under the same L1 inclusion assumption as proof landing; governance cannot authorize an unbounded maintenance window.

After proposal cutoff, anyone scans the fixed unfinalized proposal interval and pending forced interval in bounded calls. Each proposal preimage must hash to its still-addressable ring cell; each source and direct forced row supplies its canonical blob timestamp/hash slice. Streaming domain-separated accumulators bind every index, stored hash and derived

expiry in order. The scan starts from the then-current `lastFinalizedProposalId+1` and `forcedHead`, ends at the cutoff-frozen `nextProposalId` and `forcedTail`, and may include a proposal later finalized before quiescence; that harmless superset prevents a moving-start race. The deployment profile fixes `LEGACY_BLOB_RETENTION_SECONDS=1,572,864`, the conservative 4,096-epoch mainnet minimum at 32 slots per epoch and 12 seconds per slot. For each blob-bearing source the compatibility implementation derives, rather than accepts, `dataExpiry=checked(uint64(blobSlice.timestamp)+LEGACY_BLOB_RETENTION_SECONDS)`. A proposal row's expiry is the minimum of all of its blob-bearing source expiries; a forced row has exactly its stored source expiry. A row containing no blob-bearing source has expiry `UINT64_MAX`. Zero timestamp, arithmetic overflow, malformed source shape or a caller-supplied expiry rejects. Every blob row in the scanned unfinalized/pending superset contributes to `minDataExpiry`, including a row already outside that exact retention horizon. Before entering QUIESCENT the contract rechecks the fixed endpoints, complete counts/roots and requires `minDataExpiry>=resumeByTimestamp+LEGACY_RESUME_PROOF_GENERATION_MAX_SECONDS+REORG_MARGIN_SECONDS+T_INCLUDE_MAX_SECONDS`, where the reviewed release value of `LEGACY_RESUME_PROOF_GENERATION_MAX_SECONDS` is 900. The two ZK proofs may be generated in parallel, but each complete route proof must meet that bound in the shadow-fork release gate. Therefore a failed campaign cannot itself age any legacy data out of availability or resume into a suffix that the deployed no-skip protocol cannot process. A stale row rejects quiescence even if a successful migration would have abandoned it. Operators must first finalize or otherwise repair that legacy state; if the exact deployed rules provide no safe repair, in-place GENESIS_IMPORT is unsupported and a separately initialized state migration is required.

Blob availability is not the sole resume predicate. The campaign's `reviewCommitment` binds a golden `legacyResumeProfileHash` over the proxy/implementation/configuration, proof verifier and every composed sub-verifier runtime/configuration, the legacy SignalService checkpoint path, proposer checker, prover permission state, bond parameters and all time-dependent legacy branches. The design does *not* assume that the same proof bytes remain valid: deployed SGX instances expire after 90 days. A supported profile instead pins one canonical sufficient verifier route whose members ignore both `proposalAge` and `block.timestamp`. Its root is the new non-proxy, ownerless `LegacyResumeZkPairVerifierV1`; generic ComposeVerifier roots are unsupported. It accepts exactly the ascending pair `[RISC0_RETH=5,SP1_RETH=6]` and no SGX-containing or one-proof route, so fresh proof bytes may be generated after resume. The route uses reviewed non-proxy, ownerless fixed-key adapters: the RISC0 adapter accepts exactly one block image and one aggregation image, and the SP1 adapter accepts exactly one block program key and one aggregation program key. Those four keys are immutable code/configuration, not entries in the deployed mutable trust mappings. Each adapter recursively binds the exact remote verifier dispatch graph and rejects every other key. A generic owner-controlled RISC0/SP1 trust-map wrapper is not a supported resume-route member, because its mapping is not enumerable and an on-chain profile check cannot prove that no additional key was authorized.

For a supported profile, `minBond=livenessBond=0`, proving is public, there is no contest/skip deadline, a later forced-due time only strengthens the next proposal's inclusion requirement, and historical withdrawals may only mature. Therefore advancing by the bounded QUIESCENT duration cannot make the bound state transition unprovable solely because time advanced; the data-expiry check covers its per-record availability horizon. Router and proxy recheck that exact profile at campaign publication, quiescence, activation and resume. A root

verifier that requires SGX on every sufficient route (including the legacy `MainnetVerifier`), a route member that consumes proposal age or wall clock, nonzero liveness-bond deadline, challenge window, expiring claim, mutable runtime, or any other non-pause-aware validity/custody branch makes in-place GENESIS_IMPORT unsupported. The pre-campaign compatibility upgrade must point the Inbox to the exact fixed-pair root; otherwise a separately initialized state migration is required.

The checkpoint write is a separate consensus dependency. Legacy `Inbox.prove` calls `SignalService.saveCheckpoint` in the same transaction before storing its finalized core state. The bound `signalServiceCheckpointDescriptorHash` therefore commits to the exact proxy runtime, direct implementation address and runtime hash, `VERSION=1` checkpoint storage layout, implementation-embedded authorized syncer equal to the legacy Inbox, remote SignalService and pauser immutables, and the upgrade-authority/fence descriptor. A ForkRouter or any implementation with a reachable delegate target beyond the one bound UUPS proxy-to-implementation hop is unsupported; governance must first install the reviewed final direct SignalService implementation before campaign publication. This avoids making checkpoint liveness depend on an incompletely enumerated dispatch graph. Pausing is deliberately not a profile bit because the reviewed V1 `saveCheckpoint` path is not pause-gated; any replacement that changes that property has a different runtime hash. Publication, quiescence, activation and resume recheck this descriptor. Its implementation/configuration authority must be burned or share the same exact campaign fence as the Inbox and verifier graph. If the live checkpoint graph cannot be fenced, in-place migration is unsupported. For the pinned layout, `inner=H(u256(1)||u256(254))` and `record=H(u256(blockNumber)||inner)` under Solidity's mapping-key encoding; `blockHash` is storage word `record` and `stateRoot` is checked `record+1`. The layout hash below commits to that VERSION, base slot, field order and widths.

The byte-exact profile commitment is

```
legacyCampaignFenceDescriptorHash = H(
 "slot-chain-legacy-genesis-campaign-fence-v1" ||
 address20(router) || address20(legacyProxy) ||
 bytes4(0x718b2ac7) || bytes4(0x4c474331) || u16(512) || u32(100000) ||
 bytes4(0x9b698000) || bytes4(0x4c475331) || u16(512) || u32(100000) ||
 bytes4(0xef3cdce0) || bytes4(0x4c475353) || u16(608) || u32(100000))
risc0ResumeKeyPolicyHash = H(
 "slot-chain-legacy-genesis-risc0-key-policy-v1" ||
 risc0BlockImageId || risc0AggregationImageId)
sp1ResumeKeyPolicyHash = H(
 "slot-chain-legacy-genesis-sp1-key-policy-v1" ||
 sp1BlockProgramVKey || sp1AggregationProgramVKey)
risc0RethVerifierDescriptorHash = H(
 "slot-chain-legacy-genesis-risc0-verifier-v1" ||
 address20(risc0Adapter) || risc0AdapterRuntimeHash || u64(l2ChainId) ||
 address20(risc0RemoteVerifier) || risc0RemoteVerifierRuntimeHash ||
 risc0ResumeKeyPolicyHash)
sp1RethVerifierDescriptorHash = H(
 "slot-chain-legacy-genesis-sp1-verifier-v1" ||
 address20(sp1Adapter) || sp1AdapterRuntimeHash || u64(l2ChainId) ||
 address20(sp1RemoteVerifier) || sp1RemoteVerifierRuntimeHash ||
 sp1ResumeKeyPolicyHash)
```

```
proofVerifierGraphHash = H(
 "slot-chain-legacy-genesis-proof-verifier-graph-v1" ||
 address20(resumePairRoot) || resumePairRootRuntimeHash || u8(2) ||
 u8(5) || address20(risc0Adapter) || risc0RethVerifierDescriptorHash ||
 u8(6) || address20(sp1Adapter) || sp1RethVerifierDescriptorHash)
legacyResumeVerifierRouteHash = H(
 "slot-chain-legacy-genesis-resume-verifier-route-v1" ||
 proofVerifierGraphHash || u8(2) ||
 u8(5) || risc0RethVerifierDescriptorHash || risc0ResumeKeyPolicyHash ||
 u8(6) || sp1RethVerifierDescriptorHash || sp1ResumeKeyPolicyHash)
proposerCheckerDescriptorHash = H(
 "slot-chain-legacy-genesis-proposer-checker-v1" ||
 address20(proposerCheckerProxy) || proposerCheckerProxyRuntimeHash ||
 address20(proposerCheckerImplementation) ||
 proposerCheckerImplementationRuntimeHash || u16(64) ||
 legacyCampaignFenceDescriptorHash)
proverWhitelistDescriptorHash = H(
 "slot-chain-legacy-genesis-public-proving-v1" || address20(0) || u8(1))
checkpointStorageLayoutHash = H(
 "slot-chain-legacy-genesis-checkpoint-layout-v1" ||
 u256(1) || u16(254) ||
 H("CheckpointRecord(bytes32 blockHash,bytes32 stateRoot)"))
signalServiceCheckpointDescriptorHash = H(
 "slot-chain-legacy-genesis-signal-service-checkpoint-v1" ||
 address20(signalServiceProxy) || signalServiceProxyRuntimeHash ||
 address20(signalServiceImplementation) ||
 signalServiceImplementationRuntimeHash || u256(signalServiceVersion) ||
 address20(signalServiceAuthorizedSyncer) ||
 address20(remoteSignalService) || address20(signalServicePauser) ||
 checkpointStorageLayoutHash || legacyCampaignFenceDescriptorHash)
legacyResumeTimePolicyHash = H(
 "LegacyGenesisResumeTimePolicyV2(sameProofPreserved=false,"
 "ageIndependentRouteRequired=true,publicProvingRequired=true,minBond=0,"
 "livenessBond=0,noContest=true,forcedDueOnlyStrengthens=true,"
 "withdrawalOnlyMatures=true)")
legacyResumeProfileHash = H(
 "slot-chain-legacy-genesis-resume-profile-v2" ||
 legacyDeploymentHash || legacyCampaignFenceDescriptorHash ||
 proofVerifierGraphHash ||
 legacyResumeVerifierRouteHash || signalServiceCheckpointDescriptorHash ||
 proposerCheckerDescriptorHash || proverWhitelistDescriptorHash ||
 u64(minBond) || u64(livenessBond) || u64(withdrawalDelay) ||
 u64(provingWindow) || u64(permissionlessProvingDelay) ||
 u64(maxProofSubmissionDelay) || u48(ringBufferSize) ||
 u16(forcedInclusionDelay) || u64(forcedInclusionFeeInGwei) ||
 u64(forcedInclusionFeeDoubleThreshold) ||
 u16(permissionlessInclusionMultiplier) ||
 u16(maxForcedInclusionsPerProposal) ||
```

```
 u16(maxNormalBlobHashesPerProposal) ||
 u64(legacyBlobRetentionSeconds) ||
 u64(legacyResumeProofGenerationMaxSeconds) || legacyResumeTimePolicyHash)
```

Configuration reads use exactly 100,000 gas each and these canonical ABIs:

```
legacyResumeVerifierConfigV1() // selector 0x7e52cacd; 160 bytes
 returns (bytes4 magic,address risc0Adapter,address sp1Adapter,
          bytes32 risc0KeyPolicyHash,bytes32 sp1KeyPolicyHash) // LRV1
legacyResumeRisc0ConfigV1()    // selector 0xe516c43e; 192 bytes
 returns (bytes4 magic,uint64 l2ChainId,address remoteVerifier,
          bytes32 remoteRuntimeHash,bytes32 blockImageId,
          bytes32 aggregationImageId)                          // LR01
legacyResumeSp1ConfigV1()      // selector 0xe2b5d958; 192 bytes
 returns (bytes4 magic,uint64 l2ChainId,address remoteVerifier,
          bytes32 remoteRuntimeHash,bytes32 blockProgramVKey,
          bytes32 aggregationProgramVKey)                     // LSP1
legacyCheckpointConfigV1()     // selector 0x68011023; 192 bytes
 returns (bytes4 magic,address authorizedSyncer,
          address remoteSignalService,address pauser,
          bytes32 checkpointStorageLayoutHash,
          bytes32 campaignFenceDescriptorHash)                // LCK1
```

The four magics are respectively `0x4c525631`, `0x4c523031`, `0x4c535031` and `0x4c434b31`; every narrow/address/magic word has canonical left padding. The proposer checker uses exact 32-byte reads `impl()` `0x8abf6077`, `operatorCount()` `0x7c6f3158`, `getOperatorForCurrentEpoch()` `0x343f0a68` and `getOperatorForNextEpoch()` `0x72a8a551`, each with the same stipend. The SignalService proxy also returns its implementation from `impl()` and `VERSION()` `0xffa1ad74` as exact 32-byte words. Every returned field must reproduce the descriptor preimages above; revert, OOG, short/trailing data, wrong magic, padding, code hash, address or key rejects before publication or state change.

These are the complete descriptor grammars; no caller may substitute an opaque component hash. The root and both adapters are direct non-proxies. Its proof bytes are the canonical ABI encoding of exactly two `SubProof(uint8verifierId,bytesproof)` rows in IDs 5,6 order; minimal offsets, zero padding and no trailing bytes are required. The fixed RISC0 row is canonical `abi.encode(bytesseal,bytes32blockImageId,bytes32aggregationImageId)`. The fixed SP1 row is exactly `aggregationProgramVKey||blockProgramVKey||opaqueProof`, with a nonempty opaque suffix. Both adapters recompute the existing Taiko public input using their own bound address and chain ID before the remote STATICCALL. Each remote verifier is also direct, stateless, ownerless and non-delegating, contains no reachable `SSTORE/SELFDESTRUCT` or configuration mutator, and has the exact runtime hash in its leaf descriptor; any proxy, gateway with mutable dispatch or unlisted child makes the profile ineligible. The Inbox's `proverWhitelist` address must be exactly zero—a mutable zero-count whitelist is no longer accepted. The proposer checker and legacy SignalService are the only descriptor leaves allowed one UUPS hop, in addition to the source Inbox proxy itself. Their reviewed implementations apply the common fence to every operator/ejecter/configuration/ownership-affecting mutator and upgrade; at each profile checkpoint, `operatorCount` must be in 1–64 and both current- and next-epoch operator getters must return nonzero addresses.

The common fence makes exact 100,000-gas STATICCALLs to LGC1, LGS1 and LGSS and accepts only their specified lengths/magics/canonical padding. Every fenced mutation rejects while the Router retains an unexpired SCHEDULED campaign or while local LGS1 is not clean ACTIVE with EMPTY LGSS and zero cutover words. EXPIRED Router state alone is insufficient until permissionless local cleanup completes. Revert, OOG, short/trailing return or inconsistent Router/ local campaign fails closed. The compatibility Inbox, proposer checker and direct SignalService implementation use this identical descriptor; their runtime hashes prove that every authority-bearing mutator invokes it. Thus a separately worded or self-reported fence hash is never accepted.

The fixed-key route adapters and remote leaves have no mutation authority; the three campaign-aware UUPS implementations may change only outside the common fence. A proxy implementation, key, child pointer or configuration that an independent administrator can change during the fence is unsupported. The supported deployment profile additionally requires the constructor/profile values 0, 0, 604,800, 14,400, 432,000, 180, 21,600, 576, 1,000,000, 50, 160, 10, 21, 1,572,864 and 900 in the field order above. The intervening IInbox.Config-only `basefeeSharingPctg` is exactly 75; its five address words equal the descriptor leaves named above, including the zero prover whitelist, and its bond token is the reviewed deployed token. Thus the exact 17-word getter, not a list of assumed constructor inputs, determines `legacyInboxConfigurationHash`. The two source-shape bounds pin the deployed Inbox's ten-forced-source constant and the current Fulu mainnet maximum of 21 normal blob hashes. The retention value is a conservative protocol dependency, not an assertion that every archival provider retains data that long; publication must be disabled before a canonical L1 fork shortens that minimum or raises the per-block blob maximum above 21. Any other profile receives a different reviewed commitment and must independently prove resume safety.

The review artifact is byte-bound rather than referenced by prose:

```
reviewCommitment = H("slot-chain-legacy-genesis-review-v1" ||
  legacyDeploymentHash || legacyResumeProfileHash ||
  u64(targetProtocolVersion) || targetManifestHash ||
  targetRegistrationHash)
```

ProtocolVersionManager recomputes this exact value from live proxy/component descriptors and the delayed target tuple in the publication transaction; a caller-supplied review hash is never authoritative. The review is reusable across timing-equivalent campaign retries, while the separately computed `campaignId` binds nonce, generation and every campaign deadline.

The proposal preimages required by the scan are canonical `abi.encode(Proposal)` values. The `Proposed` log supplies ID, proposer, parent proposal hash, submission-window end, basefee share and every source; its containing canonical L1 header supplies the proposal timestamp, and that header's canonical parent supplies `originBlockNumber=eventBlockNumber-1` and `originBlockHash`. The stored ring hash authenticates the reconstructed whole, so the compatibility contract need not trust a caller's archive or use the 256-block `BLOCKHASH` window. Forced rows are read from proxy storage. No builder or proposer secret is required. Their availability is the same canonical-L1- log/header archive assumption already required to reconstruct legacy proofs. A missing canonical log/header copy stops the campaign before QUIESCENT; it cannot strand the proxy. The final scan and activation receipt separately bind proposal scan start, actual abandoned proposal start `lastFinalizedProposalId+1`, proposal end, both counts, forced start/end/count, both ordered roots, aggregate abandoned native wei, zero abandoned bond liability, minimum data expiry and campaign ID. A proposal final-

ized after scan begin remains in the conservative scan root but lies before the receipt's actual abandoned start and is not reported lost. Each aggregate is revalidated when QUIESCENT is entered and again before LGAR. Exceeding any hard count, byte, call or time cap rejects the campaign rather than silently weakening the permissionlessness claim. `abandonedNativeWei` sums only the exact stored fees of pending forced rows. Raw proxy balance and forced ETH are not a campaign precondition: the review runbook records an informational balance observation, but no raw-balance word is accepted by or bound into activation, and all unaccounted value remains locked. Therefore an unsolicited `SELFDESTRUCT` transfer cannot veto migration or invalidate a prepared proof. There is intentionally no protocol cap on the accounted forced fees that governance may knowingly abandon; the fixed forced-row count bounds computation, while the final review/runbook must disclose the exact amount before quiescence.

The exact campaign and scan commitments are

```
campaignId = H("slot-chain-legacy-genesis-campaign-v1" ||
  legacyDeploymentHash || u64(nonce) || u64(generation) ||
  u64(forceCutoffBlock) || u64(proposalCutoffBlock) ||
  u64(quiesceNotBeforeBlock) || u64(resumeByBlock) ||
  u64(resumeByTimestamp) || u64(reviewFinalizedByBlock) ||
  address20(targetSettlement) || u64(targetProtocolVersion) ||
  targetManifestHash || targetRegistrationHash || reviewCommitment)
scanCommitment = H("slot-chain-legacy-genesis-scan-v1" || campaignId ||
  u48(proposalScanStart) || u48(nextProposalId) || u16(proposalCount) ||
  u32(proposalBytes) || proposalRowsRoot || u48(forcedHead) ||
  u48(forcedTail) || u16(forcedCount) || u32(forcedBytes) ||
  forcedRowsRoot || u256(abandonedNativeWei) ||
  u64(minDataExpiry) || legacyResumeProfileHash)
proposalRowsRoot[0] = H("slot-chain-legacy-genesis-proposal-rows-empty-v1")
proposalRowsRoot[i+1] = H("slot-chain-legacy-genesis-proposal-row-v1" ||
  proposalRowsRoot[i] || u48(proposalId) || u32(encodedBytes) ||
  storedProposalHash || u64(dataExpiry) || u256(0))
forcedRowsRoot[0] = H("slot-chain-legacy-genesis-forced-rows-empty-v1")
forcedRowHash = H("slot-chain-legacy-genesis-forced-record-v1" ||
                  canonicalAbiEncodedForcedInclusion)
forcedRowsRoot[i+1] = H("slot-chain-legacy-genesis-forced-row-v1" ||
  forcedRowsRoot[i] || u48(forcedIndex) || u32(encodedBytes) ||
  forcedRowHash || u64(dataExpiry) || u256(feeInGwei * 1 gwei))
```

The supplied proposal preimage is canonically re-encoded and its existing legacy hash must equal `storedProposalHash`; the forced record is read from storage and canonically ABI-encoded. Checked additions derive row counts, encoded byte totals and abandoned native value. The row domains and empty roots are published as golden vectors; implementations must reproduce them from the exact legacy proposal and forced-record codecs.

Those codecs are the deployed Solidity structs, not opaque bytes:

```
BlobSlice = (bytes32[] blobHashes,uint24 offset,uint48 timestamp)
DerivationSource = (bool isForcedInclusion,BlobSlice blobSlice)
Proposal = (uint48 id,uint48 timestamp,
 uint48 endOfSubmissionWindowTimestamp,address proposer,
```

```
 bytes32 parentProposalHash,uint48 originBlockNumber,
 bytes32 originBlockHash,uint8 basefeeSharingPctg,
 DerivationSource[] sources)
ForcedInclusion = (uint64 feeInGwei,BlobSlice blobSlice)
proposalPreimage = abi.encode(Proposal)
forcedPreimage = abi.encode(ForcedInclusion)
```

Because each top-level struct is dynamic, the first ABI word is the canonical `0x20` tuple offset. Every nested offset is minimal, 32-byte aligned and relative to its ABI-defined container; tails are monotone and contiguous, all narrow words/bools/addresses have canonical padding, and trailing bytes reject. A deployed proposal has 1–11 sources: zero to ten leading forced sources, each with `isForcedInclusion=true` and exactly one nonzero blob hash, followed by exactly one normal source with `isForcedInclusion=false` and 1–21 nonzero hashes. A stored ForcedInclusion has exactly one nonzero blob hash. Each `BlobSlice.offset` is uint24-canonical and every timestamp is nonzero uint48. The compatibility implementation decodes, validates this shape, re-encodes and only then hashes; `encodedBytes` is the length of that unique encoding. The profile maxima make the publication-time row/byte bound valid; another legacy shape or L1 blob schedule requires a separately reviewed profile.

The resulting profile bounds are exact:

```
MAX_PROPOSAL_ROW_BYTES = 3,808
MAX_FORCED_ROW_BYTES = 256
MAX_PROPOSAL_BATCH_ROWS = 16
MAX_PROPOSAL_BATCH_ROW_BYTES = 60,928
MAX_PROPOSAL_SCAN_CALLDATA_BYTES = 62,084
MAX_PROFILE_TOTAL_SCAN_BYTES = 1,024 * (3,808 + 256) = 4,161,536
SCAN_BYTES_HARD_CAP = 4,194,304
GENESIS_SCAN_BATCH_GAS_RELEASE_CAP = 12,000,000
```

The calldata maximum includes selector, the three-word argument head, `bytes[]` length/offsets and each element's length word. Publication relies only on rows created by the reviewed deployed Inbox while the canonical L1 maximum was at most 21; from campaign publication, compatibility admission also rejects any source count/hash count or encoded proposal length above this profile before mutation. A decoded historical row outside the profile rejects its scan and therefore cannot enter QUIESCENT; it is evidence that this in-place profile is unsupported, not permission to use a smaller batch. The production Foundry suite must execute a 16-row maximum-shaped proposal batch from cold and warm storage, including exact LGC1/profile reads, strict decode/re-encode, roots, expiry and LGSS writes, below the 12,000,000-gas release cap. Failure blocks this profile before any campaign is published. Thus deterministic maximum-progress batches cannot concentrate the aggregate 4 MiB allowance into an unmineable transaction.

The GENESIS_IMPORT receipt's nonzero transition auxiliary is

```
legacyGenesisAbandonmentReceiptHash =
 H("slot-chain-legacy-genesis-abandonment-receipt-v1" ||
   campaignId || scanCommitment ||
   u48(proposalScanStart) || u48(abandonedProposalStart) ||
   u48(proposalEnd) || u16(scannedProposalCount) ||
```

```
u16(abandonedProposalCount) || u32(proposalBytes) ||
proposalRowsRoot || u48(forcedStart) || u48(forcedEnd) ||
u16(forcedCount) || u32(forcedBytes) || forcedRowsRoot ||
u256(abandonedNativeWei) || u256(abandonedBondLiabilityWei) ||
u64(minDataExpiry) || legacyResumeProfileHash)
```

Here `proposalEnd` is the scan's fixed proposal endpoint. The activation transaction samples the abandonment start and derives both counts as follows:

```
abandonedProposalStart = lastFinalizedProposalId + 1
scannedProposalCount = proposalEnd - proposalScanStart
abandonedProposalCount = proposalEnd - abandonedProposalStart
```

Every addition and subtraction is checked. The Router emits the following exact event in the same rollback domain as the sealed receipt:

```
event LegacyGenesisAbandonmentSealed(
 bytes32 indexed receiptId, bytes32 indexed transitionAuxiliaryHash,
 bytes32 campaignId, bytes32 scanCommitment,
 uint48 proposalScanStart, uint48 abandonedProposalStart,
 uint48 proposalEnd, uint16 scannedProposalCount,
 uint16 abandonedProposalCount, uint32 proposalBytes,
 bytes32 proposalRowsRoot, uint48 forcedStart, uint48 forcedEnd,
 uint16 forcedCount, uint32 forcedBytes, bytes32 forcedRowsRoot,
 uint256 abandonedNativeWei, uint256 abandonedBondLiabilityWei,
 uint64 minDataExpiry, bytes32 legacyResumeProfileHash);
```

The event fields must recompute the auxiliary hash embedded in the immutable ARV1 receipt; the zero bond-liability word is checked, not conventional. The raw proxy balance is deliberately absent because anyone can change it without calling the proxy. A missing/mismatched event or auxiliary write reverts before ACTIVE, and a VERSION_MIGRATION neither emits this event nor accepts a nonzero auxiliary.

At or after `quiesceNotBeforeBlock`, before either expiry bound, any caller may enter QUIESCENT only after these checks. No target proof is required for that reversible step.

Campaign publication first uses the exact read-only preflight

```
legacyGenesisPreparationV1() returns
 (bytes4 magic,uint48 nextProposalId,uint48 lastFinalizedProposalId,
  uint48 forcedHead,uint48 forcedTail,
  uint16 unfinalizedProposalCount,uint16 pendingForcedCount,
  uint32 maximumScanBytes,bytes32 legacyResumeProfileHash)
```

Its selector is `0x6880cd05`, magic is LGPR `0x4c475052`, and return length is exactly 288 bytes. Counts are checked differences of the preceding endpoints, `maximumScanBytes` is their checked product/sum under the exact profile row maxima, and the profile hash is recomputed from live code/config; none is a cached or caller-supplied readiness assertion.

The permissionless scan/quiescence surface is exactly

```
beginLegacyGenesisScanV1(uint64 generation,bytes32 campaignId)
```

```
 returns (bytes4 magic,uint48 proposalStart,uint48 proposalEnd,
          uint48 forcedStart,uint48 forcedEnd)
scanLegacyGenesisProposalsV1(
 uint64 generation,bytes32 campaignId,bytes[] proposalPreimages)
 returns (bytes4 magic,uint48 nextCursor,bytes32 rowsRoot,
          uint32 bytesScanned,uint64 minDataExpiry)
scanLegacyGenesisForcedV1(
 uint64 generation,bytes32 campaignId,uint16 maxRows)
 returns (bytes4 magic,uint48 nextCursor,bytes32 rowsRoot,
          uint32 bytesScanned,uint256 abandonedNativeWei,
          uint64 minDataExpiry)
legacyGenesisScanStateV1() returns
 (bytes4 magic,uint64 generation,bytes32 campaignId,
  uint48 proposalStart,uint48 proposalCursor,uint48 proposalEnd,
  uint16 proposalCount,uint32 proposalBytes,bytes32 proposalRoot,
  uint48 forcedStart,uint48 forcedCursor,uint48 forcedEnd,
  uint16 forcedCount,uint32 forcedBytes,bytes32 forcedRoot,
  uint256 abandonedNativeWei,uint64 minDataExpiry,
  bytes32 legacyResumeProfileHash,uint8 scanPhase)
enterLegacyGenesisQuiescenceV1(uint64 generation,bytes32 campaignId)
 returns (bytes4 magic,uint64 generation,bytes32 scanCommitment)
resumeLegacyGenesisV1(uint64 generation,bytes32 campaignId)
 returns (bytes4 magic,uint64 generation,bytes32 campaignId)
```

Their selectors are respectively `0xe9d1a07f`, `0x7da66460`, `0x032ae99e`, `0xef3cdce0`, `0x4c0ae8da` and `0x2bf6b656`. Magics are LGBS `0x4c474253`, LGSP `0x4c475350`, LGSF `0x4c475346`, LGSS `0x4c475353`, LGQS `0x4c475153` and LGRS `0x4c475253`. Fixed returns are exactly 160, 160, 192, 608, 96 and 96 bytes in that order. Scan phase is EMPTY=0, SCANNING=1 or COMPLETE=2. Begin is one-shot per campaign and snapshots the post-proposal-cutoff endpoints. Proposal input contains exactly `min(16,proposalEnd-proposalCursor)` consecutive canonical `abi.encode(Proposal)` preimages; zero remaining rows rejects. The dynamic array uses minimal monotone offsets, contiguous tails, zero right-padding and no trailing bytes. Forced scan requires `maxRows=min(16,forcedEnd-forcedCursor)>0` and reads, rather than accepts, each stored row. Thus a front-run may invalidate a stale transaction but must advance the shared cursor by the same deterministic batch size. The returned cursor/root/count/byte/value/expiry fields must equal LGSS after the call. COMPLETE requires both cursors equal their ends and exact checked count differences; no caller supplies an aggregate.

All six calls are nonpayable, only-proxy and non-reentrant but otherwise permissionless. Direct implementation calls reject. Each begins with a 100,000-gas exact LGC1 STATICCALL; profile/configuration reads at begin, quiescence and resume use their separately published fixed stipends. A wrong stage, endpoint, row index, preimage hash, offset, length, padding, campaign, profile, expiry or return value reverts the entire batch. No external call follows a scan-state write. Explicit resume accepts the exact retained campaign in SCHEDULED after either clock bound or in EXPIRED. It accepts either QUIESCENT with that campaign or local ACTIVE with that campaign's nonempty SCANNING/COMPLETE scan state, then clears every scan/campaign/cutover word and returns or retains ACTIVE without executing a proposal, proof or other business rule. ACTIVE with no matching local campaign state is

already clear; a mismatched campaign, DRAINING, READY or FROZEN rejects. Resume does not call or mutate Router. An ordinary legacy mutator performs the same local clear before continuing, then still exact-reads LGC1 under ACTIVE. Separately, anyone may call Router `expireLegacyGenesisCampaignV1(uint64,bytes32)` (selector `0xa4a37936`) after either bound; it returns the sole exact LGEX word `0x4c474558` and marks the Router campaign expired/tombstoned. Local resume and Router expiry commute, so neither is a reentrant callback and neither is required to precede the other.

The exact compatibility interface is

```
legacyGenesisStateV1() returns
 (bytes4 magic,uint8 phase,uint64 generation,bytes32 campaignId,
  bytes32 scanCommitment,uint64 targetProtocolVersion,
  bytes32 targetManifestHash,bytes32 targetRegistrationHash,bytes32 armId,
  bytes32 launchId,uint48 nextProposalId,uint48 lastFinalizedProposalId,
  bytes32 lastFinalizedBlockHash,uint48 forcedHead,uint48 forcedTail,
  bytes32 poststateCommitment)
checkpointLegacyGenesisV1(uint64 generation,bytes32 campaignId)
 returns (bytes4 magic,uint64 generation,bytes32 campaignId,bytes32 armId)
finalizeLegacyGenesisV1(uint64 generation,uint64 targetProtocolVersion,
  bytes32 targetManifestHash,bytes32 targetRegistrationHash,
  bytes32 candidateDigest,bytes32 outputCoreHash)
 returns (bytes4 magic,bytes32 launchId,bytes32 poststateCommitment)
```

The selectors are respectively `0x9b698000`, `0x8781a058` and `0xc2de6417`. State return is exactly 512 bytes; LGAR returns exactly 128 bytes and LGFN exactly 96 bytes. Magics are LGS1 `0x4c475331`, LGAR `0x4c474152` and LGFN `0x4c47464e`. Calls are nonpayable, Router-only, only-proxy, non-reentrant and fixed-gas. All narrow/address/magic padding is canonical; short, trailing, malformed or arbitrary bubbled returndata rejects.

The local phase numbers remain ACTIVE=0, DRAINING=1, QUIESCENT=2, READY=3 and FROZEN=4 for storage/ABI compatibility. DRAINING is unreachable and fails closed. QUIESCENT is a bounded, reversible campaign fence; READY exists only inside a successful activation transaction; FROZEN is permanent. The default-zero cutover region needs no initializer. Entering QUIESCENT records the exact monotone generation, campaign ID, completed scan commitment and an O(1) snapshot of `nextProposalId`, `lastFinalizedProposalId`, `lastFinalizedBlockHash`, `forcedHead` and `forcedTail`; every target, arm, launch and poststate word remains zero. While QUIESCENT all five boundary-changing selectors reject. At or after either campaign expiry bound, any caller may resume ACTIVE, and the first ordinary mutator performs that same resume before its normal checks. The resume tombstones the attempt, clears every cutover and scan word, changes no legacy core, proposal hash, queue cursor, bond or claim, and cannot run after a successful freeze.

During atomic activation LGAR authenticates Router ACTIVATING/MACT and the exact live campaign, moves QUIESCENT to READY, retains the scan and boundary, computes the campaign-bound arm ID, and returns it. Router exact-staticreads LGS1 before calling LGFN in the same transaction. LGFN writes the exact target, launch ID, FROZEN and poststate. A fault in LGAR, LGS1, LGFN or any later adoption write reverts to the pre-existing QUIESCENT snapshot; no canonical transaction can leave READY. Expiry then still restores ACTIVE if no later retry succeeds.

The snapshot may contain unfinalized proposal IDs and a nonempty native forced range.

Successful atomic activation intentionally imports only the last finalized header/checkpoint. The proposal suffix `[lastFinalizedProposalId+1,nextProposalId)` and every pending or already consumed forced input reachable only from that unfinalized suffix are non-canonical and permanently abandoned when the proxy enters FROZEN. The legacy head/tail and proposal slots are not rewritten, no refund owner is invented, forced fees and raw Inbox ETH remain forever in the legacy proxy, and no owner or governance sweep exists. This loss is an explicit genesis migration trade-off forced by the deployed record format, which stores neither a refund owner nor durable blob bytes. A deployment requiring lossless treatment of those records cannot use in-place GENESIS_IMPORT and must use a separately initialized state migration.

FROZEN permits only permanent reads and the existing legacy bond request/cancel/withdrawal paths. QUIESCENT admits reads, those same historical bond paths, campaign resume and the Router's LGAR; transaction-local READY adds only Router LGFN. Every still-escrowed legacy bond becomes requestable under its existing withdrawal delay without a runtime minimum-bond floor and without slash or relabeling. Phase 1, any change to the QUIESCENT/READY snapshot, or any stale/nonzero field outside the phase matrix rejects.

The fresh depth-64 ForcedQueue is deployed empty with root/count/cursor golden values, `dueAt(0)=UINT64_MAX`, zero liability and this exact legacy proxy as its initial active authority. Genesis QMIG therefore has a real expected old authority and atomically changes it to the target Settlement with `start=end=count=creditedWei=0`; no zero-address authority convention is accepted. The deployment configuration is not an opaque hash. Router/PVM call `getConfig()` with selector `0xc3f909d4` and exactly 100,000 gas. The return is exactly 544 bytes—the 17 static IInbox.Config words in this order—with canonical high padding:

```
(address proofVerifier,address proposerChecker,address proverWhitelist,
 address signalService,address bondToken,uint64 minBond,uint64 livenessBond,
 uint48 withdrawalDelay,uint48 provingWindow,
 uint48 permissionlessProvingDelay,uint48 maxProofSubmissionDelay,
 uint48 ringBufferSize,uint8 basefeeSharingPctg,
 uint16 forcedInclusionDelay,uint64 forcedInclusionFeeInGwei,
 uint64 forcedInclusionFeeDoubleThreshold,
 uint8 permissionlessInclusionMultiplier)
legacyInboxConfigurationHash = H(
 "slot-chain-legacy-genesis-inbox-config-v1" ||
 address20(proofVerifier) || address20(proposerChecker) ||
 address20(proverWhitelist) || address20(signalService) ||
 address20(bondToken) || u64(minBond) || u64(livenessBond) ||
 u48(withdrawalDelay) || u48(provingWindow) ||
 u48(permissionlessProvingDelay) || u48(maxProofSubmissionDelay) ||
 u48(ringBufferSize) || u8(basefeeSharingPctg) ||
 u16(forcedInclusionDelay) || u64(forcedInclusionFeeInGwei) ||
 u64(forcedInclusionFeeDoubleThreshold) ||
 u8(permissionlessInclusionMultiplier))
```

The proxy's `impl()` selector `0x8abf6077` returns exactly the implementation address as one canonical word under the same stipend. PVM checks `EXTCODEHASH` of both proxy and implementation. Storage-layout compatibility is a reviewed property of that one pinned implementation runtime against the deployed proxy layout; there is no self-reported layout hash

or second implementation accepted by consensus. The release artifact publishes the compiler storage-layout diff, and any bytecode change requires a new reviewed campaign.

The fixed-width commitments are

```
legacyDeploymentHash =
 H("slot-chain-legacy-genesis-deployment-v1" || u256(settlementChainId) ||
   address20(proxy) || proxyRuntimeHash || address20(implementation) ||
   implementationRuntimeHash || legacyInboxConfigurationHash ||
   address20(router))
boundaryHash = H("slot-chain-legacy-genesis-boundary-v1" ||
  u48(nextProposalId) || u48(lastFinalizedProposalId) ||
  lastFinalizedBlockHash || u48(forcedHead) || u48(forcedTail))
armId = H("slot-chain-legacy-genesis-arm-v1" || legacyDeploymentHash ||
          u64(generation) || campaignId || scanCommitment || boundaryHash)
launchId = H("slot-chain-legacy-genesis-launch-v1" || armId ||
             u64(targetProtocolVersion) || targetManifestHash ||
             targetRegistrationHash)
legacyPostStateCommitment =
 H("slot-chain-legacy-genesis-poststate-v1" || launchId ||
   candidateDigest || outputCoreHash || u64(0) || u64(block.number) ||
   u8(FROZEN=4) || boundaryHash)
```

For QUIESCENT/READY/FROZEN, `nextProposalId>lastFinalizedProposalId`, the finalized hash is nonzero and `forcedHead<=forcedTail`; equality is not required for either pair. ACTIVE requires generation, target tuple, both IDs and poststate all zero. QUIESCENT requires nonzero generation, campaign and scan commitments but zero target tuple, arm, launch and poststate. READY adds the recomputed arm ID but retains the zero target tuple, launch and poststate. FROZEN requires a nonzero target tuple, the same campaign, scan and arm ID, the target-derived launch ID and recomputed poststate. Phase 1 and every stale/nonzero field outside this matrix reject. The delayed campaign, scan, target registration and migration public statement bind `legacyDeploymentHash`, `armId` and `launchId`; because the arm recomputes the campaign and scan commitments, it transitively binds every stage, reviewed target and abandoned-suffix boundary. The activation context binds that statement, proxy and target registration; the receipt binds the returned legacy poststate. None is accepted from unverified caller fields. Existing Taiko bridge messages remain on the legacy SignalService/Bridge retry path and are not silently converted into kind 1. Only new, explicitly escrowed inbox-credit messages use `BridgeInboxAdapter`; its direct-read descriptor binds the existing message hash, both domains, source epoch, permanent Bridge/execution identity, source emission block, enqueue deadline, sender, fee, ownership/value and derived calldata hash/length fields, while destination invocation remains a later bridge operation.

Before the delayed arm, the legacy L2 governance path deploys the profile-exact protocol-lifetime non-proxy InboxApplyRouterV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2, then one fresh endpoint InboxCreditStoreV2, DestinationBridgeV2 and Bridge-unique QuotaManager. The global router's constructor fixes only the permanent system sender, registrar and namespace. The store fixes that router, fresh Bridge and registrar, but its local-domain slot remains zero; Bridge `v2Active`, private mappings, liability and nonce remain zero and its quota begins full. All component addresses are precomputed from the deployment coordinator's fixed CREATE nonce sequence

recorded in the manifest, so constructor references do not require an address/init-code fixed point. It also installs any profile-required AnchorV4 runtime. Legacy SignalService and the entire V1 Bridge/Vault graph remain untouched. The L1 side permissionlessly deploys the manifest-pinned fresh SourceBridgeV2/BridgeCreditRegistryV2/ QuotaManager bundle and BridgeInboxAdapter through the exact CREATE2 factory; the release descriptor references and exact-checks the already deployed root-lifetime TerminalSignalVerifier. ProtocolVersionManager directly checks the deployed L1 components 1–3 and their constructor configuration, records those exact descriptor rows in the finalized manifest, computes the acyclic combined infrastructure/domain descriptor, seals the L1 adapter's domain once and burns that setter. The L2 release proof carries those same rows; its registrar directly checks local components 4–10 and recomputes the identical combined hash. Every V2 account is immutable and non-proxy. InboxApplyRouterV2, the Store and every L2 inbound-V2 process/retry/fail/expire path require the fresh Store/Bridge one-shot seal; only proved tx0 may set it. No separate activation gate exists. This is an ordinary proved L2 state transition; no L1 transaction is claimed to mutate an imported L2 state root. Its transaction bytes, receipts and resulting account/storage values are golden profile vectors.

The delayed genesis campaign publishes one exact target tuple and final review commitment before either admission cutoff. It never moves the public gate away from the constructor-bound legacy ACTIVE branch. Once the compatibility proxy is QUIESCENT and before either expiry bound, any caller sends the canonical RLP of the stable last-finalized L2 header, the complete migration proof and the fixed-depth MPT account/storage proof witness under its state root for the exact AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2, NativeLiquidityPoolV2 and the fresh Store/QuotaManager/ DestinationBridge runtime code hashes; exact non-proxy configuration/layout and the absence of any reachable upgrade/delegate target; InboxApplyRouterV2 permanent sender/registrar/namespace, `nextQueueIndex=0` and empty routes; inbox-store router/Bridge/registrar immutables, zero domain and empty mappings; empty release-authority/registrar records; accumulator empty domain writer, count/root/frontier; Pool zero tickets/accounted balance and no active authorization, with any forced ETH classified only as surplus; fresh Bridge zero nonce/liability/private mappings and `v2Active=false`; full quota; and arbitrary authenticated Bridge surplus. The proof follows the exact immutable account and constructor-bound reference topology encoded in the profile and rejects an unlisted hop, delegate target or mutable upgrade path. Every runtime and authenticated value must equal the immutable execution profile. These values are witness to the one immutable migration-transition verifier, not the output of a separate adapter or a stored Boolean. While Router is IDLE, the public gate is legacy ACTIVE and the compatibility proxy is locally QUIESCENT. Activation validates the exact live campaign and expiry bounds, writes ACTIVATING, exact-reads LGS1, verifies the complete typed statement and requires the header hash to equal the stable legacy last-finalized hash, `number<UINT48_MAX`, and the legacy SignalService checkpoint at that exact number to equal its block hash and state root. Its typed public statement binds the header/state/checkpoint, deployment commitment, campaign/scan-bound arm ID and target-derived launch ID. The LGS1 proposal/ forced snapshot and boundary cannot change after quiescence. A substituted campaign, scan, boundary, target or launch ID rejects before LGAR. No second verifier, behavioral getter or caller-asserted readiness Boolean is accepted.

From that authenticated header and frozen execution profile, while the old Inbox remains QUIESCENT and the Queue authority, the immutable migration circuit constructs the target

proof's base fields exactly:

```
l2BlockNumber       = header.number
tipHash             = H(RLP(header))
tipSlot             = header.timestamp - GENESIS_TIMESTAMP
stateRoot           = header.stateRoot
messageCursor       = 0
winningDataCommitment = H("slot-chain-migration-data-v2" ||
                          u256(settlementChainId) || u256(l2ChainId) ||
                          tipHash || stateRoot || terminalRoot ||
                          u64(terminalCount))
nextBaseFee         = profile.nextBaseFee(header)
nextExcessBlobGas   = profile.nextExcessBlobGas(header)
terminalRoot        = authenticatedAccumulator.root
terminalCount       = authenticatedAccumulator.count
firstV2BlockNumber    = header.number + 1
```

It rejects a pre-genesis timestamp, checked-addition overflow, missing fork-required header fields or any profile mismatch. Before cutover, an untrusted prover supplies the exact migration-transition-verifier proof for the first V2 block from this base. The block number is `firstV2BlockNumber`, and the block must begin with the custom AnchorV4 activation transaction for the staged `(protocolVersion,releaseManifestHash)`. The circuit authenticates the staged manifest commitment under an already published L1 anchor, the exact imported parent, execution-profile hash, empty depth-64 queue and exact L2 component rows 4–10. Its top-level sender is the reserved Anchor-system sender. AnchorV4 first calls `ProtocolReleaseAuthorityV2.activateRelease`; the authority verifies both `msg.sender=manifest.anchorV4` and `tx.origin=reservedAnchorSystemSender`. AnchorV4 then calls `TerminalDomainRegistrarV2.activateDomain`; the latter authenticates the fresh Store/Quota/Bridge and Pool state, transfers no native liquidity, then atomically seals Store/Bridge, inbox route, Pool domain and accumulator registration before that block's InboxApplyRouterV2 batch. GENESIS_IMPORT has exactly zero Inbox rows: tx1 is the canonical empty batch with start=end=0, the statement `forceCutoff`, base/output message cursors, queue count and retirement count are all zero, and no preactivation ingress can create another value. The custom transaction type is invalid under the legacy fork and its reserved sender has no ordinary signing key, so neither a legacy block nor an EOA can preactivate the release. Any first block that omits, duplicates, calls authority directly or through Bridge, substitutes an Anchor, or mismatches this transition is unprovable. All calls occur in one EVM transaction, so any failure rolls release and registration back together. Subsequent blocks in that candidate require the exact release seal and cannot repeat activation.

The proof has no input derived from the future cutover transaction or its block hash. The L1 coordinator instead compares the proof's exact base, campaign generation/arm ID, scan, profile/manifest and queue snapshot with the exact QUIESCENT LGS1 and scheduled target immediately before adoption. The landing block remains outside the proof and enters only the activation context, poststate joins and receipt.

After internal campaign/calldata checks, Router writes ACTIVATING before its first external operation, exact-reads QUIESCENT LGS1 and all component state, reconstructs the statement, computes and stores the shared context, and verifies the profile-pinned proof by bounded STATICCALL. Only after verification does it call LGAR, exact-read the transaction-local

READY LGS1, call LGFN, call MCAN on the PREACTIVE target, call QMIG on the empty singleton Queue, perform SOURCE/TARGET/QUEUE MAPS postreads, consume the target registration, write the complete genesis receipt/index state, store `firstV2BlockNumber`, set the target/public gate ACTIVE, mark the campaign CONSUMED, clear context and write IDLE last. LGFN exact-staticreads MACT, rechecks its implementation, configuration, campaign, scan, stable READY snapshot, target-derived launch ID, boundary, candidate and output hash, then writes FROZEN/poststate without another external call. The source MAPS role is implemented by `LegacyGenesisCutoverInboxV1` and reproduces that LGFN commitment. MCAN installs the proved poststate as sequence zero with `canonicalizedAtBlock=lastCanonicalCommitBlock=block.number`. QMIG requires the legacy proxy as old authority and all empty-Queue golden values.

Failure of LGS1/verifier, LGAR, LGFN, MCAN, QMIG, a postread, decoder, receipt/index write or final switch reverts legacy FROZEN to its pre-existing QUIESCENT snapshot, restores the old Queue authority and removes all target/receipt/Router activation writes. It does not write L2 code or storage; it adopts the proved L2 poststate. If only the activation block is orphaned, all five objects roll back to the same QUIESCENT snapshot. Retry recomputes the same arm and launch IDs and may reuse the proof and candidate, but recomputes the landing-block context, legacy poststate, three MAPS commitments and receipt. If the QUIESCENT block is orphaned, its scan/fence and any proof from that orphan reject. An orphaned receipt cannot replay. FROZEN is permanent and cannot abort or upgrade. Proof failure, component mismatch or proving unavailability leaves the source QUIESCENT only until retry or hard campaign expiry, after which permissionless resume restores ACTIVE. A bad target requires expiry and a higher-nonce delayed campaign; it cannot be swapped after final review. Thus a total proving-system outage delays migration but does not permanently pause a previously processable legacy state. The campaign and operational runbook publish both admission cutoffs, exact abandoned-value aggregates and expiry bounds; production rehearsal must demonstrate two independent proving stacks and the complete resume path against the exact interface before the first cutoff. L1 kind-1 ingress is enabled only after the authenticated L2 code/authorization proofs and router-history authorization have all succeeded. That atomic cutover enables kind-0 ingress, but V2 DIRECT send and kind-1 ingress remain disabled. The old `sendMessage` selector and all L2-to-L1 sends retain exact V1 signals and events for every caller.

After activation, bounded `MptProofV1` registrar proofs remain available for destination/terminal history but do not gate the newly active source route. `sendMessageV2` and kind-1 ingress are enabled by the atomic combination of the already-reviewed `ARM_READY` package and the exact migration EVM proof of tx0 Registrar execution. No V2 Vault selector exists. Thus no source credit can target an unregistered, merely proposed or merely staged L2 endpoint.

Every later Settlement/profile migration uses a protocol-generated two-phase cutover. The delayed manifest arms a specific cutover generation and changes the gate from ACTIVE to ARMED. This is one protocol-lifetime gate word owned by the non-proxy `ActiveSettlementRouter`, not a caller-supplied quiescence Boolean or replaceable contract. Its exact state is

```
(uint64 generation, uint64 activeProtocolVersion,
 uint64 targetProtocolVersion, bytes32 targetManifestHash,
 bytes32 targetRegistrationHash, uint8 phase)
```

where phase is ACTIVE, ARMED or READY. The `activeSettlementStateV1()` getter above

returns the constructor-bound `LEGACY_BOOTSTRAP` proxy only while `genesisConsumed=false`; otherwise it derives `activeSettlement` from the append-only registration for `activeProtocolVersion`. There is no second mutable current-address slot. The legacy branch authenticates the exact deployment, campaign and scan commitments throughout QUIESCENT and activation, cannot appear in the ordinary registry, leaves the public gate ACTIVE during a genesis campaign and is consumed only by a successful atomic genesis switch. An expired genesis campaign is tombstoned and a later attempt uses a higher nonce; local resume clears QUIESCENT without changing the public gate. For every later migration, only ProtocolVersionManager may arm `generation=current+1` for the exact current version, greater target version and the consumed EXECUTING typed arm operation. The same transaction creates the exact LIVE VML1 lease above; an existing LIVE lease rejects and no entry point can extend its `abortAfterTimestamp`. Before arm, the manager checks every target address, runtime/configuration hash, object identity and L1 component that is knowable without executing the target L2 transition; stale, skipped, public or mismatched calls reject atomically. Settlement, registry, data-session, reward/preparation and both forced-ingress adapters all read that same router-owned word. Arming atomically closes new normal contexts/candidates, sessions, data posts, builder reservations and reward work; ingress returns SYNCED before retaining a deposit or queue write. Every canonical new-work entrypoint rechecks phase ACTIVE after `sync()`. The session open/post/seal sidecars are the sole exceptions: none calls sync, each checks current target and ACTIVE atomically, and the payable open/post calls revert with value on failure. Every ARMED cleanup and READY write returns a non-continuation status. Reaching READY sets `changed=true`, so no call can fall through and reopen normal work. Already-mature claim-only withdrawals remain callable.

For VERSION_MIGRATION, ARMED and READY are reversible because authority has not moved. At `block.timestamp>=abortAfterTimestamp`, anyone may call the immutable manager's permissionless expiry abort without a governance signature, queued operation or caller-supplied target. Activation rejects at and after the same bound. Either abort path requires the old Settlement and queue still authoritative, enters ABORTING before its callback, marks that generation canceled, clears the target fields, terminalizes the lease and returns the old version to ACTIVE without restoring deleted sessions or an already-settled candidate. It performs no same-transaction post-abort sync; IDLE is the final write and the next ordinary call may sync. A aborted generation can never cut over, and the next arm uses the next generation. The abort freshness watermark invalidates all previously queued arm siblings, so retry requires a newly queued operation and another full 604,800-second notice period. The fixed lease caps closed admission at 604,800 seconds plus one inclusion bound, even if the DAO and every target prover disappear. Thus a bad deployment, unavailable proof or abandoned upgrade cannot strand the protocol in ARMED or READY. Abort changes no reservation, canonical, queue, claim or liability state. Genesis campaign publication is immutable; the bounded expiry/resume path, not target substitution, is its recovery path.

The old Settlement finishes only the already-open boundary. An open normal window may commit its already-recorded best at its deadline; a newly due head cancels it. A current recovery proof may commit until that round's existing `expiresAt`; after expiry `sync()` cancels the round and is forbidden to increment its revision or open another episode. If neither is open, arm clears a merely armed normal context immediately. At and after that boundary, admissions remain closed and every cleanup call scans exactly eight consecutive physical cells from the authenticated cursor. Empty and occupied cells both consume a step; every scanned LIVE cell becomes REFUND regardless of expiry, and no owner or sink is called. At arm, the

generation freezes one objective refund deadline as saturating `uint64` addition of the 86,400-second claim window to the later of `armedAt` and the already-open normal deadline or recovery `expiresAt`. Every cell converted by that generation uses this same deadline; a keeper cannot extend it by delaying one scan. While any normal or recovery boundary remains open, no ARMED LIVE cell may be converted or claimed. A scan never converts and forfeits the same cell.

The manager's arm transaction writes the current Settlement's exact generation and this objective refund deadline before invoking any internal `sync()` that can observe the router's new ARMED phase. That write, the router phase change and the leading sync share one EVM revert domain. Thus an arm with LIVE cells and no open boundary can immediately perform its first migration scan, while any injected failure restores the pre-arm gate, deadline, cursor, cells and counters together; an ARMED scan can never observe an unset generation deadline.

The cursor advances by eight and makes `changed=true` even if the scan found no LIVE cell. Thus 128 uninterrupted migration scans cover every cell from any starting cursor; public refund maintenance cannot interleave and steer the cursor while ARMED. READY requires the source-local `liveCount` zero, not `refundCount` or `occupiedCount` zero. Claims may clear REFUND cells concurrently without changing the cursor. Once READY or FROZEN, the historical source's permissionless maintenance scans only overdue REFUND cells, and the unpausable claim/surplus surfaces remain live. Abort never restores a converted REFUND, claimed cell or forfeiture; unscanned LIVE cells remain live when ACTIVE returns, and a later generation cannot double-refund an old cell. BuilderRegistry, ScheduleOracle, generation records, tranche rings and reservations are protocol-lifetime objects shared by identity across every Settlement. Existing `RESERVED` tranches remain byte-for-byte `RESERVED`; migration never counts, scans, refunds, converts or releases them. Arming only closes creation of new reservations until ACTIVE returns. Ordinary expiry, replacement, exit, slashing and evidence-safe `RESERVED->LIABLE->RELEASED` transitions continue under their existing rules and are not migration cleanup. This keeps all already-funded current/future schedules usable after cutover and prevents a version change from manufacturing either a refund or a liability. READY is derived solely from zero live-session count, no normal boundary and no recovery; no calldata supplies those facts. The process preserves all schedule and slash/evidence state and leaves matured reward/refund claims callable on permanent claim-only surfaces. It then enters MIGRATION_READY before another `sync()` decision. A due queue head is not drained or discarded; old adapters receive SYNCED while their caller fee remains in the adapter pull-claim ledger and retry after activation.

READY is written only by the exact cross-contract handshake below:

```
migrationReadinessV1() returns
  (bytes4 magic,uint64 generation,uint64 activeProtocolVersion,
   uint64 targetProtocolVersion,bytes32 targetManifestHash,
   bytes32 targetRegistrationHash,uint8 boundaryState,
   bool localArmComplete)
markMigrationReadyV1(uint64 generation) returns (bytes4 magic)
```

The readiness magic is `0x4d525331` (ASCII `MRS1`); boundary states are exactly NONE=0, NORMAL=1 and RECOVERY=2; and its returndata is exactly 256 bytes with canonical padding. RECOVERY is derived exactly when canonical mode is RECOVERY and its exact current round exists, with no normal marker. NORMAL is derived when there is no recovery and any canonical normal boundary marker is live, including `normalArmBlockNumber`,

`normalDeadline` or `normalBest`. NONE is derived only when no recovery record and none of those normal markers exists. A conflicting mode/record/marker combination makes the view revert; it can never be reported as NONE. The view derives, rather than stores, `localArmComplete=true` exactly when the shared gate is ARMED or READY for the returned tuple, `boundaryState=NONE`, the same Settlement's local LIVE count is zero, `seatMigrationLocalGeneration=generation`, the stored exact seat-arm tuple/receipt is present and matches every returned field, `migrationRefundGeneration=generation`, the frozen migration refund deadline is nonzero, and seat closure has completed. The arm callback's stored receipt phase remains ARMED permanently; the current shared gate may instead be ARMED or READY, so receipt validation compares generation, active and target versions, target manifest hash and target registration hash but never compares phase. The arm callback's leading sync always uses `allowReady=false`; it writes the refund tuple before that sync and writes the seat-arm receipt/local generation only after all local closure succeeds. Consequently even an empty source remains ARMED after the arm callback and can become READY only on a later permissionless sync. The mark success magic is `0x4d524459` (ASCII `MRDY`) encoded as one exact 32-byte ABI word; the other 28 bytes are zero.

Only the exact currently active Settlement may call `markMigrationReadyV1` and only while the Router is ARMED for that supplied generation. Router first requires lifecycle IDLE and writes READYING. Before its single READY write, it makes separate 100,000-gas `STATICCALL`s to the caller's `migrationReadinessV1()` and `dataSessionAccountingV1()` views. Revert, OOG, wrong magic, wrong exact length, noncanonical padding or any tuple mismatch reverts. The readiness view must match the Router's generation, active/target versions, target manifest and target registration, report NONE and true; the accounting view must report LIVE count zero, NOT_ENTERED guard and the registration's exact DataSession configuration hash. The Router checked-adds the returned live and refund liabilities and requires `BALANCE(sourceSettlement)` at least that total. Only then does the Router write READY, return MRDY and write lifecycle IDLE last. Settlement treats any failed call or wrong return as a revert; a successful sync returns a non-continuation changed status. Activation repeats both exact reads from that same source while READY, before any proof adoption or authority write, and requires the identical closed/zero tuple and balance/liability lower bound. Abort writes no READY state and leaves the source as the unique current Settlement. Thus no per-session global counter or caller assertion is part of the safety argument.

While READY, any prover constructs an immutable fork-local first-V2 candidate from the frozen old Canonical and live queue snapshot. It carries the exact target release/profile, complete bounded Inbox rows and raw proof bytes. The proof executes the exact release-activation call as tx0 before tx1 InboxApply, authenticates the fresh Store/Quota/Bridge and lifetime rows, and binds both raw EIP-2718 transaction hashes. The statement binds old full core/sequence, migration generation, Queue root/count/cursor/range/descriptor commitment, candidate and beneficiary, target profile/manifest, full output core, deployment observation, exact tx0/tx1 hashes and every release effect. Proof generation does not mutate L1 or canonical L2 state.

This migration candidate is exactly one block and is subject to the release- activation geometry in the custom-system-transaction section: tx0 is index 0, tx1 is index 1, the uniquely classified maximum forced kind-0 FIFO prefix (if any) follows, and the transaction list ends there. Its discretionary count and canonical discretionary `bodyRoot` are both zero; no private tail, builder-selected coinbase, blob or second block is legal. Every included kind-0 raw transaction is recovered from its canonical L1 Queue-admission transaction calldata, checked against the leaf hash

and queue index; that published L1 calldata, not a prover-private witness, is the replay preimage. The circuit derives and binds the complete transaction root, receipt root, header/block hash and execution-output commitment. The migration statement's `candidateDigest` commits that exact single-block candidate and `outputCanonicalHash/outputCore.tipHash` commits its resulting block; substitution or omission changes the verifier input. If a required canonical L1 raw preimage is unavailable to the prover, migration cannot land and the immutable lease's permissionless expiry abort restores the old protocol instead of adopting an unreplayable state.

The L1 call uses the exact five-argument activation ABI frozen above. Calldata supplies only transition kind/generation/source and target versions/source sequence, candidate digest, output core, beneficiary, anchor pair, force cutoff, pre-Inbox plus-one value, the complete registered manifest, canonical Inbox rows, the bounded launch header when applicable and bounded proof. Router derives the base core/hash, statement digest, execution profile, Queue address/ code/config/root/count/start/end and descriptor commitment, verifier descriptor, tx0/tx1 preimages and hashes, target Settlement, domains/Bridges, registration identities and activation receipt from current storage plus those exact bytes. None is a second caller field. The call contains no Router, verifier or L2 object, capability, seal, authority handle or validity Boolean. Future landing block and timestamp are absent. Router rereads the exact READY tuple and all live L1 identities, reconstructs the statement and raw tx0/tx1 preimages, and invokes only the execution-profile-pinned immutable verifier by bounded STATICCALL before any write. Success requires exact descriptor/code/config/key/ schema, proof bounds and exactly 32 bytes equal to the reconstructed statement hash; short/long return, revert, OOG or substitution leaves all state unchanged.

Only after verification does the Router execute the exact shared-context MFRZ–MCAN–QMIG–MAPS journal defined above. Router never writes another account's storage: the immutable source, target and Queue each authenticate MACT and write only their own bounded transition. MCAN changes only the L1 target Settlement's Canonical/history bookkeeping; it never reads or writes the target Protocol, InboxApply, Registrar, Store, Pool, Bridge or another live L2 object. The execution node selects the immutable fork-local poststate only after observing this exact successful L1 transition for the same candidate, core, version, sequence and Queue range. Losing, abandoned, pre-cutover and reorged transitions cannot mutate canonical L2 state; a local replay failure is a node resync condition and cannot revert a valid L1 transition.

Any L1 late fault rolls all L1 state back. A canceled generation or changed source sequence invalidates the proof; later landing remains valid only while the complete bound tuple is unchanged. Kind-0 preimage loss retains its bounded expiry/discard path. Existing builder reservations and historical V1 bridge signals keep their original semantics; migration enumerates or converts neither.

13 Parameters and enforced geometry

Parameter	Initial value	Enforced relation
L2 slot / schedule window	1 s / 384 slots	Aligned by genesis timestamp.

L2 height / timestamp	`uint48` checkpoint bound	Launch import is strictly below `UINT48_MAX`; every later header is checked consecutive and at most that maximum. Timestamp equals genesis plus signed slot.
`N_MAX`, `Q_MAX`	64 / 76	Six addresses fill 384 tickets.
`BOND_MAX`	$2^{192} - 1$	`bond*(Q_MAX+1)<2^256`.
Entry/exit/reservation ahead; moves	8 / 268 / 16 windows; 4 per window	Liability residence < 268; capacity 4×268.
`D_SNAP`, `F_FINAL`	8 / 2 L1 epochs	Strictly exceeds scan + finality + seal margin + lookahead.
`F_FINAL_BLOCKS`	64 L1 blocks	Exact block-number alias of two 32-slot L1 epochs for campaign review/stage inequalities.
Genesis campaign preparation	1,024 proposals; 1,024 forced rows; 4 MiB	Limits apply from publication; 16 rows/call and at most 128 scan calls. Receipt binds exact counts, roots, bytes and accounted abandoned forced fees; arbitrary raw surplus is excluded and cannot veto. Bonds remain withdrawable.
Genesis scan codec/gas	3,808 B/proposal; 256 B/forced row; 62,084 B max proposal calldata; 12,000,000 gas	Exact deployed ABI shape at 10 forced plus 21 normal blob hashes. A cold maximum batch above the gas cap blocks profile release.
Genesis legacy resume proof	RISC0_RETH 5 + SP1_RETH 6; 900 s	Ordered age-independent route through immutable fixed-key adapters; generic mutable-key wrappers, SGX-required roots and unfenced remote verifier graphs are unsupported.
Genesis campaign stages	force-to-proposal ≥ 64; scan ≥ 192 blocks	Target/campaign review closes at least `F_FINAL_BLOCKS` before force cutoff; proof/finalization remains live through the scan stage.
Genesis QUIESCENT bound	600 L1 blocks / 7,200 s maximum	Entry requires at least 128 blocks and 1,800 s remaining; activation ends before either bound and resume begins at either bound. Every scanned pending/unfinalized data expiry, including an already-stale row, must exceed the time bound plus 900-s proof-generation, reorg and inclusion margins.
`H_LOOK`	768 L2 slots	Guaranteed by snapshot geometry with one-window slack.

`G_MAX`	3,600 L2 slots	Tier-1 parent gap; equals final-lag trigger and fits within the 12-window proof cap.
`W_SETTLE_SECONDS`	1,200 s	Timestamp deadline, never L1 block count.
`DELTA_TIP`	1,200 s	At least submit inclusion + skew = 144 s after the frozen escape slot.
`DELTA_FINAL_LAG`	3,600 s	At least activation + escape offset + submit + skew = 2,164 s.
`F_L1`; `T_DEPTH_MAX`	64 L1 blocks; 900 s	Stored recovery-anchor depth and assumed maximum time to reach it.
`P_PROVE_MAX`	900 s	Exact alias of `recovery.proofTimeMaxSeconds`; used by session OPEN and recovery.
`ESCAPE_OFFSET`	1,900 s	At least depth-time bound + proof bound + margin.
`FORCE_DELAY`	1,500 s	At least settlement window plus `T_INCLUDE_MAX_SECONDS=120`.
Aggregator seat geometry	1 primary + 3 standbys; 4 waiting cells; 1 stage	`pendingCount+stagedCount<=4`; every canonical seat/duty pass is fixed at four cells.
Seat runway/tenure	UNCALIBRATED	`SEAT_RUNWAY>=MIN_PRIMARY_TENURE+HANDOVER_EXECUTION_BUFFER+SLA_TAIL`; the handover buffer covers delay, grace and L1 inclusion.
Seat premium/bond	UNCALIBRATED	Claim delay covers reorg stability; SLA bond covers maximum-ask closed-tail grief and the modeled collusive-promotion subsidy. An uncalibrated profile rejects.
Forced prefix	64 items; 1,048,576 B; 20m gas	Each item at most 131,072 B and 5m accounted gas; descriptors and one boundary remain durable.
Candidate caps	4,096 blocks; 12 windows; 1 anchor; 16 sessions; 2,100 records	Also 256 forced items, 4 MiB, 80m accounted gas.
Queue / range proof	depth 64 / at most 257 hashes	`queueCount<UINT64_MAX`; final leaf unused; canonical contiguous range proof.
Same-L1 credit read	fixed-size record / bounded staticcall	Launch source equals settlement Ethereum; registry runtime, return length, record, source generation and Bridge liability are checked directly and atomically.

Bridge domain release	at most 64 new entries/version; append-only	Only atomic activation of a delayed immutable manifest registers entries; the historical registry has no global cap, delete, redirect or reinterpretation.
Source generation support	one immutable active-profile entry	The frozen endpoint tuple is usable only after `SUPPORT_FINALITY_BLOCKS`; no automatic implementation boundary exists.
V2 launch asset scope	DIRECT ETH only	ERC20/ERC721/ERC1155 and every Vault flow remain V1; a future asset kind cannot reinterpret DIRECT.
Kind-0 validity / kind-1 enqueue	604,800 / 604,800 s	Bounds missing bytes before expiry or source cancellation.
`BRIDGE_PROCESS_TTL_SECONDS`	2,592,000 s	Starts at the immutable L2 pin; afterward anyone transitions NEW/RETRIABLE to FAILED without Message bytes.
Pool read / authorization cleanup / value callback gas	50,000 / 50,000 / 100,000	Exact immutable stipends/reserve; the release benchmark covers every cold path, EIP-150 boundary and fixed return copy with at least 30% margin.
Pool Bridge-prefix / consume-call bounds	UNCALIBRATED / UNCALIBRATED	The executable profile must publish nonzero measured values; positive-gas forwarding adds the exact Message budget and preserves both outer reserves.
Terminal accumulator/proof	depth 64 / 64 siblings / 2,048 B	Checked `uint64count<UINT64_MAX`; leaf commits index, domain, Bridge, credit and terminal; storage is 64 frontier words plus root/count, and a complete leaf event supports trustless proof verification with explicit archive-availability dependence.
Settlement canonical history	256 version-sequence cells	$256 > \lceil (T_{include} + REORG\_MARGIN)/12 \rceil + 2$ with at most one canonical write per L1 block; the exact version/sequence tag prevents sparse L2 heights from choosing or churning a cell, and frozen versions never overwrite again.

Registration MPT proof	65 nodes/path; 130 nodes total; 600 B/node; 78,264 B; 11m gas	Canonical RLP and exact account/storage paths under the routed canonical state root; all bounds reject before an unbounded allocation or call. Selected nodes are fully validated; unselected inline subtrees remain authenticated opaque items so unrelated state cannot amplify traversal gas. The frozen adversarial fixture consumes 7,940,171 gas and passes the 1.30x release gate at 10,322,223 gas.
L1 migration activation	Genesis: 0 rows/2,048-B header; version: 64 rows/0-B header; at most 131,072-B proof	Let `ceil32(x)=32*ceil(x/32)` and `genesisActivationBytes= 4+82*32+32+(32+ceil32(2048))+ (32+ceil32(maximumProofBytes))` and `versionActivationBytes= 4+82*32+(32+32*64+62*(6*32) +2*(6*32+ceil32(541)))+32+ (32+ceil32(maximumProofBytes) )`. The exact bound is `maximumActivationCalldataBytes= max(genesisActivationBytes, versionActivationBytes)`. GENESIS cannot carry a row; VERSION_MIGRATION has an empty header and its 13m activation budget permits at most two 541-byte kind-1 rows. `worstCaseActivationIntrinsicGas= 21000+16*maximumActivationCalldataBytes`. Activation requires $\lceil 1.30\,(G_{intrinsic} + worstCaseActivationAdoptionGas)\rceil \leq supportedL1BlockGasLimit$. The adoption measurement already includes a verifier that consumes the full configured limit.
Body chunks	9 / 1,142,748 B	Per-body chunks / maximum discretionary bytes.
`DATA_TTL`	86,400 s	Greater than candidate, settlement, proof and reorg path.
Data refund claim window	86,400 s	Same-cell REFUND grace; at least inclusion plus reorg margin.
`REORG_MARGIN, EVIDENCE_ DELAY`	1,800 / 86,400 s	Data resubmission and finite liability retention.
`SUPPORT_FINALITY_BLOCKS`	214 L1 blocks	Historical destination/terminal proof retention after a canonical L2 registrar record is proven to L1; it does not gate source cutover.

`BRIDGE_ROUTE_ARM_REVIEW_BLOCKS`	214 L1 blocks	Begins when the exact version-keyed source package is staged after release registration; it is an L1 package review, not MPT finality.
Live windows / sessions	268 / 1,024	Fixed rings with safe-eviction predicates and bounded GC.
EIP history / max arm age	8,191 / 255 blocks	$255 + \lceil (W_{settle} + CLOCK\_SKEW + 384 + EVIDENCE\_DELAY + REORG\_MARGIN)/12 \rceil + 2 < 8,191$; the conservative profile bound remains greater than every schedule carrier path; BuilderRegistry evidence does not consume this history.

Launch tooling evaluates every inequality in both seconds and its native clock unit. Governance cannot set a value that violates a relation.

14 Production gates and verdict

The consensus, migration/control-plane and Settlement–Market transition protocols are frozen implementation inputs. BRS1/BRD1/BRC1/ABR2 reconstruct their authority from exact RTR2/BRX1/PIR2/PIM2/PIA2/BIP1/BID1 raw records and do not use a cross-contract `SettlementRegistration` witness. The final freeze model resolves every live component through an address-indexed account world, checks EXTCODEHASH plus exact configuration/immutable returns, and seals the root SourceBundleFactory, root-lifetime terminal verifier and their deployment receipt before root activation. Builder lifecycle and release-compatibility uniqueness are now expressed by bounded counters and address/version-keyed reverse indexes. ICV2 now gives DestinationBridge an exact O(1) credit-ID lookup, VML1 uses a generation reverse index, RAV2 binds the Authority watermark, and DRV2/DSV2 keep multi-hop reclaim independent of the latest active tip. The executable model's broad Python rollback snapshots are semantic test journals, not contract-gas evidence; production relies on native EVM revert and must be measured against compiled bytecode. The model includes executable raw-ABI fault corpora and cold-cache restart coverage across root deployment, REGISTER–BRD1–ARM–READY–ACTIVATE and historical release/reclamation watermarks. Those tests establish specification consistency; they do not replace compiled EVM execution evidence. The initial executable profile, compiled contracts, gas certificate and circuit release bundle are still absent. This document is therefore implementation-ready as a contract/client/circuit specification, but is not production authorization. Before deployment, one reviewed release must contain all of the following concrete artifacts and two independent implementations must reproduce every hash and execution result:

- strict canonical ABI schema, exact profile bytes and `executionProfileHash`, plus a rejecting decoder corpus;
- independently reproduced MWV1/SMI1/SLV1/SIR1/SMR1/MEC1/MHS1/MRO1 bytes and hashes; real CALL/STATICCALL rollback harnesses at every read/write/postread fault; no-op, stale-generation, migration-stage, historical-FROZEN, repeated-stage and multi-hop-rotation regressions; and EIP-150/full-stipend/ one-less-gas certificates for every seat wire and direct historical call;

- an immutable manifest binding protocol version, chain IDs, activation, profile hash, verifier address/runtime hash and verifier-key hash;
- exact predeploy and immutable topology, including both fresh V2 Bridge bundles, CREATE2 factory/salt/init-code derivation, Bridge-unique Registry and QuotaManager accounts and the non-proxy/no-owner/no-upgrade rule for every V2 endpoint; the 232-byte Settlement deployment descriptor, published target init-code artifact and CREATE2 derivation, internal DataSession custody/configuration and rejection of every other payable Settlement path, `targetRegistrationHash` inclusion in delayed arm/cancel/activation/receipt records, exact configuration-read ABI and fresh-PREACTIVE rejection corpus; the exact MACT/MFRZ/MCAN/QMIG/MAPS and legacy LGS1/LGAR/LGFN implementations, shared-context/poststate corpus, per-call stipend/EIP-150 reserve harness and Router lifecycle journal specified above; the storage-layout-compatible final legacy proxy implementation and a rehearsal proving no old proposal, forced-ingress, initializer or upgrade path bypasses its local phase; the age-independent legacy resume root, immutable fixed-key RISC0/SP1 adapters and recursively bound remote verifier graph; the final direct legacy SignalService implementation and shared campaign upgrade fence, with ForkRouter and unfenced trust-map configurations rejected; BuilderRegistry and ScheduleOracle runtime/configuration hashes and retained object identities; terminal runtime code hashes, constructor immutables and relevant storage selectors for AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2, ActiveSettlementRouter, ForcedQueue (including its depth, capacity, empty leaf, descriptor schema, router binding, accounted-liability lower bound, forced-ETH surplus behavior and `forcedQueueConfigHash`), both immutable ingress adapter runtimes/configurations and their append-only router registrations, L1/L2 legacy SignalService, fresh V2 Bridges, Bridge-unique QuotaManagers and NativeLiquidityPoolV2, including proof that no V2 upgrade selector, owner, reinitializer or delegate target is reachable, plus the exact chain placement and L1-direct/L2-direct/finalized-manifest validation trace for every infrastructure row;
- exact nonzero source/destination domain namespaces and vectors, append-only BridgeDomainRegistry authorization, BridgeCreditRegistry direct ABI/runtime/record layout, immutable source generations, Bridge aggregate and per-credit liability/status/pull-credit layouts, per-Settlement sequence-tagged canonical-history rings and fixed-depth TerminalSignalVerifier vector grammar, 64-word accumulator frontier/event layout and append gas bound, byte-exact Bridge-kernel and destination-infrastructure descriptors, registrar/adapter/store one-shot bindings, finalized L2 registration proof, stamped two-call ingress ABI, proof-first router activation ABI/invariants, exact live-lease expiry/abort transition, direct-call limits and valid/invalid fixed-return corpus;
- exact unsigned system transaction type and reserved sender addresses; chain-ID equality, caller/origin injection, initial warm set, intrinsic gas, refund quotient, zero GASPRICE, nonce/balance non-transition and exact typed transaction/receipt trie values; Store/Bridge release seal and cutover proof; system nonces; the one DIRECT-only additive BridgeV2 selector and bounded immutable inbox-store batch ABI, selectors and storage; fee handling, gas ceilings and all fork and header recurrence rules, including byte-exact first activation and steady-state type-`0x7f` transaction/receipt/header vectors; and
- complete positive and negative vectors for parent, prestate, input, transactions, receipts, headers and poststate, as listed in §12, including domain/source-generation changes,

unauthorized clones and foreign local-domain replay, enqueue/cancel and capacity races, PREACTIVE rejection, legacy preactivation attempts, first-block release activation, direct-record absence, router-SYNCED refund/retry, old-domain routing, permanent-pin independence from legacy upgrades, processing expiry, DONE release, FAILED recall, legacy SignalService pause, zero ordinary ETH quota, pooled-balance drain attempts, rejected Vault callers, endpoint authorization, exact old-selector V1 compatibility, explicit V2 opt-in, terminal-root/count public-output binding, canonical-sequence collisions, historical proof after later appends, direct-sender/Bridge/foreign-Anchor release attempts, reserved-system-sender-to-Anchor activation, old-domain terminal append after new-domain activation, mixed old/new-domain global inbox batches and domain/Bridge proof substitutions, multi-credit batches and every V1/V2 replay path, manifest/profile dependency-cycle rejection, chain-ID overflow/local mismatch, fresh/partial/reused endpoint state, singleton queue proofs at and across the 2^{32} boundary, forced-ETH surplus, stale ingress stamps, failed target proofs/components, ARMED/READY expiry abort, shared-gate identity and two complete successive migration generations.

Those values are outputs of real compiled contracts, fork code and a real circuit/verifier key. Inventing placeholders in LaTeX or Python would make the profile hash look precise while leaving clients unable to construct the same escape block. Their schema, derivation and acceptance inequalities are normative; their concrete values are implementation-produced release inputs. Consequently the production release artifact cannot be completed by document-only revision even though the design is ready to implement; implementation and circuit work must now produce the bundle. Once it exists and is re-audited, production deployment remains blocked until every additional gate below produces a versioned artifact and passes:

1. **Proof performance.** Independent tier-1, tier-2 and worst-case tier-3 proofs, including 4,096 blocks and 2,100 data records, finish within 900 seconds on the published minimum hardware. Peak RAM, recursion setup and failure recovery are measured; a miss requires changing parameters and another design review. Before the first genesis force cutoff, two independent implementations must also scan, quiesce, generate and verify the exact campaign-to-target genesis proof from a shadow-fork snapshot. Each independent pipeline must generate the complete accepted RISC0+SP1 pair; removing either proof ecosystem must block activation but must not block permissionless expiry and local resume. A drill also advances beyond SGX instance expiry and proves that freshly generated route proofs, rather than cached SGX bytes, finalize the retained legacy suffix.
2. **Contract gas.** `sealScheduleWindowV1`, `postData`, normal submit, `sync`, direct source-credit enqueue, recovery submit, slash and garbage collection fit current L1 gas limits with 30% margin. The same compiled- bytecode harness exercises cold maximum REGISTER_RELEASE with the release's actual strict canonical ABI profile and both required ingress rows, including the named-gas Router call, Market install and every bounded postread; maximum REGISTER_FORK_VERIFIER; and publication, arm and both abort branches under their 8,000,000-gas Router caps. The harness searches and publishes each cold-path minimum, proves the frozen nested cap is at least `ceil(1.30*measuredMinimumGas)`, and succeeds at that cap; the enclosing intrinsic plus execution gas also has at least 30% headroom below `supportedL1BlockGasLimit`. The published fixture binds compiler version, optimizer settings, fork rules, storage warm/cold assumptions, exact profile bytes and every component runtime/configuration hash. Any failure changes the candidate constant/profile and reopens design review before the rele-

vant typed operation can be queued. Fuzzing also includes empty registry, zero seat, full data session, full queue prefix and maximum history age.

3. **Cryptographic conformance.** Solidity, Go, Rust and circuit code match the Keccak/EIP-712 golden vectors, direct same-L1 Bridge registry ABI and runtime-hash checks, KZG Fiat–Shamir transcript and exact body manifest. Two independent implementations generate every vector.
4. **State-machine verification.** Stateful fuzzing models EVM rollback, same-block ordering, reorg replay, stale versions, activation/close coincidence, message consume-and-discard and storage reclamation. The executable models in Appendix D remain green.
5. **Economics.** Auction bond, per-window tranche, forced-envelope deposit, storage bond and proof credits are calibrated under blob/L1 fee spikes. The published liveness claim explicitly becomes conditional above fee caps.
6. **Operations.** At least two independent prover stacks and three independently administered canonical archives, plus public data-session mirrors, schedule sealers and recovery bots, complete a shadow-fork outage exercise. One archive serves a clean node, after deleting its state, body, receipt *and code* databases, from only the latest finalized package while the other two are removed. The exercise also removes every builder and seat, forces a user transaction, reorgs the first recovery attempt, and still reaches finality. Loss and restore drills verify that unavailable prestate is never misreported as bounded recovery. For every supported source domain, an analogous drill creates an old immutable credit record, removes every message-archive copy, exercises source cancellation, then recreates a message, orphans its direct enqueue transaction and replays it from the surviving record. It also exercises two separately registered frozen source generations, attempts and fails to deploy a wrong CREATE2 bundle or mutate an immutable V2 account, pauses legacy SignalService and zeros ordinary quotas, replays a pin through a second funded Bridge, expires an unprocessed pin and proves that a superseding profile cannot remove an old domain or endpoint entry. The exercise also deletes one terminal-log archive, rebuilds a historical 64-sibling proof from a second independently operated archive, and verifies it against a retained Settlement root/count.
7. a Foundry activation harness using the exact release bytecode and profile-maximum proof. It covers the largest reachable successful genesis call (zero Inbox rows, a 2,048-byte header and the maximum proof), the version- migration 62-kind-0/two-kind-1 cold path with empty header, and the configured verifier consuming its complete stipend. It measures calldata intrinsic gas separately and the complete cold Router execution as one non-overlapping bound, while also reporting decode/read/getter/verifier/call-overhead/suffix components for diagnosis; it succeeds below `supportedL1BlockGasLimit`, and demonstrates at least 30% margin by the enforced integer inequality. It separately exercises exact and one-less gas for MACT, MFRZ/LGFN, MCAN, QMIG, all three MAPS reads, the separate atomic legacy campaign scan/quiescence/resume, LGAR/LGS1 and the retained activation suffix; each injected late fault proves byte-identical rollback of source, target, Queue, Router and receipt. Any compiler, fork or storage-layout change reruns and republishes that measurement before release;
8. **External review.** Separate proof-system, Solidity, bridge and economic audits close all critical/high findings; the migration rehearsal and emergency governance path are in scope.

Verdict. *The earlier fallback architecture was not implementation-ready: it depended on a scheduled builder,*

could revert activation inside a rejected submission, made payment a commit precondition, and could not bind whole-window data with a six-blob cap. This revision removes those dependencies, fixes a single consensus path, binds bridge credits to their source application, and defines bytecode-complete recovery packages. Its bridge integration now uses fresh immutable non-proxy V2 bundles, keeps legacy callback applications on V1, binds the local destination, and makes V2 terminal proof/refund paths transitively pause- and quota-independent. Router foreign-storage assumptions are removed: source, target, Queue and legacy proxy now authenticate one context and commit their own poststate inside one rollback journal, with a complete sealed receipt. The broader design is frozen for implementation: the address-indexed component world, root Source topology, exact ABI graph and restart/serialization authority paths are now closed at the specification/model boundary. No known Critical or High design defect remains after the final adversarial review. ICV2, RAV2 and the direct- successor reclamation lifecycle are now exact at the design/model boundary. Safe in-place genesis migration is conditional on the deployed legacy Inbox accepting the exact final compatibility implementation; without that capability, in-place cutover for that deployment is impossible and must be replaced by an independently initialized state migration. The absent compiled executable profile, measured gas values, verifier/circuit artifacts, multi-language conformance and external audits remain production release gates, not unresolved document-level consensus choices. A failed compiled-size, gas, proof-performance or legacy-layout gate reopens the affected design decision; until those gates pass, "implementation-ready" must not be represented as "secure for production."

A Normative commitment encodings

All integers in packed commitments are unsigned big-endian at the named width; addresses are 20 bytes, hashes are 32 bytes, and ASCII domains have no length prefix or trailing NUL. `H` is Ethereum legacy Keccak-256. No `abi.encodePacked` call may contain a dynamic field. Raw transaction bytes are represented by `H(rawTx)` plus an explicit `u32(byteLength)`.

The canonical core/base, candidate, schedule list, sealed-session list and execution outputs use these exact encodings:

```
coreHash = H("slot-chain-core-v3" || u64(l2BlockNumber) || tipHash ||
  u64(tipSlot) || stateRoot || u64(messageCursor) || winningDataCommitment ||
  u256(nextBaseFee) || u64(nextExcessBlobGas) || terminalRoot ||
  u64(terminalCount))
baseCanonicalHash = H("slot-chain-canonical-v2" || coreHash ||
  u64(canonicalizedAtBlock))
H("slot-chain-candidate-v2" || baseCanonicalHash || u16(n) ||
  (u64(slot) || blockStructHash || blockHash || bodyRoot ||
   dataManifestRoot || u64(messageEnd))*)
H("slot-chain-schedule-list-v1" || u8(count) ||
  (u64(window) || entryRoot || seed)*)
H("slot-chain-session-list-v1" || u8(count) ||
  (sessionId || u16(recordCount) || root)*)
winningDataCommitment = H("slot-chain-winning-data-v1" ||
  candidateCommitment || sessionRefsCommitment)
H("slot-chain-outputs-v2" || stateRoot || transactionsRoot || receiptsRoot ||
  logsBloomHash || withdrawalsRoot || terminalRoot || u64(terminalCount))
```

Candidate encoding accepts exactly $1 \leq n \leq 4096$ rows, and their `uint64slot` values are strictly increasing. A schedule list contains at most 12 rows with strictly increasing `uint64window` values. A sealed-session list contains at most 16 rows with strictly increasing `bytes32sessionId` values, and every `recordCount` is at most 2,100. Every bound and addition is checked before narrowing; an empty candidate, duplicate or descending key, surplus row or width overflow rejects rather than being hashed. `blockStructHash` is the EIP-712 struct hash of the exact signed `SlotChainBlock` fields in §5.1; tier-3 recovery uses the same struct fields without a signature. These hashes are respectively the statement's `baseCanonicalHash`, `candidateCommitment`, `scheduleRootsCommitment`, `sessionRefsCommitment` and `executionOutputsCommitment`. Settlement must recompute the displayed `winningDataCommitment` before verifier dispatch and again before the canonical write; unsorted or duplicate lists reject.

The 64-cell registry leaf at index i is

```
H("slot-chain-registry-leaf-v1" || u8(i) || u8(present) ||
  address20 || u192(bond) || u64(registrationIndex) ||
  u64(effectiveL2Slot) || bytes32(trancheRoot) || u64(tombstonedAtL2Slot))
```

An empty cell has `present=0` and every following byte zero. A present cell has `present=1`; `UINT64_MAX` means not tombstoned. At tree height $h = 0, \ldots, 5$, the parent is:

```
H("slot-chain-registry-node-v1" || u8(h) || left || right)
```

There is no padding because there are exactly 64 leaves.

Admission position $i < 1,136$ is:

```
H("slot-chain-admission-leaf-v1" || u16(i) || u8(present) ||
  u8(location) || address20 || u192(bond) || u64(registrationIndex) ||
  u64(effectiveL2Slot) || u64(tombstonedAtL2Slot))
```

Location is 1 for active and 2 for liability; an empty leaf has present and all following fields zero. Positions $1,136 \ldots 2,047$ and unused positions are canonical empty leaves. Its eleven levels use `H("slot-chain-admission-node-v1"||u8(height)||left||right)`. `admissionVersion` increments before publishing the corresponding root; the pair is stored together and never reconstructed from a newer root. Tranche roots occur only in the active registry and liability storage records, not in this stable signer-identity commitment.

The omission-free ranked-entry commitment for window W has exactly 64 leaves:

```
H("slot-chain-entry-leaf-v1" || u8(rank) || u8(present) || address20 ||
  u192(bond) || u64(registrationIndex) || u64(effectiveL2Slot) ||
  u64(tombstonedAtL2Slot) || bytes32(trancheLeafHash))
```

followed by six levels of `H("slot-chain-entry-node-v1"||u8(height)||left||right)`. Empty ranks have all fields zero. The schedule circuit derives all 384 tickets from this root and seed; a schedule list cannot substitute a different set.

Each registry cell's tranche ring is a 512-leaf tree. Leaf index j is:

```
H("slot-chain-tranche-leaf-v1" || u16(j) || u64(window) || u8(state) ||
  u192(amount) || u64(liableUntil))
```

An empty leaf has `window=UINT64_MAX`, state `EMPTY=0` and remaining fields zero. Other states are `FREE=1`, `RESERVED=2`, `LIABLE=3`, `RELEASED=4` and `SLASHED=5`. A ring-index collision may be overwritten only from RELEASED or SLASHED after its absolute release predicate; the write installs the new exact window and FREE/RESERVED state atomically. `liableUntil=0` for EMPTY/FREE; reservation sets it exactly to `evidenceDeadline(window)+REORG_MARGIN_SECONDS`, and LIABILITY preserves that value. RELEASED/SLASHED preserve it until safe overwrite. Its nine node levels use `H("slot-chain-tranche-node-v1"||u8(height)||left||right)`. The snapshot supplies one inclusion proof for index `Wmod512` for every active cell, including canonical empty and `FREE` leaves.

Every fixed-tree inclusion proof has one canonical grammar. Its sibling array contains exactly one `bytes32` per level, bottom-up at heights $0, \ldots, depth - 1$. The four fixed depths are registry 6, admission 11, ranked-entry 6 and tranche 9. At height h, bit h of the separately bound leaf index alone determines whether the sibling is left or right; the proof carries no direction bit, height, depth or node-domain field. The leaf itself continues to commit that same index or rank, and verification requires both uses to agree. Domain-specific verifiers fix the depth and node domain and reject an out-of-range index, a missing or extra sibling, an incorrectly ordered sibling, a substituted leaf/root or any unused element. Thus neither an alternative proof length nor redundant metadata can encode the same proof. The executable oracle additionally pins the four canonical empty roots. They are built from all 64 position-bound empty registry leaves, all 2,048 position-bound empty admission leaves, all 64 rank-bound empty entry leaves and all 512 index-bound tranche leaves with `(window=UINT64_MAX,state=EMPTY,`

`amount=0,liableUntil=0)`. These are reviewed constants for initialization and conformance; production code must not replace them with a repeated positionless leaf or reconstruct them on a runtime path.

The two durable descriptor bodies are distinct fixed-width types named `Kind0ForcedDescriptorV2` and `Kind1ForcedDescriptorV11`. There is no universal nullable superset struct. The heterogeneous Queue/list wire retains an explicit kind byte and rejects every other kind or body length. Kind-0 forced leaf *i* is legacy Keccak over this exact packed sequence:

```
"slot-chain-force-user-v2" || u64(i) ||
address20(sender) || u64(nonce) || u256(l2ChainId) || bytes32(rawTxHash) ||
u32(byteLength) || u64(gasLimit) || u64(accountedGas) ||
u256(maxFee) || u64(validUntil) || address20(refundAddress) ||
u64(enqueuedAt) || u64(dueAt) || u256(deposit)
```

A kind-1 leaf is:

```
"slot-chain-force-bridge-v11" || u64(i) || msgHash ||
u256(srcChainId) || sourceDomainId || u64(srcEpoch) || address20(srcBridge) ||
bridgeExecutionHash || u64(emittedAtBlock) || destinationDomainId ||
u256(destChainId) || u64(enqueueBy) ||
address20(sender) || address20(srcOwner) || address20(destOwner) ||
u256(value) || u64(fee) || u64(liquidityFee) || calldataHash || u8(refundMode) ||
address20(refundVault) ||
refundCapsuleHash || escrowId ||
u32(byteLength) || u64(accountedGas) || address20(refundAddress) ||
u64(enqueuedAt) || u64(dueAt) || u256(deposit)
```

Its application has no expiry; no `validUntil` field exists. Launch V11 requires `refundMode=DIRECT`, `refundVault=address(0)` and `refundCapsuleHash=bytes32(0)`; these retained fixed-width projection fields cannot be reinterpreted as an asset mode.

For `forcedDescriptorCommitment`, `descriptorBytes` is exactly the corresponding leaf sequence beginning immediately after `u64(i)`; the list already supplies index and kind, so neither leaf ASCII domain nor index is duplicated inside it. Its fixed byte length is 220 for kind 0 and 541 for kind 1. The list encoding and optional first excluded boundary are defined in §8. The circuit prepends the correct leaf domain and index to reconstruct every Merkle leaf; any other length rejects. The checked list encoder accepts only kind 0 or 1, at most 256 consumed rows, and exactly zero or one boundary row. Row indices are the contiguous `uint64` sequence `start+i`; when present, the boundary is exactly `start+consumedCount`. Every index addition is checked in 64 bits before encoding, so a gap, duplicate, caller-supplied replacement index, narrowing or wrap rejects.

The DIRECT-only acyclic `canonicalBridgeKernelDescriptor` is exactly 115 bytes:

```
u16(kernelVersion=2) || u8(DIRECT=1) ||
u64(MAX_BRIDGE_ENQUEUE_DELAY=604800) ||
u64(BRIDGE_PROCESS_TTL_SECONDS=2592000) ||
bytes32(bridgeV2AbiHash) || bytes32(statusTransitionHash) ||
bytes32(custodyRulesHash)
```

The three final hashes commit byte-exact selector/tuple/event grammar, status transition table and liability/reserve/pause rules in the release bundle. None may contain a source/destination domain, infrastructure hash or full `executionProfileHash`. The kernel hash is

```
bridgeKernelProfileHash =
  H("slot-chain-bridge-kernel-profile-v2" || u32(115) ||
    canonicalBridgeKernelDescriptor)
```

`canonicalSourceBridgeDescriptor` has this exact 752-byte packed grammar:

```
address20(factory) || bytes32(factoryRuntimeHash) ||
bytes32(factoryConfigurationHash) || bytes32(bundleSalt) ||
bytes32(bundleInitCodeHash) || address20(bundleDeployer) ||
bytes32(bundleDeployerRuntimeHash) ||
address20(legacyV1Bridge) || address20(sourceBridge) ||
bytes32(bridgeRuntimeHash) || bytes32(bridgeConfigHash) ||
bytes32(storageLayoutHash) || address20(creditRegistry) ||
bytes32(registryRuntimeHash) || bytes32(registryConfigHash) ||
address20(quotaManager) || bytes32(quotaRuntimeHash) ||
bytes32(quotaConfigHash) || address20(supportRegistry) ||
bytes32(supportRegistryRuntimeHash) ||
bytes32(supportRegistryConfigurationHash) ||
address20(terminalVerifier) || bytes32(terminalVerifierRuntimeHash) ||
bytes32(terminalVerifierConfigHash) || address20(pauser) ||
address20(signalService) || bytes32(bridgeKernelProfileHash) || u64(srcEpoch)
```

Define, over the literal ASCII domains shown,

```
bundleSalt = H("slot-chain-source-bundle-salt-v1" ||
               bytes32(manifestNamespace) || u64(protocolVersion))
create2(factory,salt,initHash) = low20(keccak256(
  0xff || address20(factory) || bytes32(salt) || bytes32(initHash)))
createChild(bundleDeployer,nonce) = low20(keccak256(
  0xd6 || 0x94 || address20(bundleDeployer) || u8(nonce)))
bundleDeployer = create2(factory,bundleSalt,bundleInitCodeHash)
sourceBridge = createChild(bundleDeployer,1)
creditRegistry = createChild(bundleDeployer,2)
quotaManager = createChild(bundleDeployer,3)
```

Profile word 205 must equal this derived `bundleSalt`; it is not a publisher-selected entropy value. Profile word 209 is the exact existing V1 Bridge address fixed by the migration root. A mismatch in either word rejects the profile before registration. Since the immutable PVM admits only its own nonzero `manifestNamespace` and protocol versions are append-only, two registered releases cannot derive the same source bundle. Here `0xd6` and `0x94` are the canonical RLP prefixes for the 23-byte `[address,smallNonce]` preimage. The bundle init code is the release artifact that commits the generic constructor logic plus every child creation-code/configuration primitive. Its constructor computes the three addresses from `address(this)`, performs exactly those three CREATEs at nonces 1, 2 and 3, verifies their

returned addresses/configurations and returns the pinned inert nonproxy bundle-deployer runtime; any child failure reverts all three. Thus peer addresses are runtime-derived rather than embedded in the CREATE2 preimage, and a configuration change changes the bundle init hash and the entire address set without a hash fixed point or post-deployment configuration race.

Every address/hash and epoch is nonzero. The factory, bundle deployer, legacy Bridge, SourceBridge, Registry and QuotaManager are pairwise distinct; the bundle and all three children equal those derivations and have the exact current runtime and configuration hashes. Factory code/configuration likewise equals the first three descriptor fields, and the support Registry's code/configuration equals its three descriptor fields. After each runtime-hash check, every configuration-bearing source object (factory, SourceBridge, credit Registry, QuotaManager, support Registry and terminal verifier) is read only through `componentConfigHashV2()` with exactly four calldata bytes, a 50,000-gas `STATICCALL` and exactly one 32-byte return. Its selector is `0xf6c0f7d2`. Fault, empty/short/trailing return or mismatch rejects; no opaque factory receipt substitutes for these direct checks. The support registry stores the whole manifest-authorized descriptor and computes:

```
bridgeExecutionHash = H("slot-chain-source-bridge-execution-v4" ||
  u32(752) || canonicalSourceBridgeDescriptor)
```

The descriptor's `terminalVerifier`, runtime hash and configuration hash are exactly infrastructure component row 3. Its one configuration is the 21-byte `address20(activeSettlementRouter) ||u8(64)` grammar below and its getter returns that row's `configHash`. It does not fold the BridgeDomainRegistry, its own address/runtime, Router runtime/configuration or any live support record into a second verifier configuration identity; those authorizations remain independent SourceBridge/Registry checks. The credit registry exposes this fixed descriptor for the source generation; there is no block-indexed executor. Direct admission checks the immutable runtime and record on the same L1. No source state root, beacon walk or caller-chosen topology proof exists in that path.

`canonicalDestinationBridgeDescriptor` has this exact 408-byte packed grammar:

```
address20(bridge) || bytes32(runtimeHash) || bytes32(configurationHash) ||
bytes32(storageLayoutHash) || bytes32(bridgeKernelProfileHash) ||
address20(inboxCreditStoreV2) || address20(terminalAccumulatorV2) ||
address20(terminalDomainRegistrarV2) || address20(quotaManager) ||
address20(nativeLiquidityPoolV2) || address20(bridgeSurplusSink) ||
address20(forceSendHelper) || bytes32(forceSendInitcodeHash) ||
bytes32(forceSendCompilerBuildHash) || bytes32(forceSendEvmRulesHash) ||
u64(forceSendCreate2FixedGas) || u64(forceSendChildGas) ||
u64(forceSendPostcheckReserve)
```

Every field is nonzero and `bridge=destinationBridge`. The account itself must be fresh immutable non-proxy code with the exact runtime/config/layout; no topology byte, implementation slot, reuse branch or delegated getter exists. It commits destination custody, terminal handshake, quota and Pool binding separately from the source descriptor. The hash is:

```
destinationBridgeExecutionHash =
  H("slot-chain-destination-bridge-execution-v4" || u32(408) ||
    canonicalDestinationBridgeDescriptor)
```

`canonicalDestinationInfrastructureDescriptor` has exactly ten component records in this order:

```
BridgeInboxAdapter, ActiveSettlementRouter, TerminalSignalVerifier,
InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2,
TerminalDomainRegistrarV2, TerminalAccumulatorV2, NativeLiquidityPoolV2,
destination Bridge
```

Each record is

```
address20(component) || bytes32(runtimeHash) || bytes32(configHash)
```

so the descriptor is exactly 840 bytes. For kind $k = 1, \ldots, 10$ in that order,

```
configHash = H("slot-chain-component-config-v1" || u8(k) ||
               u16(byteLength) || configBytes)
```

Kinds 1–10 expose the fixed getter

```
componentConfigHashV2() view returns (bytes32)
```

A checking side requires exactly 32 bytes equal to this value under the manifest-pinned runtime hash. Empty, short, trailing or malformed returndata rejects. L1 checks rows 1–3 directly and L2 checks rows 4–10 directly; neither substitutes a Boolean result for the fixed call and hash comparison. The circuit also authenticates the fresh row-10 account/runtime/config/layout and empty private state directly under the imported state root; the getter is not a substitute for those observations. The ten `configBytes` lengths are exactly 100, 344, 21, 73, 60, 52, 80, 21, 76 and 324 bytes, respectively, and their contents are:

1. `address20(bridgeCreditRegistry) || address20(sourceBridge) || address20(activeSettlementRouter) || address20(forcedQueue) || address20(protocolVersionManager)`
2. `u256(settlementChainId) || address20(protocolVersionManager) || bytes32(protocolVersionManagerRuntimeHash) || address20(forcedQueue) || bytes32(forcedQueueRuntimeHash) || bytes32(forcedQueueConfigHash) || address20(builderRegistry) || address20(scheduleOracle) || bytes32(routerNamespace) || bytes32(inboxApplyDeploymentDescriptorHash) || u64(l1HistoryFirstSupportedBlock) || bytes32(l1HistoryReadConfigurationHash) || bytes32(legacyBootstrapDescriptorHash)`
3. `address20(activeSettlementRouter) || u8(64)`
4. `address20(terminalDomainRegistrarV2) || address20(inboxSystemSender) || bytes32(routerNamespace) || u8(64)`
5. `address20(inboxApplyRouterV2) || address20(destinationBridge) || address20(terminalDomainRegistrarV2)`
6. `address20(anchorSystemSender) || bytes32(manifestNamespace)`
7. `address20(protocolReleaseAuthorityV2) || address20(inboxApplyRouterV2) || address20(terminalAccumulatorV2) || address20(nativeLiquidityPoolV2)`

8. `address20(terminalDomainRegistrarV2) || u8(64)`
9. `address20(terminalDomainRegistrarV2) || bytes32(poolNamespace) || u64(POOL_EXTERNAL_READ_GAS) || u64(POOL_AUTH_CLEANUP_GAS) || u64(POOL_VALUE_CALLBACK_GAS)`
10. `bytes32(storageLayoutHash) || bytes32(bridgeKernelProfileHash) || address20(inboxCreditStoreV2) || address20(terminalAccumulatorV2) || address20(terminalDomainRegistrarV2) || address20(quotaManager) || address20(nativeLiquidityPoolV2) || address20(bridgeSurplusSink) || address20(forceSendHelper) || bytes32(forceSendInitcodeHash) || bytes32(forceSendCompilerBuildHash) || bytes32(forceSendEvmRulesHash) || u64(forceSendCreate2FixedGas) || u64(forceSendChildGas) || u64(forceSendPostcheckReserve)`

For kind 2 the two nested hashes are independently derived as follows:

```
inboxApplyDeploymentDescriptorHash = Keccak256(
  "slot-chain-inbox-apply-deployment-descriptor-v1" || u16(104) ||
  address20(inboxApplyRouterV2) || address20(inboxApplyRegistrar) ||
  inboxApplyRuntimeHash || inboxApplyConfigurationHash)
legacyBootstrapDescriptorHash = Keccak256(
  "slot-chain-legacy-bootstrap-descriptor-v1" || u16(136) ||
  address20(legacyProxy) || legacyProxyRuntimeHash ||
  address20(legacyImplementation) || legacyImplementationRuntimeHash ||
  legacyInboxConfigurationHash).
```

The 344-byte Router payload is acyclic: it contains no Router self-address or self-runtime, no Market identity, no activation receipt and no PVM configuration hash. Conversely the PVM configuration may commit the completed Router identity. Every mutually referenced root address is the code-independent factory/campaign/role CREATE3 address defined in Section 4.2; it MUST NOT be claimed to arise from mutually dependent component CREATE2 init-code hashes. Each role deploys in a separate bounded transaction and remains INACTIVE. The final factory call checks every predicted address, EXTCODEHASH, exact configuration getter and the acyclic cross-graph, then activates and publishes all nine in one revert domain.

The component scope is exhaustive. Kinds 1 (BridgeInboxAdapter), 5 (InboxCreditStoreV2) and 10 (DestinationBridgeV2) are release-scoped and use fresh accounts in every protocol version. The destination native QuotaManager derived from kind 10 and the source Bridge/Registry/Quota bundle are release-scoped as well. Kinds 2 (ActiveSettlementRouter), 3 (TerminalSignalVerifier), 4 (InboxApplyRouterV2), 6 (ProtocolReleaseAuthorityV2), 7 (TerminalDomainRegistrarV2), 8 (TerminalAccumulatorV2) and 9 (NativeLiquidityPoolV2) are protocol-lifetime; every successor repeats their address/runtime/configuration byte-exactly. There is no third scope and current-pointer aliases do not change ownership.

The runtime semantics for kind 1 fix its stated initializer authority and bind the protocol-lifetime Queue code/configuration into every release. Kind 3 is the immutable root-lifetime verifier derived and deployed once by ProtocolRootFactory; every release descriptor must repeat its exact address/runtime/configuration while fresh source bundles reference that shared verifier. Kind 4 is protocol-lifetime and preserves its cursor and existing routes while requiring the new route absent. Kind 5 requires a fresh endpoint domain and empty pin state. Kinds 6–9

are protocol-lifetime and preserve all old append-only release/domain/writer/ticket entries while requiring only the new release/domain registrations absent. Kind 10 requires fresh immutable code/configuration, empty private state, full initial quota and `v2Active=false`; exact reuse is forbidden. Tx0 seals kinds 5 and 10 and registers the new kind-9 Pool domain atomically. It transfers no ETH. Kind 4 contains only protocol-lifetime constants and never an endpoint Store or Bridge; endpoint-specific targets enter only through the registrar's append-only route. `configBytes` excludes the derived `destinationDomainId` itself to avoid self-reference. Every component address/runtime/config hash is nonzero and equals the separately named domain/profile field. The infrastructure commitment is:

```
destinationInfrastructureHash =
  H("slot-chain-destination-infrastructure-v3" || u32(840) ||
    canonicalDestinationInfrastructureDescriptor)
```

Rows 1–3 name L1 contracts and rows 4–10 name L2 contracts. At manifest activation, L1 ProtocolVersionManager directly checks rows 1–3 against L1 code/configuration and commits all ten rows. The finalized manifest proof authenticates those L1 rows to ProtocolReleaseAuthorityV2; TerminalDomainRegistrarV2 directly checks only rows 4–10 against L2 code/configuration. Both sides recompute the same 840-byte descriptor and hash. No contract is required or permitted to treat a cross-chain address as locally inspectable code.

Each release profile also publishes the complete bounded `ProfileIngressAuthorizationV2[]` from which the manifest's `ingressAuthorizationRoot` is recomputed. Its Solidity type is fixed:

```
struct ProfileIngressAuthorizationV2 {
  uint8 kind; address adapter;
  bytes32 adapterRuntimeHash; bytes32 adapterConfigurationHash;
  bytes32 adapterConstructorPoststateCommitment;
  address activeSettlementRouter;
  bytes32 routerRuntimeHash; bytes32 routerConfigurationHash;
  address forcedQueue;
  bytes32 queueRuntimeHash; bytes32 queueConfigurationHash;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  bytes32 sourceBridgeExecutionHash;
  uint256 destinationChainId; bytes32 destinationDomainId;
  address destinationBridge; bytes32 destinationBridgeExecutionHash;
  bytes32 destinationInfrastructureHash;
  uint256 fixedIngressWei; uint256 executionWeiPerAccountedGas;
  uint256 proofWeiPerAccountedGas; uint256 permanentWeiPerByte;
  uint256 maximumAcceptedFeeWei;
}
```

`INGRESS_AUTHORIZATION_TYPEHASH` is the Keccak hash of the exact canonical Solidity signature above with the field names and order shown. `authorizationId=keccak256(abi.encode(TYPEHASH,rowfields...))`. Kind 0 requires a nonzero `adapterConstructorPoststateCommitment` equal to KCS1 recomputed from the profile's exact constructor tail, configuration and runtime. Kind 1 requires that field to be zero because its separate SourceBundle deployment, BRC1 and one-shot seal are its constructor/lifecycle authority. Kind 0 requires its three source fields and all four destination-topology fields (domain, Bridge, Bridge execution hash and infrastructure

hash) to be zero but binds the nonzero destination chain, Router, Queue and exact kind-0 adapter. Kind 1 requires all source and destination fields nonzero and equal to the separately committed source Bridge and destination infrastructure descriptors; its adapter configuration independently binds the exact source Registry/Bridge and Router/Queue constructor graph. Both kinds use the same nonzero five-word fee schedule.

This version has exactly two rows: one kind-0 row and one kind-1 row; multiple rows of either kind reject even when their adapters differ. A future kind requires a new profile/schema version and cannot consume latent capacity or reinterpret launch roles. The decoder rejects an unknown kind, noncanonical ABI padding, duplicate adapter, duplicate authorization ID or a row inconsistent with the profile graph. IDs are sorted in ascending unsigned `bytes32` order and

```
idsHash = keccak256(sortedAuthorizationId[0] || ... ||
                    sortedAuthorizationId[count-1])
ingressAuthorizationRoot = keccak256(abi.encode(
  INGRESS_AUTHORIZATION_ROOT_TYPEHASH, uint16(count), idsHash))
```

where the root typehash is Keccak of `"IngressAuthorizationRootV2(uint16count,bytes32idsHash)"`. Concatenation contains only the fixed 32-byte IDs and has no length prefix; the typed root supplies the count. ProtocolVersionManager recomputes this value from canonical profile bytes before registration, including the derived kind-0 constructor commitment, and stores an append-only `authorizationId->row` and `adapter->authorizationId` index. No caller-supplied root, row-validity Boolean or post-cutover governance bind is accepted.

The Queue exposes this exact constructor/configuration commitment:

```
descriptorSchemaHash = H("slot-chain-force-descriptor-schema-v11")
queueConfigBytes = address20(activeSettlementRouter) || u8(64) ||
                   u64(UINT64_MAX) ||
                   H("slot-chain-force-empty-v2") || descriptorSchemaHash
forcedQueueConfigHash =
  H("slot-chain-forced-queue-config-v1" || u16(93) || queueConfigBytes)
```

Its account runtime hash and this config hash are the two row-2 fields above. The depth-64 vector empty leaf is `H("slot-chain-force-empty-v2")`. A parent at height $h = 0,\ldots,63$ is `H("slot-chain-force-node-v2"||u8(h)||left||right)`. The queue commitment is `H("slot-chain-force-root-v2"||u64(count)||treeRoot)`; leaves at indices `count..2^64-1` are empty. The canonical range proof recursively visits the smallest tree covering `[start,end]` (including a boundary leaf when `end<count`); it emits in DFS order exactly one hash for each maximal outside subtree. The verifier rejects extra/missing nodes, a nonempty padding leaf, or a reconstructed root/count mismatch. Append at old index i starts with the new leaf at height zero, repeatedly hashes `frontier[h]` on the left while bit h of i is one, stores the carry in the first zero-bit frontier position, increments count, and reconstructs all higher right subtrees from `FORCE_EMPTY[h]`. Genesis frontier words are exactly zero. After increment, the fold is

```
node = FORCE_EMPTY[0]
for h = 0..63:
  node = bit(count,h)
```

```
    ? H(D_NODE || u8(h) || frontier[h] || node)
    : H(D_NODE || u8(h) || node || FORCE_EMPTY[h])
```

Only set-bit frontier words are meaningful; stale zero-bit words are ignored. At old count `UINT64_MAX-1`, append writes height zero and produces final count `UINT64_MAX`; old count `UINT64_MAX` rejects, so leaf index `UINT64_MAX` is unused and no height-64 frontier exists. This is the same root as the full vector and uses exactly 64 frontier words. Every leaf and descriptor list uses `u64(index)`; no queue index is narrowed to 32 bits.

The terminal vector uses depth 64. Its empty leaf is `H("slot-chain-terminal-empty-v2")`; a parent at height $h = 0, \ldots, 63$ is

```
H("slot-chain-terminal-node-v2" || u8(h) || left || right)
```

and leaf index i is

```
H("slot-chain-terminal-leaf-v2" || u64(i) || destinationDomainId ||
  address20(destBridge) || creditId || u8(terminal) || liquiditySettlementHash)
```

with `terminal=1` for DONE or 2 for FAILED. FAILED requires `liquiditySettlementHash= bytes32(0)`. DONE requires the nonzero exact `H("slot-chain-liquidity-settlement-v1"|| ||ticketId||address20(l1Recipient)||u256(value+fee))` authenticated by the Pool's atomic consume return; substitution of any ticket, recipient or amount changes the leaf. The committed root is

```
H("slot-chain-terminal-root-v2" || u64(count) || treeRoot)
```

The accumulator stores only `frontier[0..63]`, wrapped root and checked `uint64count`. To append old count i, it starts with the leaf, combines `frontier[h]` on the left while bit h of i is one, stores the carry in the first zero-bit position, increments count, and reconstructs the tree root:

```
node = TERMINAL_EMPTY[0]
for h = 0..63:
  node = bit(count,h)
    ? H(D_NODE || u8(h) || frontier[h] || node)
    : H(D_NODE || u8(h) || node || TERMINAL_EMPTY[h])
```

Frontier words at zero bits are stale and ignored. The complete canonical leaf event is the proof-generation data source; any party reconstructs exactly 64 siblings in ascending height, and the L1 verifier uses bit h of the index for left/right order. It requires `index<count`, status DONE/FAILED, exactly 64 siblings and exact wrapped root/count. Append rejects old `count=UINT64_MAX`. Carry-depth and full-fold gas remain subject to the mandatory Foundry calibration in §14.

A data-chunk leaf is:

```
H("slot-chain-data-leaf-v1" || bytes32(sessionId) || u16(index) ||
  bytes32(versionedHash) || bytes32(fullBodyRoot) || u16(blockOrdinal) ||
  u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
  bytes32(chunkRoot) || address20(publisher) ||
  u64(validUntil) || u256(z) || u256(y))
```

Each cell stores fixed `bytes32peaks[12]` and `uint16count<=2100`. Slot h is meaningful exactly when bit h of count is one; zero-bit slots may be stale. Append at old count n uses leaf index n and carry=leaf. While bit h of n is one it sets `carry=H("slot-chain-data-node-v1"||u8(h)||peaks[h]||carry)`; it stores the carry at the first zero height, then increments count. Old count 2,100 rejects.

Logical peaks left-to-right have strictly decreasing height. The single root preimage bags them rightmost-to-leftmost, which means the repeated tuples are encoded at set heights in strictly *ascending* order:

```
H("slot-chain-data-bag-v1" || u16(count) || u8(popcount(count)) ||
  (u8(h) || peaks[h]) for h=0..11 ascending where bit(count,h)=1)
```

The empty MMR uses the same preimage with zero count and zero peaks. An inclusion proof requires `index<count`, derives the unique target mountain and base from count's descending set bits, and accepts exactly target-height siblings bottom-up. Sibling orientation is derived from the local leaf index, never supplied as a caller bit. It then accepts exactly `popcount(count)-1` other peaks in the ascending bag order above. A wrong height, duplicate, missing or extra sibling/peak, inconsistent count or unused proof element rejects.

The proof value is exactly `DataMmrProofV1`, containing only two `bytes32[]` sequences named `mountainSiblings` and `otherPeaks`. It has no caller-supplied target height, mountain base, peak height or orientation. Verification initializes `base=0` and visits set bits of count from height 11 down to zero. For each set height h, its mountain is `[base,base+2^h)`; the first such interval containing index uniquely fixes `targetHeight=h`, `targetBase=base`, and otherwise base advances by 2^h. It requires `mountainSiblings.length=targetHeight`, hashes them bottom-up, and derives left/right at level h only from bit h of `localIndex=index-targetBase`. It then requires `otherPeaks.length=popcount(count)-1`, consumes those hashes while visiting count's set heights from zero to 11 and skipping the target height, inserts the reconstructed target peak, and rebuilds the exact ascending-height bag. Exact array lengths plus final root equality reject wrong index, count, target mountain/base, orientation, order, value, root and every unused element. Empty MMRs have no inclusion proof. Proof identity is semantic: no signature, commitment or replay key is derived from raw ABI calldata for either array, so ABI offset placement or unrelated trailing call bytes cannot create a second protocol proof identity. Implementations decode the two arrays and apply the checks above; they must not hash the undecoded proof bytes.

A manifest leaf is:

```
H("slot-chain-manifest-leaf-v1" || u16(position) ||
  u16(blockOrdinal) || sessionId || u16(recordIndex) ||
  u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
  fullBodyRoot || chunkRoot)
```

Its next-power-of-two tree uses

```
H("slot-chain-manifest-node-v1" || u8(height) || left || right)
H("slot-chain-manifest-root-v1" || u16(count) || treeRoot)
```

for the node and final root respectively; padding uses the domain-separated empty leaf. For zero entries, `treeRoot=H("slot-chain-manifest-empty-v1")`. The zero-discretionary- transaction body is the ordinary body encoding `u32(0)` and therefore `H("slot-chain-body-v1"|`

|u32(4)||u32(0)). Tier 3 requires exactly those empty body/manifest values and the zero-count session-list commitment. This tree is instantiated separately for every block and contains at most 2,100 entries. position is exactly $0, \ldots, count - 1$ inside that block, and every entry must carry the single expected enclosing block's blockOrdinal; blockOrdinal remains the candidate-wide block index. A missing, mixed or substituted ordinal rejects. A two-block candidate therefore commits two signed manifest roots and never hashes their leaves into one shared tree. The InboxApply disposition commitment is:

```
H("slot-chain-dispositions-v1" || u64(start) || u64(end) ||
  u16(count) ||
  (u64(queueIndex)||u8(disposition)||u32(txIndex)||resultHash)*)
```

The list contains at most 64 rows, checked arithmetic requires end=start+count, and row i must carry queueIndex=start+i. A gap, duplicate, reordered index or uint64 wrap rejects. Here UINT32_MAX means no Ethereum transaction.

For a raw discretionary body chunk, the byte-exact commitment omitted by any ABI convenience wrapper is

```
chunkRoot = H("slot-chain-body-chunk-v1" || fullBodyRoot ||
  u16(blockOrdinal) || u16(chunkIndex) || u16(chunkCount) ||
  u32(chunkByteLength) || chunkBytes)
```

where chunkByteLength=chunkBytes.length; the explicit length precedes the sole dynamic suffix and trailing or differently partitioned bytes change the root.

The recovery ID is the encoding in §7. Schedule leaf/root, seed and tie hashes are as in §4; body and blob encodings are as in §6.

The implementation rounds deliberately separate encoding primitives from stateful algorithms and wire protocols. Round 2 owns checked, domain-fixed preimage encoders: width conversion, canonical validation and the individual fixed-record, bounded-list, leaf, node and root-wrapper hashes above. It does not own tree/MMR construction, frontier append/fold, proof traversal or inclusion verification; Round 3 implements those algorithms while consuming the Round-2 hash primitives. Exact external-call and returndata codecs, component ABIs, and migration or legacy-campaign wire codecs are implemented and exercised only in their later owning rounds. Treating a later algorithm or call codec as a generic Round-2 "encoding" is a scope error.

Round-2 conformance fixtures must be mutation-sensitive rather than merely representative. The oracle therefore publishes the wrapped empty data bag and a three-record MMR root whose two bag tuples are exactly height 0 followed by height 1. Reversing those tuples rejects. It also publishes deliberately asymmetric data and manifest leaves with recordIndex=7, chunkIndex=1, blockOrdinal=3 and chunkCount=9; the manifest position is 11 and the chunk bytes are the ASCII string asymmetric-data-record. The corresponding independent node fixtures use data-node height 7 with right child 0xa7...a7 and manifest-node height 5 with right child 0xb5...b5. A Round-2 conformance suite must compare these leaf and node results directly, exercise the valid two-peak ascending bag, and reject its reversed encoding. A fixture where recordIndex=chunkIndex, a one-peak-only bag, or a height-zero-only node check is insufficient because it lets a field swap, reversed peak order, or ignored height pass without changing the expected hash.

Round-3 conformance fixes proofs without redundant choices. The fixed-tree fixtures cover registry index 3, admission index 64, ranked-entry index 0 and tranche index 7, including exact proof digests and roots. The independent data MMR fixture has 15 leaves `H("slot-chain-round3-mmr-proof-fixture-v1"||u16(i))`, target index 10, target height 2 and target base 8. Its other peaks occur at heights 0, 1 and 3 in that derived ascending order. Named rejects cover wrong index, count, root, target mountain/base, sibling orientation/value/length and other-peak order/value/length. The four position-bound empty roots and direct Force-count-1, data-MMR-count-3 and Terminal-count-2 frontier digests/roots pin initialization, append and fold independently of full-history root construction.

`commitment-model.py` is the normative executable fixture. It exports all 812 records through one deterministic typed oracle, including records owned by later rounds. The Round-2 Solidity conformance test executes only an explicit reviewed allowlist for checked fixed-preimage encoders; Round 3 consumes the tree, frontier and proof records above. Still-later rounds consume their remaining records when production algorithms and ABIs exist; export is not evidence of premature implementation or coverage. The initial vectors are:

```
TYPEHASH      ee6a8c8e31e8245cd527869508f6e464d6084893991203876f734d1855aed87c
registryRoot 0bf297d7b9b6a5529a319a06cb08484923a89bab15d51f8baeaf5c30bebdf3fd
admissionRoot 3bf2dcaf78292c832108e29205bf99cc2d22137a0545e4528d8da7309d4b482b
entryRoot acee83a690b868a4a7960c55a9f7228f91cad26b704e24106d4db87e9c7a8f34
forcedRoot a54e9f797ffe7f04dd5ca7df4c858edf02ce45a81808522633fc9cee8fe72e57
emptyRegistry 2f40c290594200091bcd31881e40bf56ba1960016a24c9931cf3b1a9a8705ae8
emptyAdmission 71a511ce5247c6c3b0411e182c8e4b4dcbd0adc97163c585cac94ca3b031ac54
emptyEntry 986d3e795bd9ddfabe213b93cea0211eea5a663e895bfc112d90c5bf2fff1564
emptyTranche dee49bfb4494eee086cf6485f471866cd49591358b3490d4a156813702501768
registryProof 188d53708f68cd601ace09953d87031b89c92f5b9d7bffd847e324ab7adc8261
admissionProof 65a5501dc5440301031bc0d21ac6506ce1f98b226139c93ab54251883d5bd12d
entryProof 89655918472451054bf7362e8e7b19bbded634877c532efdfb545e0ca36667f3
trancheRoot 12ce6da6104ea0fb21edbc0e932f7500fabf18173911a48bc5e1d4d2af1cc336
trancheProof 8d56574545d36cb4cbb7f1452055275a3dbd8e5b2aa9062e717095f387a0a2d8
forceFrontier1 1ba50296675195de90944defd33d2e346c1a996aa6edbb5bfb1b187cc64b3c1d
forceFrontierRoot1 5cb37cac8283fe7742e0f8e40bacececf85c1f8c15a05c8a9c9dc2018ac532e8
 sourceDomain a955e9cbaafee3fb51bca7d966012607d8ae8354574180ecbeb6ce71ed159f26
 destDomain 123173ae26db2c61b26412a377907f65420237d7c36efc79c59710e43e6be77e
bridgeKernel a23f9994e8d1c475500768b67cf2b2d1f7a0f367df6f44bac5ccf1fa12bc1338
 bridgeExec 9cbef2cdc597906c40300da43a0b3296fb529a82094f4be32f7e8b7ed70d53af
 destBrExec b570703833da3df51773eb8d72d8d38074391da1f4f4ef2198b0d074f3c81600
 destInfra 766a9dc18c36febd06b3818232d6bd0004b16cc2b5bd1a3239585d66fb361566
 poolRowAlt 47565d6e5708e5a516daa376505c4e8db91c6dfa83127214b55a88e14fb84086
mvConfigType 0acb8f9e39dd43a4208edc38c8925bbd3433e72eb46f9e5fac584bd14a970b98
mvConfigHash 950691bac22ceef2dd834142420a79e589bc489165cb10815e889475c39d2613
mvDescType 6f1be44c261607b4c2345985bfb5a081aabd621b9168982df9b7820c7dedebd6
mvDescHash ba4e50bfb5fabf8cd253f48ed38e4cc1a4f3fc04174ae94e4dd971b1d6e36eed
mvConfigSel 476b9aef
 relTypehash e7dc28b9b7147fbe9f9b0e0bf910800983be65e9ca19020f296fb0e780a0804b
 activateSel 33f5ca80
 relManifest b6594be9afa1d7472e9d9b8b1b7027435dee9fe8b318cc5e9ce20ebe717f43ba
 succManifest 9cc41591925c1bc6845e391029d9f148e0c7d45ce0afe84ce6fac11dcc94b65f
```

```
 destReg c66eb449fc3da2146daa9e9308137bec087f4159de08ac0d220d23d4a951d560
migStmtType 832785a7f9ce32f97dfa06fb39b9c583c13e65a5967c88f7c8c3fbda94fcef2b
deployType babd40345cd96b87434cc1bcc50b0c693556c37283b9b3399abf60339dc363a0
 components 39c81654d0554b17c41192b3e5f9069488373902e9aa7236cfc76cf8a5db5408
 genesisDeploy 900ae1ae455245accc5522cfde52d3df63278baa4e6a1d30cd7abf34b8f40309
 versionDeploy 328cd08a675b046c0ba73044bf1c8cd3f16e07a78dc34fc0f17d52cbfc3c1d97
legacyCkpt 6ab84b0c77035309ac300830aa1c31d95f68c44b697d97a6836f7f3f54bf0708
 genesisStmt f98dce382f3a3733bd3a89cfe43ff07c706f116e6fe68cc4a4bd090108c9ec88
 versionStmt 769dc1d028023e5920cb898c3540c3ca3202c2af6cd774ae3ce811248ffc92d1
genBaseCore 12243b817561a9362bb03dffb36bb73f9314d47e3165673793419692cd8a8566
genBase 5081c70042287a8b7156b2626675ea4ed9623c755d5f51b5c83be94e90da3ff3
genCandidate 90949c0865a5e0226c6c0d736c0533f9e5f1d793dc9c5d21407a210ee3896338
genOutput 46ecec9370133392051393730b413c8e508bc579ec9c0ede247161132134a8
regStmtType c049f967468e58f1a5c9b9e1a147dfc233695ae69c5d4a95ec4ffb49b5687da0
regRouteType 4368ad9403b46ef3830e21af8cddcaedbd444c8c57bc3414ebc6fdd250e1e6da
 regRoute 8a92802ac27ef7bc1fe751df95c3543df477ec7e7220793bad45d7c7d3e0c05d
verifyRegSel 33639818
 regStmtHash ba6aca55dd8732be6d74273ff0eeeda4d9d4838e79d6fde760bb3928197267b5
regProof 0027acbf8d87ef5b7c901c3dc27af4b56f1e4fb64953cf87f6fcd6ee878c963c
regCfgType 38f7fbc63e45f650bd5cbaffba0d81d5ca69f21ebdddfa74280d7e6eb5c319d6
regCfgHash 015ea002adbfab085cf51db1714c889961bb1a416cd69b7d22f1f816a5787e4c
regDescType 9533e2f6ac9bcf6830306a9ef5d14e21a06008f9ceb59208d09a8d7ab1e6100f
regDescHash e3c4dd9d13404822af1360b3bc97f3d82dc1dd7124a137e900c6351270505310
regCfgSel 59bfe418
 verifyRegCall 735ba835ff03812b1cd376c5ea133ea5d3016833ad3f812ddab58fc0640ac224
verifyRegLen 516
srcCtxType 6069dff5f628f94ceff984d5ce3ef62019eaeb4efd7e0627c45a14677bd13c70
 srcCtxHash 003912c395f0a29be213bb41bfb1f171cf5da31f549f6b8bda07bfbeb67c00a4
dstCtxType b8170dbf684e1fc4dd4dae8fb78ad24984cc0aa0c03cbd1e29b1b2eb5728eefb
 dstCtxHash db2d32748d67c58393965146124540af9055cc8adc7179f934bf47d4ac1f06e3
ingAuthType 7d996545314cabc22ca6b931d7f0e88f910b93ff80840cf27b339490d35845b2
kind0Auth c825101ced1a3a9749e8923e95680e717166327dbe26572999c134a18829caad
 kind1Auth 3728bb4741d1ecd493da584356b9421882aedb92bc7dafc7ebd4961a7ec89c1e
ingRootType c7b11126d8d1984cc17cbc108be2a1be0ef9c4e8fd519fa9033c949962f1b042
 ingRoot 31612dec382343351a15ff38ea1bb5d8aab57060c9ca1abc7f614aeac3944612
sendMsgSel 9211d7e9
enqCreditSel 81805d6b
msgTuple 0e85a708462e96cbaca7158a1534011a25137c3b7aada7f381e4fc5b3afbe40d
msgData f08683775f4a25dfef721c487073fb77026d45ac57e423424290e47af9fd2835
sendMsgCall 9099cce58d3f36714ee59f37510261394e31583f8bd55da4551b653b5a060e12
sendMsgLen 516
enqCreditCall 02db9c77025d4c2bae1d3f79849020e6d8711c4b6b6c54ec45ce9181e48cfeb4
msgPreLen 576
normalizedMsg f3010702b7b7bf10b6dbfe396a4c7ab07e7c560c28ffdbf6ff8d01d9a96ea4c7
enqueueTxSel 9f06b1b4
enqueueTxCall 3e802e6f72e50c267cc18d256c2863d04446eb8f2d47bc474bbcb9c68876d19a
enqueueTxLen 196
syncSel 6c880b72
```

```
syncStamp 1fd01b194948c635358fbb51b4a5f32f8ceab4dc4153e0230215f8afc94ee434
syncChanged ef662a629ce07c9ed715124d8141a6e430d0a3065f8ce8074a7ea95e8751f184
appendSel 1927261d
appendK0 caabdaadee6df8cea48fb88dacad863e6e24e13f5b0c06f99408d5c71611788a
appendK0Len 388
 appendK1 df92daaee8cffdd28f003e0b1c748cb294297c86cd1e79de08f03ddf15c1617a
appendK1Len 708
queuedReturn 76f821bd39721ec0e26efc55d7b667d20aab74992e0feae7d7755e386ecd694d
inboxApplySel 6b326168
inboxApply 65332d0b3230b33c0ae8bddd8a1f3be8473739f865e73fc8e52f4f99cecc89d7
inboxApplyLen 14436
markBatchSel a92f72cd
 markBatch be9129ebd6027b0d846db7ace60a2a8e8c3526a407556ed7b99de4cb04092b0d
inboxMagic 49425632
routeCfgSel 4b64fa11
verifyPinSel 720f747b
 verifyPinCall 5b788adc101bbf80b73678356ace6b4aca61da85302d4747a155f2fa33adddf4
verifyPinMagic 49435632
 verifyPinRet 0e5a24d737136abf8a7a010846356b518761e7813af6495e341c961c51f5756c
verifyPinRetLen 256
liqQuoteSel 43dc48e0
fundStateSel 6a9a6c32
attemptSel 4cbe2fe2
 attemptCall 0a5967f00fb99235d29f71624346a9de42c1de0e8e1eb436a7a375e453ed276f
attemptLen 1124
finalFailSel 745dcb69
targetErrSel f9cc2b44
 attemptDigest c4f86e156a64040bee74ca23cecdb4c8e7bcae139ba95eb49003ad008a2f53ce
 nativeQuota 7364d461f03b17da03a689abfdb4e723065464c3419e3a222bd8ee29cb876ee5
appendTermSel abc194f5
termCommitSel 2c984c97
termStateSel 998c57ed
ticketId 5dc074de7029f6762c8c386b15cbd61cc7ad94b9432c25a35a5ae48e72c3bf2b
poolCfg 680054a7895708487ddd24a202ddf1069f2ef81238e909e826c368d1da396881
poolReadGas 50000
poolCleanGas 50000
poolCbGas 100000
poolDepSel eda2a3f6
poolWdrSel fe4f5ccf
poolProcSel 5fbbe107
poolRetrySel 031f93e7
 poolProcCall 47dc72deef658ef7706f048546e451720725d289fc3999c627641951e30ab9d3
poolProcLen 1028
 poolRetryCall 5684e0a109e39ad3ef09ca453f04da4cfcce30f8a0d32a6e77830e4692fa5a7d
poolRetryLen 1060
poolBridgeSel a535a986
 poolBridgeCall c72c0755e2b5341c9c5ab67a62f9bef26ad674828a837e74a9478cc6c9022c48
poolBridgeLen 1124
```

```
poolTickSel 5defa7e1
poolAcctSel f2b3441e
poolConSel 37093d2a
poolCbSel a34908bb
poolAccMagic 504c4132
poolValMagic 4e4c5632
poolResultMagic 4c415632
 poolAuth 703adb389bc4f60c23e557d7a80817bb66127f29517749c3b918aa16000614fc
 poolAccept 38b65eec03d37ae72cb7395f9d19537d9bd5356c794c5805a8136ddc4da21dcb
policyType d702a337b74fc40bfc746fb1aeeaa705e60a95947bfc3076c76222703205b4b1
policyHash 5eb7c00399d64d91e416d5a3dbe75187c39dfc2a4645b867864b5fd9e649f3e3
policySel b2d0e286
hookSel 7f07c947
policyMagic 49505632
 l2ReceiptId 9822bea9fecc7a319d988cd1885fd8ea0c02a4e7b13ff42d6971142fa9224757
 l1ReceiptId 63b846ea90c10457129a0b4259f4a022a10b99fbbff9ec421ccf4d9c04c52004
 execProfile 2d73aacf177350dd88961abf28b86dbfb9c79c0907ae916bb83f1b6d0d9a274b
factoryDeploySel 4af63f02
factoryCfg 4bf4831c096e251d0ef171bc0fd0582319fdc7ad7a2bea91a6b9b64601aa2837
marketAuthCfg e348b5cddcb8eb6da9bedb54c23e10cf9fe0c20b9f50f769258b47dfdb870c07
 deployDesc 14f68ac2868e0af536dfdd9351e4981ac4857b52cf96ea028eb1258d4a296a81
 deployDescAbi 5218d672f53f8093fd5be0522f7863fd82c2627da461b0410a13b857d2857422
deployDescLen 256
succDeployDesc 377aa582c38874cf50ef87030c628b15c5e0302e6830e51b09d8542659bedc18
 targetReg 0ce538d6056ad9a98b71822ce8e14ee74c792795ee37a5feb2d24ce534deeac2
 succTargetReg 4c025c81e10a40d73a642b2135a7cec30d62ed5b078de89c7caeaa413a1942f9
 mprHash 2d8b21e094b1ca3c4085ce8ef6dffda4e9f07a14b5e194476eab152ee9c69304
 succMprHash 7e1aec64542e3fd1dc19efa0002db7550c23115553ccd98383bbee0378ab8209
mprSel c65ff64e
mprMagic 4d505232
mprCall 3dd403c33c2e7c4e6b324efac7461b5b0cc6f3696a2a8d8c94c9cb9bafcf9c85
 mprRet e4aef6e7c28950782fe03b7d2ba982ce2db4144ae584319cc5b24e7bd33b5456
mprLen 768
opDomain edc1be882290e6241a79546c41db66a1d975095ae60cf3b2b16f1ed876f6038b
timelockDesc 442d9b608ecc43eea4009fb7f95c764c747636f42c885174c1442681cc0ab495
 pvmCfgHash c363eb350b2bf5021b6e4bc882d42d480f454c2889b6b3846d277ad985b16e76
 pvmDesc b4e57361ef06ff7db13eb812b123a3cdf89ce8a562cc138f72282a0a061e4021
soc1Sel e3d91a33
soc1Magic 534f4331
soc1Ret 8df57002258b7d91f346af8230d0ac6145972ab5d17570a9537e143f8ca7c802
soc1RetLen 576
protoDelay 604800
maxMigLive 604800
armExecWindow 604800
protoReadGas 100000
queueOpSel bd5c80a8
execOpSel 31d81ea2
cancelOpSel 5701c308
```

```
applyOpSel af3927f6
expireForkSel 379c9ecc
timelockCfgSel b80095ca
pvmCfgSel 4deb7821
armFreshSel bc4707fb
opGetSel 4b80fe68
timelockMagic 50435431
pvmMagic 50564d31
armFreshMagic 4d414631
opMagic 50434f31
applyMagic 50415031
expireForkMagic 50465831
forkChangeWindow 86400
forkQueueInclusion 900
forkRenewalRunway 1811
 regRelPayload 750cae8318e59ead577835d3b4025aa5c00ba19844706d5e0b52fd6b8486b1e9
 regRelPayloadLen 11392
regForkPayload c07ba77bf323b341a5c54ef97d01da41c78e962f9c51ab2b9644e7cd6c003012
regForkPayloadLen 576
regForkReviewEvidence 14795f387018d6c52030f07fd28f32b34abc62439c2dd8aa28785bf5156bfc68
regForkReviewCert 15a242cf0fd70257617f69c7ccd4075d46c7e76ae22377f878a0884f3cd9eafb
replaceForkPayload f9d94647d71f55db575b40a58082faf63ec67e4897b097f83d7af5195807f132
replaceForkPayloadLen 1536
replaceForkReviewEvidence 3ec105c1c8c3673411009311dfe6a66d3136ada378f65469528381b3ae0d25cd
replaceForkReviewCert 8d6189732cb1cf1bb44c997a3d0e64b70b595c1bc02dcc1297fb7c438f30ea76
splitForkReviewEvidence c05a310757688c48297f6007e5c8cd2b684f2ef2f2786a85421a64c37ea30e1c
splitForkReviewCert b148161fb4f81c944885fe38cdf2588d21ebe9ad5dcecb5d5df2804a31cfe1de
splitForkQueuedRet d9abc75048f1e1f7ecbed1986331ed37af38716dfbba28a4015c38ecd580c25b
splitForkExpiredRet 36100d0d72d0f520507901b3d3a5f50bc01d53c956252df1606bdef7debe030a
expectedPredForkReg 233d72ef0935f897e49377ce160962e2ea7ea14c37c309bb094b78e0b50bb118
expectedOldForkReg eed5e788c296e7b8449f4aacb62cd9e5b15ec488dc667f4cf3e3aa48a244c6c5
splitForkPayload 92ac8d8cd580e118ad74b14b48a8e7739594ad8e3434834619bd626e183a16ff
splitForkPayloadLen 1056
 pubGenPayload f21eb3f7bb3ccaf8b200333c469f2932d7621bc121aea52fa9cd4c01b244b798
pubGenPayloadLen 320
 pubMigPayload 286c3a7f9f6b8531df3a2bb2ba7f4bca4b94d06c331b18a207dc520bb287813e
pubMigPayloadLen 128
 regRelOp 7979b57e53962a0bb028a846da3b71efcfea05448fb1af6f8d1033e201de871c
regForkOp 59169620023947479c6848e10e72e21f6bad34f51f9b38d30c504b6c522bbf3d
replaceForkOp fe46b8930ee06263cc46f6fae27477f35528a751133875cc228e0126d8561322
splitForkOp abce70ea742b56d07e890e5425a1eb355ac2304ace53d73ed105e746ddeceede
replaceForkSel b48bbef1
replaceForkCall 7f5672e814198405380028deb977bebdad7c69273d1b9b2c72edf8ca26982aca
replaceForkCallLen 548
replaceForkRet 9b9e757b4ebf73b65db810d4bd2da658f26d189916c4b97595c5470ad7972534
replaceForkRetLen 160
splitForkSel 455b4ff5
splitForkCall 793eb33ab7e71b38a63b6b04704dbdc816de18915df79b09af7d75c4e812b41b
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splitForkCallLen 516
splitForkRet a3e070e8b9e742f15f072cb46690cf08b62bf018a99b83666455adec80e54833
splitForkRetLen 160
 pubGenOp 4d4157b55f20723f505ebb603a447a3180509fe96c56f45b39644a0a2faf21e0
 pubMigOp 91e95eef0ba0ca3edce846a4eb0a3dcc987662a760b3df6f0d566ec70db7e0a6
 queueOpCall c73900f892f4842af503c2dd847fbeb1a866ba86473fdc24c6ebe9c464a088f0
 queueOpLen 11492
execOpCall a00db7a4bd41a8e2546adec7f99463a7c370a517d1b67e34f91778bf65ffcf94
execOpLen 708
 cancelOpCall af81e8eb6e2b69325d6343e98230dcd0f2e5da91daafd61832713cb525d5e946
cancelOpLen 452
 applyOpCall c9f9a0de0fb0a078fb5ba9abb977e24253ecd22008ecad075228242605cb2852
applyOpLen 260
expireForkCall 0674d1859f790b11abfb90dbf54a4f3ceb88574b484c62199a0f0febdb0e9ec7
expireForkCallLen 36
expireForkRet a338559b4b34fc40aaaf1c9d8b3837a34e59a5c4f826e855c9e27a5dc0d04b6d
expireForkRetLen 64
timelockCfgRet 1813d922f5db2c6e82ca7be236e9cab4b8ca3c8b5f32231d60bd51b75e54b29c
 pvmCfgRet 3305018df3355e109413b0c25e69814e56a645ab71e4fda4024bb31e84feadd6
armFreshRet b7483e72711cfc12792d104e0b0bc11c51e9e51ba92ff0978675a1e336756dd1
 opGetCall 33db780ef52173d3664c9a6e9ad9e843d7c3b3835dfecae60a55ae0d7f4ae53a
 opGetRet 9048743c1e6ca5f512e6d0da01be9a5155158454a6ffebce66c8c640e7ac3ced
applyRet f9361bdc99345939494c4794891942dafae95fa74ace38bcf4430ae53ac12c4a
leaseSel aaac4c97
leaseAbortSel ea3e96c2
leaseMagic 564d4c31
 leaseArmId 8a9dd52cad8bb4b7c6e4ed6dfdf1f56c21bfff2589c87fbf873a6982dd02232c
leaseAbortAt 624800
 leaseRet 8aace3b0bf7d453d337c25aa93b3b4cf48c93691e16dbf48011b802e768d25d8
leaseLen 320
regTargetSel 9aa71eff
targetRowSel f588fec3
targetRowMagic 52545232
 regTargetCall 4ab406d631295bfab335f771d8815cc6f2c6e4b13081a53cc76335a89e2123b2
 regTargetLen 11396
 regTargetRet 79246a4401d3c56245712aee0da97f0ea06f6a76422c21908647dcf1d4ed72c9
targetRowCall 5c5db1ed7485ad89b4d9125c80db863078603f32ddd3032552a6345bd0dba5ee
 targetRowRet 7ba50fcac2193351bed4267f07eb2a974827747adfe1bff6d9987c37a1a28025
targetRowLen 512
pirSel 2d2bbe23
piaSel 2181b974
pirMagic 50495232
piaMagic 50494132
 pirRet 0c9045484a4e36879db2aac9c499b07ced0892c62cecf12b644c6397f236bae9
pia0Ret 160cb5e1f74e0e42df2a95218e0c124e7a728107de3d6530fa60513f8392205b
 pia1Ret 746910578abf39247a18029200b0690cf4d448bdccf873df424c316638835cf7
authInstallSel b1a3fef9
authGetSel 1693ae01
```

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authInstallMagic 53414931
authGetMagic 53415431
 authId 10ebe04656d3d5a4a1c4cab773e149c180473ba5a31cd193a85d313ddabce908
 authInstallCall 078bc64984dcff624d011d636a85cccc66e0fa342504feb3250f11286363ef3d
 authInstallRet e5ccb7b5d25ca7a2f734daae6a115354b91a0c6e47e0b1d319ea0757520afcbf
 authGetRet 73c9fa7c8c9db37a089b81ceb6144295848d8a9021fd7b84b7ffd1b6e2283b6d
seatStateSel cf52185b
seatTermSel 76d5ecd4
seatDutySel 9a649489
seatAuthGas 100000
forkConst c17afdd5367849af551b3a2f322a2c351321e916ad913a78b646586fe34f59fc
forkSchema 623a145841922b79691d33358ec5233f02cf68a4176d9e7c2cfafaeb357fa63a
forkCfg 9256c85ac8b81181ff7538f757939218a42c945cb7228ed5755d10f5e4ef643d
forkInstallSel 9bb6fe73
forkRowSel c614591c
forkCfgSel 44efa773
forkRouteSel 7e9f3c0d
forkVerifySel 7e981e0b
forkInstallMagic 46564931
forkRowMagic 46565231
forkCfgMagic 53465631
forkRouteMagic 46525331
forkCarrierMagic 53464331
forkInstallCall 67ab3af72e37b0366daf3887450a66395aa7d45fb4114d323d6d80a083ae6dbf
forkRowRet 252a8f9b5199a2b88f07e01d89d4fabb49903713d19500ca1cf34ad3fbf9c00f
forkCfgRet d71c98eddf7e49dfebf8278d29f45aa24a115e507fb309965fcaf8f523409dba
forkRouteState 4ada318df9f5ffeb65fa6a0ce68aa2b019746320323628be080fc142dbe255e9
forkRouteRet f4433ec078e97ab1615b273d74599b14f3458581e83e7d4dbb1f77dfc65d891e
forkRouteLen 160
carrierCall a6f4b2260cd81afd71756624c295b886d29e85c0c9e9b1cdc89f0148bbd0fb79
carrierCallLen 132
carrierStmt e5e7ef6967d544c41242fd48103d1a53c28f1d6da31a1649d34f4bd11025e123
carrierRet 3e7028adb7b81c521979a0024999718518559b1ddec5cde8e3a9379d6cef3125
publishLegacySel 5f0ed7f5
armVersionSel e3bcfcb4
abortVersionSel c4eee12d
publishLegacyMagic 4c475031
armVersionMagic 564d4131
abortVersionMagic 564d4231
 publishLegacyCall 6b6f65db1654be9c8faa63dbd6fe3c51a654ba5f03848df39962294bd0694ddd
 publishLegacyRet a546984d8fba2c96070e50dc34c550dd3689649673ccb2a6f3b3cc497bf8c314
 armVersionCall 9cbcb2ece843d71518462fef44761169a72edb47610e8518c2ed667c136c2fc8
 armVersionRet b3389d9ee6e7d3f9ffe6c66859f58543113b78b621c9abdc5f21e03cdbb7d3e9
 abortVersionCall 2a415d455bc04de9e5dddcd3369d0076ecb00cd4bbabadf29e71bea1458979c5
 abortVersionRet 649fff0dfd2aa5a43272f3c63dbb17086d8ac7084c256915ffa088834039c469
pvmRouterGas 8000000
legacyInboxCfgSel c3f909d4
legacyInboxCfg 22bc70c195669ace965e7565ae253f760d785e048965f94efa51845573b36808
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legacyInboxCfgRet 4d6b260943518f7ce988e85ddb9321b2a3111a42e9ba431f45342ff6f4f3cea7
legacyInboxCfgLen 544
legacyDeploy 98cc45a7619c75ff658f46edbe2e3c11c19e1ec424797cf04c760b9f07fd567c
legacyFence cf5cd470448a26b70eddedf8c30fa39533e66e874e2fc1aa2e310ab633c0d737
legacyDescGas 100000
legacyTimePolicy ae8419c16fd9493a76f4192c6bef1396d6237a2f2d981dc2bb55acfb3086e5d6
legacyRisc0Key 60651c05d2012e7254373988020eaf9dec6c635e84d535112e7ade81aac5584a
legacySp1Key b449ce6092b398fd4178d946077f8f19c4f2451cc88d614bb682233134162f12
legacyRisc0Desc 4805e24aa165db57d51ab99349241fd843ca03e468afb8d33781d4a08ccca55b
legacySp1Desc 523d245ce9369abd6a116dd7cc4b4a989fcfb8a9d1f8eb796e1cb35945f18bf6
legacyProofGraph 4fc1c37502713218107f6aefba90f1a5720e9ccd5a5569abb590746e85f64efa
legacyRoute 527763e6ea020b909c9a74a7aa5b7c2fdb372c5d8f283644d4d454bc63c16275
legacyProposerDesc 1bd9411f4082f5d010edfad3ba2a52ed5164a7a7ec345c1d56f11a2b3448515a
legacyPublicProving afda6179076d7f2625ac4e691fd28ff557ef3419a708f22598bc8419b6a27c05
legacyCkptRecord 1461a32136f8c498043934fce575dddec6830744145afbccde57eea1ea61c9b0
legacyCkptLayout f9e5f221ec2368348e185feb34fee0cdb27a812a2533c56fa7dfe8ccf762ffe1
legacySignalCkpt 5e10f1f156d0913967b53921f75484530facf090ceb544306381abb78894143c
legacyVerifierCfgSel 7e52cacd
legacyVerifierCfgMagic 4c525631
legacyVerifierCfgRet 76ded5a00e4182d6eb5d2d40700d85414c53a99657ebed6dc9612205c703dded
legacyVerifierCfgLen 160
legacyRisc0CfgSel e516c43e
legacyRisc0CfgMagic 4c523031
legacyRisc0CfgRet c7b1f85e99c9b82651e09ced5b86de094ca46ef38aa0f5c22d94efa1d8222906
legacyRisc0CfgLen 192
legacySp1CfgSel e2b5d958
legacySp1CfgMagic 4c535031
legacySp1CfgRet 864c870262c85ba745d71a95e0d667b76fddf17d60a5efe48512b289ae61a9b0
legacySp1CfgLen 192
legacyCkptCfgSel 68011023
legacyCkptCfgMagic 4c434b31
legacyCkptCfgRet 4ab6f89dcd323544a673c0243a0ad55779c74ae37d981d3308829092d7713534
legacyCkptCfgLen 192
legacyImplSel 8abf6077
legacyOpCountSel 7c6f3158
legacyCurOpSel 343f0a68
legacyNextOpSel 72a8a551
legacyVersionSel ffa1ad74
legacyPropImplRet 760f5b3a5acc4b21ba8112ebe315c3946dc41e502bdb323e50afeac90a42ae8b
legacySignalImplRet 8e4eff68fda06b95881cfa2df901251b566e87657e1a6ce77e69fd5546318034
legacyOpCountRet c2575a0e9e593c00f959f8c92f12db2869c3395a3b0502d05e2516446f71f85b
legacyCurOpRet 7606a1b3fbf7c1835d2aeeddf4207ed367c600f20c898605ba54d05cbbb3be39
legacyNextOpRet c7a0cb7d58c090fb9b48529de670e08db7fa5d48d0b8fb1a2c25f357ff0215ee
legacyVersionRet b10e2d527612073b26eecdfd717e6a320cf44b4afac2b0732d9fcbe2b7fa0cf6
legacyResumeProfile 6a993ebf703937ddefdd8a5ae08911f15a19286ae47b40d0b91a08824cbb2812
legacyBlob1 f0c8974a111225866a147954a2f98c29c679c1aa4680c42b15ac2eb3a1cea148
legacyBlob1Len 192
legacyBlobMax d9791c1e9f76963f86cdfe6423b8d897dcc5c0ef5ad4cc16363b2b2ab452240d
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legacyBlobMaxLen 832
legacySourceMixed 7e0699ca450524a9f618d63c9c6ac4690e44bf47ae3a24980346f4d91b3a8789
legacySourceMixedLen 288
legacyPropMixed b83ae28a39b0d773dfaccf449234cc30a737b896dc3a1a2b1d4b15bcf2ae07d4
legacyPropMixedLen 1152
legacyPropMax ac882b6809d66ab7fa52d59ef700c30ef460ae5b45db8ccdd0ea6cfdc62fb298
legacyPropMaxLen 3808
legacyForcedAbi ffad3668322a174e4a6b3180869175b6395a8657c6940353ce2cdf442fe98cd4
legacyForcedAbiLen 256
legacyMaxForced 10
legacyMaxNormalBlobs 21
legacyProofGenMax 900
legacyMaxPropBytes 3808
legacyMaxForcedBytes 256
legacyMaxScanBound 4194304
legacyMax16Raw 60928
legacyMax16CallLen 62084
legacyFullScanBytes 4161536
legacyFullScanHeadroom 32768
 legacyReview a67a67dbe8f5b3504cf4fc02b1d3478a305d14a087fdc3e1990ee37e338ea8f6
legacyBlobExpiry 1573864
legacyCampaignSel 718b2ac7
 legacyCampaignId 63f656e3634d5b498acb02d9c620360b99093bf4cb67a749a4072e2987e14d21
 legacyCampaignRet 113d1e08ee789cc650d80d5e7b60670104c20ca97d5dda155b9885defd20873f
legacyPrepSel 6880cd05
legacyPrepRet 8c869f3dfe7df0f01a65e96d7691b5b0d095590acd9f034770c33a6c75472a98
propRowsEmpty 0736a35926f1618d1fcca0fdf57530d52f3d1694eb227f31c28b158b370793d5
propRowsRoot aeacf67ac14d90928d567c659ff80316a0d780fc490ac6b4b69e9620836d2b46
forcedRowsEmpty 5631fad8f285ac1643b4e817c5454ee4f1e35174786dd255842a987b21c3fd92
forcedRecord dbda8510c73bc3eb57de26ae5c5bcc82a45bf6fedab68abf100d9baea6f4a380
forcedRowsRoot ab7f340cae89170a3a619086538cace009399987a1bffbee40c17da60c10fcc8
 legacyScanCommit 2953e2ef617a444b0161c2de908735e45dec98ca23b6394d07e2f7304fe3812b
 legacyAbandon 64ee4f382cf6779d96bb3cc7498c933f3f77c3d56bd5c6145402a419f5d3e1da
legacyAbandonTopic a3b978444273f8f347857235c224846aa45e8ce5bb73df549b9916e96371c8c9
legacyBeginSel e9d1a07f
 legacyBeginCall a7e6e3e0092a5046d16b7890f0fec8cc2ab767902c7ea589310bd6f89555b06b
legacyBeginRet add644dde24164cfb29ab6a4598cb55acb73c55dd295b0c2a227dc3264c6f65c
legacyScanPropSel 7da66460
 legacyScanProp1 02f6239a60f3e81b66d055f987febb52123f3d2aff404aec86ec21d0dd510212
 legacyScanProp16 c32f6d21687b66573e15131ed07315a1e8da72d17f74874ab61e9c86ff6ca1b7
legacyScanForcedSel 032ae99e
 legacyScanForcedCall 6af6748c2803053bd1f63dabe1e1e15dbf7a6b24a38c99918f6e0ba9aec4b09e
legacyScanForcedRet 10dcb42db706d4c4a5afe207c6327f8cecbcbc4b298c0255fe2434662e6599d6
legacyScanStateSel ef3cdce0
 legacyScanStateRet 204083d12fb27e46060855378b2282a022d7d99e15fe79c4f3ad0fb1ad02196d
legacyQuiesceSel 4c0ae8da
 legacyQuiesceCall 5c10ee87160f0b6a861d14c6dae4e912d85f027f6e2be684eb3ce19aa1339f69
 legacyQuiesceRet a43378e14b51d45554357cd4f88cba6b63d0c80a3bc713f02f581c99ab6fce91
```

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legacyResumeSel 2bf6b656
 legacyResumeCall 077167cc09de7a1815a9fb16809ca62973783e807f8ba726ec9848303f2c3ffc
 legacyResumeRet 7b78fc9f66ee28dbe333c289dd665d5445a272167de24c7580b9405980614357
 legacyArmId c986fcbbee5c1db5bb6ee5e1da22cb0d47911d1f9d86046b8334c85ee29d21aa
legacyBoundary 0215b43e3b2143b3fcac2c97bc2f8cc32ab47ac695a6fd173d919abdcc170263
 legacyLaunch cdfa857c3e0e633c7267fb55e75fc623f1ab00663352c319c9f9498da26bdf8e
 legacyPost bc95f4331d702a071b3128e76f9df510c3803d5aefa5adeb613eddc6221cde1a
legacyStateSel 9b698000
legacyArmSel 8781a058
legacyFinalSel c2de6417
 legacyStateRet 88f3d2c544effdf5d667e1c4eba8fc21c120ce5e70394aed80250aa49fe0239a
 legacyQuiescentRet 531b46cfeba52c1d7b414610616f7444327c3875996fd434765111b182f38b50
 legacyReadyRet 29e6a459688cb593a2ea84bee1fa517f36a20bd8b2ae136d4b8334def3585354
legacyStateLen 512
 legacyArmCall 371925b8292fe1d24cef06105eec2ba338d887ecc8de9737c6ca340891bd895e
legacyArmCallLen 68
 legacyFinalCall ddcee71fb07e96e5363d5c01efc76350afc5591e04d6edd73f9cfaa18114b387
legacyFinalLen 196
 legacyArmRet af4a11b19b5eb3e3979b326841ca1310d4f1ef11932a09b8cefc3109d911857c
legacyArmRetLen 128
 legacyFinalRet 2c38cd1bfa2809742c239971d9a0623c19affa4341dd68221579c24d81f346f5
arv1Sel 0a4434d0
 genReceiptCall 6b14ae2240bf97c609c7c4b52caae608ac4cb70e2c243ff083727f28da88c0a4
 genReceiptRet 061f8c9a0478d36f24ad36f016fc57d98e2f354c4120db4f9f5ea089e4f05d0c
 versionReceiptRet 06594634175e72e75e3c3d0d434e67b1b332f2dc1e2e00411e10e6e56628165a
 genCtx 8ad93ea1b308ba0c1e99acaa917befe37b2f372cedfee38c31d1dc1788a79aae
 genCtxRet 8fd31818e07215c67007d9b257d2749ad8c47189a5a79fe20c3bbc66bc633254
 genAdopt f5abfc13598584e4fae8d5bc42fa1cf7e1cc80d491717b2bfeb4de8ef7d8ef9a
 genAdoptCall 2fb507c85fc16f5b51d9804ecd625a9fe92e22c2c1484f9e2b55c12cdd922791
 genAdoptRet 10d4a9e197dd97ab9e5f01f40c733f869e815ab7eedf771b6a253e33bc2e53fc
 genQueuePost e928a4279bc6cbaf66f92127b59ebcbc9dfe239e5a21280132a4bf8d94f9f27d
 genQueueCall 143f65b5c9803b66a8ad2081f8112ee334bb881825fbdaba030bc0baed6c32e3
 genQueueRet d007b7d57cf2ea079a08806c183efce01b65bdda4e23b3fa31dd7d00767c954d
 genSrcPostRet cf7a54ae68ffa4e58038ec1127015fba51d5a97ed2c7ac8ae86f41c229b0067c
 genTgtPostRet c003db2923c3db9ac53f55e6aa28b74d01c1e74e2ebb99179f1d8c30c7c3a738
 genQPostRet 243df6ac6cb7e64ccc98cf53ef80347ae54b39094538b3792f8fa68449c60ccc
 genReceipt 10685d2bf06df00986e509986d96276f316dc7e4a8fbaa6fb85eb088724559e7
 reorgLegacyPost f9b06b8bbbe599ee8f929155b6008bf2362b42257010f4faa159a6da5ba8b744
 reorgGenCtx b41594bb91556bd09f96ec0211f3544dabf7b44786a397331de05f8537c63759
 reorgGenReceipt 6471280dc583b43061f89097c33957adc982346b03ebccc8633553b1134f8559
 actContext c14f0ba9f943005a718798dad25e4f6734432ca4b60247c78ab0928cede3b2e7
mactSel 7cf70319
 mactRet 7f939ca13a82ddcb79cd2d2daa740d672e8e212c7412855ae1e56ffb693ceebb
mactLen 320
armLifeRet 298524396c6b44dd68c8c21326e4ddb544cc51cfe8d5b89ed34bda45fa998753
mcanSel 3286443c
 mcanCall 3a568138e816eba1fa1903262e1d633c119b8349770e41514f07c398668b1969
mcanLen 580
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 mcanRet 29bbe49eb933dff169a474328388efc21a587b53b29b294fc1547134144ae2ab
 sourceFreezePost 6ff1bcb7b84ffd272868cc3f1bcaa8580d63b5846f26debb77bcb2370d4dd9de
mfrzSel 45a80913
 mfrzCall 30f0e9a8404fe31cc78997826426362b360c66df97ef2d2653133a236e866bd2
 mfrzRet 718fadb554309568b5346c194319e86b87e59b28f513bf231907b4f4666212b2
 queuePost 0df01cc54f068cfb543e177be4ff2a56c54ec9a6d8f24c6045971bf717103f6d
queueCredit 64000000000000000
qmigSel 9461f698
 qmigCall 04e3ca8a4c3b3bfe608add6435faedce8e1a312b9399f02b2bdd92dfbd491663
qmigLen 260
 qmigRet c2e7792f30055cec1dfc1280a888ebdc75a26036a8f8a4ceb66922271df8977b
 adoptPost 50e73ce0a0e4032731e98de594bf9bda1791186698546fa7e6e11cc743b2f62f
mapsSel 66e664cb
 mapsCall fcc55bc889a9b845336b77d4153fa9c2e41ef3f9ce4402e818924b6cc2668c9a
 sourceMapsRet b1468ca66bd19deb4caf9a7ec2479420d57f0719d4ea52c65262cc67ebf77315
 targetMapsRet 3a389f1049b4b00aec37a0068e4be9209485fbb69bf7f44049d8ce341f6ca411
 queueMapsRet b1129ad0e21577e428d9de12fe5ca9fb68e0aa5f6803b0180099261ee33ed2ff
activateVSel 17a548ed
genesisFixed 15236b38830f35f4df4c6bd78015c9a3b495a0f3fe08a028cdb1dd57aac970ca
versionFixed 24f54f2ca1c6731f04741371c1e2705fcd912e1b5cf2bd5d727729cc5de89a41
 genesisCall ee1babaad0f4d7c02d4c556aede841b85a13d35e2fb2d4554a47c1c50070e35d
 genesisLen 2948
 versionCall a60b0937d5a514a1bce25be2894848f0df589df5f5d3fc20d698105c4d29a9fb
 versionLen 17220
 maxGenesis e3d564883db160c03282e1c6dfe11002da217040fd5ed13cf72ca23d535f046d
 maxGenesisLen 135908
 maxVersion e9863f61a7a20480db8fbb3663623417cdae6da5850c81df3b0909319092f931
 maxVersionLen 149284
regBase 20b3dfc457e3cecf32b0c047177351f0814e426c1548e87b79f58830655810c3
regSlot dfa6283b763bbadeb604401a78e2fefeddb72000addcdb94ed2e3de5cc69846b
regTrie 200031adff46d90b1cd5c67ff8e31098235d1dddb08ec98b0d20f5f8660c0ac8
relBase a0b7a29a75032f37561036cd3741e7b375213309367f37b5ffec4ad55cf6154f
relSlot 719bb73ba856aeab1b203e322bfefc6d84a4c41a3222bcf1634b1b44e5b9aba8
relTrie dac8109059d03da2ad16ac3acc50d2e58897b8c3a7f6889ae77bfb20737e87a2
 routeConfig 9cc87d80608fb3cc866707547127aa776556ecfb64db4e398ae337620e76aac4
queueConfig 72e27c19ebfab08e1fb27feeff50609c7c4bb69f0570a7bdb72eda7736c50f4e
 dataSessCfg 14182a405d2b49b4b45bb7447ce360efecf5c9c96c3197a84ee3e88cb51cd84f
dsOpenSel 7bda4d11
dsPostSel a1cc526a
dsSealSel 340e11fb
dsMaintSel 1e7a916a
dsClaimSel fdd2b0db
dsSweepSel 9d083a2b
dsCellSel 011efada
dsByIdSel eeaad0bb
dsAcctSel e2a62969
activeSetSel 4a95c306
migReadySel b36c83ce
```

```
markReadySel e0c25827
activeSetMagic 41535231
migReadyMagic 4d525331
markReadyMagic 4d524459
dsOpenTopic a81132592bd8a549a0bfc83415ab47fbca586f0b48fff5f6cb5bfae0a9fd1f68
dsRecordTopic 30ee2de166c53a480d028e5b94d4f8759dbd84b5f7b6af1f23e0c5889ea17f8c
dsSealTopic f6d45ab0ecc3348b36ac48bc769f7391962fc2fa240c1c6bb43269ed3780078e
dsRefundTopic 086de7ad4d27cf66f63e9dc6ecb09c0c3122146dd43e7f480f3a875070eb984e
dsClaimTopic 69fe7d8d95811e3a02a4ad4b0d1a3e1360b683a31f7cb30b4c69546b258232fb
dsForfeitTopic f93f771339298d1fb502bd2c556fc18130b95d44599f35109de39d8d5731f957
dsSweepTopic 3a5f2fa0ab342d3b79fc2aa40be5e1296009e8fb31f86c3577c51839dd159a86
dsMaintTopic 920669b9670911aa86cd718dceebaa1372d224ca0fdac50a63dc1d45a53e1e89
 bridgeLeaf db1a0c935f1173a569006488882a2abeb620ee397b0d4b57ba14f479e93d4ad2
 feeLeafAlt ca3a5c71fcfd67060d91fa2dbd4f8d839d678f6fd5242cce1b7ad4fbddf6fd51
 creditId c5106651fbf5aef64142a4abdbf29dd670e2464fa1fcaee8253a23b45695835b
 feeCreditAlt 5729c4dbda07a37c0e80fe34f7c080e6f26742c2df888766e46ce8df0599d432
 escrowId 0b5dda48fd1355bc82eebd46369a48e66034fe92e15e5e49393787281782aebc
 inboxSlot afae055286a47fcc5a76c3b75bef8aea556d024f5249c28372e59672f222caf4
inboxTerms 193212327937708859556780430318453749473365 7
 terminalDone d5f92fbe75e51e2bcddee051eb01e222a101f16a9e6f9cd3a6e486e114fd5778
liquiditySet 625ff42ed879b94a7499fecb7abc07988b348300d1c4e9cc1f7596968eaf2f19
 settleLeafAlt 96baecdcde62bd5dd9d86625842471da07d3008a205ef8c79481254779e68649
 terminalFail 2cdb7e63b317779f0745088a4148bc6cdda12d8331718a6874904beb4ab15362
 terminalRoot2 8b47e5d6f541fa86cf14b6568107c0631a3df79afd50d24baeb936fdb39e687b
emptyTermRoot c5da197edc2f03c7023cc6afe137ccb77d01fc56514d322b6ba66a149315bcb0
 terminalFrontier2 582eeebaacbe2b12774eef4b3286045287fc7b5ff92dedf77b5443cfddf6723f
 terminalFrontierRoot2 8b47e5d6f541fa86cf14b6568107c0631a3df79afd50d24baeb936fdb39e687b
 bridgeResult e45204346d0dea5083fef855ff0260f7b3eb64fb02548813d09e87824f4da74f
manifestRoot 417be737a57e38eb410f2d6e65c77ee19d5c314cdaf432067861c6a36c6a990f
emptyDataBag b3caa2379816b63eebbf789e33e7d84ef29d6d350179803dd000102e8182f66a
mmrRoot3 dea2c7de9f14ce6f9c59fda963520ced2cb6e6ebef7c6abe6cb773afab888e89
dataFrontier3 7e141a72088ecc5ed590e1e619b607fb5d97aafb003c4a56230cff2832c2af01
dataFrontierRoot3 dea2c7de9f14ce6f9c59fda963520ced2cb6e6ebef7c6abe6cb773afab888e89
dataMmrCount 15
dataMmrIndex 10
dataMmrLeaf 39d13a13b5803fd94af45496a7b19e2e666957cd75780ed2508c702a703ca85b
dataMmrRoot 7c7185dfbebadab3501e051b2960e8fb4b22dd11821fc4e474ef14b2edb27b22
dataMmrSiblings 3ad2e114ce1d543a3a479e69b945dfeabdfdd0fc6df543b28c069469ea5e26fa
dataMmrPeaks d5b32274fe3e21ffa9dbaf5853b469ecaaf4ac295e788fc95c16b4220a00b196
asymDataLeaf 204849a6179174bd92e677b252370e3e2904cf4eb2a94fe1ba154e1310b9cd64
dataNodeH7 e60d61327adb017addbf3012131b62c0a5897d30e302fcfac9ffad46d0fe848a
asymManifestLeaf 92ed7b0f26f1048042ad681be312e3d79d6942ea6da03445c1cc8fd6b1eb1d6b
manifestNodeH5 9f4c64a89c253d4d65f93cba1f55e7a17a3e4027be7351650060632ed1bc70f0
normalContext 5b93691b397d8ab377682acee7cf6cc77ff2726cd83a8a7b725084f5d6f468bd
migrationData 2c36740d76ae6192335d4c603f42edace094b33e8f54e959b40241a94c1f6deb
winningData 4ae34aa9efb842528d353b175f94191d01cfd168b5ad828f64a0e7972a2ca9e3
 statementHash 14a0fba13c924b2be26b39908af2ba3cb556ab4f050398d28a9db0416fb72992
stmtRewardGas 12345678
```

```
stmtRewardBytes 9
 rewardReceipt c579a9feb5bfb979f075526c1290aa049adcb95e099c6e02726797e5ca2e201a
candidateCommitTopic 51629f6515f461b4c6f912a8eecad46d8ff89ab5e8235d95c30004be2c9ac738
rewardFundTopic d979ecc5f5821fb9e6643744111b04290fc1da57eebaba07def52e526c6eb49e
rewardClaimTopic 9f41046da255bf29062408ae9f20e06ea2453df29227caf32b709ccfed59d506
 candidateLogTopics 3bddf4ef7c357f1130e5d4dad8eb0571e0d202560c7c46574f4c88c470f00472
candidateLogTopicCount 3
 candidateLogId 14a0fba13c924b2be26b39908af2ba3cb556ab4f050398d28a9db0416fb72992
candidateLogBeneficiary 000000000000000000000000000000000000000000000000000000000000cafe
 candidateLogData 0c7970d382a72a99a7d156cc870cbf7d3728cc91da9f86422767a1420be16538
candidateLogDataLen 192
rewardFundLogTopics dec401d3769ac3d7c894189f394a778a66106bd32460f51135caea4824360efe
rewardFundLogTopicCount 3
rewardFundLogClass 0000000000000000000000000000000000000000000000000000000000000001
rewardFundLogFunder 000000000000000000000000000000000000000000000000000000000000f00d
rewardFundLogData e1aede520e7a9dc85a896fcdc9cc395ef1f2fd03d93f7e4f6acbd5fda09bf690
rewardFundLogDataLen 96
 rewardClaimLogTopics aacc32b671d1f77255783a46a8b1fcbd718c719cd826380b2e001b81ef95480e
rewardClaimLogTopicCount 3
 rewardClaimLogId 14a0fba13c924b2be26b39908af2ba3cb556ab4f050398d28a9db0416fb72992
rewardClaimLogBeneficiary 000000000000000000000000000000000000000000000000000000000000cafe
rewardClaimLogData db2be80ad7fe2991fbabee31c9917dca99f927a94e09c6a4e810cec6d4a8cbae
rewardClaimLogDataLen 64
fundRewardSel 15e08308
fundRewardCall 1086e7c6ffda5d1b4d31999795549a413160e18f638c43801cf6bd8293b430f1
fundRewardCallLen 36
fundRewardRetLen 0
claimRewardSel aa5498cb
 claimRewardCall 7574e5084380d2c9bb8b89291b5ea65d6fbd7da4802d99118c33af93c4926bf9
claimRewardCallLen 36
claimRewardPaid 1000000
claimRewardRet c1af4b94166cd32fc49b7b926cbb91ee421de2d04450e8ae57857b9b56ac7e53
claimRewardRetLen 32
rewardViewSel 3ed526c7
 rewardViewCall 6d5b9359bca2ec8f27213e903919cca58faad5a2207d8c7277b1af41a6d4fdb0
rewardViewCallLen 36
rewardViewMagic 52525631
 rewardViewHit 5df9160fcccb9142a3a383d083014765cc6cad63bb724c6807bc3dc50cd60058
rewardViewMiss 910614e273a7326ae3c583449cfdf24c64d9ef3d7d6f448b6d3042d3ba95ae80
rewardViewLen 384
 rewardCollision a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a4a492
builderCfg 5c2fe6d16935d1557c545b24c0b5982391f7a8ac702d7db3a6c55e467427ed79
rewardClassSel 3d273ee7
rewardClassCall 3717990a97396f7a0a947025434cf954d9937932f5e00fdf793c129bfbfb3006
rewardClassCallLen 36
rewardClassMagic 52435631
rewardClassRet 3d21e30cd424ac4eb52ed85673aac8960a703cce5dd930fceb901b0ce5e93490
rewardClassRetLen 224
```

```
rewardClassId 1
rewardClassGas 50000
 dsAcctEmpty 5a62fbdc4883c16fe3209e7b093da31d1656fb286f5714002e6833fbdec50c35
 dsAcctFunded 3f32ebc55d31c72b1f50c6fc26b743ad13127bc7d7324ff0a3dc165175684760
dsAcctLen 512
recoveryId fd0552b28542fa3e236c86807695f3d5a4bc0436add285daa8506d9a79511b15
bodyRoot 0f4e161a46c8b18c2a86f23a0a4e7169a838a12af8b389f65e97b547a99707e9
chunkRoot e652cb05b1f44f3c09c650870b7b9ade4132548bd0c769bdda35b5bfcac5139e
```

Release CI must reproduce every vector in Solidity, Go, Rust and the circuit.

B Normative transition ordering

Every external candidate entry point implements the following structure:

```
function armNormalContext() returns (Status) {
    if (mode == PREACTIVE) return REJECTED_PREACTIVE;
    if (sync()) return SYNCED;
    if (migrationGate.phase != ACTIVE) return MIGRATION_ARMED;
    if (mode == NORMAL_ARMED && block.number > armBlockNumber + 255) {
        armBlockNumber = block.number;
        emit NormalContextRearmed(armBlockNumber);
        return REARMED;
    }
    if (mode != NORMAL_IDLE) return IGNORED;
    armBlockNumber = block.number; // no caller-selected hash or anchor
    mode = NORMAL_ARMED;
    return ARMED;
}

function activateNormalContext(armHeaderRlp) returns (Status) {
    if (sync()) return SYNCED;
    if (migrationGate.phase != ACTIVE) return MIGRATION_ARMED;
    if (mode != NORMAL_ARMED) return IGNORED;
    age = block.number - armBlockNumber;
    require(age > 0 && age <= 255);
    armBlockHash = blockhash(armBlockNumber);
    require(keccak256(armHeaderRlp) == armBlockHash);
    freeze base, stable admission pair and parsed arm header;
    freeze minimum slot, deadline and landing horizon;
    mode = NORMAL_OPEN;
    return ACTIVATED;
}

function submit(input) returns (Status) {
    if (mode == PREACTIVE) return REJECTED_PREACTIVE;
    if (sync()) return SYNCED;     // successful transaction, never revert
    if (migrationGate.phase != ACTIVE && mode != RECOVERY)
```

```
        return MIGRATION_ARMED;
    if (mode == NORMAL_OPEN) return submitNormal(input);
    if (mode != RECOVERY) return REJECTED_NO_CONTEXT;
    return submitRecovery(input);
}

function sync() returns (bool changed) {
    if (migrationGate.phase == ARMED) {
        if (mode == RECOVERY && now > round.expiresAt) {
            cancelCurrentRecoveryWithoutRolling();
            changed = true;
        } else if (isNormalMode(mode) && forcedHeadDue &&
                    normal boundary open) {
            cancel normal window;
            changed = true;
        } else if (normal deadline matured) {
            settleOrCancelAlreadyRecordedBest();
            changed = true;
        }
        if (no boundary/recovery) {
            inspectExactly8AndConvertLiveToSameCellRefund();
            changed = true; // cursor progress counts even when all are empty
        }
        // Existing protocol-lifetime reservations are never migration work.
        if (allowReady && localLiveCount == 0 && no boundary/recovery &&
            seatMigrationLocalGeneration == migrationGate.generation &&
            exactSeatArmReceiptMatchesGate &&
            migrationRefundGeneration == migrationGate.generation &&
            migrationRefundClaimDeadline > 0 && seatClosureComplete) {
            magic = ActiveSettlementRouter.markMigrationReadyV1(
                migrationGate.generation);
            require(magic == MRDY); // Router re-STATICCALLs readiness/accounting
            changed = true;
        }
        return changed; // no context opens or rolls while armed
    }
    if (migrationGate.phase == READY) return true;
    if (mode == RECOVERY && now > round.expiresAt) {
        rollRound(); // same episode/base/cause and duty; new revision/target
        return true;
    }
    if (mode in {NORMAL_ARMED,NORMAL_OPEN} && forcedHeadDue &&
        (no deadline || normal window immature)) {
        cancel normal window;
    } else if (normal deadline matured) {
        if (best exists) {
            liveBoundaryDueAt = ForcedQueue.dueAt(best.endCursor);
            dataLive = now + REORG_MARGIN_SECONDS <= best.minDataExpiry;
```

```
        }
        if (best exists && liveBoundaryDueAt > now && dataLive)
            commit(normalBest);
        else cancel normal window;
    }
    if (isNormalMode(mode) &&
        (finalLagBreach || forcedHeadDue)) {
        openRecoveryEpisodeAndRound();
        if (finalLagBreach) run bounded duty/failover pass once;
        return true;
    }
    return changed;
}
```

Genesis and every later control-plane transition additionally follow these non-reentrant journals:

```
scheduleGenesisCampaign(campaign) {
    require(routerLifecycle == IDLE && publicGate == ACTIVE(legacy));
    require(exactDelayedReviewAndTarget(campaign));
    require(campaign.nonce == lastNonce + 1 && hardStageBounds(campaign));
    requireExactStatic(legacy.legacyGenesisPreparationV1() matches caps/profile);
    requireExactStatic(legacy.LGS1() == cleanACTIVE);
    requireExactStatic(legacy.LGSS() == cleanEMPTY);
    publishExactLGC1(campaign);                // public gate stays ACTIVE
}

scanGenesisRows(campaignId, contiguousRows) {
    require(publicGate == ACTIVE(legacy) && campaignIsLive(campaignId));
    require(block.number >= proposalCutoffBlock && beforeEitherExpiry());
    require(rowCount(contiguousRows) == min(16, end - cursor));
    legacy.scanBoundedExactRows(contiguousRows); // deterministic max progress
}

enterGenesisQuiescenceLocal(campaignId) {
    require(exactLGC1PublicGateIsActiveLegacy(campaignId));
    require(afterQuiesceNotBeforeWithMinimumWindow() && beforeEitherExpiry());
    require(completeLGSSCapsExpiryAndResumeProfile(campaignId));
    phase = QUIESCENT;                         // exact boundary; gate stays ACTIVE
}

resumeGenesisLocal(campaignId) {
    require(exactLGC1PublicGateIsActiveLegacy(campaignId));
    require(atEitherExpiry());
    require(phase == QUIESCENT ||
            (phase == ACTIVE && matchingNonemptyScanState(campaignId)));
    clearCampaignScanAndSetACTIVE();           // ordinary mutator may do same lazily
}
```

```
expireGenesisCampaignRouter(campaignId) {
    require(routerLifecycle == IDLE && publicGate == ACTIVE(legacy));
    require(atEitherExpiry() && exactCurrentCampaign(campaignId));
    tombstoneCampaign(campaignId);              // no call into legacy proxy
}

activateGenesis(proof) {
    require(routerLifecycle == IDLE && publicGate == ACTIVE(legacy));
    require(liveFinalCampaign() && legacy.phase == QUIESCENT);
    routerLifecycle = ACTIVATING;
    requireExactStatic(legacy.LGS1() matches campaignScanAndStableBoundary);
    preflightAllCampaignTargetQueueReceiptAndIndexState();
    context = computeActivationContext(block.number);
    activeActivationContextHash = context;
    requireExactStatic(profileVerifier, reconstructedStatementHash);
    requireExact(legacy.LGAR(generation,campaignId));
    requireExactStatic(legacy.LGS1() matches transactionLocalREADY);
    requireExact(legacy.LGFN(context,...));
    requireExact(target.MCAN(...));
    requireExact(queue.QMIG(context,legacyProxy,target,...));
    requireExactStatic(legacy.MAPS(context, SOURCE));
    requireExactStatic(target.MAPS(context, TARGET));
    requireExactStatic(queue.MAPS(context, QUEUE));
    recomputeAndWriteAbandonmentEventReceiptRegistrationAndIndexes();
    genesisConsumed = true;
    publicGate = ACTIVE(target);
    clearActivationContext();
    routerLifecycle = IDLE;                     // final write
}

arm(target) {
    require(routerLifecycle == IDLE && publicGate == ACTIVE);
    routerLifecycle = ARMING;                   // before first external call
    publicGate = ARMED(targetManifestHash, targetRegistrationHash);
    requireExact(source.completeArm(target));
    routerLifecycle = IDLE;                     // final write
}

markReady(generation) {
    require(routerLifecycle == IDLE && publicGate == ARMED);
    routerLifecycle = READYING;
    requireExactStatic(source readiness/accounting);
    publicGate = READY;
    routerLifecycle = IDLE;                     // final write
}

abort(target) {
    require(routerLifecycle == IDLE && publicGate in {ARMED,READY});
```

```
    routerLifecycle = ABORTING;
    requireExact(source.completeAbort(target));
    recordCanceledAndClearTarget();
    publicGate = ACTIVE(source);
    routerLifecycle = IDLE;                    // final write; no sync callback
}

activateVersion(proof) {
    require(routerLifecycle == IDLE && publicGate == READY);
    preflightAllSourceTargetQueueReceiptAndIndexState();
    requireExactStatic(profileVerifier, reconstructedStatementHash);
    context = computeActivationContext(block.number);
    routerLifecycle = ACTIVATING;
    activeActivationContextHash = context;
    requireExact(source.MFRZ(context));
    requireExact(target.MCAN(...));
    requireExact(queue.QMIG(context,...));
    requireExactStatic(source.MAPS(context, SOURCE));
    requireExactStatic(target.MAPS(context, TARGET));
    requireExactStatic(queue.MAPS(context, QUEUE));
    recomputeAndWriteRegistrationReceiptAndIndexes();
    publicGate = ACTIVE(target);
    clearActivationContext();
    routerLifecycle = IDLE;                    // final write
}
```

All callback errors, exact-decoder failures, one-less-gas boundaries and late internal collisions revert the complete outer transaction. No mutating external call is made after QMIG; only the three listed STATICCALL postreads may precede the internal suffix.

No transfer, reward calculation, unbounded loop or caller-controlled external call occurs in canonical settlement. Its sole optional reward work is the bounded best-effort receipt write above; calculation, funding and transfer occur only through later reward entry-point transactions on the same immutable Settlement.

C Rejected alternatives

1. **Heaviest fallback.** Rejected because an observer cannot prove it has seen the globally heaviest off-chain tail; transparent races can prevent a stable target. Recovery is episode-bound first-valid.
2. **Scheduled-builder-only fallback.** Rejected because the same cartel that stalls normal production also controls recovery. Tier 3 is deterministic and unsigned.
3. **Promote the pre-activation normal best.** Rejected because it was accepted under different cutoff/version semantics. Only a mature best that covers the required forced prefix closes; immature state is canceled.
4. **Atomic reward-funded commit.** Rejected because an empty seat or vault becomes a consensus deadlock. Reward materialization is a separate non-authoritative transaction.
5. **One same-transaction blob set.** Rejected because Ethereum exposes at most a handful

of blobs per transaction while a candidate spans thousands of blocks. Publisher-owned authenticated sessions separate posting from proof.

6. **One KZG opening at a prover-chosen point.** Rejected because it does not soundly tie declared bodies to the blob. The challenge commits to both blob hash and body root, and the circuit recomputes the evaluation and body root.
7. **Arithmetic snapshot slot without EIP-4788 parent semantics.** Rejected. A bounded forward carrier search authenticates the parent source or fails closed.
8. **Bond as insurance.** Rejected without an on-chain reservation ledger. The tranche is deterrence only.
9. **Block-indexed delegatecall Bridge executors.** Rejected because an executor shares a proxy's storage context and can overwrite implementation, liability or status. V2 instead uses fresh immutable non-proxy Bridge bundles; new semantics require a fresh endpoint and domain.
10. **Legacy SignalService as the V2 terminal verifier.** Rejected because its verification entry points are guardian-pausable and versioned. The V2 pin store is separate and immutable; the terminal verifier proves a stable application-level vector leaf against a proved canonical root.
11. **Implicit V2 behind the legacy selector.** Rejected because it silently changes events, signals, relayer/status APIs and EOA behavior, while a vault-wide interface probe cannot choose V1 per asset. `sendMessage` is always exact V1; launch V2 exposes one separate DIRECT-only selector and rejects every Vault caller.
12. **External checkpoint copier or L2-height-indexed publication ring.** Rejected because a caller can front-run or omit the copy, while a candidate may advance 4,096 L2 blocks and collide modulo 256 in one L1 publication. Each immutable Settlement writes its full canonical tuple internally under a checked version/sequence; no external publication call exists.
13. **Frozen EVM account/storage terminal proof.** Rejected because a future state-proof format or verifier defect can strand already-queued credits. Canonical settlement instead carries a profile-independent depth-64 terminal vector root/count, while governance remains only the pre-existing validity- verifier safety root.

D Executable consensus models

`lookahead-model.py` reports 38 assertions over legacy Keccak, EIP-4788 carrier/parent selection, immutable seal deadlines, post-seal tombstones, partial/empty registry sealing, eligibility-before-ranking, capped D'Hondt and feasible placement, including absolute clock conversion, execution-block finality depth, version-independent protocol-lifetime seed and complete-schedule vectors.

`settlement-window-model.py` reports 186 assertions over PREACTIVE/migration, finite staged legacy genesis campaigns, capped permissionless scan, resume-safe QUIESCENT expiry and proof-before-LGAR atomic launch/cutover, delayed exact-target abort for later migrations, early-return semantics, force-due precedence, renewable recovery, no-seat escape, exact slot/header time, durable-descriptor prefix predicates, immutable anchor chronology, O(1) due boundaries, late close, sealed data sessions, frozen admission pairs, bounded replacement/liability generations, tranche release units, stamped direct bridge enqueue, forced-ETH-tolerant fee custody and capacity races, persistent SYNCED refund/retry, enqueue/cancel

ordering, source-generation/domain support, immutable destination pins, processing expiry, DIRECT ETH liability, quota-independent refunds, immutable authorization versus mutable liability, fresh V2 Bridge bundles, exact legacy-selector V1 compatibility and explicit DIRECT-only V2 opt-in, immutable pin storage, finalized registrar authorization, authenticated router switching, local destination binding, frontier/event terminal proofs, pause-independent escape, L2 release sealing, replay and timing geometry. Cryptographic booleans in that model are not cryptographic evidence. The companion `test-settlement-window.py` suite runs 296 adversarial regressions, including authorization substitution, callback/OOG, competing LP, owner-mode, target/quota/pull/terminal rollback, withdrawal/recreation and cross-version preservation cases for the atomic Pool path. The same suite exercises exact RootMigrationExecutor/Factory construction, staging, role-by-role activation, confirmation and abort rollback, plus SOC1–FRS1–FVR1– SFV1 pretransition joins and dirty successful post-FVR1/FRS1 rollback across Schedule append, replacement and split.

`commitment-model.py` reports 812 golden vectors and 1610 executable assertion sites for exact EIP-712 domain/struct/digest, canonical and statement hashes, registry/admission/entry/tranche commitments, forced descriptors, vector/range proofs, session MMR and per-block manifests. It also covers source/destination domains, acyclic Bridge-kernel/source-bundle, complete destination-Bridge and ten-component infrastructure descriptors, bounded chain IDs, acyclic profile-to-manifest dependencies and authenticated release manifests, the acyclic migration/registration-verifier configuration/descriptor hashes, exact configuration-getter selectors, MessageV1 and ContextV2, ingress and Store/Bridge/Pool/accumulator/policy interfaces, the static migration public input and five-argument L1 activation ABI, target-deployment registration, MACT/MFRZ/MCAN/QMIG/MAPS codecs and commitments, both genesis and later-version callback journals, and the legacy deployment/arm/boundary/launch/freeze codecs, terminal frontier/root vectors, historical 64-sibling verification and complete credit leaves, credit/escrow/inbox IDs, terminal leaves/vector roots and atomic ordered batches, dispositions, recovery ID, Fiat–Shamir input, body chunks and blob codec. The valid KZG-opening, signature-recovery, independent-implementation and proof gates remain mandatory.

`seat-market-model.py` and its companion `test-seat-market.py` suite run 114 adversarial regressions over the four-cell perpetual reverse auction, offer ranking and maturity, staged handover, premium-reserve custody, pull credits, bond terminalization and release rotation and strict MWV1/SMI1/SLV1/SIR1/SMR1/MEC1/MHS1/MRO1 codecs. The model is the executable state-machine and byte-level reference for Market custody and its Settlement/Router boundaries.

`economic-profile-model.py` and its companion `test-economic-profile.py` suite run 38 schema and checked-arithmetic regressions over the versioned economic profile, published parameter relations, overflow boundaries and rejection of incomplete or uncalibrated profiles. Passing those tests demonstrates internal consistency only; the measured production constants remain release artifacts subject to the gates in §14.

E Security invariants checklist

1. Canonical state changes only after one valid proof and only from its stored base tuple.
2. A sync transition cannot be rolled back by validation of the same call's candidate.
3. No payment, seat or standby is required for canonical commit.
4. Recovery candidates bind one live episode/revision, the current base tuple, one stored prior-block anchor at required depth and one frozen queue target. Expired rounds are permissionlessly renew-

able without creating a second duty, fault disposition or seat penalty.

5. A due forced head advances in the next canonical block; appends omitted by a frozen recovery round cannot become due until that round expires. Prefix count, bytes and accounted gas are independently bounded.
6. Tier 3 has no builder signature, discretionary transaction, arbitrary coinbase or blob body.
7. Every block header timestamp equals genesis plus its signed slot. Every tier-1/2 block has a valid scheduled signature under the frozen admission version/root pair, strict slot progress and the candidate's one authenticated anchor no later than its L2 time.
8. Every discretionary body byte is covered exactly once by a unique, unexpired authenticated data record.
9. Schedule inputs share one authenticated execution payload and use exact Keccak encodings.
10. A healthy displaced generation remains non-tombstoned in liability so an already-sealed ticket survives the replacement boundary; removal from the active table prevents new reservations and snapshots. A proven slash tombstones either an active or liability generation at the current L2 slot. A tombstoned generation cannot re-enter while retained; safe later address reuse receives a fresh registration index only after every evidence deadline. Each window has an independent slashable tranche and finite release state.
11. Candidate, window, data-session, Merkle range proof, queue count/bytes/gas, history and storage-retention work is bounded by consensus constants and safe eviction.
12. L1 reorg replay removes canonical state, penalties and credits from orphaned transactions together.
13. An armed later migration shares one generation gate across ingress, Settlement and transient ledgers, opens no new context, ends the current boundary without recovery renewal and reaches READY after at most 128 bounded cleanup calls. While old authority remains intact, a target proof executes the exact release activation from that frozen base; cutover verifies it before writing ACTIVATING/context, then executes source MFRZ, target MCAN, kind-0 binding, SourceBridge activation, source-index install, BRC1 consumption, destination seal, Bridge binding, Queue QMIG, all three MAPS joins, receipt/index/registration writes and the final READY-to-ACTIVE switch in one rollback domain. A fault also restores every prepared adapter's prior seal, the target SourceBridge and both ingress/profile maps. Context clears before Router IDLE; the exact terminal VMC1 call is the sole post-QMIG mutating external call and its canonical post-read restores PVM IDLE. No other mutating external call follows QMIG. The immutable LIVE lease's permissionless hard-expiry abort runs under ABORTING and restores old ACTIVE operation from ARMED or READY if no cutover occurred, with no post-abort sync callback and no governance dependency. The immutable non-proxy ForcedQueue, registry, schedule oracle and L1 migration gate are shared by object identity across both Settlements; incumbent reservations remain RESERVED throughout. At launch, V2 L2 entry points remain disabled under legacy rules; the first custom Anchor transaction is proved before cutover and atomically registers and seals the fresh destination bundle and route in the adopted L2 poststate. The old legacy proxy may participate only through the exact final compatibility implementation. A delayed finite campaign stages forced/proposal cutoffs, capped exact scans and a resume-profile-checked QUIESCENT proof base. Bad targets require expiry and a higher-nonce delayed campaign; proof verification, LGAR, LGFN and all adoption/publication writes share one atomic transaction. If no proof lands, hard block/time expiry permissionlessly restores ACTIVE. The fresh Queue initially names that proxy as authority, and FROZEN is permanent. If that implementation cannot be installed, safe in-place genesis migration is not claimed. Source V2 instead waits for the pre-activation 214-block `ARM_READY` package review and the exact activation proof; successful cutover consumes both in one journal.
14. A kind-1 credit is enqueued only by direct same-L1 comparison with an append-only CreditRecord, immutable source generation, source-approved destination domain and live liability at the immutable source Bridge endpoint. Enqueue and queue deposit are atomic; a reorg removes both observation and append. Launch V2 admits DIRECT ETH only; the retained legacy refund-mode, refund-vault and capsule fields are fixed to their canonical DIRECT zero values and cannot select another path. Missing Message bytes cause bounded source cancellation before enqueue or

destination expiry after the immutable pin. The exact destination domain, Bridge and terminal accumulator leaf bind recall; the complete canonical append-event history reconstructs every old proof and is retained by at least two independently operated, replaceable archives; every destination action checks its local domain and `address(this)`, and no L2 nonempty source-proof path exists. Terminal proof and reserved refund paths have no transitive guardian-pause or ordinary-quota dependency.

15. Each V2 credit keeps `value+fee+liquidityFee` as exact source liability until a pull succeeds. A depositor-owned ticket is debited only inside the external only-self attempt that also commits DONE; Pool balance covers aggregate available ticket funds. Every target/tail failure restores the debit, while FAILED/expiry never touch the Pool and DONE commits the actual Pool-returned settlement tuple. Retired Bridge reclaim requires pull and all pin/terminal gates to be zero and moves only unaccounted surplus.
16. A migration proof adopts exactly one replayable activation block with zero discretionary body. Its system transactions are reconstructed from activation calldata and every included kind-0 transaction from canonical L1 admission calldata; the proof binds complete transaction/receipt roots, header, block hash and output core. Missing preimages cause proof failure/lease abort, not adoption of a private poststate.